

Physiology AIIMS 2017

Question 1:

Given below is the demonstration of tetanization in the gastrocnemius muscle of a frog. Identify the tetanising frequency.



- a) 25-20 Hz
- b) 30-35 Hz
- c) 15-20 Hz
- d) 10-15 Hz

Question 2:

The oxygen carrying capacity of a 18 year old boy with haemoglobin of 14g/dl is

- a) 22
- b) 16
- c) 14
- d) 18

Question 3:

A 30-year-old patient with respiratory volume of 500 mL. The intrapleural pressure measured -4 before inspiration and -8 at the end of inspiration. The pulmonary compliance of the patient is

- a) 125 ml/cm H₂O
- b) 150 ml/cm H₂O
- c) 50 ml/cm H₂O
- d) 62.5 ml/cm H₂O

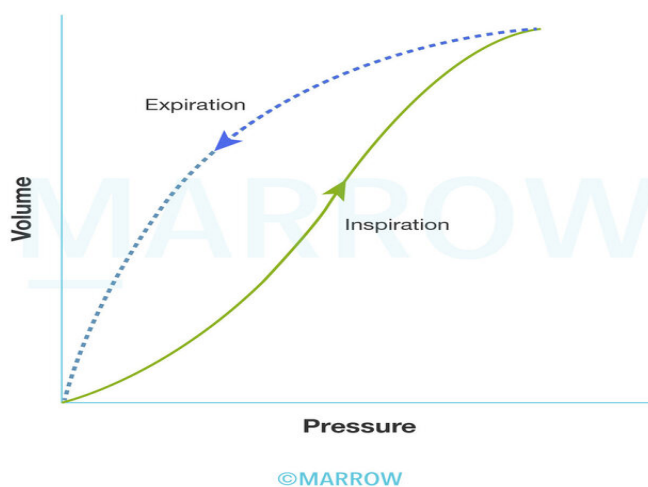
Question 4:

The function of the Golgi tendon organ is to _____

- a) Sense changes in muscle direction
- b) Sense changes in muscle tension
- c) Sense changes in position of the muscle
- d) Sense changes in length of the muscle

Question 5:

Difference in trajectory between inspiratory loop and the expiratory loop in the curve is due to:



- a) Difference in alveolar pressure during inspiration and expiration
- b) Difference in conc. of surfactant during inspiration and expiration

- c) Difference in airway resistance during inspiration and expiration
- d) Inspiration is active and expiration is passive

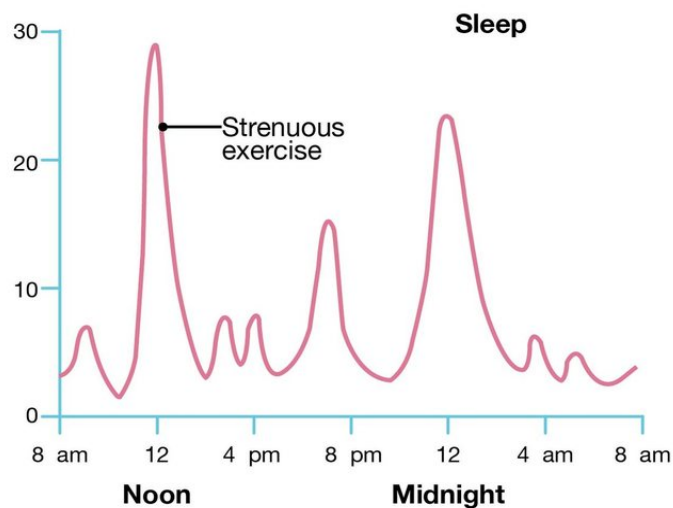
Question 6:

Reflex responsible for tachycardia during right atrial distension is:

- a) Bezold-Jarisch reflex
- b) Bainbridge reflex
- c) Cushing reflex
- d) J reflex

Question 7:

Identify the hormone from the picture



- a) Growth hormone
- b) Cortisol
- c) Estrogen
- d) Insulin

Question 8:

Feed forward control system is employed during the regulation of

- a) Blood volume
- b) pH
- c) Temperature
- d) Blood pressure

Question 9:

Cellular junctions present in a cardiac muscle are all except:

- a) Zonula occludens
- b) Fascia adherens
- c) Gap junctions
- d) Macula adherens

Question 10:

Efferent arteriolar constriction causes all except:

- a) Decrease in GFR
- b) Decreases renal blood flow
- c) Decreases oncotic pressure in glomerular capillaries
- d) Increases hydrostatic pressure in glomerular capillaries

Question 11:

Absolute refractory period is due to:

- a) Opening of calcium channels
- b) Closure of potassium channels
- c) Closure of active gates of sodium channel
- d) Closure of inactive gates of sodium channel

Question 12:

X, Y, Z are three permeable ions. $X = -50$ and $Y = -30$. At RMP, if there is no net electrogenic transfer, what is the value of Z?

- a) +20
- b) -20
- c) +80
- d) -80

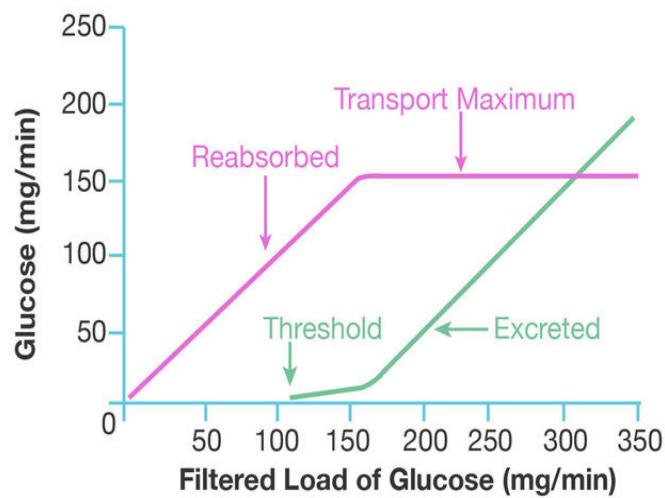
Question 13:

Slow waves are generated by:

- a) Myenteric neurons
- b) Smooth muscle cells
- c) Interstitial cells of Cajal
- d) Parasympathetic neurons

Question 14:

A patient has a blood glucose level of 200 mg/dL and GFR of 90 ml/min. What is the amount of glucose excreted if the transport maximum of the patient is as shown below?



- a) 80 mg/min
- b) 50 mg/min
- c) 40 mg/min
- d) 30 mg/min

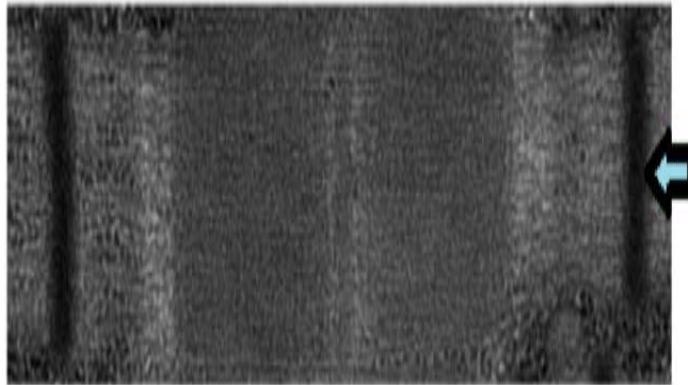
Question 15:

Which of the following can be responsible for Bezold-Jarisch reflex?

- a) Serotonin
- b) Histamine
- c) Prostaglandin
- d) Angiotensin

Question 16:

Which of the following is the constituent of the marked area in the electron microscopic picture of muscle?



- a) Alpha actinin
- b) Actin
- c) Titin
- d) Tropomyosin

Question 17:

The fourth heart sound is caused by:

- a) Closure of aortic and pulmonary valves
- b) Vibrations in the ventricular wall during systole

- c) Ventricular filling
- d) Closure of mitral and tricuspid valves

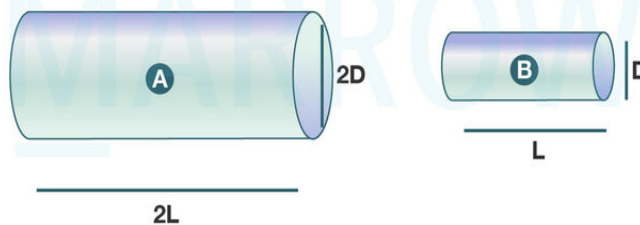
Question 18:

“a” wave on JVP is for

- a) Atrial contraction
- b) Atrial filling
- c) Ventricular relaxation
- d) Ventricular filling

Question 19:

In the 2 vessels shown below, assume the pressure along both the vessels is the same and both follow linear flow pattern. What will be the amount of blood flow in A compared to B



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- a) 4 times
- b) 32 times
- c) 8 times
- d) 16 times

Question 20:

A 13-year-old boy presents with involuntary dancing movements of his right hand. He was diagnosed with rheumatic fever. What constitutes the ganglia involved in the presenting complication?

- a) Caudate nucleus and globus pallidus
- b) Globus pallidus and putamen
- c) Globus pallidus and lentiform nucleus
- d) Caudate nucleus and putamen

Answer Key

Question No.	Correct Option
1	b
2	d
3	a
4	b
5	b
6	b
7	a
8	c
9	a
10	c
11	d
12	c
13	c
14	d
15	a
16	a
17	c
18	a
19	c
20	d

Detailed Explanations

Solution to Question 1:

Tetanizing frequency demonstrated by the muscle is 33Hz. In the above graph, we know,

- The speed of the paper is 1000mm/sec.
- The y-axis represents millivolts, the x-axis represents time.
- The scale on the x-axis is 10mm.
- The area under the graph represents the muscle contraction.

Contraction period (latent period to maximum force of contraction) includes 3 scales = 30 mm.

As paper speed is 1000mm/sec,

Contraction period = 30mm = $(30/1000) = 30$ msec.

Tetanizing frequency = $1/\text{contraction period} = 1000/30 = 33\text{Hz}$.

Tetanzation

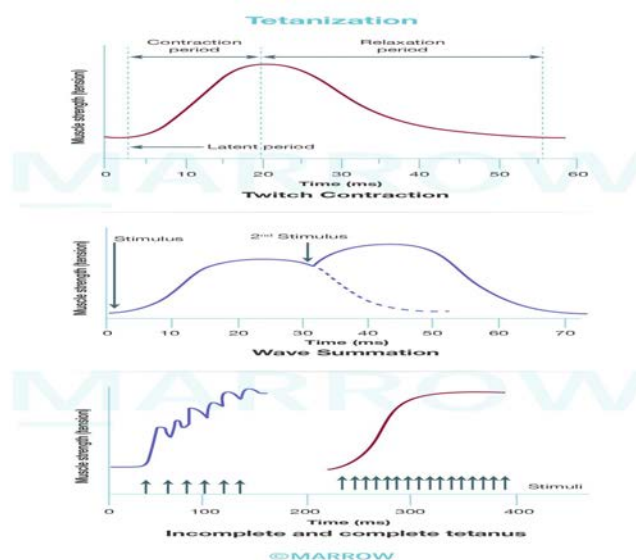
The features of muscle contraction can be demonstrated by eliciting single muscle twitches. The adding together of individual twitch contractions to increase the intensity of overall muscle contraction is Summation. Summation occurs in two ways:

- by increasing the number of motor units contracting simultaneously, which is multiple fiber summations.
- by increasing the frequency of contraction, which is called frequency summation and can lead to tetanization.

When the frequency of a contraction reaches a critical level, the successive contractions eventually fuse together, and the whole muscle contraction appears to be completely smooth and continuous, this is called tetanization.

Tetanization occurs in two ways:

- Complete tetanus occurs when no relaxation occurs between stimuli.
- Incomplete tetanus when periods of incomplete relaxation take place between the summated stimuli.



Solution to Question 2:

The oxygen carrying capacity of a boy with haemoglobin 14g/dl is 18.

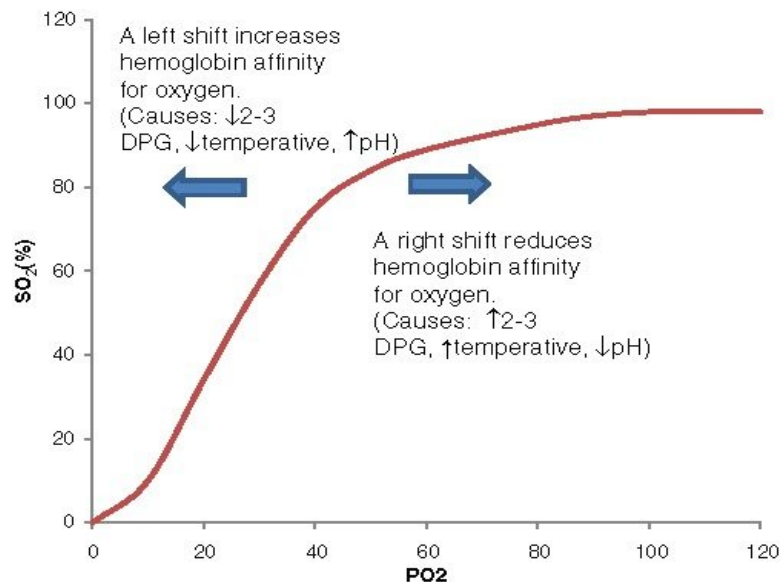
When blood is saturated with 100% O₂, the haemoglobin is also 100% saturated and each gram of normal haemoglobin contains 1.39 mL of O₂.

However, blood normally contains some inactive haemoglobin derivatives, and the measured value in vivo is thus slightly lower hence, each gram of haemoglobin in vivo contains 1.31 mL of O₂ (Hufner's constant).

According to the above question, when Hb=14g/dl, O₂= 1.31*14= 18.34

The oxygen-hemoglobin dissociation curve

- Demonstrates a progressive increase in the percentage of haemoglobin bound with oxygen as blood Po₂ increases, which is called the percent saturation of haemoglobin.
- Blood leaving the lungs and entering the systemic arteries usually has a Po₂ of about 95 mm Hg, the usual oxygen saturation of systemic arterial blood averages 97 per cent.
- Conversely, in normal venous blood returning from the peripheral tissues, the Po₂ is about 40 mm Hg, and the saturation of haemoglobin averages 75 per cent.



Solution to Question 3:

Pulmonary compliance of this patient is 125 mL/cm of H₂O.

Compliance is developed due to the tendency for tissue to resume its original position after an applied force has been removed.

After an expiration during quiet breathing, the lungs have a tendency to collapse and the chest wall has a tendency to expand. The interaction between the recoil of the lungs and recoil of the chest can be measured through a spirometer and a curve of airway pressure plotted against volume is the pressure-volume curve of the total respiratory system (Ptr).

Compliance of the lung and chest wall is measured as the slope of the Ptr curve, or, as a change in lung volume per unit change in airway pressure ($\Delta V/\Delta P$).

Compliance = Change in lung volume / Change in pressure

Compliance = $500/4$

Compliance = 125 ml/cm H₂O

Solution to Question 4:

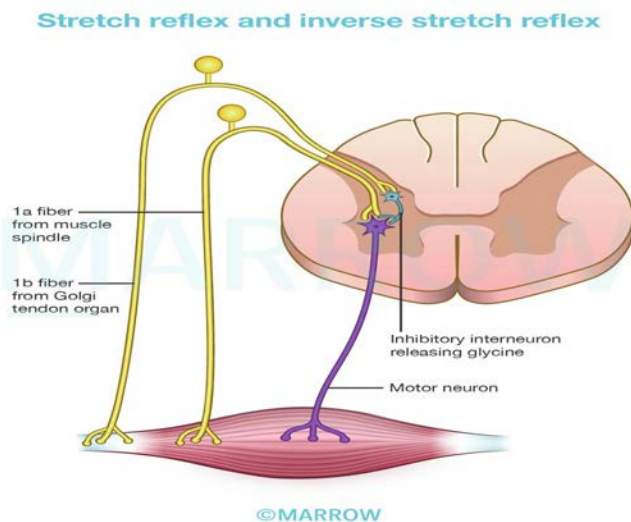
Golgi tendon organ sense changes in muscle tension.

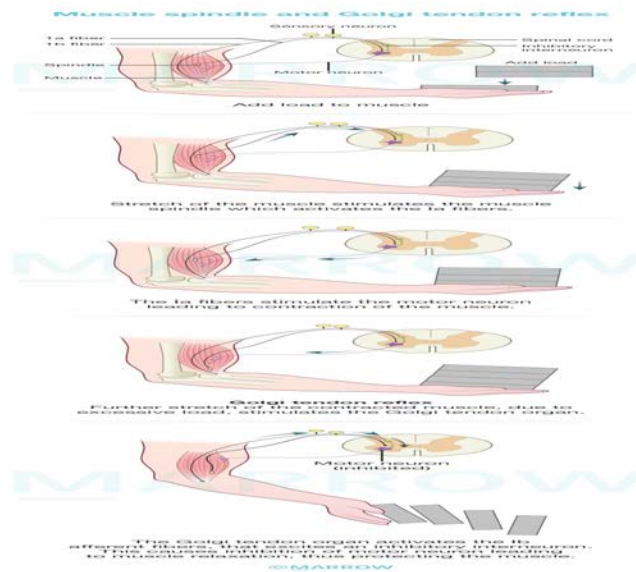
Golgi tendon reflex (Inverse myotatic/ Inverse stretch reflex) is disynaptic, is the opposite of the stretch reflex.

When the tension in a contracted muscle is significantly elevated, the inhibitory effect from the Golgi tendon organ leads to instantaneous relaxation of the entire muscle. This effect is called the lengthening reaction.

It is a protective mechanism to prevent tearing of the muscle or avulsion of the tendon from its attachments to the bone.

Note: Muscle spindles sense changes in muscle length (stretch).





Solution to Question 5:

The difference between the inflation and the deflation paths is called hysteresis.

It exists because a greater transpulmonary pressure is required to open a previously closed airway, due to a deficit of surfactant at the air-water interface, than to keep an open airway from closing, as then it has abundant surfactant.

Solution to Question 6:

Reflex responsible for tachycardia during right atrial distension is Bainbridge Reflex.

The stretch receptors of the atria that elicit the Bainbridge reflex transmit their afferent signals through the vagus nerves to the medulla of the brain. Then efferent signals are transmitted back through vagal and sympathetic nerves to increase heart rate and strength of heart contraction. Thus, this reflex helps prevent damming of blood in the veins, atria, and pulmonary circulation.

Other options:

Option A: Bezold–Jarisch Reflex: Activation of chemosensitive vagal C fibers in the cardiopulmonary region causes profound bradycardia, hypotension and a brief period of apnoea followed by rapid shallow breathing.

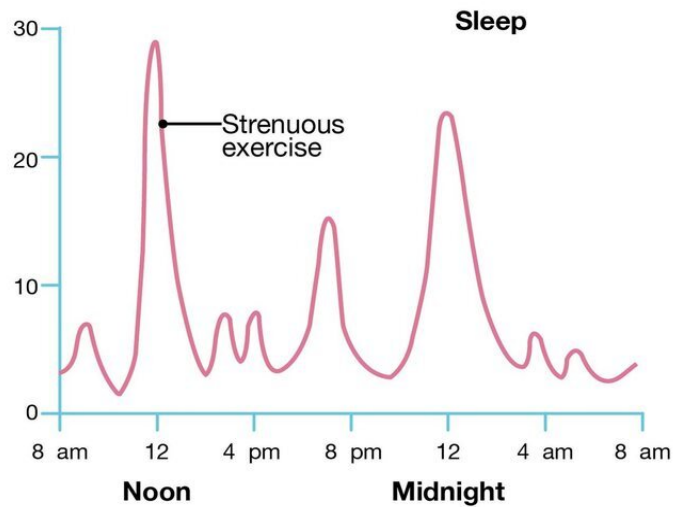
Option C: Cushing Reflex: When intracranial pressure is increased, the blood supply to Rostral Vento-Lateral Medulla (RVLM) neurons is compromised. The local hypoxia and hypercapnia cause an increase in discharge from the neurons, producing hypertension, bradycardia, and irregular respiration.

Option D: J Reflex: The stimulation of the J-receptors (C receptors/juxta-pulmonary capillary receptors) causes apnea followed by rapid breathing, bradycardia, and hypotension.

Solution to Question 7:

The image depicts the fluctuations of secretion of growth hormone throughout the day.

- Growth hormone is found at relatively low levels during the day unless specific triggers such as strenuous exercise are present.
- During sleep, on the other hand, large pulsatile bursts of growth hormone secretion occur.



Stimuli that Affect Growth Hormone Secretion in Humans

Note:

The early morning surge of Growth Hormone during sleep, as seen in the image, plays a role in the Dawn Phenomenon

Stimuli that Increase Secretion	Stimuli that Decrease Secretion
Exercise Going to sleep Fasting/Hypoglycemia Stressful stimuli (including various psychological stresses) High Protein Meal Glucagon Estrogens and androgens Pyrogen	Glucose Cortisol FFA Medroxy progesterone Growth hormone and IGF-I

Solution to Question 8:

Feed forward control system is employed during regulation of temperature.

Feed forward control

There is a control system in our body when no stimulus is required but still the system anticipates and makes corrective changes.

Such a system is called Feed forward or Anticipatory or Adaptive Control.

Examples include

- Temperature control – Thermoregulatory responses are initiated by hypothalamus BEFORE the changes in environmental temperature have succeeded in changing the body's core body temperature
- Cephalic phase of gastric secretion- Just thinking about food increases gastric acid production
- Thinking about exercise itself increases heart rate & respiratory rate
- Role of cerebellum in motor coordination

Solution to Question 9:

Zonula occludens is the other name for the tight junction. It is absent in a cardiac muscle.

Cardiac muscle cells are connected by Fascia adherens (a structural analog of zonula adherens), Gap junctions and Macula adherens.

Cardiac muscle fibers are made up of many individual cells connected in series and parallel to one another. The areas where the end of one muscle fiber abuts on another are called intercalated disks. At each intercalated disc, the plasma membrane fuses with one another forming gap junction which provides low resistance bridges for the spread of excitation from one fiber to another permitting the cardiac muscle to function as a syncytium.

Solution to Question 10:

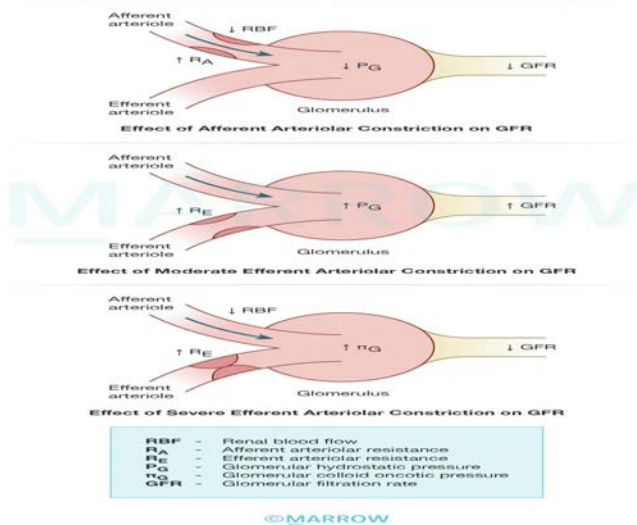
Efferent arteriolar constriction increases oncotic pressure in glomerular capillaries.

Efferent arteriolar constriction has a biphasic response on GFR depending on the severity of resistance.

Moderate efferent arteriolar constriction causes increased glomerular hydrostatic pressure and slightly increased GFR.

However, in cases of severe efferent arteriolar constriction (more than threefold increase in efferent arteriolar resistance), the renal blood flow is significantly reduced. This results in increased plasma protein concentration, thereby increasing the glomerular colloid oncotic pressure. As the increase in colloid oncotic pressure exceeds the increase in hydrostatic pressure, the net force for filtration decreases, resulting in decreased GFR.

Effects of Arteriolar Constriction on GFR



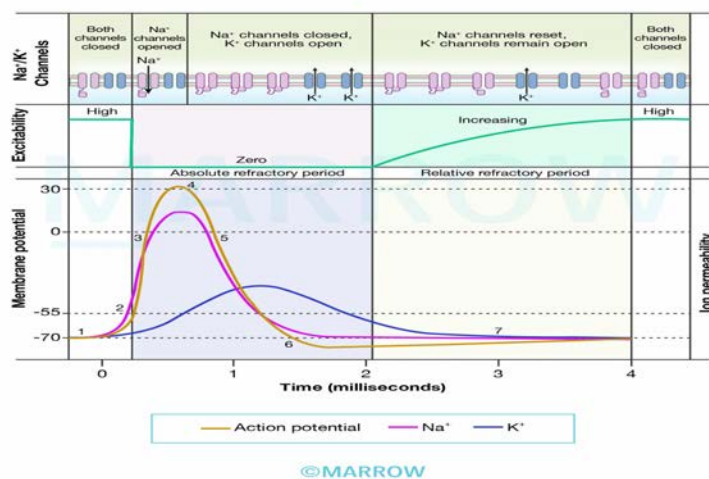
Solution to Question 11:

Absolute refractory period is due to closure of inactive gates of sodium channel.

Voltage-gated sodium channels have 3 states that you can find them in. They can either be closed, open, or inactive.

When the stimulus causes activation of the open gates, sodium rushes into the cell causing the membrane potential to become more positive. After 0.5 to 1 millisecond, the inactive gate automatically closes and no stimulus, no matter how strong, can excite the nerve. This is called the absolute refractory period.

Changes in membrane potential during an action potential



Solution to Question 12:

No net electrogenic transfer at RMP means $X + Y + Z = 0$

Since $X = -50$ and $Y = -30$, then Z must be $+80$ since $(-50 - 30 + 80 = 0)$

Solution to Question 13:

The interstitial cells of Cajal undergo cyclic changes in membrane potential due to unique ion channels that periodically open and produce inward (pacemaker) currents that generate slow-wave activity.

Interstitial cells of Cajal are the pacemaker cells of the GIT that initiate the basic electrical rhythm (BER). These are stellate mesenchymal pacemaker cells with smooth muscle-like features.

In the stomach and the small intestine, these cells are located in the circular smooth muscle layer near the myenteric plexus. In the colon, they are at the submucosal border of the circular muscle layer.

The slow waves usually do not by themselves cause muscle contraction in most parts of the gastrointestinal tract, except perhaps in the stomach. Instead, they mainly excite the appearance of intermittent spike potentials, and the spike potentials in turn actually excite the muscle contraction.

Note: Gastrointestinal stromal tumors arise from the interstitial cells of Cajal.

Solution to Question 14:

The amount of glucose excreted in the above scenario is 30 mg/min.

Calculation:

Step 1: Calculate total filtered glucose:

In this question,

- $GFR = 90 \text{ ml/min}$
- Plasma glucose concentration = $200 \text{ mg/dL} = 2 \text{ mg/ml}$ ($1 \text{ dL} = 100 \text{ ml}$).
- Total filtered glucose = $GFR \times \text{Plasma glucose concentration}$.

$$= 90 \text{ ml/min} \times 2 \text{ mg/ml}$$

$$= 180 \text{ mg/min.}$$

Step 2: Check transport maximum:

In the above question transport maximum is 150 mg/min .

Step 3: Calculate the excretion:

Excretion = Total filtered glucose - Total reabsorbed glucose (Transport Maximum)

$$= 180 - 150 \text{ mg/min}$$

= 30 mg/min

Solution to Question 15:

Serotonin can be responsible for Bezold–Jarisch reflex.

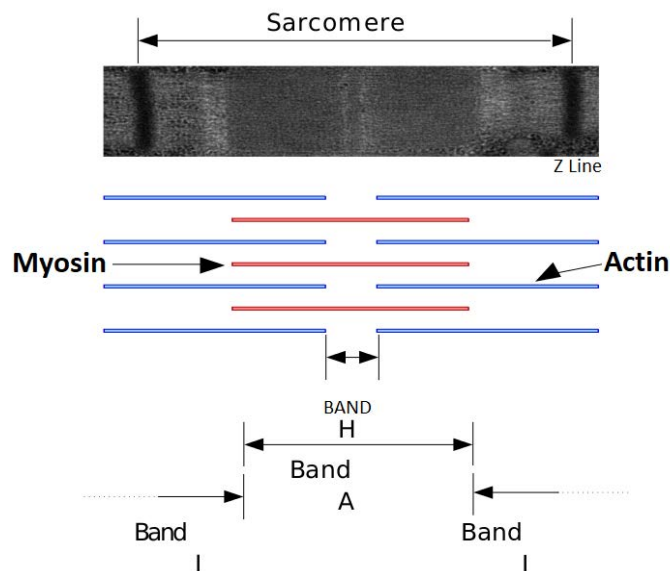
Activation of chemosensitive vagal C fibers located in the cardiopulmonary region (ex: ventricles, atria, great veins, pulmonary artery, juxtacapillary region of alveoli) causes profound bradycardia, hypotension, and apnea followed by rapid shallow breathing. This response is called Bezold-Jarisch reflex.

Other substances that can cause Bezold–Jarisch reflex are capsaicin, phenyl biguanide, and veratridine.

Solution to Question 16:

Alpha actinin is the constituent of the marked area.

The image depicts sarcomere and the arrow points towards Z-line. Alpha actinin binds actin to Z-lines.



The orderly arrangement of actin, myosin, and related proteins that produces this pattern. The area between two Z-lines is called sarcomere.

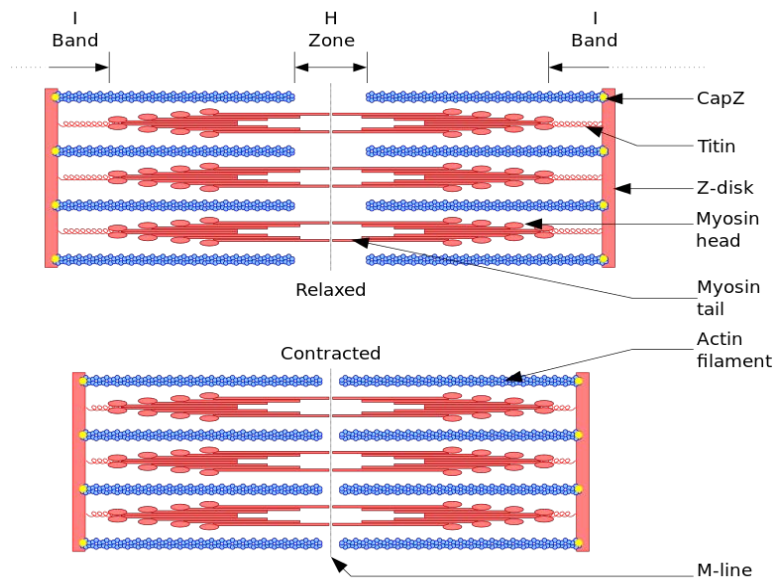
The thick filaments, which are about twice the diameter of the thin filaments, are made up of myosin, which are lined to form A bands; the thin filaments are made up of actin, tropomyosin, and troponin. The array of thin filaments extends out of the A band and into the less dense staining I bands.

The lighter H bands in the center of the A bands are the regions where, when the muscle is relaxed, the thin filaments do not overlap the thick filaments. The Z lines allow for anchoring of

the thin filaments.

Actinin, titin and desmin are additional structural proteins. Titin connects Z-lines to the M line and provides the scaffolding to the sarcomere. Desmin adds structure to Z-lines by connecting Z-line to plasma membrane.

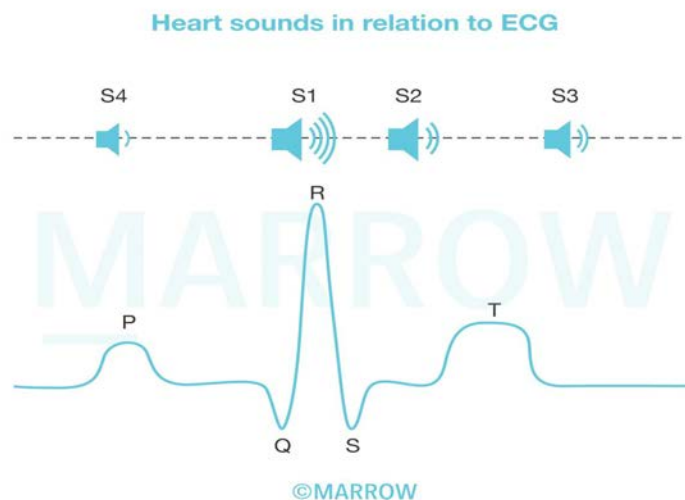
Note: Some sources say that even titin is a component of Z-disc so if alpha actinin is not given in the options titin is the second best answer.



Solution to Question 17:

The fourth heart sound is caused by ventricular filling.

A fourth sound is heard immediately before the first sound when atrial pressure is high or the ventricle is stiff in conditions such as ventricular hypertrophy. It is due to ventricular filling and is rarely heard in normal adults.

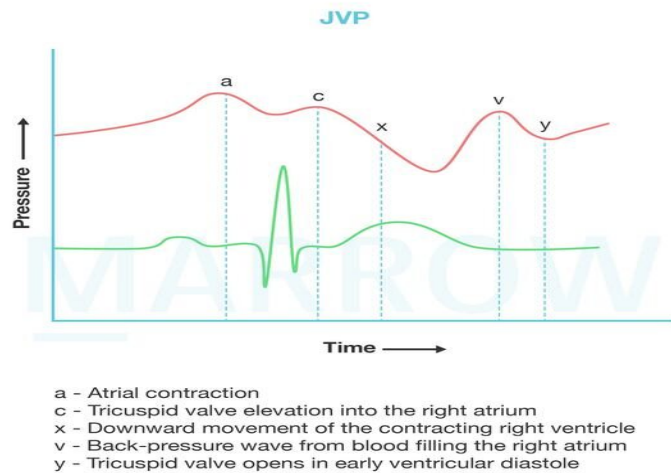


Solution to Question 18:

"a" wave in JVP represents atrial contraction.

Waves of JVP :

- a- atrial systole pushing the blood to the ventricles.
- c- isovolumetric contraction of ventricles causing the tricuspid valve to bulge.
- x'- fall in atrial pressure during ventricular systole due to pull of valve leaflets.
- v- venous filling which creates an increase in atrial pressure.
- y- passive filling of ventricles by a passive flow of the blood from the right atrium.



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Solution to Question 19:

The relationship between the flow in a long narrow tube, the viscosity of the fluid, and the radius of the tube is expressed mathematically in the formula called as Poiseuille's law :

Poiseuille's law

$$Q = \frac{\Delta P \pi r^4}{8 \eta l}$$

Q - Volume flux	r - Pipe or vessel radius
ΔP - Change in pressure	η - Viscosity
	l - Pipe or vessel length

Blood flow (F) inside the vessel depends on:

- Pressure difference: directly proportional
- Viscosity: inversely proportional
- Radius: directly proportional to the fourth power
- Length of the vessel: inversely proportional

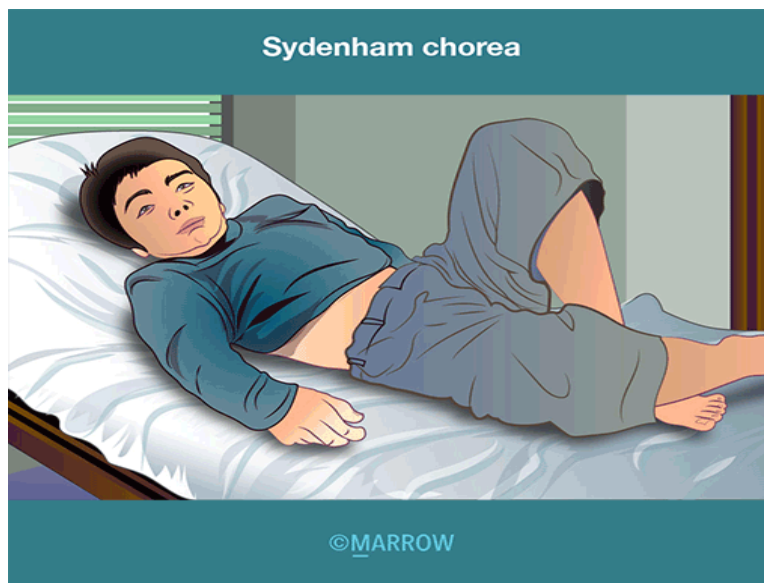
Here, the radius is doubled so the flow increases by 2 to the power of 4 = 16.

the length of the vessel is also doubled, so the flow decreases by half ($\div 2$) = 8

Solution to Question 20:

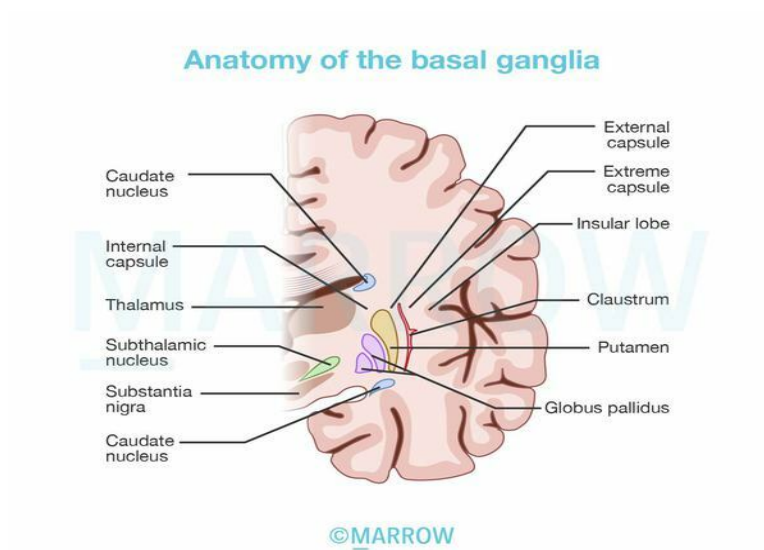
The given clinical scenario is suggestive of Sydenham's chorea, a complication of rheumatic fever. The choreic movements are caused by the destruction of D2 neurons of the striatum. The neostriatum consists of the caudate nucleus and putamen.

The image given below shows Sydenham's chorea in a patient.

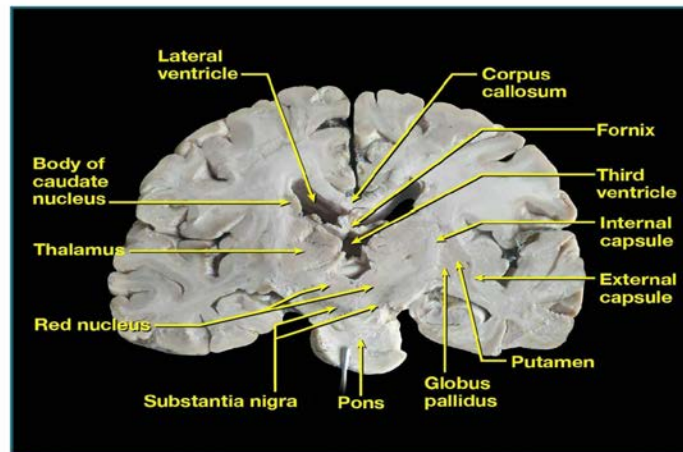


Nomenclature of basal ganglia:

- Corpus striatum: Caudate nucleus + putamen + globus pallidus
- Lentiform nucleus: Putamen + globus pallidus
- Neostriatum: Caudate nucleus + putamen



Coronal section of human brain



Physiology AIIMS 2018

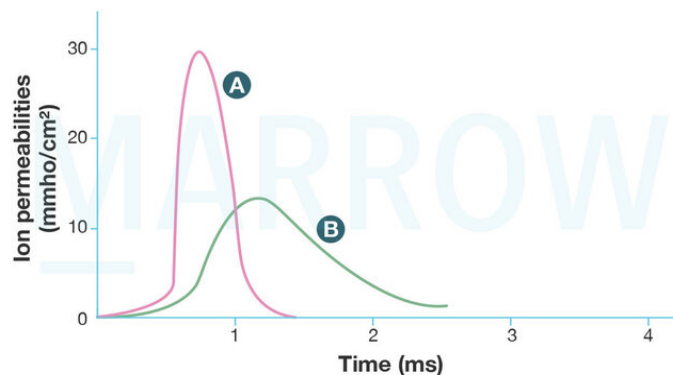
Question 1:

Which of the following is the diluting segment of the kidney?

- a) PCT
- b) Collecting duct
- c) Ascending thick loop of Henle
- d) Descending loop of Henle

Question 2:

Identify the ions A and B based on conductance graph of a neuron shown below:



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- a) Na, K
- b) K, Na
- c) Na, Ca
- d) Ca, Na

Question 3:

How is an electrical synapse different from a chemical synapse?

- a) No cytoplasmic continuity between presynaptic and postsynaptic membrane
- b) Direction of impulse is unidirectional
- c) Connexons are involved in electrical synapse
- d) Electrical synapse has more latent period than chemical synapse

Question 4:

Hormone degradation as an important mechanism for regulating local concentrations is seen in _____.

- a) 11β -hydroxysteroid dehydrogenase inactivation of glucocorticoids
- b) Thyroid hormone deiodinases inactivating T₃
- c) Degradation of retinoic acid by Cyp26b1 in primordial germ cells of males
- d) All of the above

Question 5:

The factors that can cause a person to drink water due to stimulation of thirst mechanism is/are _____.

- a) Dehydration
- b) Hemorrhage
- c) Prandial Drinking
- d) All of the above

Question 6:

On performing a histamine test on an individual with brachial plexus trunk injury, you can expect _____.

- a) Central capillary dilatation → wheal → Surrounding flare
- b) Central capillary dilatation → wheal without surrounding flare
- c) Central capillary dilatation → wheal with exaggerated surrounding flare
- d) Absent triple response

Question 7:

GFR is increased by all except

- a) Increased renal blood flow
- b) Efferent arteriolar constriction
- c) Ureteric stone
- d) Decreased oncotic pressure

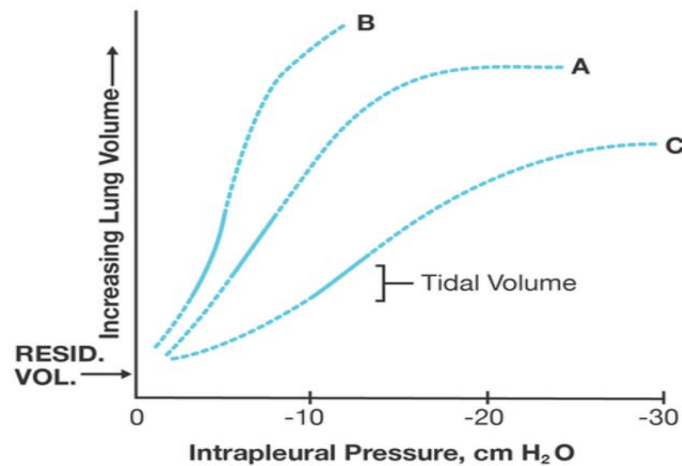
Question 8:

Baroreceptor provides _____

- a) Negative feedback
- b) Positive feedback
- c) Both negative and positive feedback
- d) Feedforward control

Question 9:

The diagram below shows the pressure-volume curves for 3 different lungs. Which of the following is the best representation of compliance between the three lungs?

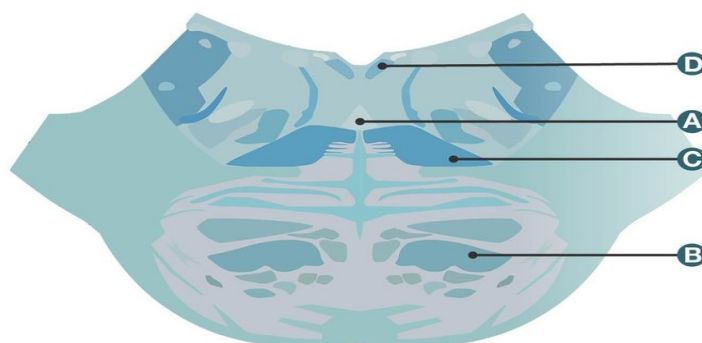


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- a) A<B<C
- b) B>A>C
- c) B>C>A
- d) C>A>B

Question 10:

Identify the site which is responsible for relieving pain when pressure is applied to the painful area.



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- a) A
- b) B
- c) C
- d) D

Question 11:

Inhibitory control by the hypothalamus is exerted on:

- a) Prolactin only
- b) Prolactin and growth hormone
- c) Growth hormone only
- d) Neither

Question 12:

Which of the following hormones play a permissive role in the attainment of puberty?

- a) GnRH

- b) Leptin
- c) Insulin
- d) GH

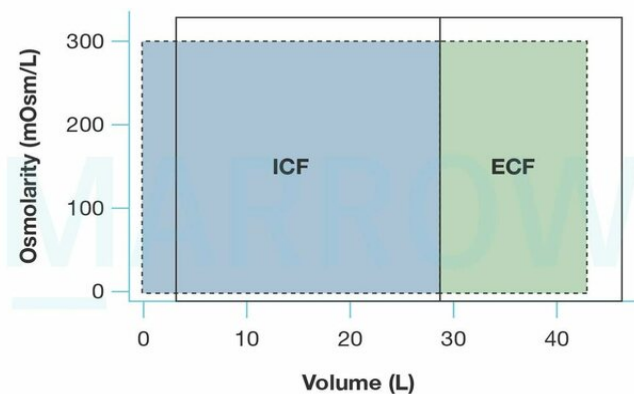
Question 13:

Weber-Fechner law relates magnitude of sensation to ____.

- a) Strength of muscle contraction
- b) Log of Intensity of stimulus
- c) Amplitude of action potential
- d) Duration of stimulus

Question 14:

The following Darrow Yannet diagram is suggestive of which condition? Note : The solid line is the original fluid compartment and the colour shaded compartment within the dotted line denotes the change caused by the condition.



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- a) Excessive sweating
- b) Adrenal Insufficiency
- c) Diarrhoea
- d) SIADH

Question 15:

The mechanism by which glucose is absorbed in the intestine is:

- a) Primary active transport
- b) Secondary active transport
- c) Facilitated diffusion
- d) Any of the above

Question 16:

The earliest stage at which erythropoietin is effective in red blood cell formation is?

- a) Burst forming units
- b) Colony forming units
- c) Late erythroblast stage
- d) Normoblast stage

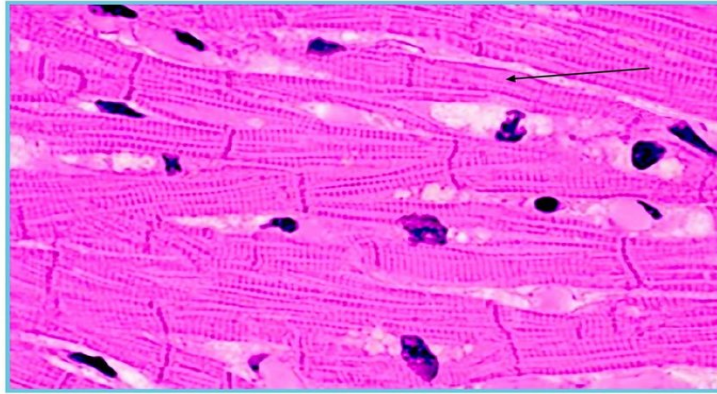
Question 17:

Integrin connects with:

- a) Cadherin
- b) Vitronectin
- c) Fibronectin
- d) Selectin

Question 18:

Which of the following organelles is absent in the area of the cardiac muscle marked:



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- a) Mitochondria
- b) Lysosome
- c) Golgi bodies
- d) Endoplasmic reticulum

Question 19:

Which of the following fibres reach the Purkinje cell directly?

- a) Inferior olivary nucleus
- b) Raphe nucleus
- c) Vestibular nucleus
- d) Locus coeruleus

Question 20:

Calculate the net filtration pressure across the GBM for a patient with Glomerular ultrafiltration coefficient = 6 Glomerular hydrostatic pressure = 45 Glomerular capillary oncotic pressure = 20 Bowman's capsule hydrostatic pressure = 10 Bowman's capsule oncotic pressure = 0

- a) 90 mmHg
- b) 60 mmHg
- c) 30 mmHg

d) 15 mmHg

Question 21:

In a 100-m dash, most of the energy consumed by skeletal muscles to replenish ATP is derived from _____

- a) Phosphocreatine
- b) Aerobic glycolysis
- c) Oxidation of Fatty acids
- d) None of the above

Question 22:

In the homunculus, which of the following has maximum representation?

- a) Low control and skilled movements
- b) More muscle fibres are present
- c) Increased preciseness of movements
- d) None of the above

Question 23:

What is the usual position of the Golgi apparatus in serous acinar cells?

- a) Supranuclear
- b) Basal
- c) Adjacent to nucleus
- d) Apical

Answer Key

Question No.	Correct Option
1	c
2	a

3	c
4	d
5	d
6	b
7	c
8	a
9	b
10	a
11	b
12	b
13	b
14	b
15	b
16	a
17	c
18	b
19	a
20	d
21	a
22	c
23	a

Detailed Explanations

Solution to Question 1:

The ascending thick limb of the loop of Henle is called the diluting segment.

In the ascending limb, there is the movement of Na^+ and Cl^- out of the tubular lumen, and the tubular fluid becomes more dilute.

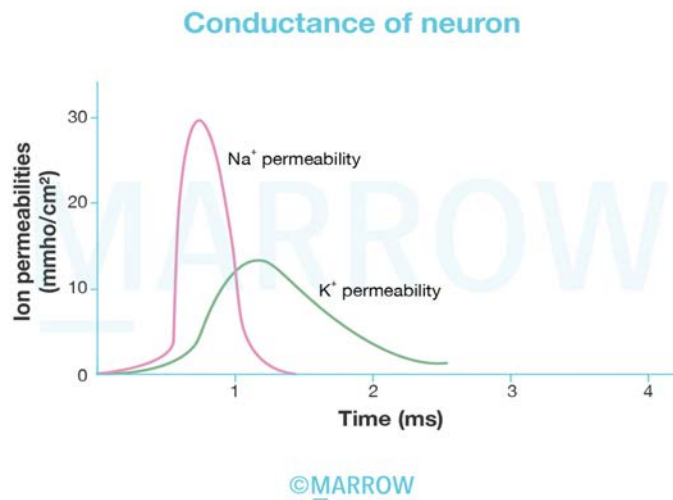
When the fluid reaches the top of the ascending limb, it becomes hypotonic to plasma.

Solution to Question 2:

Channel A: Na^+ , Channel B: K^+

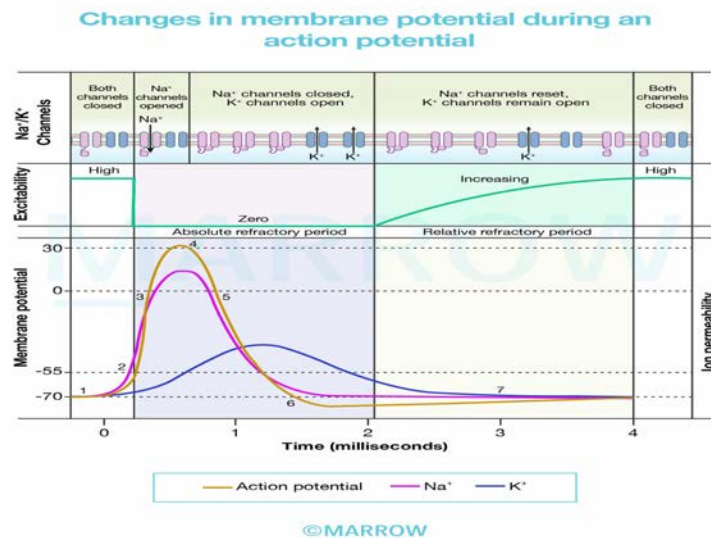
Na^+ channels are fast activating and are responsible for depolarization phase of action potential.

K⁺ channels are slow activating and open for prolonged duration than the sodium channels. They are responsible for repolarization phase of action potential. Also, the potassium channels return slowly to their closed state, which is responsible for after-hyperpolarization.



The sequence of events during the action potential occur in 7 steps as described below.

1. Resting state - Both channels are closed
2. Na⁺ channels open in response to depolarizing stimulus. Sodium enters into the cell. The membrane reaches threshold potential.
3. The sodium influx causes opening of more sodium channels through positive feedback mechanism. Rapid depolarization occurs.
4. Sodium conductance decreases.
5. Na⁺ channels close and K⁺ channels open. Repolarization occurs.
6. Sodium channels are reset, and are ready to respond to another depolarizing stimulus (relative refractory period). During this phase, the potassium channels remain open and slowly return to closed state causing after-hyperpolarization.
7. Both channels are closed. The membrane returns to RMP.



Solution to Question 3:

In electrical synapses, the cytoplasm of adjacent cells is directly connected by gap junctions (connexons) that allow free movement of ions from the interior of one cell to the interior of the next cell.

The transmission in electrical synapses is bidirectional and there is no synaptic delay. Unlike chemical synapses, electrical synapse has cytoplasmic continuity.

Note:

Most of the synapses used for signal transmission in the central nervous system are chemical synapses.

Solution to Question 4:

Hormone degradation as an important mechanism for regulating local concentrations is seen in all of the above.

11 β -hydroxysteroid dehydrogenase inactivates glucocorticoids in renal tubular cells, preventing actions through the mineralocorticoid receptor.

During development, degradation of retinoic acid by Cyp26b1 prevents the primordial germ cells in the male from entering meiosis, which occurs in the female ovary.

The thyroid hormone deiodinases convert T₄ to T₃ and can inactivate T₃. All the three are hence examples of local hormone degradation being utilized as a method for regulation of hormone concentration locally.

Solution to Question 5:

Dehydration, hemorrhage, and prandial drinking: All three can cause increased thirst resulting in water drinking.

Increased osmolality acts via osmoreceptors in the anterior hypothalamus.

The decreased plasma volume, as in hemorrhage, acts via

1. Renin-angiotensin system
2. Baroreceptors in heart and blood vessels.

The intake of liquids during a meal is known as prandial drinking. The increase is known as learned or habit response but is not understood completely.

The following factors are thought to play a role

1. Increase in plasma osmolality due to food absorption
2. Action on hypothalamus by one or more gastrointestinal hormones

Solution to Question 6:

Brachial plexus trunk injury leads to anesthetized skin over the region. On performing a histamine test on such an individual, one can expect central capillary dilatation → wheal without surrounding flare. The incomplete triple response over the anesthetic skin is characteristic of postganglionic lesions.

Intradermal injection of histamine usually causes a triple response in the surrounding skin i.e, red reaction → flare → wheal

- Red reaction is independent of innervation and is due to histamine-mediated relaxation of the precapillary sphincters
- Flare reaction (Flush, Axon reflex) is due to Substance P mediated arteriolar dilatation (antidromic conduction of afferent impulses via collaterals of the sensory nerve fiber that lead to the release of vasoactive mediators)
- Wheal (skin edema) is due to histamine-mediated increased capillary permeability
- The triple response persists after total sympathectomy
- Flare is absent in locally anesthetized skin

Solution to Question 7:

Ureteric stone causes a reduction in glomerular filtration rate (GFR).

Following ureteral obstruction, there is a progressive fall in glomerular filtration rate (GFR) due to a reduction in a single-nephron glomerular filtration rate (SNGFR) and a reduced number of filtering nephrons.

All the other factors/options increase GFR.

Solution to Question 8:

Baroreceptors provide negative feedback in response to an increase in blood pressure.

Increased baroreceptor discharge inhibits the tonic discharge of sympathetic nerves and excites the vagal innervation of the heart.

Baroreceptor reflex

- Increased BP → stimulation of baroreceptors → stimulation of NTS → inhibition of RVLM → inhibition of sympathetic system
- Stimulation of NTS also stimulates CVC → parasympathetic stimulation
- Net effect = sympathetic inhibition and parasympathetic stimulation → decreased HR and BP

NTS - Nucleus tractus solitarius

CVC - Cardiovascular center

RVLM - Rostral ventrolateral medulla

HR- Heart rate

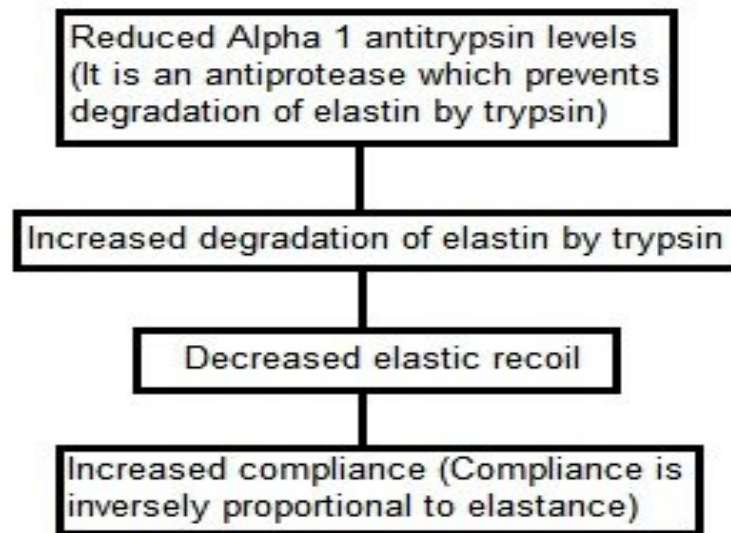
BP - Blood pressure

Solution to Question 9:

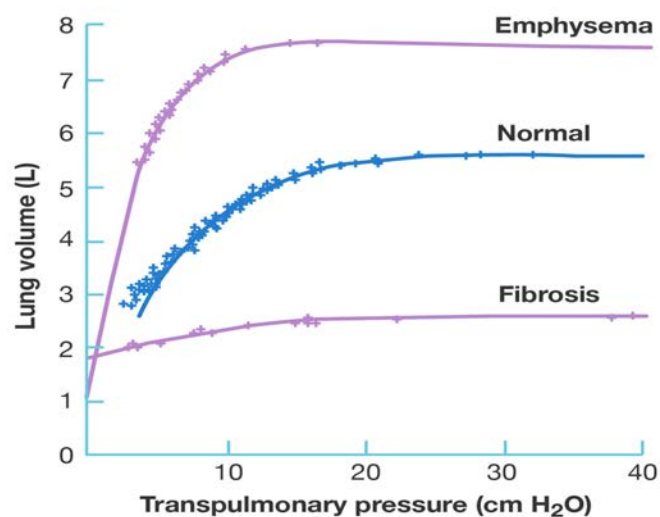
- The pressure-volume relationship is the fundamental relation between the work of breathing and lung mechanics.
- The diagram depicts the change in volume with respect to the change in intrapleural pressure, which is compliance. Here, the change is more (steep curve) with B and least (flat curve) with C.
- The “stiffer” lung displays a more horizontal curve (C), meaning that it takes a greater pressure to achieve the same inflation volume compared with a normal lung (A). These lungs are called less compliant.
- Curve (B) is more compliant as it takes lesser pressure to achieve the same inflation volume compared to normal (A).

Applied aspect:

Compliance is increased in emphysema. The mechanism is shown in the flow chart below.



Compliance is decreased in case of pulmonary edema or fibrosis.



Solution to Question 10:

It is a cross section of brain stem at pons.

- A is a periaqueductal gray matter which also consists of raphe nucleus.
- Descending pathways from the mesencephalic periaqueductal gray matter inhibit transmission in nociceptive pathways. This descending pathway includes a synapse in the ventromedial medulla (Raphe nucleus) and the release of endogenous opiates. So periaqueductal gray matter releases endogenous opiates.

Other marked structures are

B: Corticospinal tract

C: Medial lemniscus

D: Medial longitudinal fasciculus

Solution to Question 11:

Inhibitory control by the hypothalamus is exerted mainly on prolactin and growth hormone.

The major hypothalamic releasing hormones are:

- Thyrotropin releasing hormone (TRH)
- Corticotropin releasing hormone (CRH)
- Growth hormone releasing hormone (GHRH)
- Gonadotropin releasing hormone (GnRH)

The major hypothalamic inhibitory hormones are:

- Prolactin inhibitory hormone or dopamine
- Growth hormone inhibitory hormone (GHIH), also called Somatostatin.

Solution to Question 12:

The hormone that plays a permissive role in the attainment of puberty is leptin.

It now appears that leptin may be the link between body weight and puberty. Leptin is a satiety-producing hormone secreted by adipocytes. High leptin levels reduce appetite and increase energy expenditure.

Evidence for which is given by:

- Obese ob/ob mice that cannot make leptin are infertile, and their fertility is restored by injections of leptin.
- Leptin treatment also induces precocious puberty in immature female mice.
- However, the way that leptin fits into the overall control of puberty remains to be determined.

Leptin also accelerates GnRH pulsatility in neurons, fastening puberty.

Solution to Question 13:

Weber-Fechner Law states that the magnitude of sensation felt is proportionate to the log of the intensity of the stimulus.

Other Laws in Physiology:

- Doctrine of specific nerve energies - When the nerve pathways from a particular sense organ are stimulated, the sensation evoked is that for which the receptor is specialized no matter how or

where along the pathway the activity is initiated.

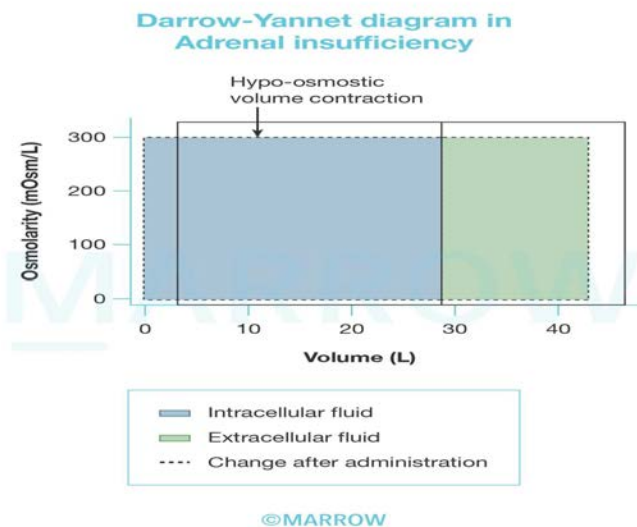
- Law of projection - No matter where a particular sensory pathway is stimulated along its course to the cortex, the conscious sensation produced is referred to the location of the receptor. This principle is called the law of projection.

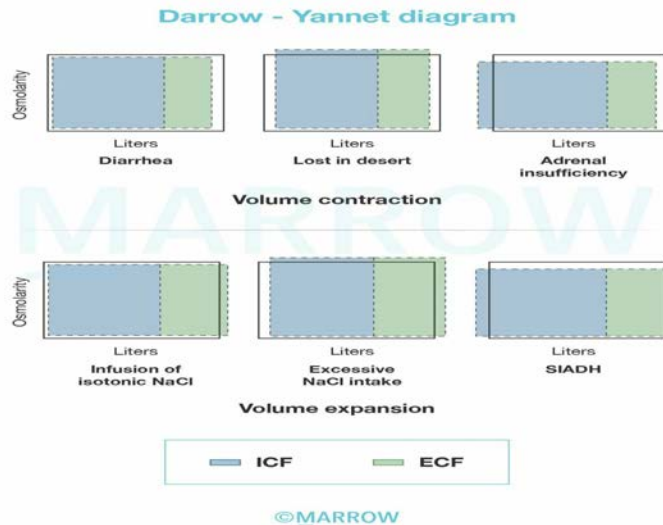
Solution to Question 14:

In adrenal insufficiency there is hypo-osmotic volume contraction. Aldosterone deficiency leads to increased sodium loss (loss of hyperosmotic fluid).

- The osmolarity of ECF decreases.
- ECF volume decreases because of the loss of volume.
- Water shifts into ICF; as a result of the shift, ICF osmolarity decreases until it is equal to ECF osmolarity, and ICF volume increases.
- So, loss of hypertonic fluid causes hypotonic volume contraction.

Please note that in the diagram, the horizontal axis represents volume and vertical axis represents osmolarity.





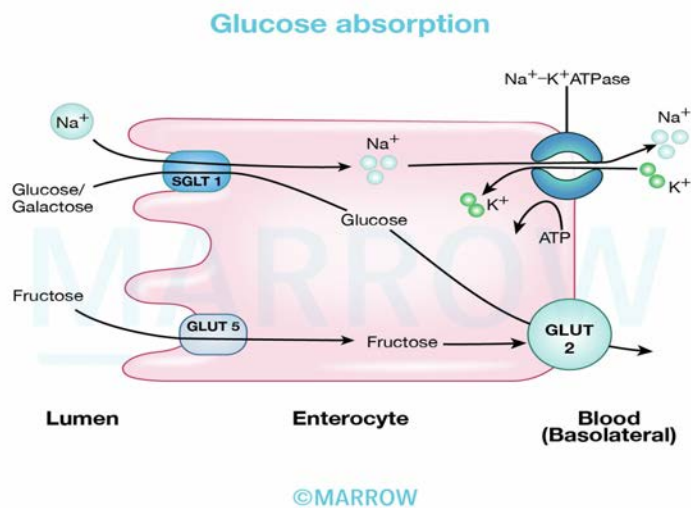
The Darrow Yannet diagrams of various conditions are shown above.

Solution to Question 15:

Glucose is absorbed in the intestine by secondary active transport.

On the basolateral side, there is active transport of sodium ions out of the cell into the interstitial fluid. This depletes sodium inside the cell. Because of this Na^+ concentration difference, more Na^+ binds to the SGLT on the luminal side. This increases the affinity of SGLT for glucose and glucose binds to it.

Then, both the Na^+ and glucose are transported together to the interior of the cell. Thus, glucose is absorbed in the intestine by secondary active transport. The energy for glucose transport is provided indirectly, by the active transport of Na^+ out of the cell. The glucose moves out of the cell into the blood across the basolateral membrane by facilitated diffusion mediated via GLUT.



Solution to Question 16:

BFU is the earliest stage of hematopoiesis where erythropoietin (EPO) acts

Under the influence of EPO, both BFU and CFU grow in semisolid culture media and give rise to colonies of well-hemoglobinized erythroblasts.

Erythroid burst-forming unit (BFU-E) is the most primitive single lineage committed erythroid progenitor. In addition to EPO, they form the sub-population of Erythroid Colony Forming Units (CFU-E).

CFU is derived from burst-forming units.

Note: While CFUs have the highest number of receptors for EPO, the earliest stage at which it acts is on the BFU.

Solution to Question 17:

Integrin connects with fibronectin.

Integrins are transmembrane glycoproteins (integral proteins) that allow cells to attach to extracellular matrix constituents such as laminin, fibronectin and collagen. It helps in linking the intracellular cytoskeleton functionally and structurally with the outside world.

Solution to Question 18:

The area of the cardiac muscle marked is devoid of nuclei. Lysosomes are commonly observed in perinuclear areas. Other organelles are present in abundance throughout the cell.

Solution to Question 19:

The fibres of inferior olivary nucleus (climbing fibres) reach the purkinje cell directly.

Note:

The input from the Purkinje cells to the deep nuclei is inhibitory.

However, the deep nuclei output is always excitatory.

Solution to Question 20:

Net filtration pressure = Hydrostatic pressure in the glomerular capillaries - Pressure in Bowman's capsule - Oncotic pressure in the glomerular capillaries.

$$\text{Net filtration pressure} = [P_G - P_B - \pi_G + \pi_B]$$

Therefore,

$$\begin{aligned}\text{Net filtration pressure} &= (45 - 10 - 20 + 0) \\ &= 25 - 10 \\ &= 15 \text{ mmHg}\end{aligned}$$

Note:

Here, just the net filtration pressure is asked and not the GFR.

$$\text{GFR} = K_f [P_G - P_B - \pi_G + \pi_B]$$

Solution to Question 21:

In a 100-m dash that takes 10 s, 85% of the energy consumed is derived anaerobically by phosphocreatine, which is hydrolyzed to creatine and phosphate groups with the release of considerable energy.

The ATP concentration in a muscle is low enough (4 millimolar) to sustain contractions for 1-2 seconds after which ATP is dephosphorylated into ADP. The ADP is re-phosphorylated by phosphocreatine for the next 5 to 8 seconds.

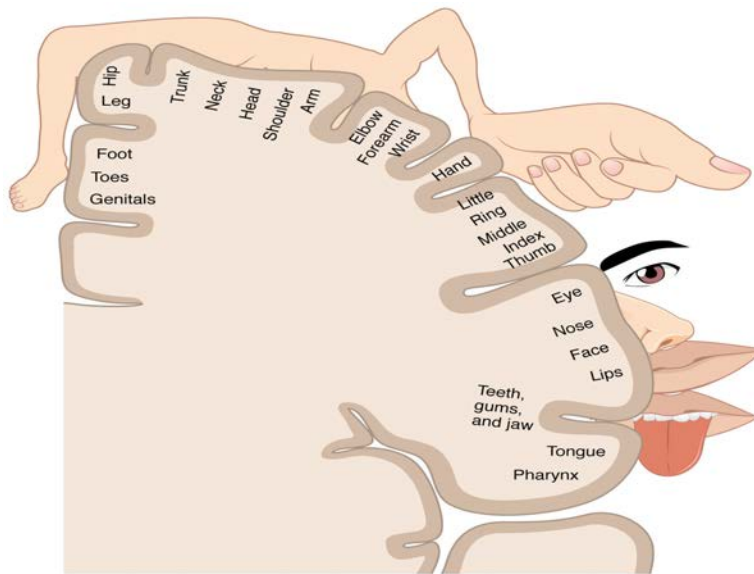
ADP is re-phosphorylated by 3 sources.

- Phosphocreatine after 2 seconds upto 10 seconds.
- Glycolysis (anaerobic) after 10 seconds upto a minute
- Oxidative metabolism - after a minute upto 2-4 hours.

Solution to Question 22:

In a homunculus, the more the representation of an area, the more precise the movements are.

- The relative sizes of the cortical receiving areas of the sensory homunculus are proportionate to the use of the part.
- The sizes of these areas are directly proportional to the number of specialized sensory receptors in each respective peripheral area of the body.
- Some body parts are represented by large areas in the somatic cortex—the lips the greatest of all, followed by the face and thumb, whereas the trunk and lower part of the body are represented by relatively small areas.



Solution to Question 23:

Golgi complex is located in supranuclear cytoplasm in serous acinar cells. Serous acinar cells are generally found in pancreas and the submandibular salivary glands.

Physiology AIIMS 2019 (May & Nov)

Question 1:

The function of the Golgi tendon organ is to _____

- a) Sense changes in muscle direction
- b) Sense changes in muscle tension
- c) Sense changes in position of the muscle
- d) Sense changes in length of the muscle

Question 2:

Which is the first-order neuron in the optic pathway?

- a) Rods and cones
- b) Ganglion cells
- c) Cells in lateral geniculate body
- d) Bipolar cells

Question 3:

Pacemaker generating the rhythm for breathing is _____

- a) Pneumotaxic centre
- b) Dorsal group of neurons in the medulla
- c) Pre-Bötzinger complex
- d) Apneustic centre

Question 4:

Which of the following is not an example of feed forward control mechanism?

- a) Temperature Regulation
- b) Cephalic phase of gastric secretion
- c) Cerebellar action in motor coordination

- d) Exercise resulting in heart rate increase

Question 5:

Single breath nitrogen method is used for measuring _____

- a) Dead space volume
- b) FRC
- c) Tidal volume
- d) Diffusion capacity

Question 6:

I-125 labelled albumin is used for measurement of:

- a) Plasma volume
- b) Interstitial fluid volume
- c) ICF
- d) ECF

Question 7:

Which of the following hormones is exclusively secreted by adrenal gland but not by the ovaries of females?

- a) Progesterone
- b) DHEA
- c) Estrogen
- d) DHEA-S

Question 8:

Which hormone crosses the plasma membrane for its action?

- a) Epinephrine
- b) Thyroxine
- c) Glucagon

d) Insulin

Question 9:

Afferent fibers from baroreceptors for blood pressure control goes to?

- a) Nucleus ambiguus**
- b) Rostral-ventro lateral medulla**
- c) Nucleus Tractus Solitarius**
- d) Caudal ventrolateral medulla**

Question 10:

If the lung is allowed to undergo elastic recoil without the chest wall, the resultant volume of the lung would be?

- a) Minimum volume**
- b) Zero**
- c) Functional residual capacity**
- d) Residual volume**

Question 11:

Myosin head detaches from actin by which of the following mechanism?

- a) Binding of ATP**
- b) Entry of calcium into sarcoplasmic reticulum**
- c) Change in troponin C configuration**
- d) Release of ADP and Pi**

Question 12:

You are an intern and want to determine the quantity of IV fluids needed for a patient. How much amount of transient loss/insensible loss per day would you account for in your calculations?

- a) 150-200 ml**
- b) 500-600 ml**

- c) 200-350 ml
- d) 350-400 ml

Question 13:

A soldier posted in Siachen is brought to you in the ER. In order to rule out hypothermia, how do you best measure his temperature?

- a) Oral
- b) Rectal
- c) Infrared tympanic thermometer
- d) Axillary

Question 14:

"Vasopressin escape" in SIADH results due to:

- a) Downregulation of aquaporin 2 production
- b) Increase in urine osmolarity
- c) Decrease in ADH secretion
- d) Upregulation of aquaporin 2 production

Question 15:

Which of the following hormones do not act on the pancreas?

- a) Secretin
- b) Gastrin
- c) Cholecystokinin
- d) GIP

Answer Key

Question No.	Correct Option
1	b

2	a
3	c
4	d
5	a
6	a
7	d
8	b
9	c
10	a
11	a
12	b
13	b
14	a
15	b

Detailed Explanations

Solution to Question 1:

Golgi tendon organ sense changes in muscle tension.

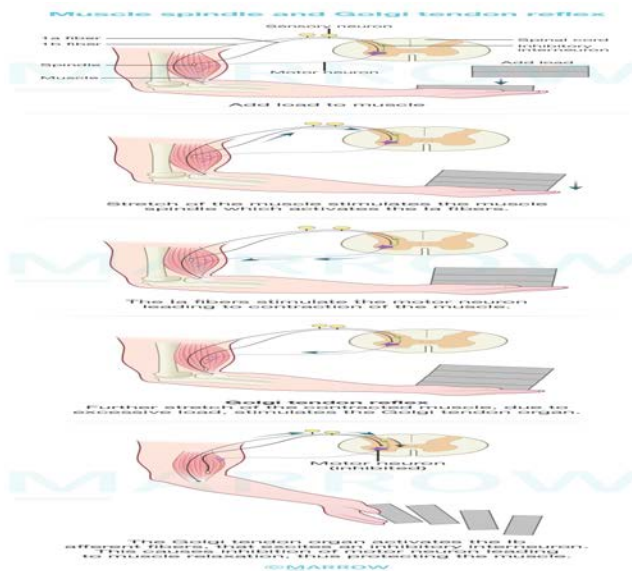
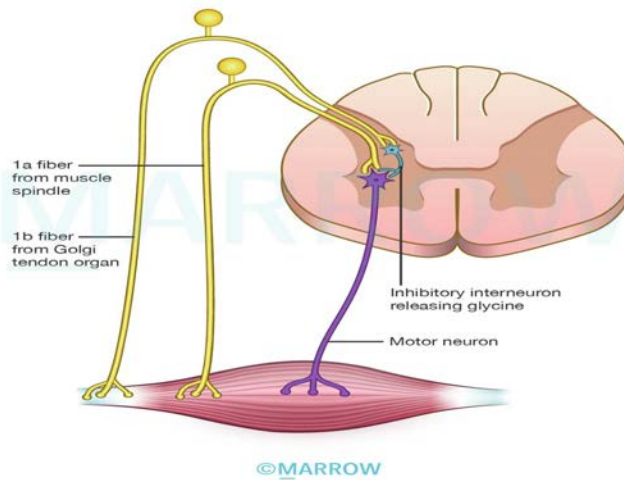
Golgi tendon reflex (Inverse myotatic/ Inverse stretch reflex) is disynaptic, is the opposite of the stretch reflex.

When the tension in a contracted muscle is significantly elevated, the inhibitory effect from the Golgi tendon organ leads to instantaneous relaxation of the entire muscle. This effect is called the lengthening reaction.

It is a protective mechanism to prevent tearing of the muscle or avulsion of the tendon from its attachments to the bone.

Note: Muscle spindles senses changes in muscle length (stretch).

Stretch reflex and inverse stretch reflex



Solution to Question 2:

Photoreceptors - rods and cones are the first-order neurons in the visual pathway.

Solution to Question 3:

Pre-Botzinger complex generates the rhythm for automated breathing.

- These are also known as pacemaker cells located in the medulla.
- These are the first set of neurons that initiate inspiration.

- These neurons discharge rhythmically, and they produce rhythmic discharges in the phrenic motor neurons.

Breathing is under the control of both the following:

- Voluntary control: Located in the cerebral cortex (via corticospinal tract)
- Automatic control: Driven by pacemaker cells in the medulla (Pre-Botzinger complex). These activate motor neurons via cervical (Phrenic nerve innervating diaphragm) and thoracic spinal cord (external intercostal muscle).

There is a reciprocal innervation to expiratory and inspiratory muscles in that motor neurons supplying expiratory muscles are inactive when motor neurons supplying inspiratory muscles are active, and vice versa.

Inspiration	
Initiation	Pre-Botzinger complex (Pacemaker cells) in Medulla
Maintenance	Apneustic center in lower Pons stimulates Dorsal group of neurons in the medulla
Termination	Apneustic center inhibited by Vagus and Pneumotaxic center

Regulation of expiration: During quiet breathing, expiration is a passive process; there is no firing of neurons. Ventral respiratory group of neurons are involved in forceful expiration like during exercise.

Mnemonic DIVE: Dorsal respiratory group for Inspiration and Ventral respiratory group for Expiration.

Solution to Question 4:

Exercise resulting in heart rate increase is not an example of feedforward control mechanism. However, thinking about exercise and it resulting in a heart rate increase would be an example of feedforward control mechanism.

Feedforward control is that which makes an anticipatory or adaptive response without a specific stimulus.

Temperature control - Thermoregulatory responses are initiated by the hypothalamus before the changes in environmental temperature changes the core body temperature. Thus, shivering in response to cold environment would be an example of feedforward control.

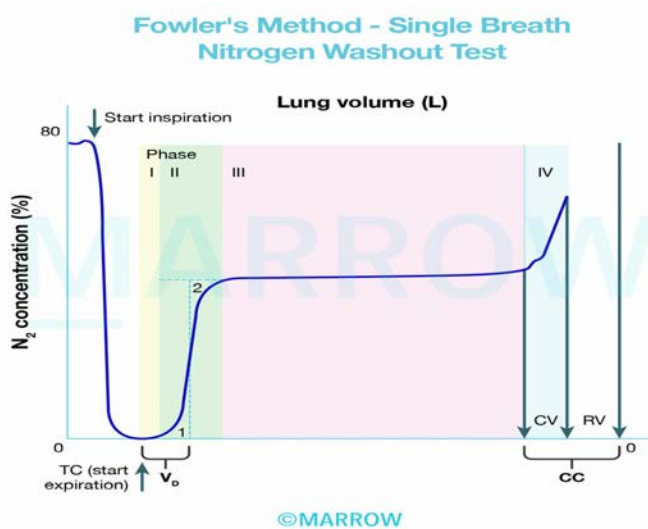
In the cephalic phase of gastric secretion, the thought of food itself and not food per se produces a gastric secretion.

The cerebellum has a feed-forward inhibition that limits the duration of the excitation produced by any given afferent volley.

Solution to Question 5:

Anatomical dead space can be measured by the single-breath N₂ curve (Fowler's method).

- Anatomical dead space is the respiratory system volume exclusive of alveoli.
- Alveolar dead space is the volume of gas not equilibrating with blood i.e. wasted ventilation.
- Physiological/total dead space is the sum of anatomical and alveolar dead space.
- The equation used for the calculation of total/physiological dead space is the Bohr's equation.



The patient takes a deep breath of O₂ From mid-inspiration, then exhales steadily. The changes in the N₂ concentration of expired gas during expiration are plotted.

Region I - Dead Space (DS)

From I–II is a mixture of DS and alveolar gas.

The transition from III-IV is the closing volume (CV).

End of IV - Residual Volume (RV).

Factors affecting dead space:

Option B: Functional Residual Capacity (FRC) is measured using body plethysmography.

Option C: Tidal volume (TV) is measured using spirometry.

Option D: Diffusion Capacity is measured using carbon monoxide diffusion capacity (DLCO).

Factor	Effect
Posture: Upright/Supine	Increase/Decrease
Position of airway: Neck Extension/Neck Flexion	Increase/Decrease

Factor	Effect
Age	Increase
Artificial airway	Decrease
Positive pressure ventilation	Increase
Drug anti-cholinergic	Increase
Pulmonary perfusion:Pulmonary embolismHypotension	IncreaseIncrease
Pulmonary disease:Emphysema	Increase

Solution to Question 6:

I-125 labelled albumin is the most commonly used substance for the measurement of plasma volume.

Solution to Question 7:

DHEAS is exclusively secreted from the adrenals but not by ovaries because ovaries lack the enzyme dehydroepiandrosterone-sulphotransferase needed to form DHEAS.

90% of DHEA is synthesised by the adrenals and 10% comes from the ovaries.

Solution to Question 8:

Thyroxine crosses the plasma membrane for its action.

Thyroid hormones enter the nucleus and bind to thyroid hormone receptors. The hormone-receptor complex in the nucleus binds to DNA via zinc fingers. This increases the expression of a variety of different genes that code for proteins regulating cell function. Although thyroxine is an AA derivative, there are transporters to move it into the cell, hence they are intracellular receptors acting on the nucleus.

Epinephrine, glucagon, and insulin bind to cell surface receptors.

Solution to Question 9:

Afferent fiber for blood pressure control goes to the Nucleus Tractus Solitarius (NTS).

Increased BP stimulates the baroreceptors which provide negative feedback in response to the increase in blood pressure.

Baroreceptor reflex:

- Increased BP → stimulation of baroreceptors → stimulation of NTS → excitation of CVLM → inhibition of RVLM → inhibition of sympathetic system
- Stimulation of NTS also stimulates CVC → parasympathetic stimulation
- Net effect = sympathetic inhibition and parasympathetic stimulation → decreased HR and BP

NTS - Nucleus tractus solitarius

CVC - Cardiovagal centre

CVLM - Caudal ventrolateral medulla

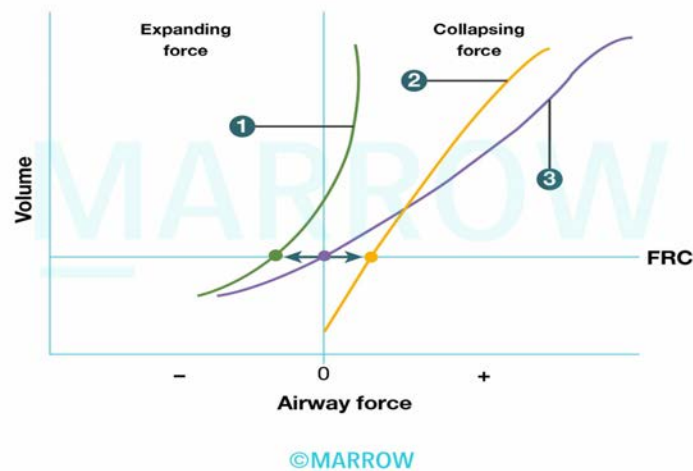
RVLM - Rostral ventrolateral medulla

HR - Heart rate

BP - Blood pressure

Solution to Question 10:

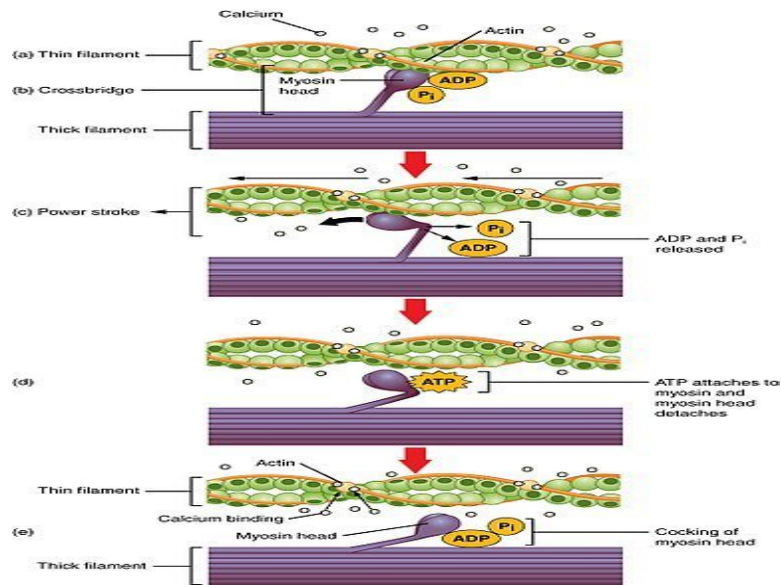
If lung is allowed to undergo elastic recoil without the chest wall, the resultant volume of the lung would be minimum volume as the opposing force by the chest wall would be non-existent.



The yellow line in the above graph denotes the compliance of lungs alone. According to the graph, when the pressure (x axis) inside the lung is zero, there is some volume of air lesser than FRC still present (y axis) inside the lungs which is less than the functional residual capacity and residual volume which can be called as minimum volume.

Solution to Question 11:

Binding of ATP on the free site of myosin leads to the detachment of myosin head from the thin filament.



Solution to Question 12:

500-600 ml of transient loss/insensible loss per day would have to be accounted for in the calculation.

Although water is lost through sweating especially in hot climates or during physical exertion, we tend to lose water at rest as well. Physiologically, water evaporates from the skin and lungs at a rate of about 600 to 700 ml/day. This is known as insensible water loss.

Solution to Question 13:

In the given clinical scenario rectal temperature would be the preferred site to measure the core body temperature.

During low or high temperatures like in Siachen, the axillary and the oral temperatures are highly misleading and should not be used to measure the core body temperatures.

Infrared tympanic thermometer measures the temperature of the eardrum which is representative of core body temperature. However, more often the thermometer is not inserted adequately or for safety reasons, they are manufactured in such a way that they do not reach deep into the ear canal, or the ear canal contains cerumen which can disturb the measurement. Hence the infrared thermometer more commonly delivers the temperature of ear canal cerumen and hence not a reliable method of measuring the core body temperature.

Solution to Question 14:

Vasopressin escape in SIADH (Syndrome of inappropriate ADH secretion) results due to downregulation of aquaporin 2 production.

Prolonged exposure to elevated levels of vasopressin (ADH) leads eventually to downregulation of aquaporin-2 receptors. This prevents excessive reabsorption of water, despite high serum ADH levels and hence maintains euolemia. This is known as vasopressin escape. This helps prevent excessive dilutional hyponatremia in SIADH.

Solution to Question 15:

Gastrin does not act upon the pancreas. Gastrin stimulates gastric acid secretion and gastric mucosal growth.

GI hormones acting on the pancreas include:

GIP is also known as Glucose-dependent insulintropic peptide.

	CCK(Cholecystokinin)	Secretin	GIP(Gastric inhibitory peptide)
Stimulates	Pancreatic enzyme secretion Pancreatic bicarbonate secretion Gallbladder contraction Growth of exocrine pancreas	Pepsin secretion Pancreatic bicarbonate secretion Biliary bicarbonate secretion Growth of exocrine pancreas	Insulin release
Inhibits	Gastric emptying	Gastric acid secretion	Gastric acid secretion

Physiology AIIMS 2020 (May and Nov INI-CET)

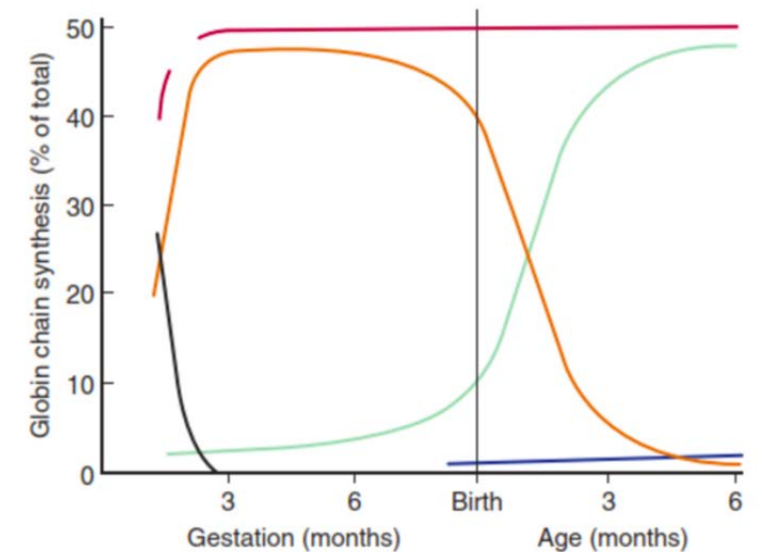
Question 1:

A 65-year old patient visits his family practitioner for a yearly examination. On examination, his BP was 190/100 mm Hg. His pulse pressure was 90 mm Hg. What would have decreased resulting in this increase in pulse pressure?

- a) Myocardial contractility
- b) Stroke volume
- c) Arterial wall compliance
- d) Cardiac output

Question 2:

Which of the following curves represents gamma-globin chain production?



- a) Green Curve
- b) Black Curve
- c) Blue Curve
- d) Orange curve

Question 3:

Transcutaneous electric nerve stimulation is used to reduce pain. What is the mechanism behind this?

- a) Central inhibition
- b) Desensitization
- c) Gate control theory
- d) Descending inhibition

Question 4:

Which of the following is not required for calculation of creatinine clearance?

- a) Urine creatinine
- b) Serum creatinine
- c) 24 hour urine output
- d) Volume of serum passing through kidney in 24 hours

Question 5:

A neuron is receiving 400 inputs from 150 sensory neurons. This is an example of?

- a) Divergence
- b) Convergence
- c) Feedback inhibition
- d) Feedforward excitation

Question 6:

Calculate the ECF volume in a volunteer in whom 10 gm of mannitol was injected in venous blood. At equilibrium urinary excretion of mannitol was found to be 10% and the plasma concentration of mannitol was 65 mg/100 ml.

- a) 14L
- b) 10L
- c) 28L
- d) 20L

Question 7:

Gibbs-Donnan equilibrium is mainly due to?

- a) Phosphate
- b) Bicarbonate in the intracellular compartment
- c) Calcium in the extracellular compartment
- d) Intracellular proteins

Question 8:

Which of these hormones have intracellular receptors?

- a) Thyroxine
- b) Epinephrine
- c) Glucagon
- d) Parathormone

Question 9:

Intercellular communication in eukaryotes is by?

- a) Lysosome
- b) Peroxisome
- c) Exosome
- d) Genome

Question 10:

In skeletal muscle, dihydropyridine and ryanodine receptors are coupled by?

- a) Chemically
- b) Electrically
- c) Chemically and then electrically
- d) Mechanically

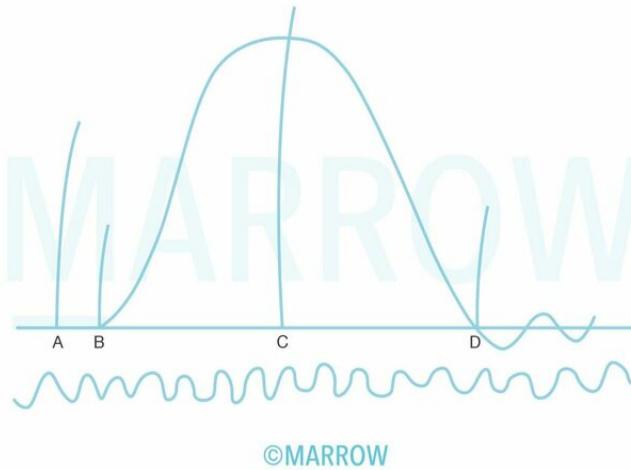
Question 11:

In negative feedback, feedback gain of infinity is seen in which of the following?

- a) Blood volume control by kidney
- b) Temperature control in hypothalamus
- c) BP control by baroreceptors
- d) Infinite gain is not possible

Question 12:

The image below shows simple muscle twitch along with the time trace, recorded in a frog gastrocnemius muscle. Calculate the minimum tetanizing frequency. (The time tracer was made with a 100Hz tuning fork).



- a) 25 per second
- b) 15 per second
- c) 17 per second
- d) 20 per second

Question 13:

Least frequency of slow waves is seen in:

- a) Duodenum
- b) Ileum
- c) Jejunum

d) Stomach

Question 14:

Macula densa is formed by which part of the nephron?

- a) PCT
- b) DCT
- c) Afferent arteriole
- d) Bowman's capsule

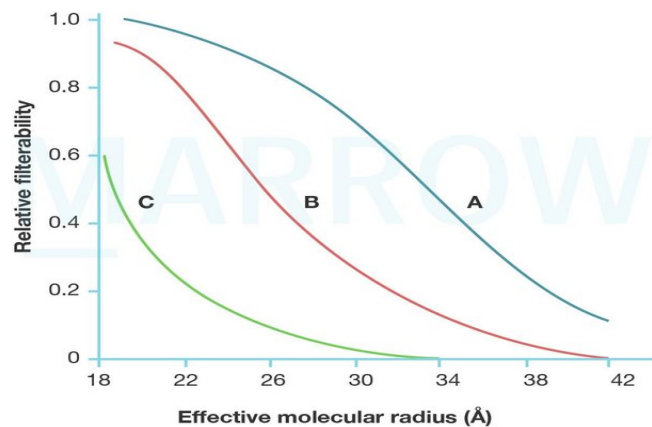
Question 15:

As filtrate flows through PCT, the concentration of all of the following decreases except:

- a) Aminoacid
- b) Glucose
- c) Chloride
- d) Bicarbonate

Question 16:

This is the graph depicting the filterability of dextran through the kidney, 1 representing complete filtration and 0 representing no filtration. Choose the substance A, B, C respectively.



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- a) Polycationic, Neutral, Polyanionic
- b) Polyanionic, Neutral, Polycationic
- c) Neutral, Polycationic, Polyanionic
- d) Polyanionic, Polycationic, Neutral

Question 17:

Calculate the respiratory quotient (RQ) in a patient with 50kg body weight, if the CO₂ exhaled is 200ml/min and O₂ consumed is 250 ml/min.

- a) 0.8
- b) 1
- c) 1.25
- d) 1.5

Question 18:

Which of the following types of amino acids are present in the transmembranous part of the membrane receptor?

- a) Amphipathic
- b) Hydrophobic
- c) Hydrophilic
- d) None of the above

Question 19:

Which of the following is not a cause of secondary hyperparathyroidism?

- a) Parathyroid adenoma
- b) Lithium intake
- c) Vitamin D deficiency
- d) Chronic kidney disease

Question 20:

Which of the following events occur immediately when a person changes his/her posture from supine to standing?

- a) Decrease in firing from baroreceptor
- b) Vagal output to cardiac fibres increase
- c) Mesenteric Vasoconstriction
- d) Increase cardiac contractility

Question 21:

Which of the following statements is true about skeletal muscle contraction?

- a) 2, 3, 4
- b) 3 only
- c) 1, 2, 4
- d) 1, 3, 4

Question 22:

Which of the following statements are correct regarding mechanisms of depolarisation, repolarisation, and hyperpolarisation in a skeletal muscle action potential?

- a) 2 and 4 are correct
- b) 1, 3, and 5 are correct
- c) 1, 4, and 5 are correct
- d) 2, 4, and 5 are correct

Question 23:

Arrange the following events in the sequence in which they occur, after 'p wave' in ECG:

- a) 1-2-3-4
- b) 1-2-4-3
- c) 3-1-2-4
- d) 1-3-2-4

Question 24:

Which of the following increases in COPD?

- a) 1 only
- b) 1 and 4
- c) 1 and 3
- d) 1 and 2

Answer Key

Question No.	Correct Option
1	c
2	d
3	c
4	d
5	b
6	a
7	d
8	a
9	c
10	d
11	a
12	d
13	d
14	b
15	c
16	a
17	a
18	b
19	a
20	a
21	b
22	b
23	b
24	d

Detailed Explanations

Solution to Question 1:

Pulse pressure increases with age due to decreased arterial wall compliance.

With advancing age, there is an increase in systolic blood pressure. Whilst diastolic blood pressure falls after 60 years of age due to increased arterial stiffness (decreased compliance). As a result of these two changes, pulse pressure increases ($\uparrow PP = \uparrow SBP - \downarrow DBP$).

Solution to Question 2:

In the given image, the Orange curve represents the gamma-globin chain production. Gamma globin chain is produced in the fetal period and drops to undetectable levels 6 months postpartum which is the pattern represented by the Orange curve.

Hemoglobin in blood has different variants throughout life, in different proportions. The type of hemoglobin essentially depends on the combination of globin chains. Adult hemoglobin is made of 2 alpha and 2 beta chains, whereas fetal hemoglobin is made of 2 alpha and 2 gamma-globin chains. Other combinations of chains can be found in embryonic hemoglobin.

Here,

- Gamma chain - Orange curve
- Alpha chain - Red curve
- Beta chain - Green curve
- Epsilon chain - Black curve
- Delta chain - Blue curve

Solution to Question 3:

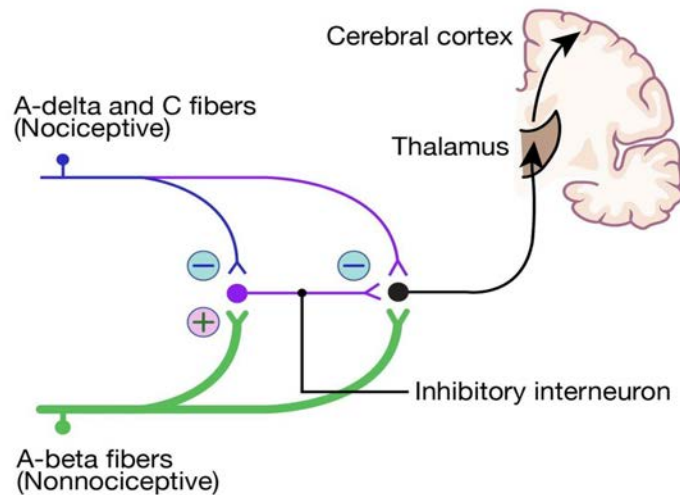
Transcutaneous electrical nerve stimulation (TENS) used for pain relief is based on the gate theory of pain. This method uses electrodes to activate $A\alpha$ and $A\beta$ fibers near the injury to relieve pain.

Gate control theory is based on the inhibition of pain pathway in the spinal cord. Various sensory inputs from a particular dermatome reach the dorsal horn of the spinal cord. The pain is transmitted via A-delta and C fibers, whereas, A-beta fibers carry tactile sensations (touch, pressure) and A-alpha fibers carry proprioception.

When these large-diameter A-beta/A-alpha fibers are stimulated, they inhibit the pain pathway in the spinal cord. This inhibition is thought to result from the local lateral inhibition of interneurons in the spinal cord.

Gate control theory operates at the substantia gelatinosa of the dorsal horn of the spinal cord.

The given image depicts the gate control theory of pain:



Solution to Question 4:

The volume of plasma passing through the kidney in 24 hours is not used in the calculation of creatinine clearance.

The parameters used to calculate creatinine clearance (CCr) are:

- Urine creatinine
- Serum creatinine
- 24hr urine output

The formula used to calculate creatinine clearance (CCr) is:

$UCr \times V / PCr$ where,

UCr - Urine creatinine(mg/dl)

PCr - Serum creatinine(mg/dl)

V - Urine volume(ml/24 hrs)

Creatinine is a metabolite of muscle metabolism and is cleared from the body mostly by glomerular filtration. Therefore, the clearance of creatinine is used to assess GFR.

Urine collection for the measurement of creatinine clearance may be subject to practical constraints. Therefore, approximate changes in GFR can be obtained by measuring the serum creatinine concentration, which is inversely proportional to GFR.

Solution to Question 5:

A single neuron receiving 400 inputs from 150 sensory neurons is an example of convergence.

Convergence means signals from multiple inputs uniting to excite a single neuron. It can be of 2 types:

- Convergence from a single source
- Convergence from multiple sources

The importance of convergence is that neurons are almost never excited by an action potential from a single input terminal. However, action potentials converging on the neuron from multiple terminals provide enough spatial summation to bring the neuron to the threshold required for discharge.

Solution to Question 6:

ECF volume of the above volunteer is calculated to be 14 L.

The ECF volume can be calculated using the formula:

Where,

- V = ECF volume
- Amount injected = 10 gm
- Amount of mannitol excreted = 10% of 10 gm = 1gm
- Plasma concentration of mannitol = 65 mg/100mL = 0.65 gm/L

Calculation:

$$\begin{aligned}\text{ECF Volume} &= (10-1)/0.65 \\ &= 13.8 \text{ L} = \text{approximately } 14 \text{ L}\end{aligned}$$

Mannitol cannot cross cell membranes, hence, it will be distributed in ECF but not in ICF. Thus mannitol is a marker for ECF volume and therefore calculating the volume of distribution of mannitol gives the value of ECF volume.

Solution to Question 7:

Gibbs-Donnan equilibrium is mainly due to intracellular proteins.

Donnan and Gibbs showed that "When an ion on one side of a membrane cannot diffuse through the membrane, the distribution of other ions to which the membrane is permeable is affected in a predictable way".

According to this, in the presence of intracellular proteins, which are negatively charged non-diffusible ions, the distribution of other ions to which the membrane is permeable is affected in a predictable way (Donnan Effect) so that at equilibrium their concentration ratios are equal.

Solution to Question 8:

Thyroxine hormone act by binding to intracellular nuclear receptors.

All other hormones mentioned in the options bind to receptors on the cell membrane.

Solution to Question 9:

Intercellular communication in eukaryotes is by Exosome.

In the last decade, a class of small, heterogeneous, secreted vesicles, broadly termed as extracellular vesicles have been identified.

These extracellular vesicles have been implicated as a new and important mediator of cell-cell communication that likely contributes importantly to both normal and pathologic physiology.

These vesicles are enclosed by a lipid bilayer.

They are heterogeneous in size (30-2000 nm diameter), and are generated by at least two distinct mechanisms:

- Microvesicles are generated by budding from the plasma membrane of a source cell.
- Exosomes are generated from the multivesicular body (MVB).

Note: The terms microvesicle and exosome are often lumped together as simply exosomes.

Solution to Question 10:

In skeletal muscle, dihydropyridine and ryanodine receptors show mechanical coupling involving structural interaction between the two.

Dihydropyridine receptors (DHPRs) are voltage-gated Ca^{2+} channels in the T tubule membrane.

The ryanodine receptor (RyR) is a ligand-gated Ca^{2+} channel with Ca^{2+} as its natural ligand.

Skeletal muscle excitation-contraction (E-C) coupling is thought to occur by a "mechanical coupling" mechanism involving protein-protein interactions that lead to activation of the sarcoplasmic reticulum ryanodine receptor by the voltage-sensing transverse (T-) tubule dihydropyridine receptor.

Solution to Question 11:

In negative feedback, feedback gain of infinity could be seen in blood volume control by the kidney.

When the blood volume falls below the equilibrium, the renal-body fluid mechanism causes the blood volume returns back to the equilibrium point resulting in zero error and hence the gain of kidney in its ability to regulate blood volume is infinity (∞).

The degree of effectiveness with which a regulatory system maintains constant conditions is determined by the gain of the negative feedback system.

Gain= Correction/Error

Example: If an external or internal factor led to a decrease in mean blood pressure from 100 mm Hg to 80 mm Hg. Activation of a negative feedback mechanism i.e. baroreflex leads to a correction of blood pressure to 96 mm Hg. So in this case,

- Correction = 16 mm Hg
- Error = 4 mm Hg
- Gain = Correction/Error = $16/4 = 4$

Note: A gain of 0 would imply a total failure of the regulatory system and a gain of infinity (∞) would indicate complete restoration of the regulated variable.

Solution to Question 12:

The tetanizing frequency of the muscle, based on the given diagram, is 20 per second.

Tetanizing frequency = $1/\text{contraction period}$. It is the frequency at which individual muscle fibres twitch that fuse and cause sustained muscle contraction, where the contraction period is the time taken from latent period to maximum force of contraction.

In this case, the time tracer was taken using a 100Hz tuning fork. Which means the tracer recorded 100 waves per second.

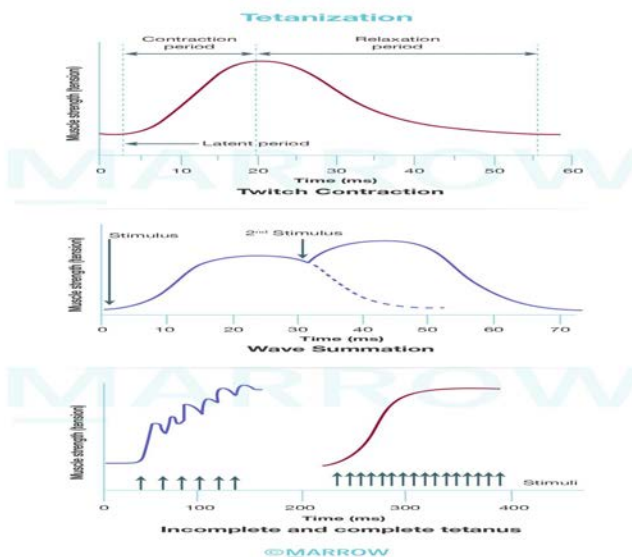
Therefore, the duration of each wave in time tracer = 0.01 seconds.

In the image given in the question, between the point B (end of latent period) to point C (maximum force of contraction), there are about 5 waves.

Hence, the duration of contraction = $5 \times 0.01 = 0.05$ seconds.

Substituting the value in the formula, we get:

Tetanizing frequency = $1/0.05 \text{ sec} = 100/5 = 20/\text{sec}$.



Tetanization:

The features of muscle contraction can be demonstrated by eliciting single muscle twitches. The adding together of individual twitch contractions to increase the intensity of overall muscle contraction is Summation. Summation occurs in two ways:

- by increasing the number of motor units contracting simultaneously, which is multiple fiber summations.
- by increasing the frequency of contraction, which is called frequency summation and can lead to tetanization.

When the frequency of a contraction reaches a critical level, the successive contractions eventually fuse together, and the whole muscle contraction appears to be completely smooth and continuous, this is called tetanization. Beyond this point, a further increase in the frequency, will not result in increased force of contraction.

Tetanization occurs in two ways:

- Complete tetanus occurs when no relaxation occurs between stimuli.
- Incomplete tetanus when periods of incomplete relaxation take place between the summated stimuli.

Solution to Question 13:

Among the given options, the least frequency of slow waves is seen in the stomach (4/min).

The Basal electrical rhythm (BER) also called slow-wave, is a rhythmic fluctuation of membrane potential in the gastrointestinal smooth muscle. It is initiated by the interstitial cells of Cajal. The membrane potential fluctuates between -65mV and -45mV.

The BER is seen in all sites of gastrointestinal smooth muscle except in the esophagus and proximal stomach.

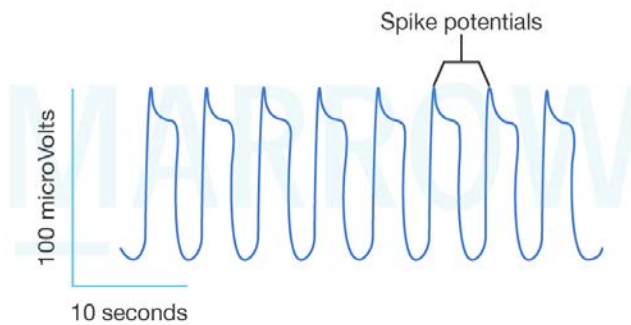
The rate of BER in various sites is as follows:

- Stomach: 4/min
- Duodenum: 12/min
- Distal ileum: 8/min
- Cecum: 2/min
- Sigmoid colon: 6/min

The BER does not cause contraction of the smooth muscle. However, when the BER is superimposed by the spike potentials, it increases muscle tension.

Acetylcholine causes an increase in the number of spike potentials and muscle tension. Epinephrine causes a decrease in the number of spike potentials and muscle tension.

Basal electrical rhythm of the GIT



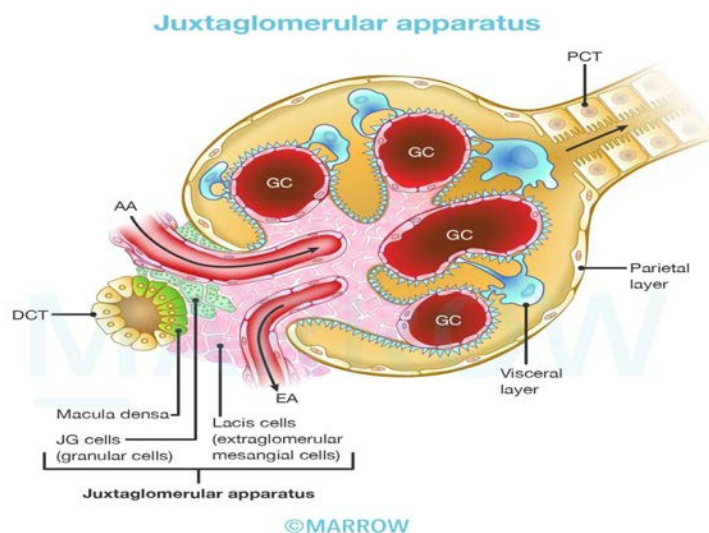
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Muscle contraction can occur only when the motor stimulus is provided during the depolarisation part of the BER.

Thus, BER does not cause muscle contraction by itself, but controls the rate and organization sites of the muscle contractions, as in coordination of the peristaltic activity and setting the rhythm of segmentation.

Solution to Question 14:

Macula densa is formed by specialized tubular epithelial cells of the distal convoluted tubule.



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The juxtaglomerular apparatus consists of the following parts:

- Macula densa: At the end of the thick ascending limb of Henle, and at the beginning of the distal tubule, the tubular epithelial cells are specialized to form the macula densa. They detect sodium

concentration of the fluid in the tubule (chemoreceptor). In response to elevated sodium, the macula densa cells trigger the contraction of the afferent arteriole, reducing the flow of blood to the glomerulus and the glomerular filtration rate. This happens as follows: When there is increased sodium, there is increased sodium-potassium ATPase activity which leads to increased ATP hydrolysis causing more adenosine formation. The adenosine receptors on the macula densa cells stimulate it to release more calcium, so that there is vasoconstriction of the afferent arteriole, thereby decreasing the GFR.

- Lacis cells: Extraglomerular mesangial cells situated between afferent and efferent arteriole.
- Granular cells/JG cells: Modified smooth muscle cells of the afferent arteriole, they secrete renin. Fall in pressure in afferent arteriole stimulates renin release (mechanoreceptor), which, in turn, increases blood pressure.

Solution to Question 15:

As the filtrate flows through the proximal convoluted tubule, the concentration of the chloride increases.

- The concentration of chloride in the early part of the pCT is ≈ 105 mEq/L.
- As the sodium is reabsorbed, it preferentially carries glucose, amino acids, and bicarbonate.
- This leaves behind the solute, that has a higher concentration of the chloride, which as it reaches the second half of the PCT is ≈ 140 mEq/L.

Solution to Question 16:

In the given graph, the order of filterability is $A > B > C$, which likely indicates that the substances are Polycationic, Neutral, Polyanionic.

The glomerular filtration barrier is negatively charged, which will repel most of the negative charged (anionic substances). Whereas, the cationic substances are easily filtered.

Hence, the order of filterability would be: Polycationic $>$ Neutral $>$ Polyanionic, represented by A, B, and C in the graph shown.

The glomerular filtration barrier consists of:

- Endothelium of the capillary
- Basement membrane:
- Lamina densa - electron-dense layer. It contains type IV collagen
- Lamina rara interna - contains glycosaminoglycan heparan sulfate which has a negative charge
- Lamina rara externa - contains glycosaminoglycan heparan sulfate which has a negative charge
- Layer of epithelial cells (podocytes) surrounding the outer surface of the capillary basement membrane

The capillary endothelium is perforated by thousands of small holes called fenestrae, which measure from 4 nm up to 8nm. Therefore, neutral substances with molecular diameters of less than 4 nm are freely filtered, and filtration of those with diameters of more than 8 nm approaches zero.

However, in addition to the size of the particles, the filtration also depends mainly on the charge of the particle being filtered. The glomerular capillary wall consists of highly negatively charged proteins called sialoproteins, which repel the negatively charged substances (anions). The property can be studied by using charged dextran.

When the filtration of the negatively and positively charged dextrans was studied, the results indicated that the negatively charged dextrans in the blood are repelled, with the result that filtration of anionic substances 4 nm in diameter is less than half that of neutral substances of the same size.

This explains, why albumin despite having an effective molecular diameter of approximately 7 nm, has a very low filtration rate.

Solution to Question 17:

The respiratory quotient is 0.8.

The respiratory quotient is the ratio of carbon dioxide production to oxygen utilization.

In the given question, the CO₂ production is 200mL/min, while the O₂ consumption is 250mL/min.

Therefore,

$$\text{Respiratory quotient} = 200/250 = 0.8$$

The respiratory quotient (RQ) is used for determining the metabolic utilization of various foods.

- When carbohydrates are metabolized with one molecule of oxygen, one carbon dioxide molecule is released. Therefore, the RQ of carbohydrates = 1.
- When fats are oxidized, for every 100 molecules of oxygen consumed, 70 molecules of CO₂ are released. Therefore, the RQ of fats is 0.70.
- Similarly, the respiratory quotient for proteins is 0.80.

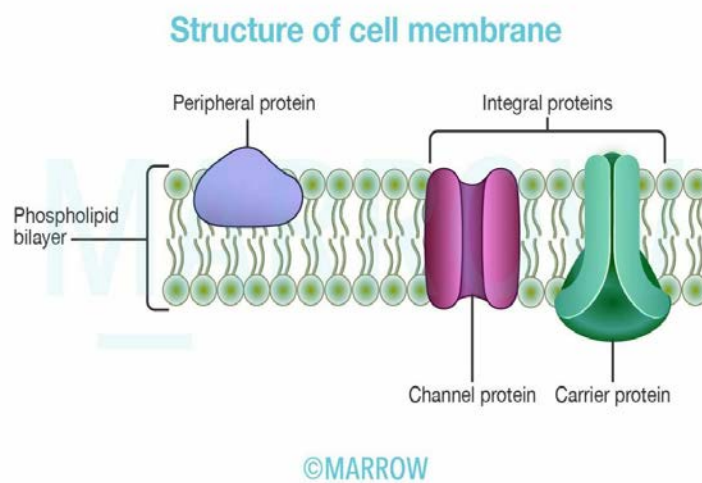
Solution to Question 18:

The transmembranous part of the membrane receptor consists of hydrophobic amino acids.

A membrane protein can be either integral protein (spans the thickness of the cell membrane) or peripheral protein (attached to the peripheral side of the membrane). The membrane-spanning segment of integral proteins are usually hydrophobic alpha-helices.

The central core of the cell membrane is hydrophobic. Integral proteins have their hydrophobic portions usually in the interior of the membrane and the hydrophilic portions towards the surfaces. The presence of these hydrophobic amino acids in the interior is a special adaptation

that allows the protein to remain stable in the hydrophobic milieu of the cell membrane.



Solution to Question 19:

Parathyroid adenoma is a benign tumor of the parathyroid gland, which causes primary hyperparathyroidism (not secondary).

Secondary hyperparathyroidism is characterized by high levels of parathyroid hormone, to compensate for the low levels of serum calcium, due to some other underlying condition.

Causes of secondary hyperparathyroidism, include:

- Chronic kidney disease: The damaged kidneys are unable to produce sufficient amounts of the active form of vitamin D, 1,25-dihydroxycholecalciferol. This further causes hypocalcemia and triggers PTH secretion.
- Vitamin D deficiency: Causes hypocalcemia, which triggers the increased secretion of PTH.
- Lithium: May cause an increased set-point of calcium for parathyroid hormone suppression, which leads to parathyroid hyperplasia.

Solution to Question 20:

There is a decreased firing from the baroreceptors.

- When a person stands up from a lying down position, it causes pooling of the blood in the legs, due to gravity. This reduces the venous return and causes a decrease in cardiac output.
- Reduced cardiac output, leads to a fall in blood pressure, which in turn causes reduced firing from baroreceptors.
- As a result, the vagal tone is suppressed leading to increased heart rate.

Solution to Question 21:

During the contraction of the skeletal muscle, the Ca^{2+} is released from the sarcoplasmic reticulum. Only statement 3 is correct.

Excitation contraction coupling refers to the process by which the membrane depolarization of the muscle fibre (excitation) leads to the contraction of the muscle.

The steps of excitation-contraction coupling in skeletal muscle are as follows:

- Motor-neuron discharge causes the release of acetylcholine at the motor endplate, which binds to the nicotinic acetylcholine receptors.
- This causes increased Na^{+} and K^{+} conductance in the endplate membrane, leading to the generation of end-plate potential, which is a local potential.
- When the local endplate potential reaches a threshold for the muscle fibre, and an action potential is generated.
- The generated action potential passes inward through the T tubules, leading to the spread of depolarisation.
- The depolarization of the muscle fibre causes the release of Ca^{2+} from terminal cisterns of sarcoplasmic reticulum and diffusion to thick and thin filaments.
- The calcium released, binds to troponin C, uncovering myosin-binding sites on actin.
- The formation of cross-linkages between actin and myosin and sliding of thin on thick filaments causes contraction of the skeletal muscle.

Steps in relaxation:

- The cytoplasmic calcium is pumped back into the sarcoplasmic reticulum, by the action of SERCA (sarcoplasmic endoplasmic reticulum calcium ATPase).
- Once the calcium concentration is lowered, the interaction between actin and myosin ceases, and the muscle relaxes.

Other options:

Statement 1: Incorrect - Extracellular calcium levels play an important role in muscle contraction. When there is a decrease in extracellular calcium as in hypocalcemia, the voltage-gated sodium channels are activated, which causes sodium influx. The increased intracellular sodium concentration causes sustained depolarization, which may lead to tetany.

Statement 2: Incorrect - Ca^{2+} is not released from mitochondria. It is released from the sarcoplasmic reticulum.

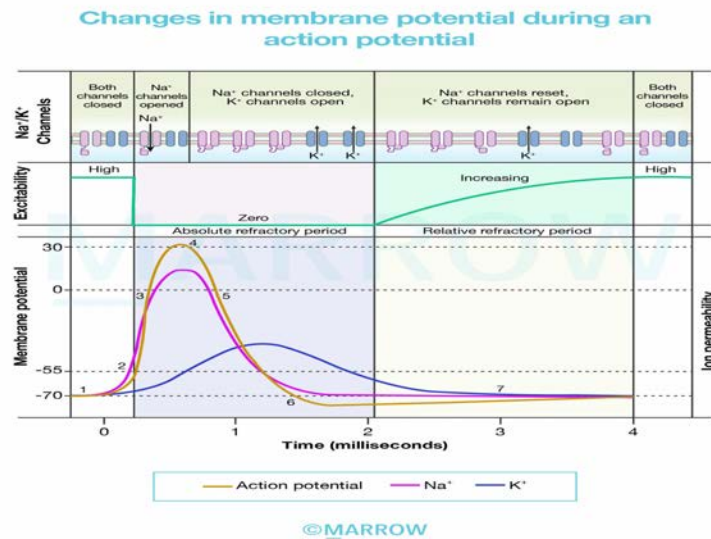
Statement 4: Voltage-gated calcium channels on T-tubules (part of sarcolemma) are activated by a depolarizing stimulus, which causes the release of calcium from the sarcoplasmic reticulum. The increased calcium concentration will eventually lead to muscle contraction.

Solution to Question 22:

Statements 1, 3, and 5 are correct.

Action potential in skeletal muscle:

- Motor-neuron discharge causes end-plate potential, which when reaches to a threshold value, causes action potential in the muscle fibre.
- The depolarization of the muscle fibre is caused by the activation of the voltage-gated sodium channels, which causes the influx of sodium.
- During repolarization, the sodium channels are closed, and voltage-gated potassium channels open, which causes efflux of potassium.
- The voltage-gated potassium channels close slowly. As a result, the potassium efflux continues, leading to hyperpolarization of the membrane.



The steps in an action potential in muscle fibre and the neuron, are similar except for the quantitative differences, such as:

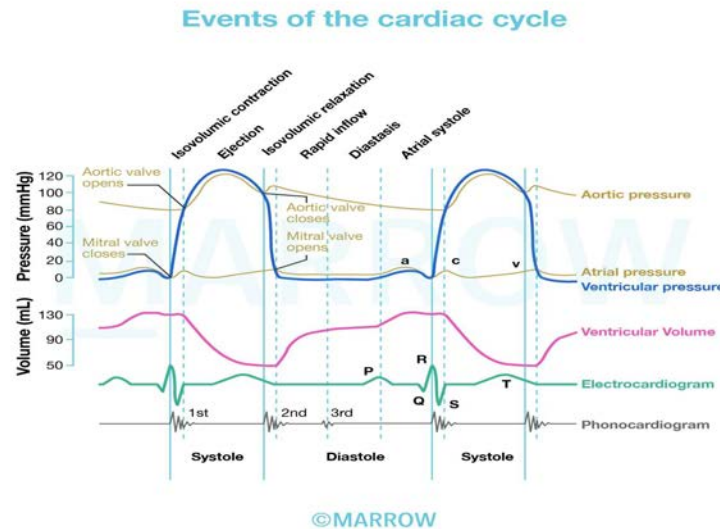
- RMP is -80 to -90 mV in skeletal muscle fibres, while it is about -70mV in neurons.
- The duration of the action potential in skeletal muscle is 1 to 5 msec, which is about 5 times longer than in the largest myelinated nerve fibre.
- The velocity of conduction is about 3-5m/s, which is about 1/13th of the velocity of the nerve fibre.

Solution to Question 23:

The correct sequence of events that occur after the p-wave is 1-2-4-3, ie:-

a wave in JVP → First heart sound → T wave in ECG → Rapid filling of ventricles.

The sequence of the events in this question can be better understood with the following diagram:



- The p wave in the ECG represents atrial depolarization, which causes contraction of the atria, represented by the 'a' wave in the JVP. Atrial contraction causes final filling of the ventricles (fourth phase of ventricular diastole)
- Atrial contraction is followed by the closure of the mitral valve, which causes the first-heart sound.
- This is followed by the isovolumetric contraction of the ventricle, which causes c wave in JVP, due to bulging of the tricuspid valve. This is followed by the rapid ejection phase of the ventricle.
- The T wave of ECG occurs during ventricular repolarization, which is followed by the opening of the mitral valve, followed by rapid filling phase of the ventricle (second phase of ventricular diastole).

Note: Passive rapid filling occurs as the mitral valve opens just before atrial contraction occurs.

Solution to Question 24:

In COPD, the functional residual capacity increases. Total lung capacity may be increased or normal. Hence, both 1 and 2 are correct.

- In obstructive lung diseases, both FEV₁ and FVC reduces, where the decrease in FEV₁ is more than the reduction in FVC. As a result, the ratio of FEV₁/FVC also decreases.
- The total lung capacity is usually normal in chronic bronchitis, while increased in emphysema.
- Functional Residual Capacity (FRC) is the volume of air present in the lungs at the end of passive expiration. In COPD, FRC is increased.

Lung Capacities:

The total lung capacity can be broken down into alternative capacities that help to define functioning lungs.

- The vital lung capacity refers to the maximum amount of air expired from the fully inflated lung or maximum inspiratory level.
- The inspiratory capacity is the maximum amount of air inspired from the end-expiratory level.
- The functional residual capacity represents the volume of the air remaining in the lungs after expiration of a normal breath. Functional Residual Capacity (FRC) is expiratory reserve volume + residual volume.

Physiology NEET 2018

Question 1:

The predominant organ producing insulin-like growth factor I is

- a) Liver
- b) Pituitary gland
- c) Pancreas
- d) Adrenal glands

Question 2:

What is Prosopagnosia?

- a) Impairment of consciousness
- b) Being unaware of one's problems
- c) Difficulty in identifying known faces
- d) Failure to identify objects

Question 3:

All of the following are components responsible for the countercurrent mechanism in the kidney except _____.

- a) Sodium outflow in thick ascending limb
- b) Water outflow in thin descending limb
- c) Sodium outflow in thin ascending limb
- d) Flow of tubular fluid from PCT to DCT

Question 4:

Glucose is absorbed in intestine by?

- a) Primary active transport
- b) Facilitated diffusion

- c) Simple diffusion
- d) Secondary active transport

Question 5:

In which of the following conditions is oxygen therapy not useful?

- a) Asthma
- b) Acute myocardial infarction
- c) Pulmonary edema
- d) Central respiratory depression

Question 6:

$V_a/Q = \text{infinity}$ means _____.

- a) Partial pressure of O_2 becomes zero
- b) No exchange of O_2 & CO_2
- c) CO_2 alone equilibrates with the venous blood
- d) Partial pressure of both CO_2 and O_2 remain normal

Question 7:

'c' wave in JVP is seen in

- a) Iso-volumetric contraction
- b) Slow filling at end of diastole
- c) End of systole
- d) Start of diastole

Question 8:

All of the following are true regarding the actions of endothelin-1 except _____

- a) Bronchodilation
- b) Vasoconstriction
- c) Decreased GFR

- d) Has inotropic effect

Question 9:

Aldosterone secretion is maximally stimulated by_____

- a) Hyperkalemia
- b) ACTH
- c) Hypernatremia
- d) Exogenous steroid

Question 10:

cGMP is the second messenger of

- a) Somatostatin
- b) Angiotensin II
- c) ADH
- d) NO (nitric oxide)

Question 11:

Transport molecule for Fe^{2+} into the enterocytes is

- a) Ferroportin 1
- b) Hephaestin
- c) Divalent Metal Transporter 1
- d) Divalent Metal Transporter 2

Question 12:

Nuclear signalling pathways are all except

- a) Caveolins
- b) NF-kB
- c) Importins
- d) Rat proteins

Question 13:

Which of the following statements is false about Cuthbertson's metabolic ebb and flow responses to trauma?

- a) The "ebb" phase is associated with a decline in body temperature and oxygen consumption.
- b) The "flow" phase occurs after resuscitation from a state of shock.
- c) The "ebb" phase is aimed at reducing posttraumatic energy depletion.
- d) The "ebb" phase leads to activation of the innate immune system and induction of the hepatic acute-phase response.

Question 14:

Kidney responds to shock by _____.

- a) Increasing renal blood flow
- b) Increasing afferent arteriole resistance
- c) Increasing GFR
- d) Both B and C

Answer Key

Question No.	Correct Option
1	a
2	c
3	c
4	d
5	d
6	b
7	a
8	a
9	a
10	d
11	c

12	d
13	d
14	b

Detailed Explanations

Solution to Question 1:

Growth hormone acts on the liver to form small proteins called “somatomedins”

- Somatomedins increases bone growth in all aspect
- Effects are similar to insulin, hence referred to as insulin-like growth factor (IGF).
- Four types available:
- Most important is somatomedin C, specifically referred to as “insulin-like growth factor-1 (IGF-1)
- Binds to carrier protein in blood
- Hence, longer duration of action than growth hormone.
- Half-life - about 20 Hrs (compared to growth hormone - 20 mins)
- Blood concentration of IGF-1, follows the levels of growth hormone.
- Growth effects of GH are mostly attributed to somatomedin (rather than its direct effect on bones & peripheral tissues)

Solution to Question 2:

Prosopagnosia is difficulty in identifying known faces.

In humans, storage and recognition of faces is more strongly represented in the right inferior temporal lobe in right-handed individuals, though the left lobe is also active. Damage to this area can cause prosopagnosia.

Patients with this abnormality can recognize forms and reproduce them. They can recognize people by their voices, and many of them show autonomic responses when they see familiar as opposed to unfamiliar faces. However, they cannot identify the familiar faces they see. The left hemisphere is also involved, but the role of the right hemisphere is primary.

The presence of autonomic response to a familiar face in the absence of recognition has been explained by postulating the existence of a separate dorsal pathway for processing information about faces that leads to recognition at only a subconscious level.

Note:

- Autotopagnosia - inability to recognize or to orient any part of one's own body
- Simultagnosia - inability to recognize multiple elements in a visual presentation, one object or some elements of a scene can be appreciated but not the display as a whole

- Somatagnosia - inability to identify any part of the body, either one's own or another's body
- Tactile agnosia (astereognosis) - inability to recognize objects by touch, in the presence of intact cutaneous and proprioceptive hand sensation; caused by a lesion in the contralateral parietal lobe

Solution to Question 3:

Thin ascending limb is not a component of the countercurrent mechanism.

The loop of Henle function as countercurrent multipliers and vasa recta as countercurrent exchangers.

The functioning of the loop of Henle depends upon:

- High permeability of the thin descending limb to water (via aquaporin-1)
- Active transport of Na⁺ and Cl⁻ from the thick ascending limb
- Inflow of tubular fluid from proximal tubule and outflow to the distal tubule.

Note: The thin ascending limbs of juxtamedullary nephrons are permeable to Na⁺ and Cl⁻ and hence do contribute passively to countercurrent multiplication. However, among the options provided, Option C is the best answer as the other options are directly responsible for the countercurrent mechanism.

Solution to Question 4:

Glucose is absorbed in the intestine by secondary active transport (sodium co-transport).

The energy for glucose transport is provided indirectly, by the active transport of Na⁺ out of the cell. This maintains the concentration gradient across the luminal border of the cell so that more Na⁺ and consequently more glucose enter.

Though both secondary active transport and facilitated diffusion play a role in glucose absorption, it is the initial active transport of sodium through the basolateral membranes of the intestinal epithelial cells that provide the eventual force for moving glucose through the membranes as well. So, secondary active transport is the single best answer for this question.

Solution to Question 5:

Oxygen therapy is not beneficial in central respiratory depression.

High dose oxygen reduces the output of peripheral chemoreceptors. So, if given to patients who already have central respiratory depression, it will indirectly reduce the hypoxic drive to breathe even further, leading to a reduction in respiratory frequency and minute ventilation.

Note: In acute severe asthma, a high concentration of oxygen should be given by face mask to achieve an oxygen saturation of >90%.

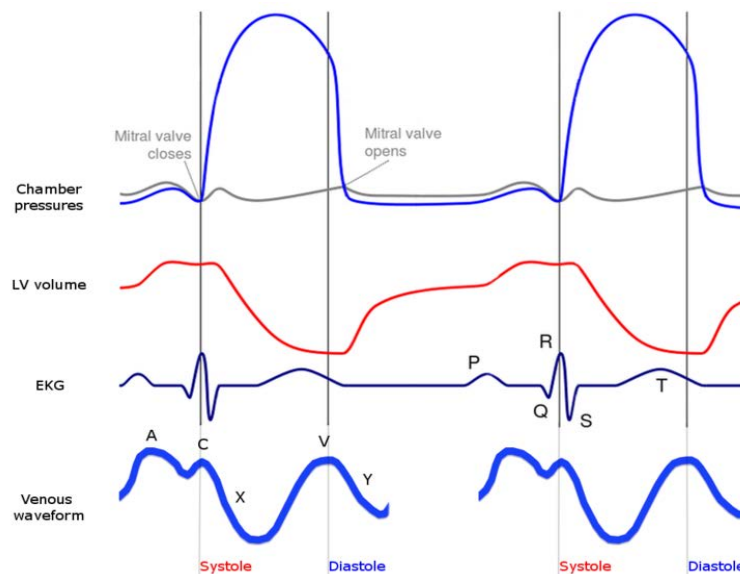
Solution to Question 6:

$V_a / Q = \text{infinity}$ means No exchange of O_2 & CO_2 .

- V_a represents the ventilation in alveoli.
- Q represents the blood flow through the alveolus.
- The ratio of V_a and Q explains the respiratory exchange when there is an imbalance between alveolar ventilation and alveolar blood flow.
- Ventilation-perfusion ratio is considered normal if there is an equal amount of blood flow & ventilation through alveolus.
- When V_a / Q ratio is infinity,
- There is adequate ventilation but no perfusion.
- No exchange of gases through the respiratory membranes of corresponding alveoli.
- The alveolar air equilibrates the humidified inspired air
- No exchange of oxygen and carbon-di-oxide occurs.
- Partial pressures of the O_2 and CO_2 are 149 mmHg & 0 mm Hg respectively

Solution to Question 7:

'c' waves are produced by bulging of the tricuspid valve into the right atrium during the Iso-volumetric contraction of the right ventricle.



Solution to Question 8:

Endothelin-1

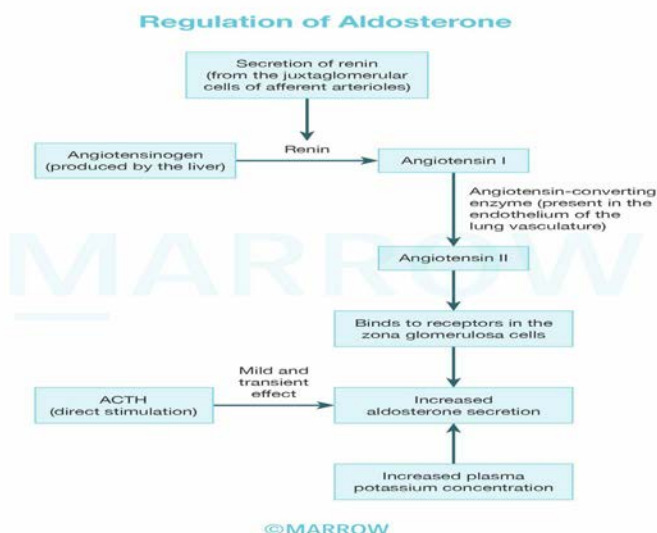
- Potent vasoconstrictor agent.
- Produced by endothelial cells.
- They are secreted locally and act in a paracrine fashion.

Actions of Endothelin-1 (ET-1)

- Potent vasoconstrictor.
- Platelet activation.
- Inotropic effect.
- Closure of ductus arteriosus.
- Have a role in the pathogenesis of hypertension, myocardial infarction, and congestive heart failure.
- In the brain, regulate the transport of substances across blood brain barrier (BBB).
- In the kidneys, they are involved in tubuloglomerular feedback → ↓ GFR.
- In the lungs, it causes bronchoconstriction (mediated via ET-B receptors).
- In GIT, they are involved in the migration of myenteric plexus to distal colon (failure leads to Hirschsprung's/megacolon).
- Deletion of both alleles of ET-1 gene → craniofacial abnormalities and respiratory failure at birth.

Solution to Question 9:

Hyperkalemia and activation of renin-angiotensin system are the most potent stimulators of aldosterone secretion.



Solution to Question 10:

Vasodilatory actions of NO are mediated by cGMP.

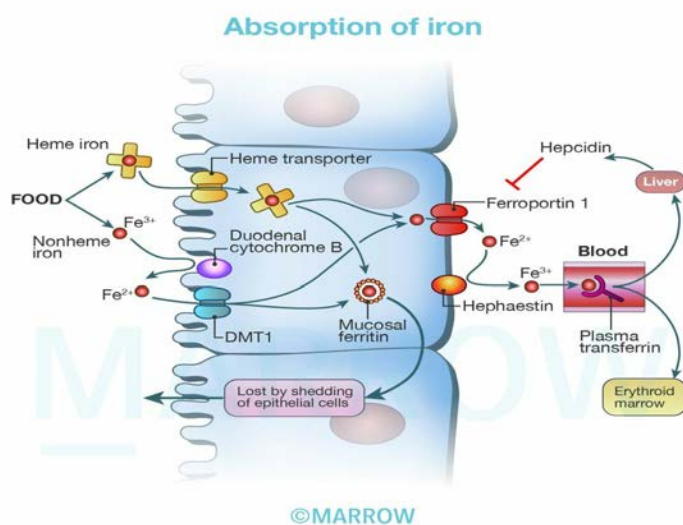
NO is a hormone derived from the amino acid arginine. It activates guanylyl cyclases in vascular smooth muscle cells resulting in the conversion of cGTP to cGMP (2nd messenger). cGMP activates cGMP dependent protein kinase, which causes blood vessels to relax.

cGMP is the second messenger for nitric oxide and atrial natriuretic factor.

Solution to Question 11:

Transport of Fe^{2+} into the enterocytes occurs via divalent metal transporter 1 (DMT1), through an active transport process.

Protein DMT-1 (Divalent Metal Transporter-1) is expressed on the apical surface of enterocytes in the initial part of the duodenum. DMT-1 is not specific to iron & transports other substances like zinc, copper, cobalt & manganese.



Solution to Question 12:

Rat proteins are not nuclear signalling pathways. Recombinant rat IFN- is a bioactive protein intended for use in cell culture applications.

Importin:

- Transporting protein molecules into nucleus by binding to specific recognition sequences, called Nuclear Localization Sequences (NLS).

- 2 subunits, importin α and importin β .

Caveolins:

- Family of integral membrane proteins that are principal components of caveolae membranes
- Involved in receptor-independent endocytosis.
- Acts as scaffolding proteins within caveolar membranes by compartmentalizing and concentrating signaling molecules.

NF- κ B

- Part of a prototypical proinflammatory signaling pathway
- based on the role of NF- κ B in the expression of proinflammatory genes including cytokines, chemokines, and adhesion molecules

Solution to Question 13:

The "flow" phase (not "ebb") leads to activation of the innate immune system and induction of the hepatic acute-phase response.

Cuthbertson's ebb and flow response to trauma:

The "ebb" phase lasts for a brief duration and is associated with a decline in body temperature and oxygen consumption, aimed at reducing post traumatic energy depletion.

The "flow" phase occurs after resuscitation from a state of shock, which leads to an increased metabolic turnover, activation of the innate immune system and induction of the hepatic acute-phase response. This hypercatabolic condition leads to a significantly increased oxygen consumption and energy expenditure.

Solution to Question 14:

Kidney responds to shock by increasing afferent arteriolar resistance.

During shock, both afferent and efferent arterioles in kidney are constricted, however, the efferent vessels are constricted much more.

The glomerular filtration rate and the plasma renal flow is decreased during shock. As plasma renal flow is decreased to a greater extent, the filtration fraction increases. (Option 1 & 3)

Physiology NEET 2019

Question 1:

According to myogenic hypothesis of renal autoregulation the afferent arterioles contract in response to stretch-induced by_____

- a) NO
- b) Noradrenaline
- c) Opening of calcium channels
- d) Adenosine

Question 2:

According to Herrington classification, the decerebrate rigidity is characterized by all except _____

- a) Rigidity occurs in all the muscles of the body
- b) Increase in the rate of discharge of the ■ efferent neuron
- c) Increased excitability of the motor neuron pool
- d) Decerebration produces no phenomenon akin to spinal shock

Question 3:

Which of the following is true for decorticate rigidity?

- a) It is produced by the removal of cerebral cortex and basal ganglia
- b) Flexion of lower limbs and extension of upper limbs occurs
- c) Rigidity is pronounced
- d) It is characterized by flexion of upper limbs and extension of lower limbs

Question 4:

Acid Citrate Dextrose (ACD), used to store blood, preserves the RBCs for a period of 21 days. The storage period when adding phosphate alone and when adding adenine and phosphate together to ACD would respectively be?

- a) 35 days and 42 days
- b) 21 days and 35 days
- c) 28 days and 45 days
- d) 35 days and 28 days

Question 5:

Functional residual capacity is:

- a) Volume expired after normal inspiration
- b) Volume that remains after forced expiration
- c) Expiratory reserve volume + residual volume
- d) Tidal volume + inspiratory reserve volume

Question 6:

Which of the following is not a dietary fiber?

- a) Cellulose
- b) Inulin
- c) Pectin
- d) Gums

Question 7:

Lesion in the medial lemniscus at the level of pons causes which of the following?

- a) Pain and temperature loss in the same side
- b) Pain and temperature loss in the opposite side
- c) Loss of tactile and proprioception on the same side
- d) Loss of tactile and proprioception on the opposite side

Question 8:

Which neurotransmitter is responsible for anterograde dilatation of the small intestine?

- a) VIP

- b) Acetylcholine
- c) Substance P
- d) Serotonin

Question 9:

Reflex responsible for tachycardia during right atrial distension is:

- a) J reflex
- b) Cushing reflex
- c) Bezold jarisch reflex
- d) Bainbridge reflex

Question 10:

Testosterone production is mainly contributed by _____

- a) Leydig cells
- b) Sertoli cells
- c) Seminiferous tubules
- d) Epidydmis

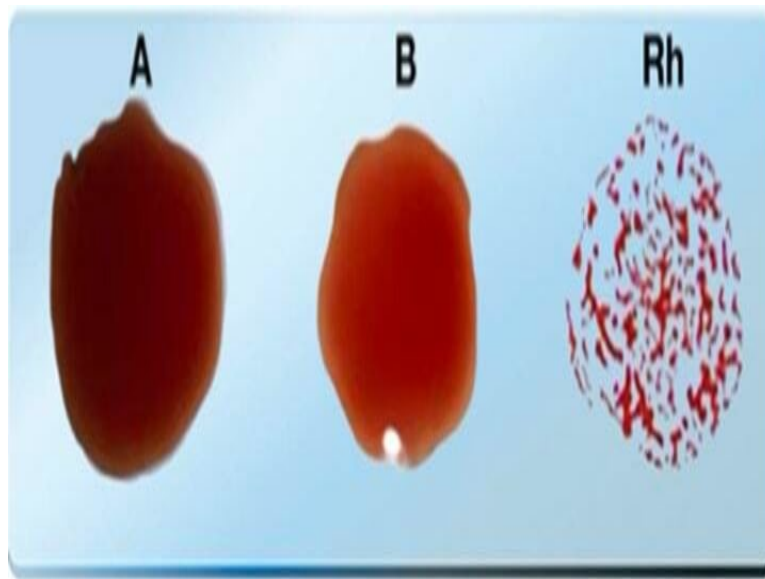
Question 11:

What is the function of tubuloglomerular feedback?

- a) Regulate GFR
- b) Decrease plasma sodium
- c) Increase plasma volume
- d) Decrease tubular secretion

Question 12:

Based on the agglutination test shown below, what is the blood group?



- a) B positive
- b) A positive
- c) O positive
- d) AB positive

Question 13:

Which of the following neurons predominantly fire during forceful expiration?

- a) Dorsal group of neurons
- b) Ventral group of neurons
- c) Pneumotaxic center
- d) Chemoreceptors

Question 14:

ADH acts through which aquaporin channel?

- a) 1
- b) 2
- c) 3
- d) 4

Question 15:

Which of the following has the least conduction velocity?

- a) Postganglionic sympathetic fibers
- b) Somatic motor fibers
- c) Pre-ganglionic autonomic fibers
- d) Touch and pressure fibers

Question 16:

Which part of the liver stores vitamin A?

- a) Ito cells
- b) Hepatocytes
- c) Endothelial cells
- d) Kupffer cells

Question 17:

Liver synthesizes all of the following except:

- a) Albumin
- b) Coagulation factors
- c) Acute phase proteins
- d) Immunoglobulins

Question 18:

Which of the following ions is involved in peripheral oxygen sensing chemoreceptors?

- a) Potassium
- b) Calcium
- c) Sodium
- d) Chlorine

Question 19:

Brown sequard syndrome is characterized by _____

- a) Fine touch lost on the opposite side
- b) Pain and temperature lost on the same side
- c) Pain and temperature loss on the opposite side
- d) Only pain lost on the same side

Answer Key

Question No.	Correct Option
1	c
2	a
3	d
4	b
5	c
6	d
7	d
8	a
9	d
10	a
11	a
12	c
13	b
14	b
15	a
16	a
17	d
18	a
19	c

Detailed Explanations

Solution to Question 1:

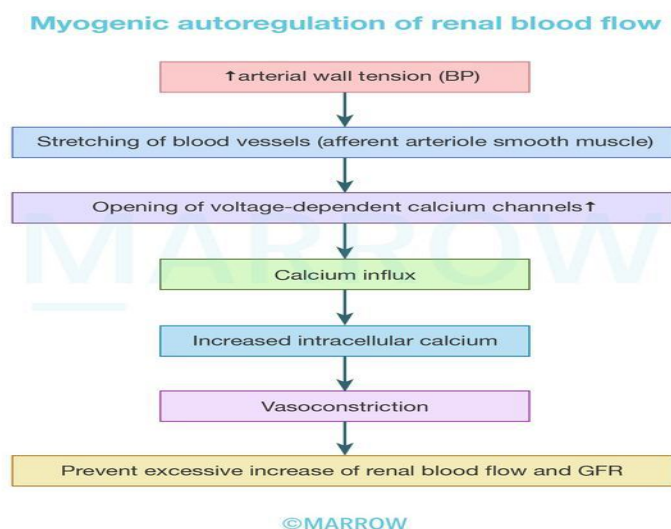
The afferent arterioles contract in response to stretch induced by the opening of calcium channels. The mechanism is called myogenic autoregulation of renal blood flow and GFR.

The stretch of the vascular wall allows increased movement of calcium ions from the extracellular fluid into the cells, causing them to contract. This contraction prevents excessive stretch of the vessel. This prevents excessive increase of renal blood flow and GFR by raising vascular resistance.

This mechanism may be more important in protecting the kidney from hypertension-induced injury.

Autoregulation has two main mechanisms:

- Metabolic mechanism: Tissues produce vasodilatory metabolites that regulate flow e.g. prostaglandins, NO, CO₂, H⁺ and K⁺
- Myogenic mechanism



Solution to Question 2:

In decerebrate rigidity, there is rigidity in extensor muscles of all the four extremities (not all muscles of the body).

Decerebration

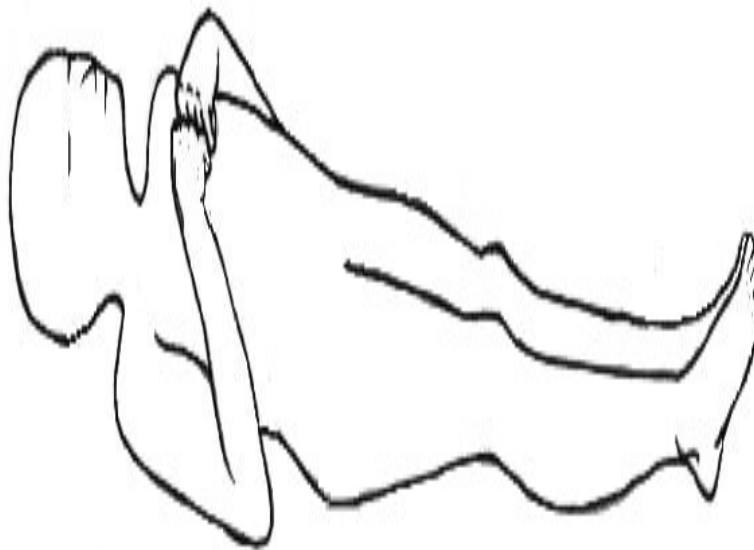
- Complete transection of the brain stem between the superior and inferior colliculi permits the brain stem pathways to function independently of their input from higher brain structures. This is called a mid-collicular decerebration.
- This lesion interrupts all input from the cortex (corticospinal and corticobulbar tracts) and red nucleus (rubrospinal tract), primarily to distal muscles of the extremities.
- The excitatory and inhibitory reticulospinal pathways (primarily to postural extensor muscles) remain intact.

- The dominance of drive from ascending sensory pathways to the excitatory reticulospinal pathway leads to hyperactivity in extensor muscles in all four extremities which are called decerebrate rigidity.
- The excitatory input from the reticulospinal pathway activates γ -motor neurons which indirectly activate α -motor neurons (via Ia spindle afferent activity). This is called the gamma loop.

Option 4. Spinal shock is different from decerebrate rigidity as it is a transient state that can occur immediately after a spinal cord injury. It is characterized by a temporary loss or depression of spinal reflexes below the level of the injury.

Solution to Question 3:

- Removal of the cerebral cortex (decortication) produces decorticate rigidity.
- Characterized by flexion of the upper extremities at the elbow and extensor hyperactivity in the lower extremities.
- The flexion can be explained by active rubrospinal excitation of flexor muscles in the upper extremities; the hyperextension of lower extremities is due to the same changes that occur after midcollicular decerebration.



Solution to Question 4:

The storage period when adding phosphate alone to ACD would be 21 days and when adding adenine and phosphate together would be 35 days.

When adding phosphate to ACD, Citrate Phosphate Dextrose (CPD) is formed – It has a storage period of 21 days. There is no change in the shelf life as phosphate only increases the 2,3 DPG levels of the stored blood.

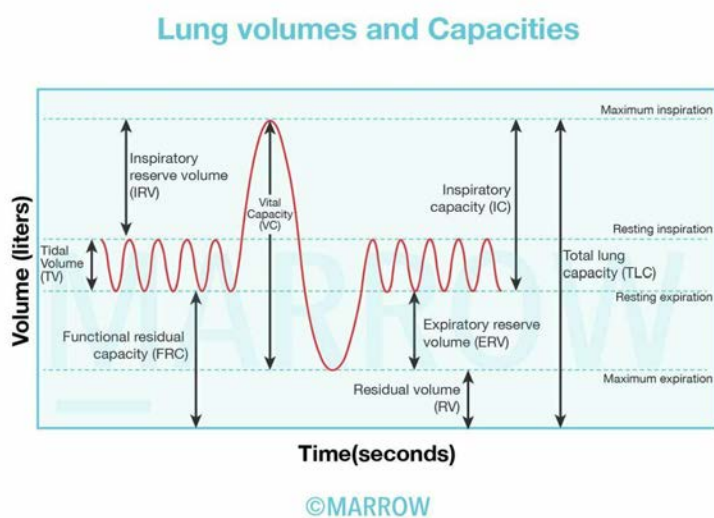
When adding both phosphate and adenine to ACD, Citrate Phosphate Dextrose Adenine (CPDA-1) is formed. It has a storage period of 35 days. It increases the shelf life by improving ATP synthesis of RBCs.

Using additive solutions containing saline, adenine, and glucose, with or without mannitol (SAGM) decreases RBC hemolysis, thus increasing the storage period to 42 days.

Solution to Question 5:

Functional Residual Capacity (FRC) is expiratory reserve volume + residual volume.

From the picture given below, ($FRC = ERV + RV$).



Lung Capacities:

The total lung capacity can be broken down into alternative capacities that help to define functioning lungs.

- The vital lung capacity refers to the maximum amount of air expired from the fully inflated lung or maximum inspiratory level.
- The inspiratory capacity is the maximum amount of air inspired from the end-expiratory level.
- The functional residual capacity represents the volume of the air remaining in the lungs after expiration of a normal breath.

Solution to Question 6:

Gum is the best answer among the given options. Even though all of the given options are dietary fibers, the consumption of gums through diet is very low and hence, it is the best answer.

Dietary fibers consist of both soluble and insoluble forms and include lignin, pectin, inulin, cellulose, hemicellulose, etc.

Advantages of fibers:

- Forms bulk of the stool and reduces stool transit time
- Reduces incidence of carcinoma colon, DM, CAD, diverticulitis

Solution to Question 7:

Lesion in the medial lemniscus at the level of pons causes loss of tactile and proprioception on the opposite side.

Fibers of medial lemniscus ascend ipsilaterally in the dorsal columns of the spinal cord to the medulla, where they synapse in the gracilis and cuneate nuclei.

The second-order neurons from these nuclei cross the midline and ascend in the medial lemniscus to end in the contralateral ventral posterior lateral (VPL) nucleus and related specific sensory relay nuclei of the thalamus.

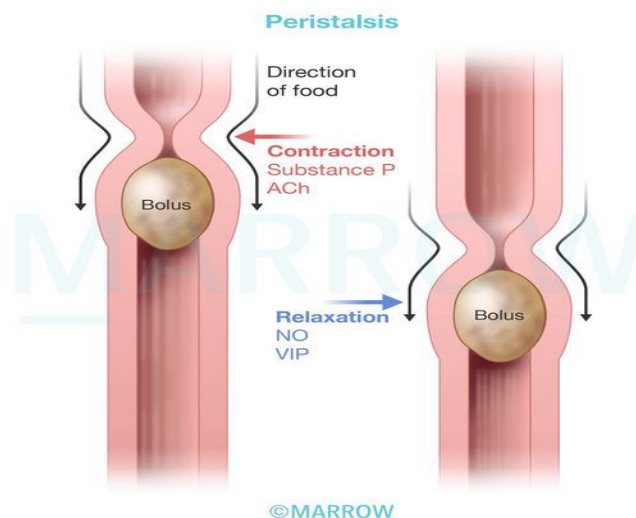
Solution to Question 8:

VIP is responsible for antegrade dilatation of the small intestine.

Peristalsis involves the contraction of a segment of intestine behind the bolus of food and relaxation of a segment of the intestine in front of the bolus of food.

Cholinergic neurons passing in antegrade direction produce NO and VIP which cause relaxation of a segment of the intestine in front of the bolus.

Cholinergic neurons passing in retrograde direction produce substance P and acetylcholine which cause contraction of a segment of the intestine behind the bolus.



Note: In achalasia cardia, there is decreased release of NO and VIP in the distal part of the esophagus.

Solution to Question 9:

Reflex responsible for tachycardia during right atrial distension is Bainbridge reflex.

Atrial Reflex Control of Heart Rate: The Bainbridge Reflex

- The stretch receptors of the atria that elicit the Bainbridge reflex transmit their afferent signals through the vagus nerves to the medulla of the brain.
- Then efferent signals are transmitted back through vagal and sympathetic nerves to increase heart rate and strength of heart contraction.

Bezold-Jarisch Reflex:

- Activation of chemosensitive vagal C fibres in the cardiopulmonary region (e.g. Juxta capillary region of alveoli, ventricles, atria, great veins, and pulmonary artery) causes profound bradycardia, hypotension and a brief period of apnea followed by rapid shallow breathing.

Cushing Reflex:

- When ICP is increased, the blood supply to RVLM neurons is compromised.
- The local hypoxia and hypercapnia causes an increase in discharge from the neurons, producing hypertension, bradycardia and irregular respiration.
- The resultant rise in systemic arterial pressure tends to restore the blood flow to the medulla.

Solution to Question 10:

Testosterone production is mainly contributed by Leydig cells.

Testosterone is synthesized from cholesterol in the Leydig cells and also from androstenedione secreted by the adrenal cortex.

The diagnosis of testosterone deficiency should be made only on the basis of clinical signs and symptoms consistent with androgen deficiency, and consistently subnormal serum testosterone concentrations at 8 to 10 AM on three occasions.

Testosterone has been used to treat primary or secondary hypogonadism in adult men.

Solution to Question 11:

Tubuloglomerular feedback regulates glomerular filtration rate (GFR).

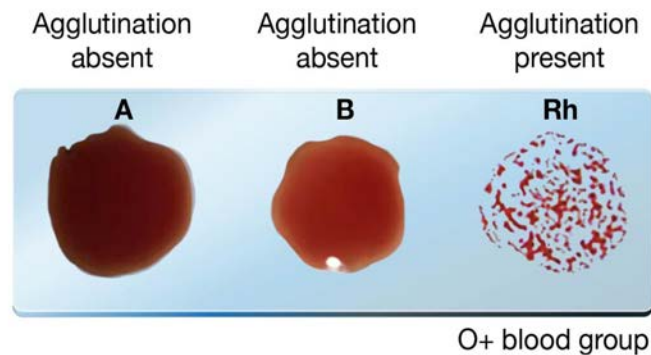
As the rate of flow through the ascending limb of the loop of Henle and first part of the distal tubule increases, glomerular filtration in the same nephron decreases, and, conversely, a decrease

in ■low increases the GFR.

Solution to Question 12:

Based on the agglutination test shown in the image, the blood group is O positive.

As agglutination is present only in 'Rh' and not with 'A' and 'B' hence the blood group is O Rh positive.



©MARROW

Solution to Question 13:

Ventral groups of neurons fire, when forceful expiration is needed like during heavy exercise.

Dorsal group of neurons control the normal inspiration and expiration.

Pneumotaxic center limits the duration of inspiration.

Solution to Question 14:

The antidiuretic hormone acts through aquaporin channel 2.

ADH binds to V2 receptors in the late distal tubules, collecting tubules, and collecting ducts, and stimulates the movement of an intracellular protein, called aquaporin-2 (AQP-2), to the luminal side of the cell membranes.

Solution to Question 15:

Postganglionic sympathetic (type C) has the least conduction velocity among the given options.

Types of nerve fibers:

Fiber type	Function	Fiber diameter (μ m)	Conduction velocity (m/s)
A α	Proprioception, somatic motor	12-20	70-120
A β	Touch, pressure	5-12	30-70
A γ	Motor to muscle spindles	3-6	15-30
A δ	Pain, temperature	2-5	12-30
B	Preganglionic autonomic	<3	3-15
C, Dorsal root	Pain, temperature	0.4-1.2	0.5-2
C, Sympathetic	Post ganglionic sympathetic	0.3-1.3	0.7-2.3

Solution to Question 16:

Vitamin A is stored in the liver perisinusoidal stellate (Ito) cells as retinyl ester (retinol palmitate).

Kupffer cells, also known as stellate macrophages and Kupffer-Borowicz cells; are specialized macrophages located in the liver that form part of the mononuclear phagocyte system. These are present, lining the walls of the sinusoids, not the perisinusoidal space (Space of Disse).

Solution to Question 17:

Immunoglobulins are not synthesized by the liver. They are produced by plasma cells.

Plasma proteins synthesized by the liver are:

- Albumin
- Coagulation/clotting factors
- Acute-phase proteins
- Steroid binding and other hormone-binding proteins

Solution to Question 18:

Potassium is involved in peripheral oxygen sensing chemoreceptors.

Type I glomus cells (carotid and aortic body) have O₂-sensitive K⁺ channels, whose conductance is reduced in proportion to the degree of hypoxia.

This reduces the K^+ efflux, depolarizing the cell and causing Ca^{2+} influx.

The Ca^{2+} influx triggers action potentials and consequent excitation of the afferent nerve endings.

Solution to Question 19:

Hemisection of the spinal cord causes damage to ascending sensory (dorsal column pathway, ventrolateral spinothalamic tract) and descending motor (corticospinal tract) pathways, which is called the Brown- Séquard syndrome.

The lesion to fasciculus gracilis or fasciculus cuneatus leads to ipsilateral loss of discriminative touch, vibration, and proprioception below the level of the lesion. (fibers from the sacral cord are positioned most medially and those from the cervical cord are positioned most laterally.)

The loss of the spinothalamic tract leads to contralateral loss of pain and temperature sensation beginning one or two segments below the lesion.

Damage to the corticospinal tract produces weakness and spasticity in certain muscle groups on the same side of the body.

Physiology NEET 2020

Question 1:

A person develops headache and breathlessness on a trekking expedition following rapid ascent to over 3000 metres above the sea level. Which of the following is not used in the treatment of this condition?

- a) IV digoxin
- b) Descent to lower altitude
- c) Administration of oxygen
- d) Tablet acetazolamide

Question 2:

$V/Q = \text{infinity}$ holds true in which of the following conditions?

- a) Foreign body obstruction in the bronchus
- b) Equal to dead space
- c) When O_2 and CO_2 ratio is equal
- d) The PO_2 of alveolar air is 159 mmHg and PCO_2 is 40 mmHg

Question 3:

Prolactin level is highest during :

- a) 24 hours after ovulation
- b) 24 hours after delivery
- c) REM Sleep
- d) After running for 1 hour

Question 4:

C wave of JVP represents:

- a) Tricuspid bulge into right atrium during isovolumetric ventricular contraction
- b) Contraction of right atrium

- c) Passive filling of right atrium
- d) Passive emptying of right atrium

Question 5:

The protein in the glomerular basement membrane responsible for charge-dependent filtration is?

- a) Fibronectin
- b) Heparan sulfate
- c) Collagen Type IV
- d) Entactin

Question 6:

Which of the following clotting factors would have decreased gamma carboxyglutamate residue in a patient on warfarin therapy?

- a) Factor 2
- b) Factor 11
- c) Tissue factor
- d) Factor 5

Question 7:

Which of the following conditions would have the highest levels of ACTH and cortisol?

- a) Early morning
- b) Normal individuals with dexamethasone administration
- c) Addison's disease
- d) Cushing's disease

Question 8:

Following complete ileal and partial jejunal resection, the patient is most likely to have

- a) Folic acid deficiency

- b) Vitamin B12 deficiency
- c) Gastric ulcer
- d) Constipation

Question 9:

Respiratory distress syndrome is due to defective

- a) Type 1 pneumocytes
- b) Type 2 pneumocytes
- c) Alveolar capillary endothelial cells
- d) Bronchial mucosal epithelial cells

Question 10:

Which of the following hormones use receptor tyrosine kinase signalling?

- a) Insulin
- b) TSH
- c) TRH
- d) MSH

Question 11:

What is the site of lesion in Wernicke's hemianopic pupil?

- a) Optic nerve
- b) Optic tract
- c) Optic radiation
- d) Optic chiasma

Question 12:

In which of the following reflexes, gastric emptying is delayed when acidic content enters the duodenum?

- a) Gastrocolic

- b) Enterocolic
- c) Gastroileal
- d) Enterogastric

Question 13:

A lady presents with vascular injury to the inferior frontal gyrus. Which functional area is likely to be affected?

- a) Motor speech area
- b) Wernicke's area
- c) Auditory area
- d) Visual area

Question 14:

A man falls into a deep sleep with one arm under his head. He finds he is not able to move this arm when he awakens. However, it tingles and pain sensation in the arm is still intact. What is the reason for the loss of motor function without loss of pain sensation is that in the nerves to his arm?

- a) B fibers are more susceptible to pressure than A fibers
- b) A fibers are more sensitive to pressure than C fibers
- c) C fibers are more sensitive to pressure than A fibers
- d) Motor nerves are more affected by sleep than sensory nerves

Question 15:

Which of the following has the highest oxygen concentration in fetal circulation?

- a) Inferior vena cava
- b) Right atrium
- c) Superior vena cava
- d) Ascending aorta

Question 16:

Respiratory distress syndrome is due to the deficiency of

- a) Dipalmitoyl Phosphatidylcholine
- b) Phosphatidylserine
- c) Phosphatidylethanolamine
- d) Phosphatidylinositol

Question 17:

What makes CPDA superior than ACD for blood storage:

- a) The fall in 2, 3 DPG is less
- b) Decreased release of O₂
- c) It has less P₅₀
- d) It is less acidic

Answer Key

Question No.	Correct Option
1	a
2	b
3	b
4	a
5	b
6	a
7	d
8	b
9	b
10	a
11	b
12	d
13	a
14	b
15	a
16	a

Detailed Explanations

Solution to Question 1:

The symptoms of headache and breathlessness following rapid ascent indicate acute mountain/altitude sickness and IV digoxin is not used to treat this condition.

On ascent to high altitudes, the low partial pressure of oxygen leads to hypoxia and systemic arteriolar dilatation. Consequently, the capillary pressure rises with increased transudation of fluid into the brain tissue and the following symptoms are seen:

- Headache
- Irritation
- Insomnia
- Breathlessness
- Nausea and vomiting

Treatment involves administration of oxygen, descent to a lower altitude and administration of acetazolamide. Acetazolamide counteracts hypoxia by stimulating respiration and also reduces the formation of cerebrospinal fluid, thereby reducing intracranial tension.

Portable hyperbaric chambers are increasingly being used in the treatment of high-altitude illnesses when rapid descent is not possible. These chambers are available in armed forces hospitals in high altitude areas

Solution to Question 2:

$VA/Q = \infty$ means that there is no exchange of O_2 & CO_2 , which means the volume of the lung equals to dead space and the physiological effort of breathing is wasted.

VA represents the ventilation in alveoli and Q represents the blood flow through the alveolus. The ratio of VA and Q explains the respiratory exchange when there is an imbalance between alveolar ventilation and alveolar blood flow. The ventilation-perfusion ratio is considered normal if there is an equal amount of blood flow & ventilation through the alveolus.

VA/Q ratio is infinity when:

- There is adequate ventilation but no perfusion.
- No exchange of gases (O_2 and CO_2) through the respiratory membranes of corresponding alveoli.
- The alveolar air equilibrates the humidified inspired air.
- Partial pressures of the O_2 and CO_2 are 149 mmHg & 0 mm Hg respectively.

Solution to Question 3:

Among the given options, the prolactin level is highest 24 hours after delivery.

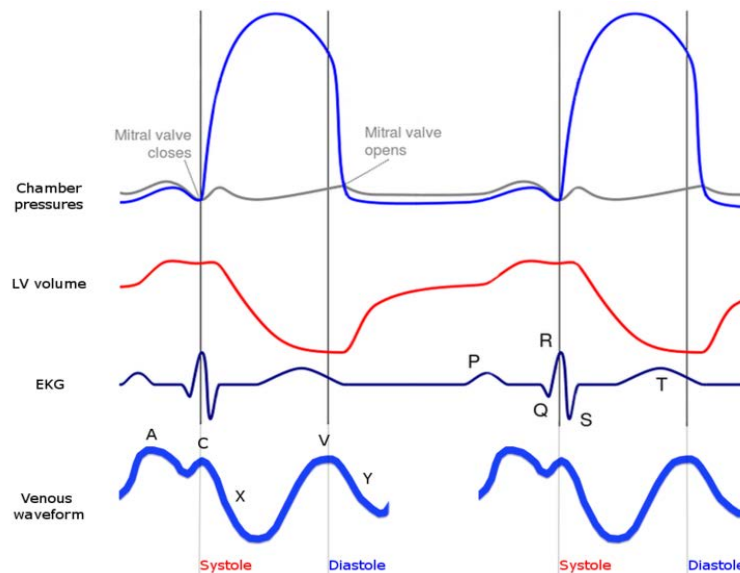
The maximum level of prolactin is seen during pregnancy and in the immediate postpartum period. During pregnancy, the secretion is increased reaching a peak at the time of delivery. After delivery, the plasma concentration falls to nonpregnant levels in a few weeks. Suckling produces surges of prolactin release.

Solution to Question 4:

'C' wave in JVP represents tricuspid bulge into right atrium during isovolumetric ventricular contraction.

Characteristics of JVP :

- a wave corresponds to atrial systole which causes regurgitation of some blood into the great veins.
- c wave occurs due to the increase in atrial pressure caused by bulging of the tricuspid valve into the right atrium during isovolumetric ventricular contraction.
- x descent occurs due to the fall in atrial pressure during ventricular systole due to pulling of valve leaflets.
- v wave is seen at the end of ventricular contraction and occurs due to the slow venous filling which causes an increase in atrial pressure.
- y descent occurs due to the passive emptying of the right atrium into the right ventricle.



Solution to Question 5:

The protein in the glomerular basement membrane responsible for charge-dependent filtration is heparan sulfate. It prevents filtration by repelling negatively charged proteins.

Collagen type IV, fibronectin, entactin are structural components of basement membrane.

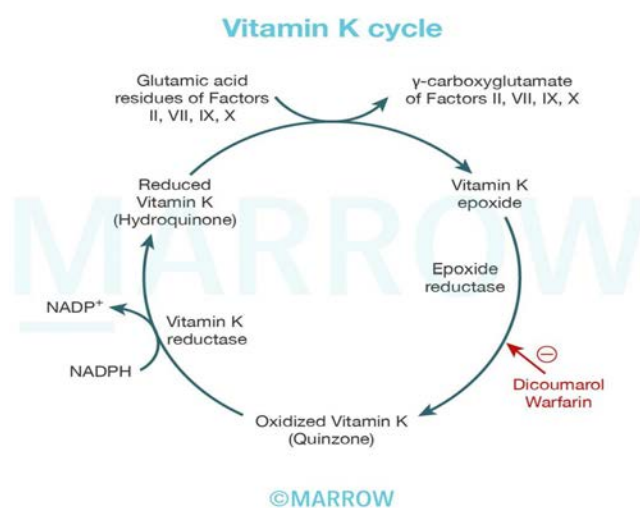
Solution to Question 6:

Factor 2 would have decreased gamma carboxyglutamate residue in a patient on warfarin therapy.

Activation of coagulation factors II, VII, IX, X is done by carboxylation of their glutamate residues. This is done by the enzyme γ -glutamyl carboxylase with the reduced form of vitamin K as a co-factor.

This reduced form of vitamin K needs to be constantly recycled back from its oxidized form for the reaction to continue. This recycling is done by the enzyme vitamin K epoxide reductase. This enzyme is inhibited by warfarin.

Hence, warfarin indirectly inhibits the γ -carboxylation of vitamin K dependent clotting factors - Factors II, VII, IX, X by directly inhibiting vitamin K epoxide reductase. Thus leading to decreased gamma carboxyglutamate residue in these factors.

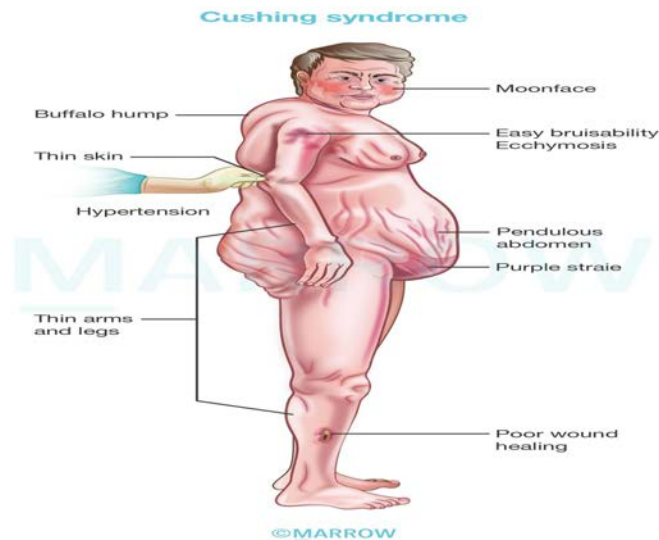


Solution to Question 7:

Among the given conditions, the levels of ACTH and cortisol will be highest in Cushing's disease.

Cushing's disease is caused by an ACTH secreting pituitary adenoma. The increased ACTH causes increased cortisol production.

Note: In normal physiologic state, ACTH levels are higher in the early morning but not as high as in Cushing's disease.



Solution to Question 8:

After complete ileal and partial jejunal resection, the patient is likely to have vitamin B12 deficiency.

Vitamin B12 is absorbed in the ileum. Vitamin B12 binds to intrinsic factor, a protein secreted by the parietal cells of the stomach, and the complex is absorbed across the ileal mucosa. Hence after ileal resection, there is impaired absorption of vitamin B12.

Solution to Question 9:

Respiratory distress syndrome is due to defective type 2 pneumocytes which produce lung surfactants. The surfactants reduce the surface tension in the alveoli. A defect in type 2 pneumocytes or surfactant deficiency can lead to respiratory distress syndrome.

The primary alveoli develop and the production of the surfactant begins by week 20-24. However, it does not reach the lungs until later. The surfactant appears in the amniotic fluid between 28 and 32 weeks of gestation. Mature levels of the pulmonary surfactant are present usually after 35 weeks of gestation.

Antenatal corticosteroids are given to increase production of surfactant.

Solution to Question 10:

Insulin uses the receptor tyrosine kinase signalling to exert its functions.

Other hormones acting in a similar fashion includes

- Growth hormone

- Growth factors like VEGF, PDGF, FGF, NGF
- Prolactin
- Leptin

Solution to Question 11:

The site of lesion in Wernicke's hemianopic pupil is optic tract.

Wernicke hemianopic pupil is the light reflex, where, illumination of the blind half of the retina shows no pupillary reaction while stimulating the normal side of the retina shows a pupillary reaction. It is seen in lesions involving the optic tract. It is also seen in cases such as TB, syphilis and tumours of the optic thalamus.

Homonymous hemianopia (HH) involves loss of vision on the same side of the visual field in both eyes. HH is characteristically seen in lesions involving the visual pathway posterior to the optic chiasma.

Solution to Question 12:

In the enterogastric reflex, gastric emptying is delayed when acidic content enters the duodenum.

When the food enters the duodenum, multiple reflexes are initiated from the duodenal wall which passes back to the stomach. These reflexes slow or stop the stomach emptying if the volume of the chyme in the duodenum becomes too much. These reflexes are called enterogastric reflexes.

This reflex can be triggered due to the following reasons:

- Presence of irritants and acids in duodenal chyme (most sensitive factor)
- Distension of duodenum due to the presence of an excessive volume of chyme
- Rapid entry of hypertonic or hypotonic fluids into the duodenum
- Presence of breakdown products of protein digestion and sometimes fat

Solution to Question 13:

Broca's motor speech area is likely to be affected in the vascular injury of the inferior frontal gyrus

A lesion in this region leads to motor/Broca's aphasia. Speech is non-fluent and interrupted by many word-finding pauses. The speech is impoverished in function words that contribute to the syntax but is enriched in meaningful nouns and verbs. However, comprehension of speech is intact and the patient knows what he wants to say.

For example, a patient with Broca's aphasia may say "walk dog" instead of "I want to take the dog for a walk".

Solution to Question 14:

The reason for the loss of motor function without loss of pain sensation is that in the nerves to his arm, A-fibers are more sensitive to pressure than C fibers.

Sleeping for a long period with the arms under the head puts pressure on a nerve. It leads to the loss of conduction in large-diameter motor, touch, and pressure fibers, while pain sensation remains relatively intact. A-fibers (includes the motor fibers) are affected.

The susceptibility of the nerve fibers to pressure is $A > B > C$.

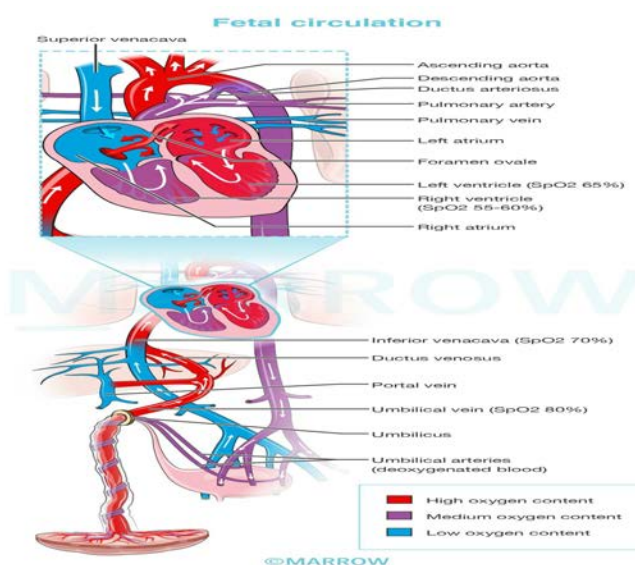
This is responsible for Saturday night or Sunday morning paralysis.

Solution to Question 15:

Inferior vena cava has the highest oxygen concentration in fetal circulation.

The blood gets oxygenated in the placenta and this passes to the umbilical vein. From the umbilical vein, 50% of the blood enters the hepatic circulation and 50% bypasses the liver to enter the inferior vena cava. From here, the blood enters the heart via the right atrium. The right atrium also receives deoxygenated blood from the superior vena cava. This mixed blood is distributed to the body via aorta.

Therefore, inferior vena cava has the highest concentration of oxygen in the fetal circulation.



Solution to Question 16:

Respiratory distress syndrome is due to the deficiency of dipalmitoyl-phosphatidyl-choline (DPPC) /dipalmitoyl lecithin. It is the major pulmonary surfactant.

Clinical relevance- Test to assess fetal lung maturity is L/S ratio - lecithin/sphingomyelin ratio. A ratio of ≥ 2 is considered normal.

Solution to Question 17:

The fall in 2, 3 DPG is less when the blood is stored in CPDA compared with ACD.

2, 3 DPG (2,3-diphosphoglycerate) binds to deoxygenated haemoglobin. Therefore, an increase in 2,3 DPG causes increased release of O₂ to the tissues. The equilibrium between Hb and O₂ can be understood from the following equation:



In stored blood, the 2,3 DPG levels usually fall. As a result, stored blood releases less oxygen to the tissues than the normal circulating blood, thus limiting its use in hypoxic patients (Eg: anaemia).

However, such a fall in 2,3 DPG is less when it is stored in a phosphate-containing solution like CPDA (Citrate, phosphate, dextrose, adenine), rather than ACD (Acid citrate dextrose). Therefore, the blood stored in CPDA, in comparison with blood stored in ACD has:

- Higher 2,3 DPG concentration
- More amount of deoxygenated blood bound to 2,3 DPG
- Decreased affinity of Hb to the oxygen.
- Increased release of oxygen to the tissues.
- Raised P₅₀ (PO₂ at which Hb is half saturated)
- Shift in O₂-Hb dissociation curve to the right

Physiology INI-CET 2021

Question 1:

Which of the following is not involved in renal auto-regulation?

- a) Tubuloglomerular feedback
- b) Myogenic reflex
- c) Renin-angiotensin system
- d) Renal sympathetic system fibres

Question 2:

The formula for urea clearance is $U \cdot V / P$, where U is

- a) Urine volume
- b) Serum urea in mg /100ml
- c) Urine urea in g/24h
- d) Urine urea in mg /100ml

Question 3:

The relaxation of the intestinal segment distal to the segment with the bolus of food during peristalsis is because of

- a) VIP
- b) Substance P
- c) Dopamine from adrenergic fibres
- d) Acetylcholine from myenteric fibres

Question 4:

A patient presented with severe watery diarrhea resembling that of rice gruel stool. Damage of which structure is likely to have resulted in this diarrhea?

- a) Zona occludens
- b) Gap junction

- c) Adherent junction
- d) Hemidesmosomes

Question 5:

Beta hCG is structurally similar to which biochemical moiety?

- a) LH
- b) FSH
- c) ACTH
- d) Oxytocin

Question 6:

What is the duration of short term memory?

- a) 3-6 Hours
- b) 12-18 hours
- c) 30-300 seconds
- d) 3-4 seconds

Question 7:

The stroke volume depends on end-diastolic volume. This is best represented as

- a) Length-force relationship
- b) Length- velocity relationship
- c) Length-tension relationship
- d) Length-viscosity relationship

Question 8:

Which of the following ions in the thick ascending limb of the loop of Henle is responsible for the resorption of divalent ions?

- a) H⁺
- b) HCO₃⁻

- c) K^+
- d) Na^+

Question 9:

Which of the following sensory receptors are responsible for tactile discrimination?

- a) Merkel disc
- b) Ruffini end organ
- c) Pacinian corpuscle
- d) Meissner's corpuscle

Question 10:

All of the following increases gastric acid secretion except:

- a) Gastrin
- b) Histamine
- c) Acetylcholine
- d) Serotonin

Question 11:

Pacinian corpuscle is stimulated by which of the following?

- a) Pain
- b) Temperature
- c) Touch
- d) Pressure

Question 12:

Which of the following jugular venous pressure (JVP) waves corresponds to the isovolumetric contraction phase of the cardiac cycle?

- a) a wave
- b) c wave

- c) x descent
- d) y descent

Question 13:

The extracellular concentration of chloride is 100mmol/L and the intracellular concentration is 10 mmol/L. What is the equilibrium potential of chloride?

- a) + 10 mV
- b) - 10 mV
- c) - 60 mV
- d) + 60 mV

Question 14:

After reaching an equilibrium state during exercising, it was found that the body was consuming oxygen at the rate of 1500 ml/min. Calculate the MET score if the baseline oxygen consumption at rest is 250 ml/min.

- a) 5
- b) 6
- c) 1/5
- d) 1/6

Question 15:

All of the following cause a right shift of the oxygen-hemoglobin dissociation curve, except ____.

- a) Increased HbF
- b) Raised 2,3 BPG
- c) Acidosis
- d) Increased CO₂

Question 16:

The latent period, contraction period, and relaxation period of a skeletal muscle are 10 ms, 40 ms, and 50 ms respectively. Calculate the tetanizing frequency.

- a) 100 per second
- b) 50 per second
- c) 20 per second
- d) 25 per second

Question 17:

An individual is in an environment of a temperature of 47°C. What is the mechanism of heat loss?

- a) Radiation
- b) Conduction
- c) Convection
- d) Sweating

Question 18:

Which among the following has the highest airway resistance?

- a) Alveolar duct
- b) Bronchi
- c) Respiratory bronchioles
- d) Small bronchioles

Question 19:

If the contractility of the heart is decreased, which of the following is seen

- a) End diastolic volume and pressure curve shifts to left without change in slope
- b) Slope changes in ESV pressure curve
- c) End systolic volume and pressure curve shifts to left without change in slope
- d) Slope changes in EDV and pressure curve

Question 20:

A lesion in which lobe will lead to impairment of hand-eye coordination?

- a) Parietal lobe
- b) Frontal lobe
- c) Temporal lobe
- d) Occipital lobe

Question 21:

All of the following ions have decreased concentration on the luminal side of proximal convoluted tubule except :

- a) Amino acids
- b) Glucose
- c) Chloride
- d) Bicarbonate

Question 22:

Sodium iodide symporter is present in all of the following except:

- a) Thyroid gland
- b) Placenta
- c) Salivary gland
- d) Pituitary gland

Answer Key

Question No.	Correct Option
1	d
2	d
3	a
4	a
5	a
6	c

7	c
8	c
9	d
10	d
11	d
12	b
13	c
14	b
15	a
16	d
17	d
18	b
19	b
20	a
21	c
22	d

Detailed Explanations

Solution to Question 1:

Renal sympathetic system fibres is not involved in the autoregulation of renal blood flow, because a denervated kidney also maintains the auto regulation of renal blood flow as that of a normal kidney.

Tubulo glomerular feedback, myogenic reflex, and renin-angiotensin system are all involved in the autoregulation of renal blood flow.

Tubulo glomerular feedback:

- When the renal artery pressure increases, the renal blood flow and glomerular filtration rate (GFR) increases
- This increased GFR results in increased delivery of solute and water to the early distal tubule, particularly the macula densa
- The macula dense, then secretes a vasoactive substance that constricts afferent arteriole via a paracrine mechanism
- This vasoconstriction then reduces renal blood flow and GFR back to normal

Myogenic reflex:

- The increase in renal arterial pressure stretches the walls of the afferent arterioles, which then contracts

- This contraction increases the afferent arteriolar resistance
- This increase in resistance then compensates for the increase in blood pressure, thereby keeping the renal blood flow constant.

Renin-angiotensin system:

- Angiotensin II is a potent vasoconstrictor of both afferent and efferent arterioles
- However efferent arterioles are more sensitive to angiotensin II than afferent arteriole
- Thus low levels of angiotensin II produce an increase in GFR by constricting efferent arterioles
- High levels of angiotensin II decreases GFR by constricting afferent and efferent arterioles, thereby decreasing the renal blood flow

Solution to Question 2:

'U' in the urea clearance stands for urine concentration of urea in mg /100ml.

The formula for renal clearance of a substance is,

$$\text{Clearance (C)} = U \times V / P,$$

where

U = Urine concentration of the substance in mg / ml

V = Urine flow rate in ml/min

P = Plasma concentration of substance in mg /ml.

Renal clearance of a substance is the volume of plasma that is completely cleared of the substance by the kidneys per unit of time.

Solution to Question 3:

The relaxation of the intestinal segment distal to the segment of a bolus of food during peristalsis is due to Vasoactive Intestinal Peptide (VIP).

Peristaltic contractions in the small intestine are designed to propel the food towards the large intestine.

The food bolus in the small intestine is sensed by enterochromaffin cells of the intestinal mucosa, which release 5-hydroxytryptamine (5-HT)(serotonin). The 5-HT then binds to the receptors on intrinsic primary afferent neurons (IPANs), which when activated, initiate the peristaltic reflex in that segment of the small intestine. Behind the bolus, excitatory transmitters like acetylcholine, Substance P, neuropeptide Y are released in the circular muscle, while these pathways are inhibited in the longitudinal muscle; thus this segment of small intestine (behind the bolus) contracts and lengthens.

(Recall that contraction of circular muscle decreases the diameter of that small intestinal segment whereas contraction of the longitudinal muscle decreases the length of that segment. To prevent the conflict that would occur if circular and longitudinal are contracted at the same time, they are

reciprocally innervated)

In front of the bolus, inhibitory pathways like vasoactive intestinal peptide (VIP) and nitric oxide (NO) are activated in the circular muscle, while excitatory pathways are activated in the longitudinal muscle; thereby relaxing and shortening this segment of small intestine.

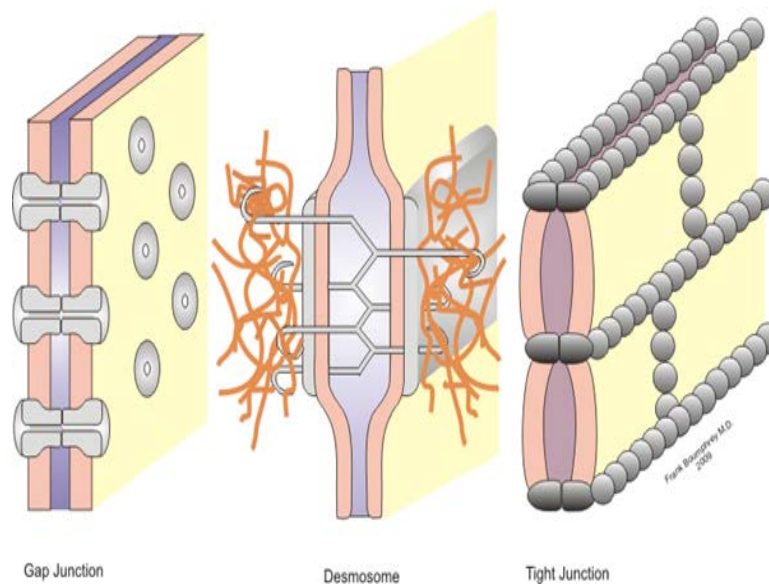
Solution to Question 4:

The given clinical picture of a patient having rice gruel stools is suggestive of the diagnosis of 'cholera'. *Vibrio cholerae* in addition to the cholera toxin (CT) produces some additional toxins such as,

- Zonula occludens toxin, which alters the intestinal permeability by acting on intestinal epithelial cell tight junctions (which are also called zonula occludens)
- Accessory cholera enterotoxin.

Zonula occludens:

- Also known as tight junctions
- Characteristically surround the apical margins of the cells in epithelia such as the intestinal mucosa, the walls of the renal tubules, and the choroid plexus.
- Also important to the endothelial barrier function and endothelium-dependent vasodilation.
- Made up of ridges- half from one cell and a half from another cell- which adhere so strongly that there is no space between the cells.
- Three families of transmembrane proteins are involved:
- Occludin
- Junctional adhesion molecule
- Claudins
- Paracellular pathway - these tight junctions permit the passage of ions, solutes, and intracellular signaling molecules in between adjacent cells.



Other options:

B) Gap junctions:

- The gap between the two cells in these types of junctions is about 4nm.
- These junctions are formed by units called connexons, which are present in the membrane of each cell.
- Each connexon is made up of six protein subunits called connexins.
- They permit substances to pass between the cells, without entering the ECF.
- The pore diameter is around 0.8 to 1.4 nm, which permits the passage of ions, sugars, amino acids, and other solutes which weigh up to 1000 Da.
- They also permit rapid propagation of electrical activity from cell to cell, as well as the exchange of various chemical messengers and intracellular signaling molecules.

C) Adherent junctions:

- Adherent junctions or Zonula adherens is a continuous structure on the basal side of the zonula occludens.
- It contains cadherins.

D) Hemidesmosomes:

- They look like half-desmosomes.
- They attach cells to the underlying basal lamina and are connected intracellularly to intermediate filaments.
- They contain integrins.

Solution to Question 5:

Beta hCG (human chorionic gonadotropin) has biological activity similar to that of luteinizing hormone (LH).

hCG is a glycoprotein produced by syncytiotrophoblasts. It is made up of α and β subunits. Though the α subunit of hCG is similar to the α subunits of LH, FSH, and TSH, the β subunit has structural similarity to LH only. Hence, LH is a better answer than FSH.

hCG is primarily luteinizing and luteotrophic. It appears to act on the same receptor as that of LH. The function of hCG is to prevent regression of the corpus luteum. It can be detected in blood as early as 6 days after conception and forms the basis of various tests used to identify pregnancy.

Its level peaks at gestational week 9 and then declines gradually but is present in lower concentration till the end of pregnancy. But hCG is not specific for pregnancy as it is secreted in a variety of gastrointestinal and genital tumors in both males and females.

Solution to Question 6:

Short-term memory lasts for seconds to minutes, hence of the given 4 options, 30-300 seconds is the most appropriate answer.

Memory is usually divided into two forms: Explicit or declarative memory which is associated with consciousness or awareness. Implicit or nondeclarative memory does not involve awareness and its retention does not usually involve processing in the hippocampus.

Explicit and many forms of implicit memory involve

- Short-term memory and
- Long-term memory.

Short-term memory:

It lasts seconds to minutes; the encoding process for short-term explicit memory involves the hippocampus. It lasts only till the person continues to think about the number or facts.

Working memory is a form of short-term memory that keeps information available, usually for very short periods of time, until the individual plans action based on it.

Long-term memory:

It stores memories that can be recollected after years or even for a lifetime. These memories are stored in various parts of the neocortex. The various parts of the memories such as, visual, auditory, olfactory, etc., are localised to the cortical regions concerned with these functions. All these parts of the memories are brought together by long-term changes in the strength of transmission at relevant synaptic junctions so that all the components are brought to consciousness when that particular memory is recalled.

Neural basis of memory:

The basis of memory is the alteration in the strength of the selected synaptic connections. Second messenger systems contribute to the changes in the neural circuits which are required for learning and memory. Alterations in the cellular membrane channels also contribute to it.

Solution to Question 7:

The relationship between end-diastolic volume and stroke volume is best represented by length-tension relationship.

This is explained by the 'Frank-Starling law of heart' which states that 'the volume of blood ejected by the ventricle depends on the volume present in the ventricle at the end of the diastole'. The length of the myocytes depend on the volume (and hence tension) present at the end of the diastole, which in turn, depends on the volume returned to the heart, or the venous return.

- The Frank-Starling law governs the normal ventricular function and ensures that the cardiac output equals venous return, within physiological limits.
- There is a curvilinear relationship between stroke volume or cardiac output and the ventricular end-diastolic volume.
- Agents that increase the contractility of the cardiac muscle have a positive inotropic effect and produce increased stroke volume and thereby increased cardiac output for a given end-diastolic volume. (Eg: Digoxin)
- Agents that decrease the contractility of the cardiac muscle have a negative inotropic effect and produce decreased stroke volume (and cardiac output) for a given end-diastolic volume.

Solution to Question 8:

Potassium (K^+) in the thick ascending limb of loop of Henle is responsible for the resorption of divalent ions such as, calcium, magnesium, etc.

This resorption is driven by the lumen-positive potential difference across the cells of the thick ascending limb, created by the potassium leak into the lumen.

Mechanism of resorption in thick ascending limb:

- The luminal membrane contains a $Na^+-K^+-2Cl^-$ cotransporter.
- There is net resorption of Na^+ , K^+ and Cl^- in the thick ascending limb as follows:
- All three ions are transported into the cell on the cotransporter.
- Na^+ is extruded from the cell by the Na^+-K^+ ATPase in the basolateral membrane.
- K^+ and Cl^- diffuse through channels in the basolateral membrane, down their respective electrochemical gradients.
- Most of the K^+ leaves the cell across the basolateral membrane; the remaining portion of K^+ however diffuses back into the lumen.
- This K^+ leak into the lumen generates a positive electrochemical potential difference in the lumen.
- This lumen-positive potential difference drives the resorption of ions such as sodium(Na^+), magnesium (Mg^{2+}), calcium (Ca^{+}), etc.,

Solution to Question 9:

Meissner's Corpuscles are responsible for discriminative touch.

Meissner's corpuscles are encapsulated receptors found in the dermis of the non-hairy skin, most commonly on the fingertips, lips, and other locations where tactile discrimination is very good. Their field of reception is very small and hence used for two-point discrimination. These are rapidly adapting receptors that encode

- Point discrimination
- Precise location
- Tapping
- Flutter

Solution to Question 10:

All of the above increase gastric acid secretion except serotonin.

Serotonin is localized in the enterochromaffin cells of the gastrointestinal (GI) mucosa and within neurons of the GI tract. Serotonin inhibits gastric acid secretion. It stimulates the production and release of gastric and colonic mucus. Serotonin acts as an enteric neurotransmitter and increases intestinal motility.

The main factors that stimulate gastric secretion are acetylcholine, gastrin, and histamine. Parasympathetic stimulation releases Ach which stimulates the secretion of

- pepsinogen by peptic cells
- hydrochloric acid by parietal cells
- mucus by mucous cells.

Both gastrin and histamine strongly stimulate acid secretion by parietal cells but have little effect on other cells.

The stomach mucosa has two important types of tubular glands:

- The oxyntic (gastric) glands secrete hydrochloric acid, pepsinogen, intrinsic factor, and mucus.
- The pyloric glands secrete mainly mucus for protection of the pyloric mucosa from the stomach acid. They also secrete the hormone gastrin.

Note- Serotonin released from the small intestine triggers vomiting by stimulation of 5HT₃ receptors. Therefore, 5HT₃ antagonists like ondansetron are used as antiemetics.

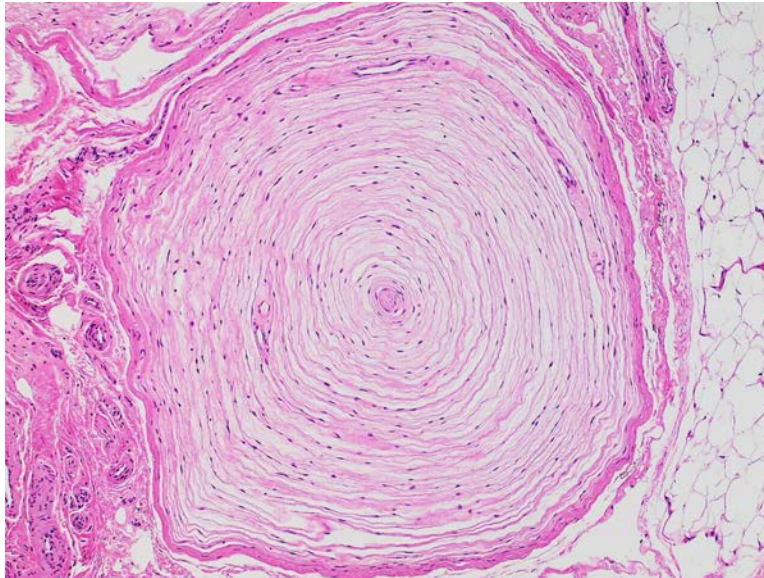
Solution to Question 11:

Pacinian corpuscle is stimulated by pressure.

Pacinian corpuscles are the largest cutaneous mechanoreceptors. They are present in the dermis of glabrous and hairy skin. They are composed of a nerve ending which is encapsulated by

concentric layers of connective tissue that give it an onion-like appearance. These receptors respond to fast vibration and deep pressure and adapt very rapidly. The sensory nerves from cutaneous mechanoreceptors are myelinated A α and A β fibers.

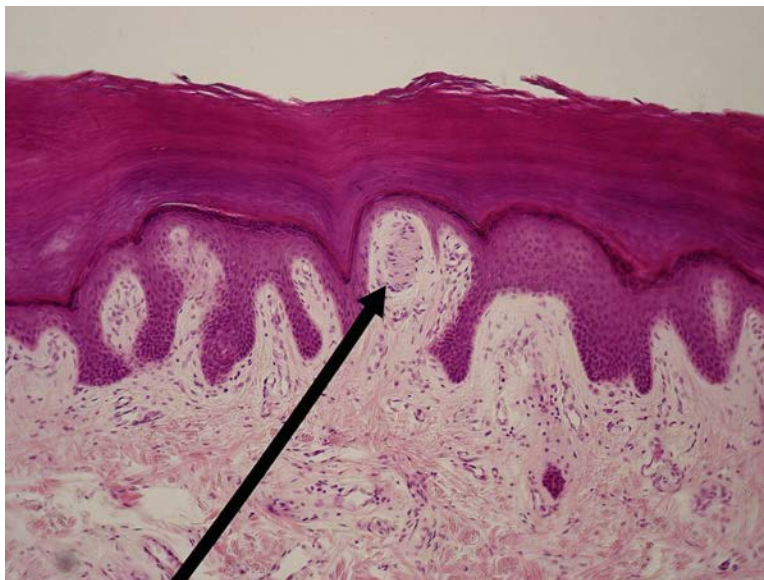
Image shows histology of Pacinian corpuscle.



Other mechanoreceptors:

- Meissner's corpuscles are present mainly in the dermis of non-hairy skin particularly in the fingertips, lips, and other locations where tactile discrimination is well developed. They have small receptive fields and can be used for two point discrimination. Meissner corpuscles are rapidly adapting receptors that sense point discrimination, precise location, tapping, and flutter. These are encapsulated nerve endings of a large myelinated sensory nerve (A β type).

The image shows the histology of Meissner's corpuscle in the dermis.



- Merkel discs are seen in the epidermis of glabrous skin. They are slowly adapting receptors and have very small receptive fields. These receptors detect vertical indentations of the skin. They

are expanded dendritic endings.

- Ruffini corpuscles are capsulated, enlarged, dendritic endings. They are located in the dermis of glabrous and hairy skin, and in joint capsules. They are slowly adapting receptors. Ruffini corpuscles detect stretch and joint rotation.

Pain and temperature sensations arise from receptors located on unmyelinated sensory neurons present throughout the glabrous skin, hairy skin, and deep tissue. Impulses from nociceptors are transmitted via thinly myelinated A δ fibers and unmyelinated C fibers.

Activation of A δ fibers is responsible for first pain (also called fast pain). It is a rapid response and helps in the discrimination of pain or the ability to localize the site and intensity of the noxious stimulus. It is mediated by the release of glutamate. An example is pinprick.

Activation of C fibers is responsible for delayed second pain(also called slow pain). It is a dull, intense, diffuse, and unpleasant feeling associated with a noxious stimulus. . Second pain is mediated by the release of a combination of glutamate and substance P. Example is visceral pain.

Types of nociceptors :

- Mechanical nociceptors respond to strong pressure (eg, from a sharp object).
- Thermal nociceptors are activated by skin temperatures above 45°C or by severe cold (<20°C).
- Chemically sensitive nociceptors respond to chemicals such as bradykinin, histamine, high acidity, and environmental irritants.
- Polymodal nociceptors respond to combinations of these stimuli.

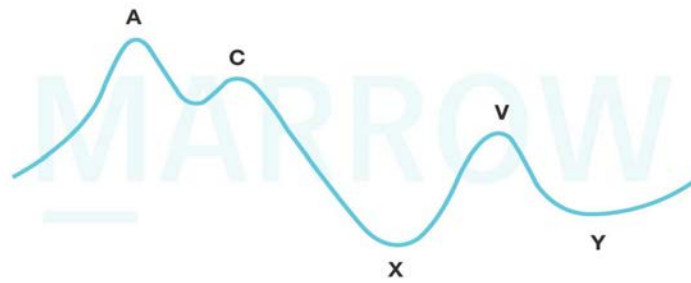
Solution to Question 12:

The c wave of jugular venous pressure wave corresponds to the isovolumetric contraction phase in the cardiac cycle.

During isovolumetric contraction, the ventricles start contracting with valves closed. This leads to bulging of the tricuspid valve. This bulging increases pressure in the right atrium which is transmitted to the right internal jugular vein. This appears as a c wave in jugular venous pressure (JVP) tracing.

JVP measured in the internal jugular vein (IJV) is a reflection of the right atrial pressure. IJV is preferred because it is not valved and is in direct line with superior vena cava and right atrium.

Jugular venous pressure



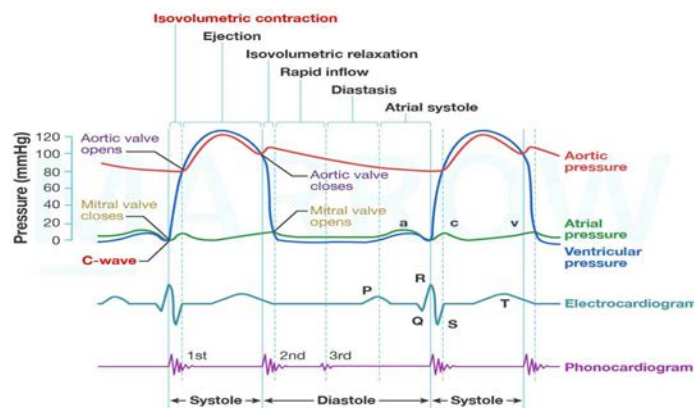
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The waveforms seen in a normal JVP are:

- a wave corresponds to atrial contraction and occurs just after the P wave (on ECG) and just before the first heart sound.
- c wave represents the bulging of the tricuspid valve into the right atrium during ventricular systole.
- x descent corresponds to the atrial relaxation during atrial diastole.
- v wave represents atrial filling during ventricular systole and peaks at the second heart sound.
- y descent represents passive emptying of the right atrium into the right ventricle after tricuspid valve opening.

The Wiggers diagram given below correlates JVP with other aspects of cardiac cycle.

Wiggers diagram - Cardiac cycle



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Solution to Question 13:

The equilibrium potential of chloride in the given question is -60 mV.

Solution to Question 14:

The MET score in the given scenario is 6.

One metabolic equivalent (MET) is defined as the amount of oxygen consumed while sitting at rest.

Normally, one MET= 3.5 ml O₂ /kg body weight/min

In the above question, the baseline O₂ consumption at rest is given as 250 ml/min. According to the definition of one MET, we can deduce that the given 250ml/min is one MET.

O₂ consumption at reaching equilibrium during exercise= 1500ml/min

MET score= O₂ consumption during exercise/ one MET

$$= 1500/250$$

$$= 6$$

The MET concept represents a simple, practical, and easily understandable procedure for expressing the energy cost of physical activities as a multiple of the resting metabolic rate. If the MET score is low, it means that the activity requires less oxygen compared to an activity that has a higher MET score. Eg. Gardening requires around 3-7 METS compared to that jogging that may need around 9-11 METS. We can classify activities as light, moderate or heavy, depending on MET scores. It can also be used to identify the functional capacity of a patient.

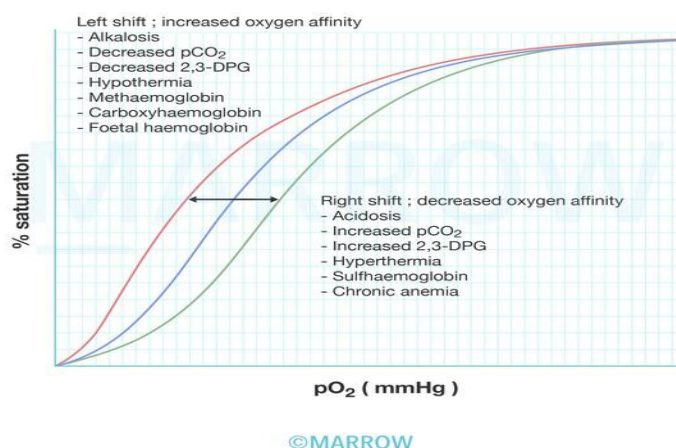
Solution to Question 15:

Increased HbF (fetal hemoglobin) causes a left shift of the oxygen-hemoglobin dissociation curve.

Oxygen-hemoglobin dissociation curve:

This is a plot of the partial pressure of O₂ and the percentage saturation of Hb with O₂. It is normally sigmoid in shape. If the oxygen-hemoglobin dissociation curve is shifted to the right, it means that the affinity of Hb for O₂ has become less, which favors O₂ delivery. This means a higher P₅₀, which implies that a higher PO₂ is required to maintain 50% saturation (R - Right shift Releases oxygen). If the curve is shifted to the left, it means that affinity has increased for O₂ (which favors O₂ binding) i.e. lower P₅₀ value (L-Left shift Leashes oxygen).

Oxygen-hemoglobin dissociation curve



Causes of the right shift of the oxygen-hemoglobin dissociation curve:

- Hypoxia and Hypercarbia - metabolically active tissues utilize O_2 and tissues produce CO_2
- $\downarrow$ pH - Secondary to increased CO_2 production
- $\uparrow$ in temperature
- $\uparrow$ in 2,3-DPG - increased anaerobic glycolysis
- Growth hormone, thyroid hormone, androgen - by increasing 2,3 DPG
- Sickle cell anemia - reduced affinity for O_2
- Exercise

Causes of the left shift of oxygen-hemoglobin dissociation curve:

- $\uparrow$ pH (alkalosis)
- $\downarrow$ PCO_2
- $\downarrow$ 2,3-DPG
- $\downarrow$ Temperature
- HbF
- Carboxyhemoglobin

Bohr's effect- The decrease in the O_2 affinity of hemoglobin when the pH of the blood falls

Solution to Question 16:

The tetanizing frequency of the given skeletal muscle is 25 per second.

In the question, it is given that the contraction period is 40 msec (40 msec is 0.04 seconds)

Tetanizing frequency = $1/\text{contraction period}$

Therefore, tetanizing frequency = $1/40$ millisecond = $1/0.04$ per second = $1000/40$ per second = 25 per second.

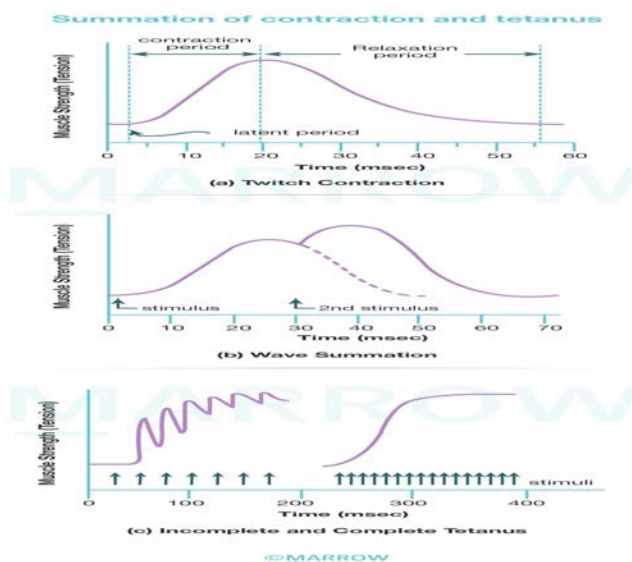
When there is a repeated stimulus to a muscle fiber before relaxation, it produces an additional contraction of the muscle. The second contraction adds to the first contraction and the total strength of contraction increases. This phenomenon is known as a summation of contractions.

With repeated stimulation, contraction occurs repeatedly before any relaxation has occurred. This causes the individual responses to fuse into one continuous contraction. Such a response is called tetanus (tetanic contraction). ■■■■

Tetanizing frequency is the minimum frequency required for producing tetanus.

Tetanus is of two types.

- Complete tetanus- when no relaxation occurs between stimuli
- Incomplete tetanus- when periods of incomplete relaxation occur between the summated stimuli.



Solution to Question 17:

The mechanism of heat loss in the given case where the surrounding temperatures are higher than the body temperature is sweating.

Different methods of heat loss:

- Radiation is heat loss in the form of infrared rays.
- Conduction is heat loss that occurs by direct contact with an object.
- Convection is heat loss resulting from air movement.

When the temperature of the surroundings is more than that of the skin, the body gains heat by both radiation and conduction. Under these conditions, the heat can be lost from the body only by evaporation or sweating.

When the body temperature is greater than the temperature of the surroundings, heat can be lost by radiation and conduction.

Solution to Question 18:

Among the given options, bronchi (medium-sized airway) offer maximum resistance.

The best answer here is bronchi, as all the other options suggest smaller airways, while bronchi are the only medium-sized airway.

Airflow can be derived from Poiseuille-Hagen formula:

$$Q = \pi \Delta P r^4 / 8 \eta L$$

Where

- Q is the flow rate
- π is pie symbol
- ΔP is the pressure difference
- r is radius
- η is viscosity
- L is length

Also,

Flow rate (Q) = pressure difference (ΔP) / resistance (R).

Therefore, substituting this in Poiseuille- Hagen formula,

$$\Delta P / R = \pi \Delta P r^4 / 8 \eta l$$

$$1/R = \pi r^4 / 8 \eta l$$

$$R = 8 \eta l / \pi r^4$$

Therefore it can be derived that airway resistance is inversely proportional to the radius by the power of four.

With Poiseuille- Hagen formula we might feel that the smallest airways are the major site of resistance. However, the contribution by the smallest airways is very little to the overall total resistance. This is because, the airway branches in each generation exist in parallel rather than in series. As the parallel branching increases, the resistance decreases. (Total cross-sectional diameter of small airways put together is high). Thus the overall resistance of the small airways is very small. Also, airflow velocity decreases substantially as the effective cross-sectional area increases (i.e., the flow becomes laminar).

Other factors affecting airway resistance :

- Lung volume, as lung volume increases the resistance decreases and vice versa. Lung volume increases the transairway pressure thus increasing the caliber of the airway.
- Airway mucus, edema, and contraction of bronchial smooth muscle increase the airway resistance. Its clinical application is seen in asthma, bronchitis, and emphysema

- The density and viscosity of inspired gas also contribute to airway resistance. A gas with more density imparts more resistance.

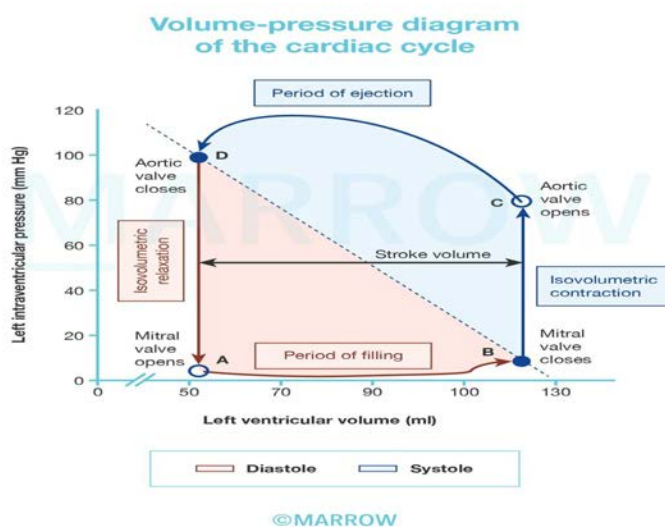
Solution to Question 19:

If the contractility of the heart is decreased, slope changes in the ESV pressure curve are seen.

To answer this question, we will first understand the volume-pressure diagram and the end-systolic pressure-volume relationship (ESPVR) during cardiac cycle.

Volume-pressure diagram during the cardiac cycle

The volume pressure diagram plots the changes in the pressure and volume in the left ventricle that occurs during the cardiac cycle. The x-axis shows the changes in ventricular volume, while the y-axis shows the changes in ventricular pressure.



It has the following phases.

Phase I- This phase is called the period of filling and extends between points A and B. Point A is at the end of systole when the ventricular volume is about 50 mL. This volume is called end-systolic volume. The pressure at this point is about 2 to 3 mm Hg. These volume and pressure go on increasing as the blood from the atrium flows to the ventricle during diastole. At the end of diastole, marked by point B, the ventricle will have a volume of around 120 mL. This volume is called end-diastolic volume. The pressure is slightly raised to 5-7 mm Hg.

Phase II- It is the phase of isovolumetric contraction, and extends between points B and C. At the end of diastole (point B), a slight increase in the ventricular pressure closes the mitral valve, and contraction starts. The ventricle contracts while both mitral and aortic valves are closed. Therefore, the pressure within the ventricle slowly rises without any changes in the volume.

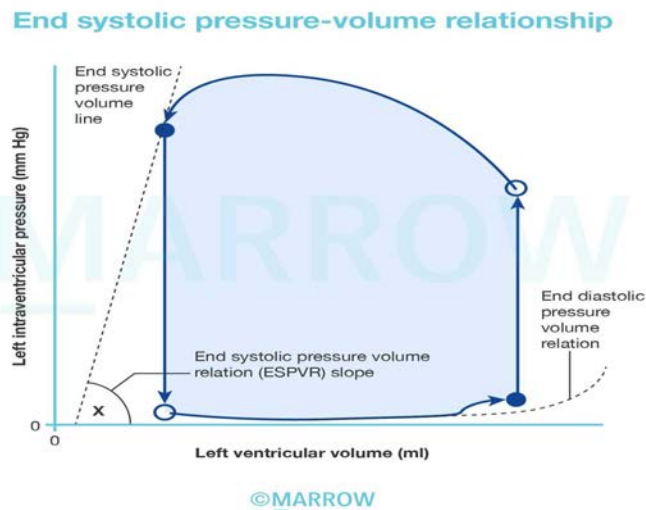
Phase III- It is the phase of ejection and extends between point C and point D. Point C is the place where the aortic valve opens. The pressure within the left ventricle rises slowly during isovolumetric contraction until it is sufficient to open the aortic valve. Once the aortic valve opens, blood within the left ventricle is ejected into the aorta. This decreases the volume of the left

ventricle, while increasing the pressure within the ventricle.

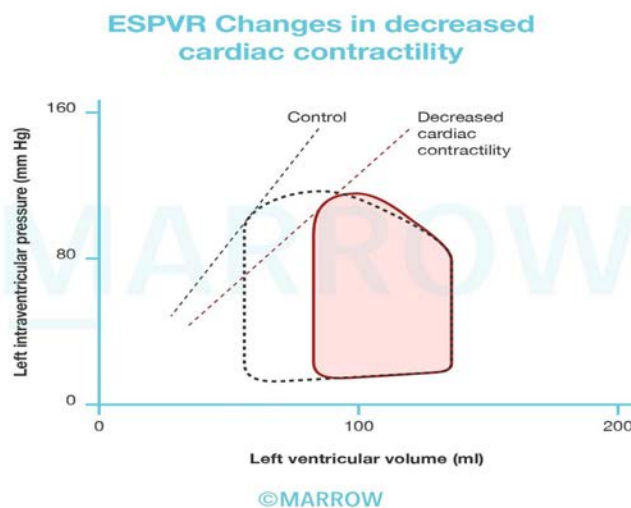
Phase IV- It is the period of isovolumetric relaxation. It extends between points D and A. At the end of the ejection (Point D), the aortic valve closes. However, the ventricular relaxation continues despite closed aortic and mitral valves. Therefore, the ventricular pressure continues to fall while the volume remains the same. The ventricle returns to its starting point, with about 50 ml of blood left in the ventricle and pressure of about 2 to 3 mm Hg

End-systolic pressure-volume relationship (ESPVR)

ESPVR is the maximum pressure that can be developed by the ventricle at the end-systolic volume. It is determined by the line intersecting the upper left corners of the loops. This line can be extended down to the x-axis. The slope of this line is called as ESPVR slope.



Now coming to the question, when the contractility of the heart is decreased, the force of ejection decreases leading to decreased stroke volume, and increased end-systolic volume. Therefore, the volume-pressure diagram shifts to the right. If the diagram shifts to the right, then ESPVR also changes, leading to a change in the slope of the ESPVR curve.



Other options:

Options A and D- End diastolic volume and pressure curve (EDPVR) mainly assesses the compliance of the ventricle, rather than contractility. The slope of this curve increases when the compliance decreases, and vice versa.

Option C- ESPVR shifts to right with change in slope.

Solution to Question 20:

A lesion in the parietal lobe will lead to impairment of hand-eye coordination.

The somatic sensory area and related parts of the posterior parietal lobe project to the premotor cortex. Some of these neurons are concerned with aiming and manipulating the hands at an object, whereas other neurons are concerned with hand-eye coordination. Neurons in this posterior parietal cortex contribute to the descending pathways involved in motor control.

There are two parietal lobes, the dominant side, and the non-dominant side.

- The functions of the dominant side of the parietal lobe are calculation, language, planned movement, appreciation of size, shape, weight, and texture. A lesion here causes dyscalculia, dysphasia, dyslexia, apraxia, agnosia, homonymous hemianopia.
- The function of the non-dominant side is spatial orientation and constructional skills. Its lesion causes spatial disorientation, constructional apraxia, neglect of the non-dominant side, dressing apraxia, and homonymous hemianopia.

Other options:

Option B: The frontal lobe is responsible for personality, emotional response, social behavior. Any lesion in the frontal lobe causes disinhibition, antisocial behavior, impaired memory, incontinence, anosmia.

Option C: The temporal lobe has two lobes and is divided as dominant and non-dominant.

- The dominant temporal lobe has important functions like auditory perception, speech, language, verbal memory, smell. Lesions lead to dysphasia, dyslexia, poor memory, complex hallucinations (smell, sound, vision), impaired face recognition (Prosopagnosia) and homonymous hemianopia.
- The nondominant temporal lobe auditory perception, music, tone sequences, non-verbal memory (faces, shapes, music). Lesions here leads to poor non-verbal memory, loss of musical skills, complex hallucinations, and homonymous hemianopia.

Option D: The occipital lobe is responsible for the analysis of vision. A lesion in occipital lobe mainly affects vision leading to homonymous hemianopia, visual agnosia, impaired face recognition (prosopagnosia), visual hallucinations (lights, lines and zigzags).

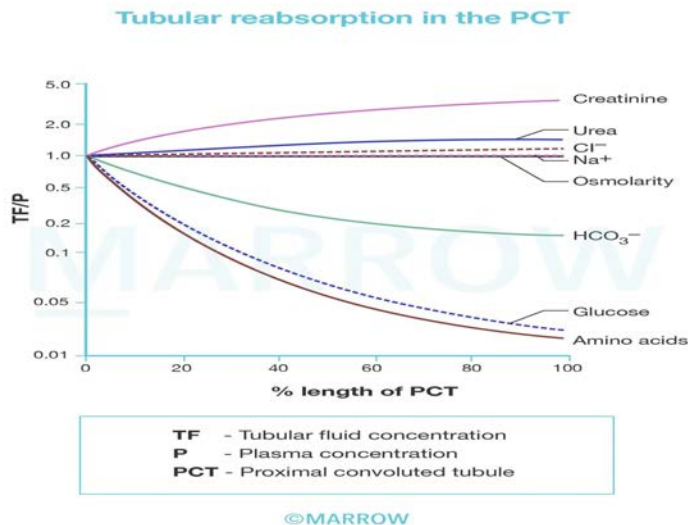
Solution to Question 21:

The concentration of chloride on the luminal side is more when compared to sodium, glucose, and amino acids.

In the first half of the proximal tubule, sodium is reabsorbed by co-transport along with glucose, amino acids, and other solutes. As sodium is reabsorbed in the proximal tubule, it preferentially carries glucose, amino acids, bicarbonate along with it. Thus, their concentrations decrease markedly along the length of the proximal tubule, and a solution having a higher concentration of chloride is left behind. The concentration of chloride here is around 105 mEq/L.

In the second half of the proximal tubule, sodium gets re-absorbed along with chloride since there is only a little glucose and amino acids yet to be re-absorbed. The second half of the proximal tubule has a concentration of chloride ≈ 140 mEq/L. This higher chloride concentration favors its own diffusion from the lumen into the renal interstitial fluid.

Although the amount of sodium in the tubular fluid decreases markedly along the proximal tubule, the concentration of sodium and the total osmolarity remain relatively constant. This is because the proximal tubule is highly permeable to water and water reabsorption increases to keep pace with the sodium reabsorption.



Solution to Question 22:

Sodium iodide symporter is present in all of the options given above except the pituitary gland.

Sodium iodide symporter is expressed mainly in the thyroid gland and is responsible for iodine uptake in the follicular cells of the thyroid gland.

Sodium iodide symporter (NIS)

The NIS is mainly present in the basolateral membranes of the thyrocytes. The NIS transports two Na⁺ ions and one I⁻ ion into the cell against the concentration gradient. The energy for this transport is derived from the Na⁺-K⁺ ATPase pump. Na⁺-K⁺ ATPase pumps sodium outside the cell and causes low intracellular sodium concentration. This causes sodium to enter the cell by facilitated diffusion. Along with sodium, iodide ions also enter the cell. Since the energy to

transport sodium and iodide ions in NIS is derived from Na⁺-K⁺ ATPase pump, this is called secondary active transport.

The NIS is also expressed in:

- Salivary glands
- Gastric mucosa
- Placenta
- Ciliary body of the eye
- Choroid plexus
- Mammary glands
- Certain cancers are derived from these tissues.

The significance of these extrathyroidal NIS is not known.

Physiology NEET 2021

Question 1:

A 30-year-old man weighing 70 kg had a sodium level of 120 mEq/L. Calculate the sodium deficit.

- a) 280 mEq
- b) 480 mEq
- c) 840 mEq
- d) 1400 mEq

Question 2:

Beta 2 receptor action leading to hyperglycemia/ hypoglycemia is mediated by which of the following enzymes?

- a) Adenylyl cyclase
- b) Guanylyl cyclase
- c) Phospholipase C
- d) Phospholipase A2

Question 3:

A body fluid sample is being studied and has Na: 10 mEq/L, K: 140 mEq/L, Cl: 4 mEq/L. Identify the compartment from which the sample has been obtained.

- a) Interstitial
- b) Intracellular fluid
- c) Extracellular fluid
- d) Plasma

Question 4:

A middle-aged female patient presented with slowly progressive motor weakness, nuchal rigidity, and difficulty in swallowing and carrying out day-to-day activities. A provisional

diagnosis of multiple sclerosis is suspected. Which of the following cells are affected in multiple sclerosis?

- a) Microglia
- b) Oligodendrocytes
- c) Astrocytes
- d) Ependymal cells

Question 5:

A child with loose stools for one day presented to your OPD. On examination, signs of dehydration were present and the child was given ORS. It facilitates glucose absorption through which of the following receptors?

- a) SGLT1
- b) SGLT2
- c) GLUT4
- d) Na- Ca transporter

Answer Key

Question No.	Correct Option
1	c
2	a
3	b
4	b
5	a

Detailed Explanations

Solution to Question 1:

The sodium deficit in the given scenario is 840 mEq.

Sodium deficit is calculated as $0.6 \times \text{body weight} \times (\text{target plasma sodium concentration} - \text{starting plasma sodium concentration})$.

Substituting the given values in the formula :

$$\text{Sodium deficit} = 0.6 \times 70 \times (140 - 120) = 840 \text{ mEq.}$$

Solution to Question 2:

Beta 2 receptor action leading to hypoglycemia/ hyperglycemia is mediated by adenylyl cyclase. Catecholamines act through beta 2 receptors and they use adenylyl cyclase - cAMP second messenger system to stimulate the target tissues.

Other hormones that use adenylyl cyclase - cAMP second messenger system are :

- Adrenocorticotrophic hormone
- Angiotensin II
- Corticotropin-releasing hormone
- Calcitonin
- Follicle-stimulating hormone
- Glucagon
- Growth hormone-releasing hormone
- Human chorionic gonadotropin
- Luteinizing hormone
- Parathyroid hormone
- Secretin
- Somatostatin
- Thyroid-stimulating hormone
- Vasopressin

Hormones exert their actions by stimulating cAMP production within the cells. When the hormone binds to the membrane receptor, the receptor is coupled to the G protein. This G protein can either stimulate (Gs protein) or inhibit (Gi protein) adenylyl cyclase that controls the conversion of cytoplasmic adenosine triphosphate to adenosine monophosphate (cAMP). The protein kinases dependent on cAMP are thus activated. This leads to a series of biochemical reactions enabling the hormone to exert its actions.

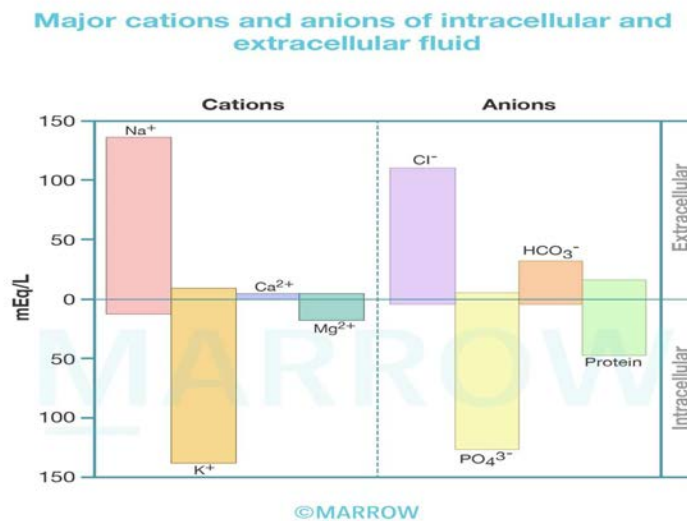
Guanylyl cyclase is the enzyme that catalyzes the formation of cGMP.

Solution to Question 3:

The sample obtained is from the intracellular fluid (ICF) compartment. The ICF contains large quantities of potassium (K⁺) ions - 140 mEq/L. The ICF also has lesser amounts of sodium (Na⁺) and Chloride (Cl⁻) ions.

The extracellular fluid (ECF) compartment includes interstitial fluids and plasma. In contrast to the ICF, the ECF has an abundance of sodium, chloride, calcium and bicarbonate ions, and lesser amounts of potassium, phosphate and magnesium ions.

The image below depicts the proportion of the important anions and cations present in the intracellular and extracellular fluids.



Solution to Question 4:

Oligodendrocytes are affected in multiple sclerosis.

In multiple sclerosis, there is patchy destruction of myelin in the central nervous system and the axons are demyelinated. This causes loss of conduction or delayed conduction in the demyelinated axons. Oligodendrocytes and Schwann cells are involved in the formation of myelin. Hence, they are affected by multiple sclerosis.

Other options

Option A: Microglia are a part of the immune system. Microglia are scavenger cells that aid in the removal of cellular debris formed due to various diseases. They originate from macrophages outside of the nervous system and are unrelated to the neural cell types.

Option C: There are three types of macroglia - astrocytes, oligodendrocytes, and Schwann cells. Astrocytes are found throughout the brain and they induce capillaries to form the blood-brain barrier. The astrocytes may be fibrous or protoplasmic.

Option D: Ependymal cells line the ventricular system. They are in contact with the cerebrospinal fluid and play a role in fluid homeostasis.

Solution to Question 5:

ORS facilitates glucose absorption through SGLT 1 (Sodium-dependent glucose co-transporter - 1). In a child with diarrhea, when oral rehydration solution is given, there is coupled absorption of sodium and glucose through SGLT -1.

Glucose transport may be active or passive.

Active transport of glucose takes place through Sodium dependent glucose transporter (SGLT 1 and 2). SGLT 1 helps in glucose absorption in the small intestine and renal tubules. SGLT 2 primarily facilitates glucose absorption in the renal tubules.

Passive absorption of glucose occurs through facilitated diffusion. This is carried out by Glucose transporters (GLUT) 1 -7.

Physiology INI-CET 2022

Question 1:

Which physiological adaptation does not happen at high altitudes?

- a) Pulmonary vasoconstriction
- b) Polycythemia
- c) Respiratory acidosis
- d) Hypoxia

Question 2:

Countercurrent mechanism is present in which of the following parts of the body?

- a) 1 and 2 are correct
- b) 1,2,3 are correct
- c) 1,2,3,4,5 are correct
- d) 2,3,4,5 are correct

Question 3:

Match the following molecules with their carriers in plasma:

- a) 1-b, 2-c, 3-d, 4-a
- b) 1-a, 2-b, 3-c 4-d
- c) 1-a, 2-d, 3-b 4-c
- d) 1-b, 2-c, 3-a, 4-d

Question 4:

Myosin ceases to be in contact with actin filaments as a result of:

- a) Dissociation of ADP and Pi
- b) Cessation of calcium release from sarcoplasm
- c) Conformational change in Troponin T

- d) Attachment of ATP to myosin head

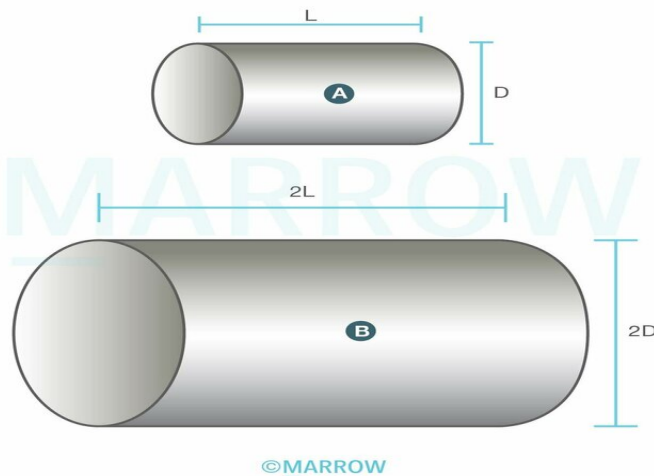
Question 5:

Transport of iron into enterocytes is done by which of the following proteins:

- a) Transferrin
- b) Ferroportin
- c) Heparin
- d) Divalent metal transporter - 1

Question 6:

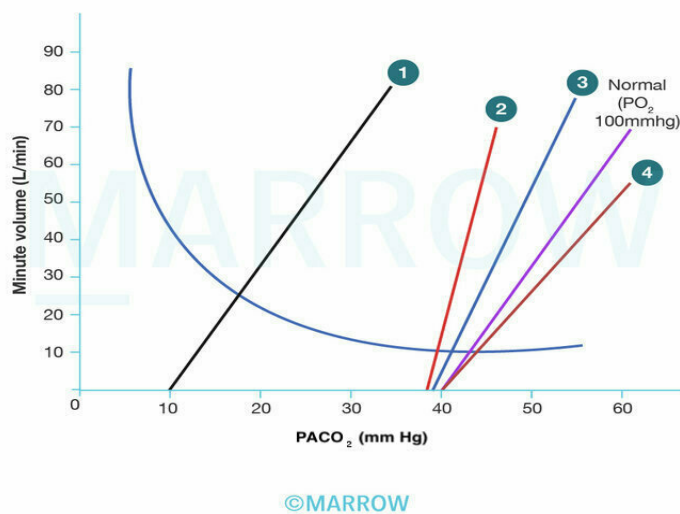
Vessel A has a length of L and diameter of D and vessel B has $2L$ and $2D$. Considering the flow is laminar through both the cylinders, the flow through B is how many times as much as the flow through A?



- a) 4 times
- b) 8 times
- c) 16 times
- d) 32 times

Question 7:

If the pH is reduced from 7.4 to 7.3 with the PO_2 being constant at 100 mmHg, which line would PCO_2 follow?



- a) 1
- b) 2
- c) 3
- d) 4

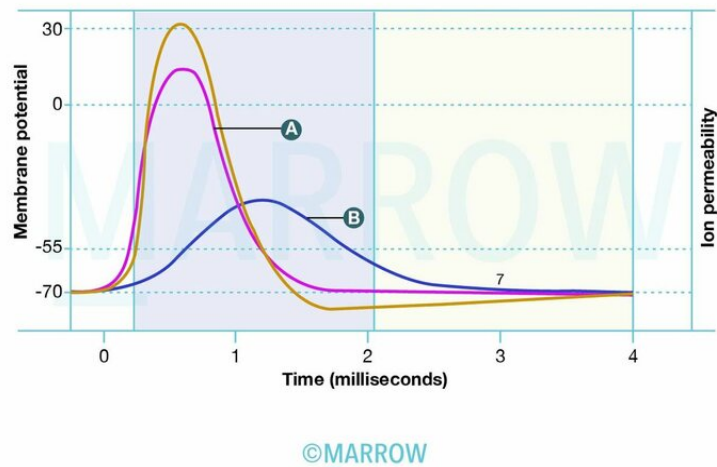
Question 8:

Substances and the type of transport across a membrane are given below, choose the correct combination:

- a) 1 and 2 are correct
- b) 1, 2, and 4 are correct
- c) 1, 2, 3, 4, and 5 are correct
- d) 1, 2, and 5 are correct

Question 9:

A and B in the graph below represent which of the following ions, in terms of permeability for action potential generation?



- a) A is Na^+ , B is K^+
- b) A is Na^+ , B is Ca^{2+}
- c) A is K^+ , B is Ca^{2+}
- d) A is Ca^{2+} , B is K^+

Question 10:

Transcutaneous electrical nerve stimulation (TENS) is used to relieve pain. What is the mechanism behind it?

- a) Gate control theory
- b) Conditioned pain modulation
- c) Descending inhibition
- d) Central inhibition

Question 11:

The parents notice that their child is tired and that he has shortness of breath on climbing stairs and while lying down. What is the first investigation that you would do?

- a) Echocardiogram
- b) Chest x-ray
- c) Spirometry
- d) MRI to rule out enlarged heart

Question 12:

Match the following molecules with their carriers in plasma:

- a) 1-b, 2-c, 3-d, 4-a
- b) 1-a, 2-b, 3-c 4-d
- c) 1-a, 2-d, 3-b 4-c
- d) 1-b, 2-c, 3-a, 4-d

Question 13:

Arrange the following steps of ventricular action potential in the correct order.

- a) B, A, C, D
- b) D, C, B, A
- c) C, D, A, B
- d) A, B, C, D

Question 14:

What does the zero pressure indicate in the pressure-volume curve?

- a) Residual volume
- b) Functional residual capacity
- c) Tidal volume
- d) Inspiratory reserve volume

Question 15:

How is cerebral perfusion pressure calculated?

- a) Systolic blood pressure + Intracranial pressure
- b) Systolic blood pressure - intracranial pressure
- c) Mean arterial pressure/intracranial pressure
- d) Mean arterial pressure - intracranial pressure

Question 16:

Fluid leaving the proximal convoluted tubule in the absence of antidiuretic hormone is:

- a) Isotonic
- b) Hypotonic
- c) Hypertonic
- d) Depends on fluid intake

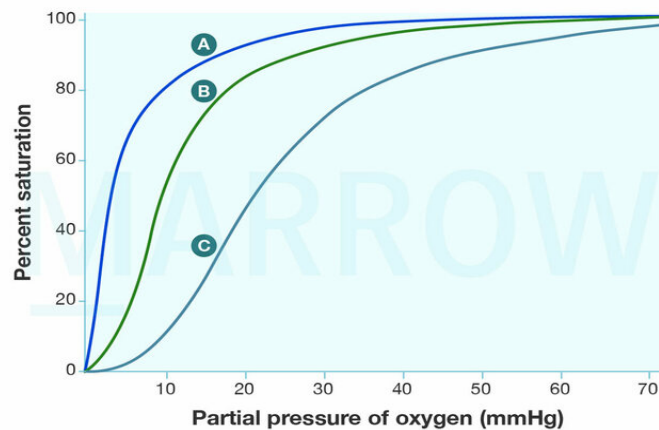
Question 17:

2,3-BPG causes the unloading of oxygen to tissues by _____.

- a) Increasing ATP production
- b) Binding to beta subunit of Hb
- c) Inhibiting taut structure
- d) Decreasing ATP production

Question 18:

The graph below shows oxygen dissociation curves. What does the curve marked as 'A' indicate?



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- a) Myoglobin
- b) Carboxyhemoglobinemia

- c) Methemoglobin
- d) Fetal hemoglobin

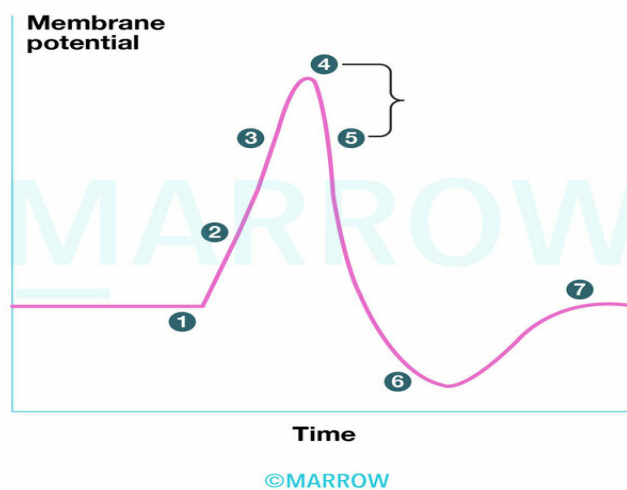
Question 19:

Which among the following organs has the least arteriovenous oxygen difference?

- a) Brain
- b) Kidney
- c) Skin
- d) Liver

Question 20:

The image shows changes in membrane potential during an action potential. The region marked between 4 to 5 is due to _____.



- a) Influx of Na^+
- b) Efflux of Na^+
- c) Influx of K^+
- d) Efflux of K^+

Question 21:

What is the value of normal tissue oncotic pressure?

- a) 12 mmHg
- b) 25 mmHg
- c) 32 mmHg
- d) 54 mmHg

Question 22:

Where is the signaling sequence in a type 2 membrane protein located?

- a) Inside the membrane at the N-terminal end
- b) Inside the membrane at the C-terminal end
- c) Outside the membrane at the N-terminal end
- d) Outside the membrane at the C-terminal end

Question 23:

Which of the following primary second messengers is involved in the contraction of ciliary muscles while focusing on near objects?

- a) cAMP
- b) cGMP
- c) Na⁺ inflow through a channel
- d) IP₃

Question 24:

Which of the following tissues are dependent on insulin for glucose uptake?

- a) Brain
- b) Adipose tissue
- c) Pancreas
- d) Muscle

Question 25:

The second phase of saliva secretion involves:

- a) Active Na^+ reabsorption, passive Cl^- reabsorption, passive K^+ secretion, active secretion of HCO_3^-
- b) Passive Na^+ reabsorption, active Cl^- reabsorption, passive K^+ secretion, active secretion of HCO_3^-
- c) Passive Na^+ secretion, active Cl^- reabsorption, active K^+ secretion, passive secretion of HCO_3^-
- d) Active Na^+ reabsorption, passive Cl^- reabsorption, active K^+ secretion, passive secretion of HCO_3^-

Question 26:

Macula densa is made of specialized cells located at:

- a) Junction of thick descending limb and DCT
- b) Junction of thick descending limb and PCT
- c) Junction of thick ascending limb and DCT
- d) Junction of DCT and collecting duct

Question 27:

Which of the following can be used to calculate the cardiac output?

- a) A, B, C, D
- b) A, D
- c) B, C, D
- d) A, C, D

Question 28:

Match the given cellular components with their respective marker enzymes.

- a) 1-D, 2-A, 3-C, 4-B
- b) 1-D, 2-A, 3-B, 4-C
- c) 1-C, 2-B, 3-A, 4-D
- d) 1-B, 2-C, 3-D, 4-A

Question 29:

Consider the following statements about sympathetic, parasympathetic, and motor nerves. Which of the following statements are true?

- a) 1,3
- b) 1,4
- c) 2,3
- d) 2,4

Question 30:

What is the normal insensible water loss?

- a) 100 mL/hour
- b) 150 mL/hour
- c) 50 mL/hour
- d) 200 mL/hour

Question 31:

A woman has vomiting whenever she eats spicy food. Arrange the sequence of events during vomiting.

- a) 6, 4, 2, 5, 1, 3
- b) 6, 4, 2, 3, 1, 5
- c) 4, 6, 2, 1, 3, 5
- d) 4, 6, 2, 5, 3, 1

Answer Key

Question No.	Correct Option
1	c
2	d
3	a
4	d

5	d
6	b
7	a
8	c
9	a
10	a
11	b
12	a
13	c
14	b
15	d
16	a
17	b
18	a
19	b
20	d
21	a
22	a
23	d
24	b
25	d
26	c
27	d
28	b
29	b
30	c
31	b

Detailed Explanations

Solution to Question 1:

The physiological adaptation that does not happen at high altitudes is respiratory acidosis. Respiratory alkalosis is the physiological adaptation seen at high altitudes due to hyperventilation. It is a method of acclimatization. Compensatory mechanisms that are carried out to increase altitude tolerance are called acclimatization.

Following are the physiological changes that occur during acclimatization:

- Hyperventilation leads to respiratory alkalosis and shift of the oxygen–hemoglobin dissociation curve to the left (and shifts to right after acclimatization).
- Increased erythropoietin secretion, leading to an increase in circulating red blood cells
- Increased number of mitochondria
- Increased myoglobin to facilitate the movement of O₂ into the tissues.
- Increased tissue cytochrome oxidase
- Increased sympathetic tone
- Renal suppression of carbonic anhydrase and excretion of dilute alkaline urine brings the pH of the blood to normal.
- Decreased maximal exercise capacity (by ~1% for every 100 meters gained above 1500 meters)

Other options:

Option A- In response to the high altitude hypoxia, pulmonary vasoconstriction occurs due to which blood flow is distributed where it is most effective, leading to distribution of blood flow to the pulmonary areas in proportion to their alveolar O₂ pressures.

Option B- Polycythemia - Erythropoietin secretion increases on the ascent to high altitude which results in polycythemia

Option D- Hypoxia - Total barometric pressure decreases as altitude increases, resulting in decreased P_{O2} (partial pressure of oxygen) leading to hypoxic hypoxia.

Solution to Question 2:

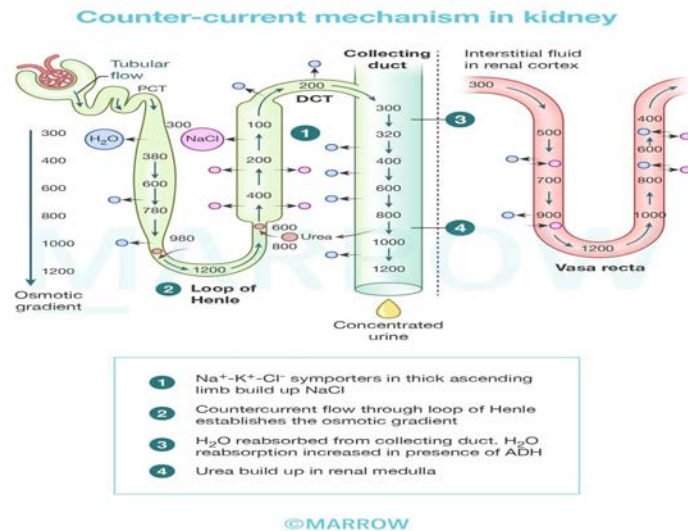
The countercurrent mechanism is present in the testis, kidney, gut, and lungs. When the inflow runs parallel to, counter to, and in close proximity to the outflow for some distance it is called as the counter current mechanism.

Spermatogenesis requires a temperature considerably lower than that of the interior of the body. The testes are kept cool by heat exchange in a countercurrent manner between the spermatic arteries and veins.

The concentration of the urine by the kidney requires an osmolality gradient. This gradient is produced by the loop of Henle as counter-current multipliers and maintained by vasa recta as counter-current exchangers.

The functioning of the loop of Henle depends upon:

- The high permeability of the thin descending limb to water (via aquaporin-1)
- Active transport of Na⁺ and Cl⁻ from the thick ascending limb, thereby diluting the tubular fluid.
- The inflow of tubular fluid from proximal tubule and outflow to the distal tubule.



The countercurrent mechanism is used in the villi to decrease the oxygen availability for local metabolic reactions of the gut. Hence this will increase the blood flow in the intestine during gastrointestinal activity.

Countercurrent exchange of oxygen is seen in the lungs as well.

Solution to Question 3:

Plasma contains various components like ions, inorganic and organic molecules which help in the transport of substances across various parts of the body. Albumin, globulin, and fibrinogen fractions are important plasma proteins. Some of the plasma proteins act as antibodies and are also concerned with blood clotting. Where some others are non-specific carriers of hormones, drugs, and other solutes. Majority of plasma proteins are synthesized in the liver. The above table shows some of the proteins synthesized by the liver and the molecules they carry.

Thyroxine is carried by thyroid-binding pre-albumin which is also known as transthyretin.

Albumin is a carrier protein as well as an osmotic regulator which carries hormones, amino acids, vitamins, and steroids too.

Hemopexin binds with heme, particularly for heme recycling.

Haptoglobin binds to free hemoglobin and prevents its excretion in urine. Hence, haptoglobin levels decrease in the presence of free hemoglobin as seen in hemolytic anemia.

Molecules	Carriers
Thyroxine	Prealbumin
Free fatty acid	Albumin

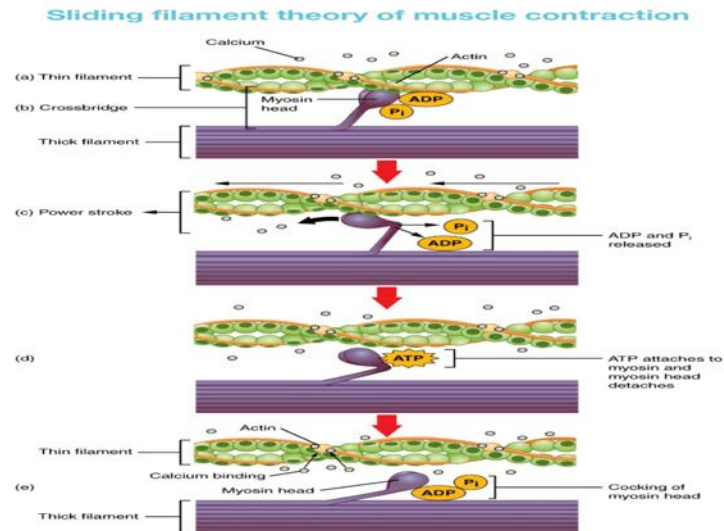
Molecules	Carriers
Hemoglobin	Haptoglobin
Heme	Hemopexin

Solution to Question 4:

Myosin ceases to be in contact with actin filaments as a result of the attachment of ATP to the myosin head.

Muscle contraction occurs by sliding of thin filament over thick filament. The sequence of events are as follows:

- Troponin I is bound to actin and tropomyosin and covers the sites where actin interacts with myosin heads in a resting muscle. The myosin head contains tightly bound adenosine phosphate (ADP) and inorganic phosphate (Pi).
- Following an action potential, cytosolic Ca^{2+} is increased due to the Ca^{2+} released from the sarcoplasmic reticulum. Simultaneously Ca^{2+} influx happens through voltage-gated Ca^{2+} channels too.
- The free Ca^{2+} binds with troponin C. This binding will weaken the troponin I interaction with actin and exposes the actin-binding site for myosin.
- This results in the formation of myosin-actin cross-bridges.
- The cross-bridge between the head of the myosin and the active site of the actin causes a conformational change in the head and the power stroke occurs, moving the thin filament relative to the thick filament.
- Once the head of the cross-bridge tilts, the ADP and phosphate ion that were previously attached to the head are released and a new ATP molecule can bind in that place.
- The binding of the new ATP molecule to the free site on the myosin causes detachment of the head of the myosin from the actin.
- ATP is hydrolyzed and Pi released, causing a re-cocking of the myosin head and completing the cycle.



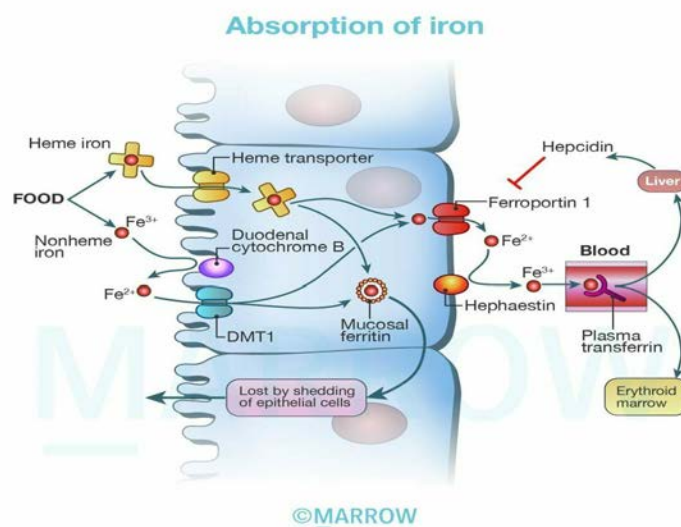
Solution to Question 5:

Transport of iron into enterocytes is done by divalent metal transporter-1 protein.

Iron absorption occurs in the duodenum. Iron in our gut exists in 2 forms, they are nonheme and heme iron.

Luminal nonheme iron is mostly in the Fe^{3+} (ferric) state and is reduced to Fe^{2+} (ferrous) iron by ferrireductases. Fe^{2+} iron is then transported across the apical membrane by divalent metal transporter 1 (DMT1).

Given below is an image of iron absorption that shows divalent metal transporter-1 protein in intestinal mucosal cells.



Heme iron is moved across the apical membrane into the cytoplasm through heme transporters.

In the duodenal cells, Fe^{2+} iron can follow one of two pathways: transport to the blood or storage as mucosal iron. The nonstorage form of Fe^{2+} iron is transported from the cytoplasm into circulation by the protein ferroportin (Option B).

Once in circulation, Fe^{2+} is again oxidized back to Fe^{3+} iron by hephaestin and ceruloplasmin. The newly absorbed Fe^{3+} iron binds rapidly to transferrin (Option A), which delivers iron to red cell progenitors in the marrow.

Hepcidin (Option C), is a protein secreted from the liver that plays a major role in the regulation of iron absorption. When the body's iron stores are adequate, it downregulates the expression of ferroportin and thereby decreases iron absorption.

Solution to Question 6:

Flow through B will be 8 times more than the flow through A.

According to Poiseuille Hagen formula,

$$F = \frac{\Delta P \times \pi}{8 \times \eta \times L} \times r^4$$

F = flow

ΔP = pressure difference between two ends of the tube

η = viscosity

r = radius of the tube

L = length of the tube

Hence flow directly varies to the 4th power of radius and inversely varies to length.

Flow through B = $(2r)^4/2L = 8$ times

Solution to Question 7:

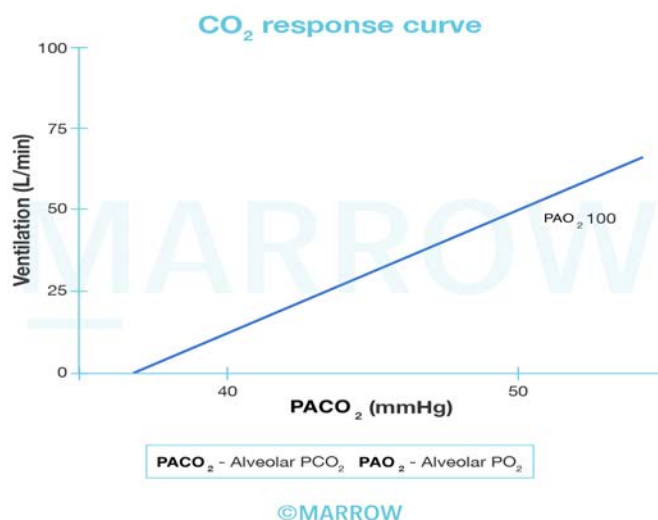
If the pH is reduced from 7.4 to 7.3 with the alveolar PO_2 being constant at 100 mmHg, PCO_2 would follow line 1.

This concept can be understood in three steps.

Step 1- Effect of CO_2 on ventilation, when the alveolar PO_2 is kept constant.

When the alveolar PO_2 (PaO_2) is kept constant, an increased amount of inspired CO_2 leads to increase in the ventilation. The relation is a straight line.

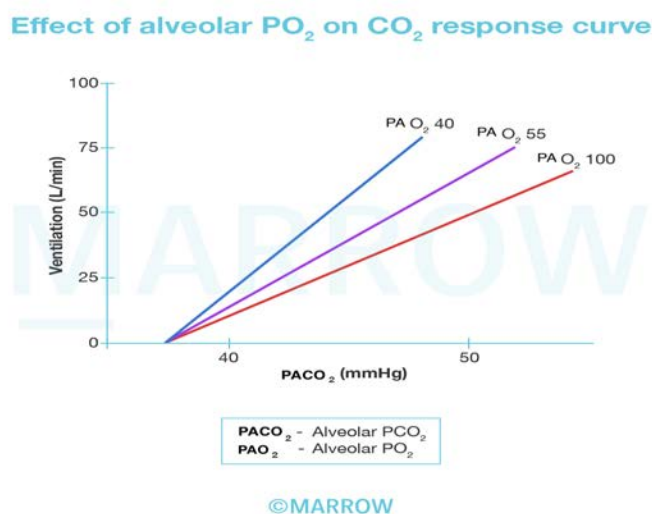
The below graph shows the effect of CO_2 on ventilation when alveolar PaO_2 is kept constant at 100mm Hg.



Step 2. The shift of the above graph when PO₂ changes.

When PO₂ changes, the relation between the amount of CO₂ inhaled and ventilation still remains in a straight line. However, the slopes of the lines change. The slope increases when alveolar PO₂ decreases and vice versa. In other words, the lines move towards the left when PO₂ decreases. However, the point of intersection of all these lines remains the same.

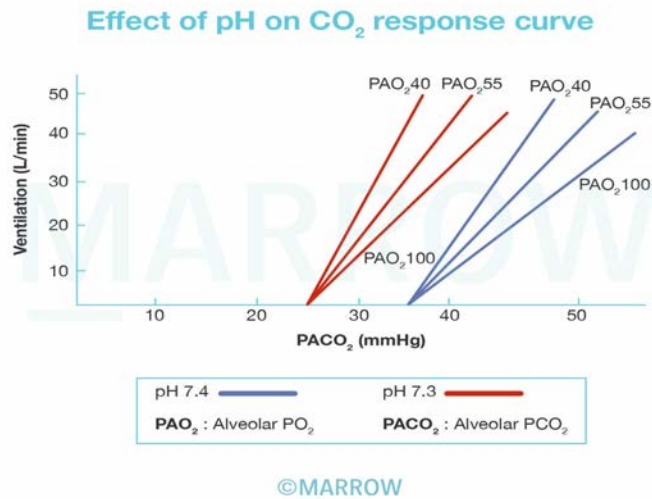
The below graph shows the relation between inspired CO₂ and ventilation when alveolar PO₂ is maintained at 40mm Hg, 55mm Hg, and 100 mm Hg. As the alveolar PO₂ has decreased, the lines have shifted to the left. But they are all intersecting at the same point on the x-axis.



Step 3. The shift of the above graph when pH changes.

When pH decreases, as in acidosis, the CO₂ response curve is similar to the above. The only difference is they are shifted to the left and have different points on the intersection on x- axis.

The below graphs show CO₂ response curves when pH is 7.4, 7.3, and PO₂ is 40mm Hg, 55mm Hg, and 100mm Hg. The lines have shifted to the left and have a different point of intersection.

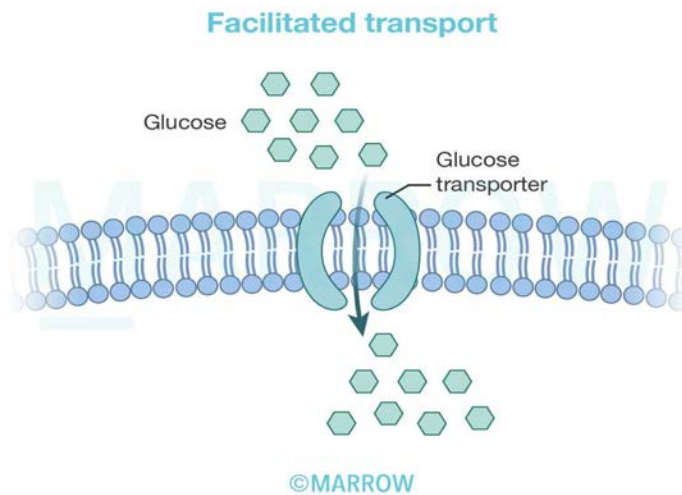


In the question image, only line 1 has been shifted to the left and has a different point of intersection on x-axis compared to other lines. Therefore, line 1 is the answer.

Solution to Question 8:

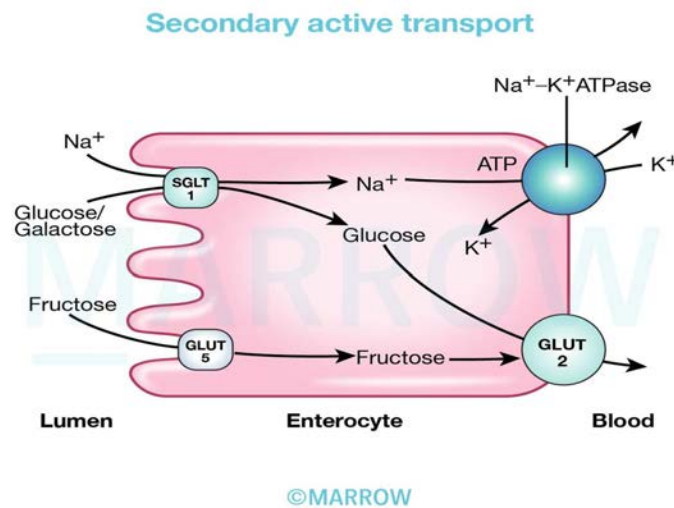
All the given combinations 1, 2, 3, 4, and 5 are correct.

- 1) Oxygen is a non-polar molecule and hence can diffuse across the lipid membrane of the cells.
- 2) Facilitated diffusion - Transporting the molecule from one side of the cell membrane to the other is done with the help of transport proteins called carriers. Molecules move from areas of high concentration to areas of low concentration. No energy input is required when carrier proteins move substances in the direction of their gradients. Glucose transport by the glucose transporter is an example, which moves glucose down its concentration gradient from the ECF to the cytoplasm of the cell.



3) Voltage-gated channels are simple ion channels acting as transport proteins, gated by membrane potential alterations. Sodium transported via voltage-gated channels is an example.

4) Transporting more than one substance together using a single transport protein against the gradient is known as secondary active transport. An example is SGLT 1, the cotransporter of sodium and glucose from the intestinal lumen into mucosal cells.



5) Active transport - Transporting of substances against the gradients, and requires energy. Ca^{2+} ATPase that pumps out calcium out of the cells is an example.

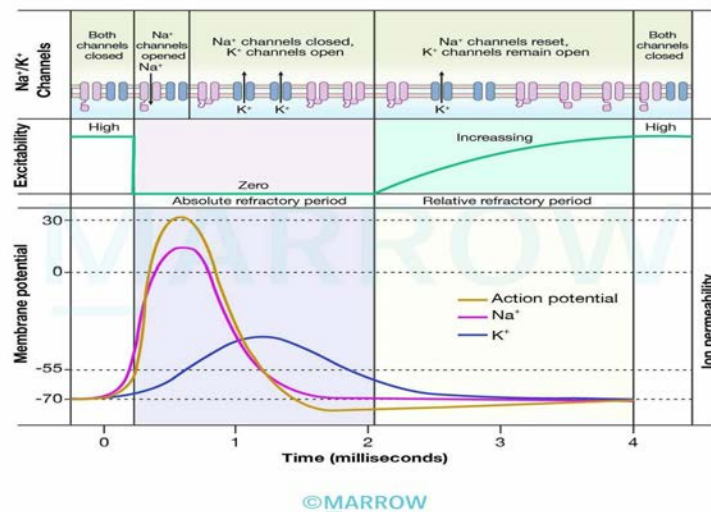
Solution to Question 9:

In the given graph, showing action potential generation in terms of ion permeability, wave A represents Na^+ and B represents K^+ .

The sequence of events during the action potential occurs in 7 steps as described below.

1. Resting-state - Both channels are closed
2. Na^+ channels open in response to depolarizing stimulus. Sodium enters the cell. The membrane reaches threshold potential.
3. The sodium influx causes the opening of more Na^+ channels through a positive feedback mechanism. Rapid depolarization occurs.
4. Sodium conductance decreases.
5. Na^+ channels close and K^+ channels open. Repolarization occurs.
6. Sodium channels are reset, and are ready to respond to another depolarizing stimulus (relative refractory period). During this phase, the potassium channels remain open and slowly return to closed state causing after-hyperpolarization.

7. Both channels are closed. The membrane returns to RMP.



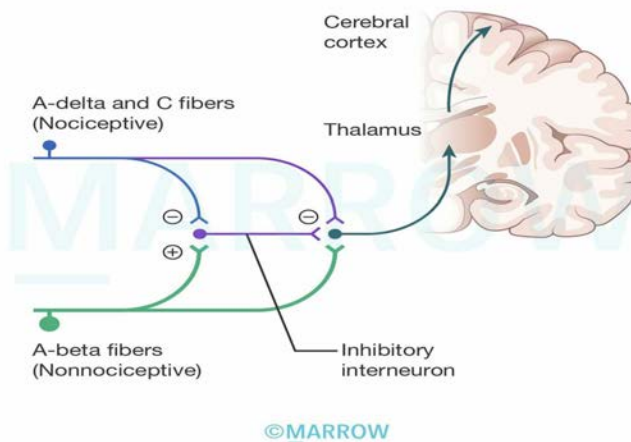
Solution to Question 10:

Gate theory of pain is the mechanism behind transcutaneous electrical nerve stimulation (TENS). This method uses electrodes to activate A α and A β fibres near the injury to relieve pain.

At the site of sensory afferent termination, transmission in nociceptive pathways can be interrupted. This is done by the actions within the dorsal horn of the spinal cord. Pain subsiding when the site of injury is rubbed or shook is an example. Afferents of innocuous cutaneous mechanoreceptors emit collaterals that terminate in the dorsal horn. The responsiveness of dorsal horn neurons to their input from nociceptive afferent terminals is reduced by these afferents. This is called the gate-control mechanism of pain modulation.

The given image depicts the gate control theory of pain:

Gate control theory of pain



Transcutaneous electrical nerve stimulation (TENS)



Solution to Question 11:

The first investigation to be done for a child who feels shortness of breath on climbing stairs and in the supine position is a chest x-ray.

The child having shortness of breath on climbing stairs is known as dyspnea on exertion. The most common cause is congestive heart failure. The child feeling shortness of breath in the supine position is orthopnea. It is relieved on sitting or standing.

Respiratory causes include asthma, chronic obstructive pulmonary disorder (COPD), pneumonia, and pulmonary embolism. Cardiovascular causes include congestive heart failure (CHF), pulmonary edema, valvular heart defect, and pulmonary hypertension.

Evaluation begins with assessing the airway, breathing, and circulation of the patient. Oxygen saturation can appear normal at rest, and hence oxygen saturation during physical activity should

be obtained. A chest x-ray is the first diagnostic test which can show pulmonary pathology or pulmonary edema and other signs of CHF that could help in identifying the cause and further evaluation.

Other options:

Option A: An echocardiogram can be done if the chest X-ray is abnormal. It is done to evaluate cardiac function, valvular function, and pericardial space.

Option C: Spirometry is done when the chest x-ray is normal. It is done to determine lung function. Abnormal spirometry can indicate either an obstructive pathology such as asthma, or COPD or a restrictive disease such as interstitial fibrosis.

Option D: Cardiac MRI (Magnetic resonance imaging) is an emerging modality done to look for cardiomegaly. It is for the accurate evaluation of left and right ventricular mass, size, and function. It is also used for localizing ischemic and non-ischemic causes such as myocarditis.

Solution to Question 12:

The correct pairs are as follows:

1. Thyroxine - B. Pre-albumin
2. Vitamin B12 - C. Transcobalamin
3. Hemoglobin - D. Haptoglobin
4. Heme - A. Hemopexin

Thyroxine is carried by thyroid-binding prealbumin, which is also known as transthyretin.

Vitamin B12 binds to transcobalamin II in the ileal cells and is secreted into the plasma.

Transcobalamin II delivers vitamin B12 to the liver and other cells of the body, including rapidly proliferating cells in the bone marrow and the gastrointestinal tract.

Hemopexin binds to porphyrins, particularly heme, for heme recycling.

Haptoglobin binds to free hemoglobin and prevents its excretion in urine.

Solution to Question 13:

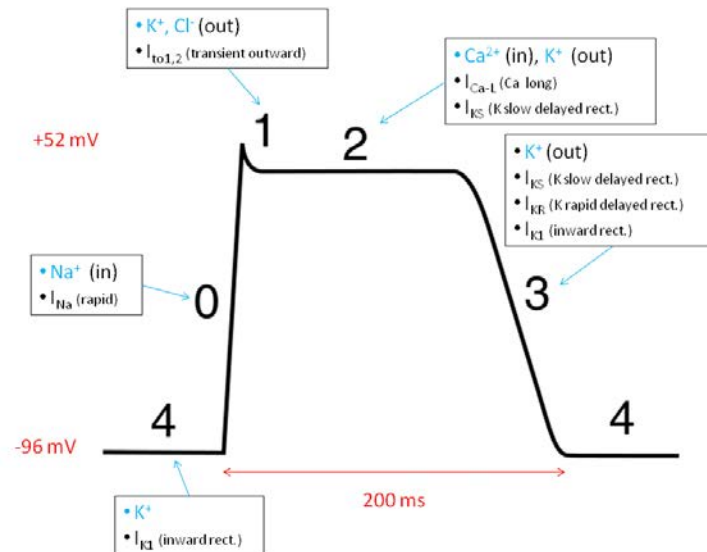
The correct sequence for ventricular action potential is as follows:

- C. Opening of voltage-gated sodium channels
- D. Transient outward flow of potassium
- A. Opening of voltage-gated calcium channels
- B. Repolarizing K⁺ currents

The events of action potential occur in the following sequence:

- Phase 0: Voltage-gated Na⁺ channels open in response to depolarizing stimulus (rapid depolarization phase)

- Phase 1: Inactivation of voltage-gated Na^+ channels and transient outward flow of K^+ causes rapid repolarization (rapid repolarization phase)
- Phase 2: Opening of slow voltage-gated Ca^{2+} channels causes the influx of calcium into the cells and results in the plateau phase (plateau phase)
- Phase 3: Opening of various K^+ channels results in the efflux of K^+ from the cells and results in slow repolarization (slow repolarization phase).
- Phase 4: Return to resting membrane potential



Solution to Question 14:

The zero pressure in the pressure-volume curve indicates the functional residual capacity.

The pressure-volume relationship is basic to the understanding of the fundamental relationship between the compliance of the lungs and the chest wall. It is measured by a spirometer.

The functional residual capacity is the volume of the air that remains in the lungs after normal expiration. The normal value is about 2.5 liters.

Other options:

Option A: Residual volume is the air left in the lungs after a maximal expiratory effort.

Option C: Tidal volume is the amount of air that moves in during each inspiration or the amount of air that moves out with each expiration, during quiet breathing. The normal tidal volume is 500–750 mL.

Option D: Inspiratory reserve volume is the air inspired with a maximal inspiratory effort in addition to the tidal volume.

Solution to Question 15:

Cerebral perfusion pressure (CPP) is calculated as mean arterial pressure (MAP) - intracranial pressure (ICP).

However, ICP values are not readily available. In this case, CPP may be calculated as MAP - central venous pressure (CVP), and if CVP > ICP.

The brain depends on continuous perfusion for oxygen and glucose delivery, thus its survival. The blood flow to the brain is normally about 55 mL per 100 g of brain tissue. If the cerebral flow reduces to less than 20 mL per minute, the brain undergoes ischemia.

Blood flow depends on cerebral perfusion pressure (CPP). Normally the CPP is 75-105 mmHg, MAP is 90-110 mmHg and ICP is 5-15 mmHg.

According to Monro Kellie doctrine, the cranium is a rigid box containing a nearly incompressible brain. In cases of head trauma, there is an expansion of cranial content, in the form of hematomas and brain swelling. This increase in cranial pressure is accommodated initially by the exclusion of fluid components, venous blood, and cerebrospinal fluid. Further increase in the cranial pressure leads to an uncontrolled rise in ICP resulting in a compromise in cerebral perfusion and eventually cerebral herniation.

Acute rise in ICP causes Cushing reflex, a triad of widened pulse pressure (increasing systolic, decreasing diastolic) bradycardia, and irregular respirations.

Thus, in cases of head injury with intracranial bleeding, cranial decompression is performed to avoid further complications due to the raised ICP.

Solution to Question 16:

Fluid leaving the proximal convoluted tubule (PCT) in the absence of antidiuretic hormone (ADH) is isotonic.

The PCT is the site of maximum water reabsorption under normal physiological conditions. It reabsorbs around 60% of the filtered NaCl and water and other nutrients such as glucose and amino acids. Isosmotic reabsorption occurs in the PCT, so the amount of solute and water reabsorbed are coupled and proportional to each other. Here, sodium is reabsorbed along with the water. The large amounts of sodium and water reabsorption are essential in maintaining the ECF volume.

In addition to the isosmotic reabsorption, the water reabsorption in the PCT also occurs through aquaporin-1 which are active water channels. Reabsorption of sodium and water in PCT is not affected by ADH levels. Thus even in the absence of ADH, fluid is isotonic in PCT.

Solution to Question 17:

2,3-Bisphosphoglycerate (BPG) causes the unloading of oxygen to tissues by binding to the beta subunit of Hb.

2,3 BPG is formed from glycolysis. It is found in high amounts in red blood cells and in trace amounts in most other cells. 2,3 BPG facilitates increased oxygen delivery by reducing the affinity of hemoglobin for oxygen.

Hemoglobin exists in 2 forms in the blood. The deoxyhemoglobin form is called the taut or T form. This hemoglobin has a low affinity for oxygen. The oxyhemoglobin form is called the relaxed or R form. This hemoglobin has a high affinity for oxygen.

2,3 BPG binds more selectively to deoxyhemoglobin and stabilizes the taut conformation of hemoglobin (Option C).

Other options:

Options A and D: The process of unloading oxygen by 2,3 BPG is independent of ATP production.

Solution to Question 18:

The curve 'A' shows a hyperbolic curve that is characteristic of oxygen dissociation in myoglobin.

Myoglobin is an iron-containing component of the skeletal muscle. Myoglobin can bind only to one oxygen molecule and hence lacks co-operative binding. It has a higher affinity to oxygen than hemoglobin and hence releases oxygen only at low PO_2 values. This leads to a leftward shift in the myoglobin oxygen dissociation curve.

Myoglobin is present in large quantities in muscles that are commonly used for sustained contraction. During such contractions, muscle blood supply is decreased due to compression. Myoglobin facilitates oxygen supply in these conditions.

Other options:

Option B: Carbon monoxide (CO) reacts with hemoglobin to form carbon monoxyhemoglobin (carboxyhemoglobin or COHb). The affinity of hemoglobin for CO is 210 times its affinity for O_2 , and COHb liberates CO very slowly. An additional difficulty is that when COHb is present, the dissociation curve of the remaining HbO_2 shifts to the left, decreasing the amount of O_2 released.

Option C: Hemoglobin normally has ferrous ion. When it is oxidized to ferric iron, it is called methemoglobin. It causes dark-coloured blood.

Option D: Fetal hemoglobin (HbF) is structurally similar to hemoglobin A except for the gamma chains. It binds less avidly to 2,3 bisphosphoglycerate and hence has higher affinity for oxygen leading to left ward shift in the dissociation curve.

Solution to Question 19:

The organ with the least arteriovenous oxygen difference is the kidney.

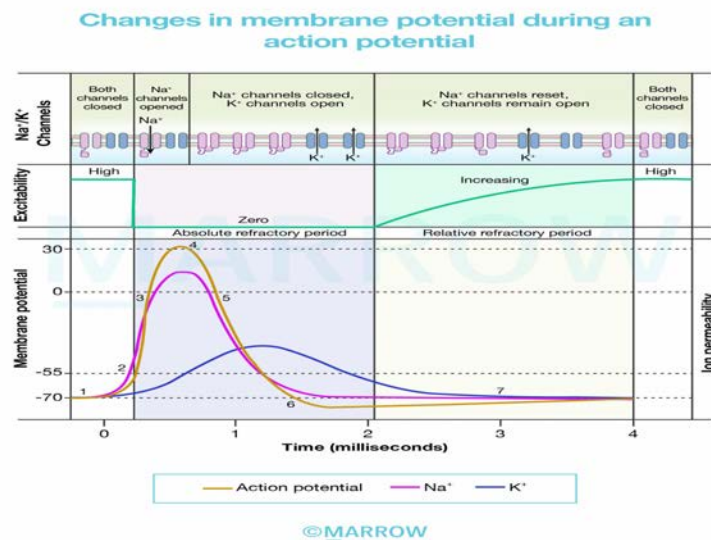
The arteriovenous oxygen difference is the amount of oxygen the tissues take up from the blood. If the amount of oxygen utilized by the tissues is more, the greater the arteriovenous oxygen difference.

The renal cortical blood flow is comparatively greater as the main function is to filter large volumes of blood through the glomeruli. Here, the oxygen extracted from the blood is relatively less. The arteriovenous oxygen difference for the kidney is only 14 mL/L of blood, which is very less when compared to the brain and heart.

Organs	Arteriovenous Oxygen Difference (mL/L)
Brain	62
Kidney	14
Skin	25
Liver	34
Heart muscles	114

Solution to Question 20:

The region marked between 4 and 5 in the action potential curve indicates repolarization due to the efflux of K^+ ions across the cell membrane.



The sequence of events during the action potential occurs in the following sequence.

1. Resting state: Both channels are closed.
2. Na^+ channels open in response to depolarizing stimulus. Sodium enters the cell. The membrane reaches threshold potential.
3. The sodium influx causes the opening of more Na^+ channels through a positive feedback mechanism. Rapid depolarization occurs.
4. Sodium conductance decreases.
5. Na^+ channels close and K^+ channels open. Due to K^+ efflux, repolarization occurs.
6. Na^+ channels are reset and are ready to respond to another depolarizing stimulus (relative refractory period). During this phase, the K^+ channels remain open and slowly return to the closed state causing after-hyperpolarization.

7. Both channels are closed. The membrane returns to RMP.

Solution to Question 21:

The value of normal tissue oncotic pressure is 12 mmHg. It refers to the interstitial osmotic pressure.

The interstitial oncotic pressure, while very small, is the pressure exerted by a small amount of protein present in the interstitium that tends to pull fluids out of the capillaries. It is different for each organ based on the level of protein components. It is mostly negligible, but sometimes can be as high as 6 mmHg in the brain. The values are also positive in the liver and kidneys.

The colloid osmotic pressure or oncotic pressure is the osmotic pressure exerted due to the presence of colloidal proteins like albumin and globulin within the plasma. The colloidal proteins are impermeable to the semi-permeable cell membrane and cannot cross the membrane. Thus, it exerts osmotic pressure, which pulls fluid into the capillaries.

The lowest number in the given options is 12 mmHg and, hence, is the best possible answer to this question.

Solution to Question 22:

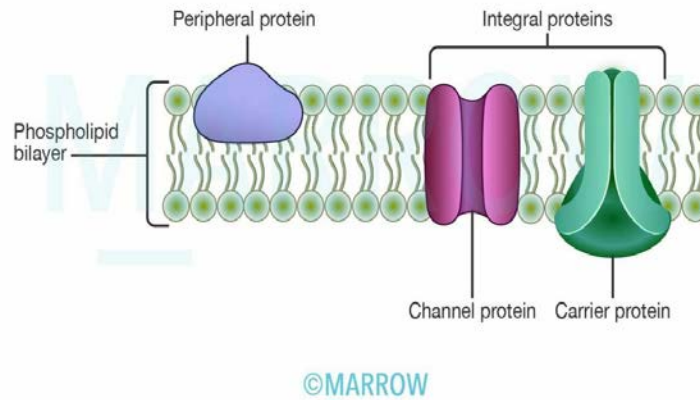
The signaling sequence in a type 2 membrane protein is located inside the membrane (transmembrane region).

A membrane protein can be either integral protein (spans the thickness of the cell membrane) or peripheral protein (attached to the peripheral side of the membrane).

Integral membrane proteins or transmembrane proteins are embedded in the cell membrane, spanning through the entire thickness. They are anchored by hydrophobic interactions and require the use of detergent for solubility. Examples are ligand-binding receptors for hormones or neurotransmitters, transport proteins like Na⁺-K⁺ ATPase, pores, ion channels, cell adhesion molecules, and GTP-binding proteins (G proteins).

Peripheral proteins are attached only to one surface of the cell membrane and do not penetrate all the way through. They are often attached to integral proteins. Examples are cell surface receptors, enzymes, ankyrin.

Structure of cell membrane



Signaling sequences are 16 to 30 amino acid residues in length and located at the N-terminal extension of the membrane proteins. After an appropriate trigger, they are either cleaved (and bind to MHC class I of the endoplasmic reticulum) or act as a membrane anchor (transporting proteins across the membrane).

Integral proteins can be divided into:

- Type I integral proteins have a cleavable N-terminal signal sequence (towards the cytosol). Examples include histocompatibility antigens, glycophorin, and membrane immunoglobulin.
- Type II integral proteins have a non-cleavable N terminal combined signal/anchor sequence. Examples include transferrin receptors and glycosyl transferases.

Solution to Question 23:

IP₃ is the primary second messenger involved in the contraction of ciliary muscles to focus on a near object.

On focusing on a near object, stimulation of the parasympathetic nerve fibers leads to the contraction of the bilateral ciliary muscle, via short ciliary nerves, which relaxes the lens ligaments. The lens becomes thicker and the increase in the refractive index allows the eye to focus on the near object. This is known as the accommodation reflex.

In the ciliary muscles of the eye, contraction is mediated via M₃ > M₂ receptors, while relaxation is mediated via the β₂ receptor.

M₃ receptors act via the formation of inositol 1,4,5-trisphosphate (IP₃) and diacylglycerol (DAG), leading to an increase in intracellular Ca²⁺ and activation of protein kinase C, respectively. The increase in intracellular Ca²⁺ sets up a chain reaction which eventually leads to contraction of the ciliary smooth muscle of the eye.

Thus, in the contraction of the ciliary muscle during focusing on a near object, the second messenger involved is the IP₃-DAG pathway.

Solution to Question 24:

Adipose tissue is dependent on insulin for glucose uptake.

The uptake of glucose in skeletal muscles is also dependent on insulin but may be mediated via non-insulin-dependent mechanisms as well (such as muscle contractions). Therefore the best answer out of the given options is adipose tissue.

Glucose uptake by skeletal muscle, cardiac muscle, and adipose tissues is via GLUT-4 which is insulin-dependent. In the presence of insulin, GLUT-4 receptors translocate to the cell membrane and hence, glucose uptake by the cell occurs.

Most of the other GLUT transporters are not insulin sensitive and are constitutively expressed in the cell membrane. Ex: GLUT-1 in RBCs, brain and kidneys. (GLUT-3 is also present in the brain and kidneys)

Solution to Question 25:

During the second phase of salivary secretion, Na^+ is actively reabsorbed, Cl^- is reabsorbed passively, K^+ is secreted actively and HCO_3^- is secreted passively (part of the secretion may be active as well, but among the given options, option D is the most accurate with respect to all the ions).

Salivary secretion occurs in two stages:

The first stage involves acini, which secrete ptyalin and/or mucin together with some ions. The composition of the ions is similar to that of the extra-cellular fluid.

The second stage involves the ducts. As the primary salivary secretion flows through the ducts, two active transport processes occur:

- Na^+ is actively re-absorbed and K^+ is actively secreted, in exchange for Na^+ .
- HCO_3^- is secreted by the ductal epithelium into the lumen of the salivary ducts. Some of this secretion occurs by passive bicarbonate exchange for Cl^- and partly active secretion.

As a result, the Na^+ and Cl^- concentrations are lower than that of plasma, while K^+ and HCO_3^- concentrations are higher than that of plasma.

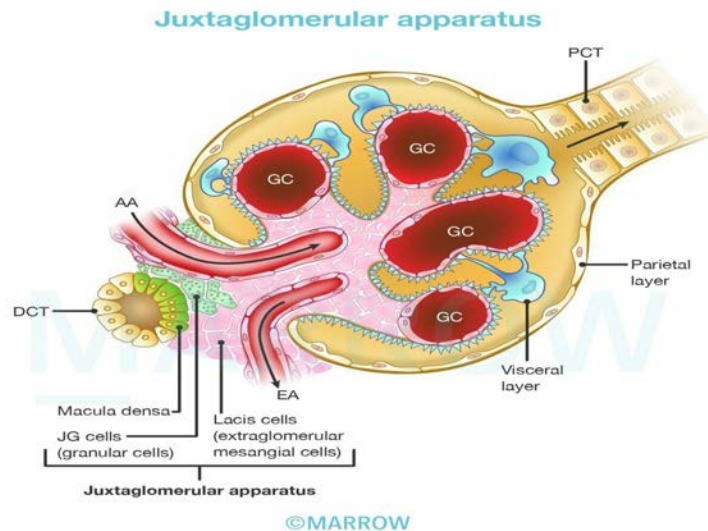
Composition of ions:

- Under resting conditions, the Na^+ and Cl^- concentrations in saliva are about 15 mEq/L each, which is 1/7th to 1/10th of the plasma concentrations.
- K^+ concentration is about 30 mEq/L, which is seven times higher than that of plasma.
- HCO_3^- concentration is 50 to 70 mEq/L, which is two to three times higher than that of the plasma.

Solution to Question 26:

Macula densa are the specialized cells present at the distal end of the thick ascending loop of Henle, where it reaches the glomerulus. Distal convoluted tubule starts from macula densa.

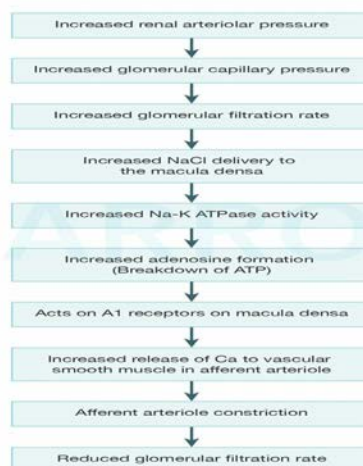
Macula forms the part of the juxtaglomerular apparatus.



Signals from the renal tubules affect filtration in its glomerulus. As the rate of flow through the ascending limb of the loop of Henle and the first part of the distal tubule increases, glomerular filtration in the same nephron decreases, and vice versa. This is known as tubuloglomerular feedback which maintains the constancy of the load delivered to the distal tubule.

The sensor is the macula densa. The Na^+ and Cl^- in the fluid enter the macula densa cells via the $\text{Na}^+-\text{K}^+-2\text{Cl}^-$ cotransporter in their apical membranes. The increased Na^+ increases the Na^+-K^+ ATPase activity causing ATP hydrolysis and the formation of adenosine. This acts on adenosine A_1 receptors on the macula densa cells, causing the release of Ca^{2+} into the vascular smooth muscle in the afferent arterioles, leading to vasoconstriction and decreased GFR.

Tubuloglomerular feedback



Solution to Question 27:

Fick principle, thermodilution, and ECHO can be used to measure cardiac output.

Cardiac output is the amount of blood pumped by the heart into the aorta every minute. The average cardiac output is 5L/min and can be measured by various indirect methods.

Fick's principle uses the total amount of oxygen absorbed from the air by the lungs and the arteriovenous oxygen difference of blood to calculate the cardiac output using the following formula:

Cardiac output (L/min) = (O₂ absorbed per minute by the lungs (mL/min)) / (Arterio-venous O₂ difference(mL/L of blood))

Thermodilution method employs cold saline which is infused into the right atrium through a double-lumen Swan–Ganz thermodilution catheter and the temperature change is measured at the pulmonary artery using a thermistor at the end of the catheter. The mean decrease in the temperature of blood is inversely proportional to the cardiac output.

An ECHO uses ultrasound waves from a transducer placed on the chest wall to measure the velocity of blood flowing from the left ventricle into the aorta. Stroke volume can be calculated from this. Cardiac Output = Heart Rate X Stroke Volume.

The other methods that can be used to measure cardiac output include impedance cardiography a.k.a. thoracic electrical bioimpedance, electromagnetic or ultrasonic flowmeter, and indicator dilution method.

Ventilation-perfusion scan is not a method of cardiac output measurement. It is used to assess pulmonary circulation.

Solution to Question 28:

The correctly matched answer is 1-D, 2-A, 3-B, 4-C.

Each organelle has some specific enzymes are called marker enzymes. After centrifugation, the separated organelles are identified by the detection of marker enzymes in the sample.

Cellular components	Marker enzymes
1. Mitochondria	D. ATP synthase
2. Peroxisomes	A. Catalase
3. Golgi complex	B. Galactosyl transferase
4. Endoplasmic reticulum	C. Glucose-6-phosphatase

Solution to Question 29:

The true statements are 1,4.

Sympathetic efferents are distributed extensively all over the body whereas parasympathetic distribution is limited to head, neck, and trunk regions.

Sympathetic fibers travel short distances to the sympathetic ganglionic chain, before finally synapsing and then have a long course to supply the target organs. Parasympathetic fibers travel a longer distance to their respective ganglia which are located close to the target organs.

Motor neurons are located in the ventral horn of the spinal cord from where the axons come out and divide into several branches, each of which supplies an individual muscle fiber. A motor unit comprises a motor neuron and a group of skeletal muscle fibers it innervates.

Solution to Question 30:

The normal insensible water loss is 50 mL/hour.

In human beings, there is a loss of water through vaporization from the respiratory tract (300–400 mL) and diffusion from the skin (300–400 mL). This is called insensible water loss. It is about 50 mL/hour. This insensible water loss via the skin is independent of sweating and occurs in people without sweat glands as well.

Solution to Question 31:

The correct sequence of vomiting is 6, 4, 2, 3, 1, 5.

6. Reverse peristalsis in the small intestine.
4. Relaxation of the pyloric sphincter.
2. Strong contractions in the stomach.
3. Inspiration against a closed glottis.
1. LES is open and UES is closed.
5. LES opens and UES opens.

The sensory signals that initiate vomiting are located in the pharynx, esophagus, stomach, and upper portions of the small intestines. Excessive distention or irritation of the duodenum provides a strong stimulus for vomiting. Area postrema constitutes the vomiting centre.

In the early stage, reverse peristalsis begins often minutes before vomiting. It begins far down in the ileum and travels up the intestine. This process pushes a large share of lower small intestine contents to the duodenum and stomach. Relaxation of the pyloric sphincter occurs during this process.

At the onset of vomiting, strong intrinsic contractions are seen in the duodenum and stomach. Once the vomiting centre has been stimulated, inspiration occurs against the closed glottis. It prevents the flow of vomitus into the lungs. The lower esophageal sphincter (LES) is open and

the upper esophageal sphincter (UES) is closed at this stage.

The muscles of the abdominal wall contract, building intra-gastric pressure. Finally LES relaxes completely and UES also relaxes allowing the expulsion of the gastric contents.

Note: The esophageal smooth muscle functions as LES at the gastro-esophageal junction. It normally remains tonically constricted. Receptive relaxation of LES occurs when the peristaltic wave passes down the esophagus.

Physiology NEET 2022

Question 1:

A person after sleeping overnight with the arm under his head now experiences paresis but no numbness in the morning. Which of the following is the best explanation for it?

- a) C fibers are more sensitive to pressure than A fibers.
- b) A fibers are more sensitive to hypoxia than B fibers.
- c) A fibers are more susceptible to pressure changes than C fibers.
- d) A fibers are more susceptible to hypoxia than C fibers.

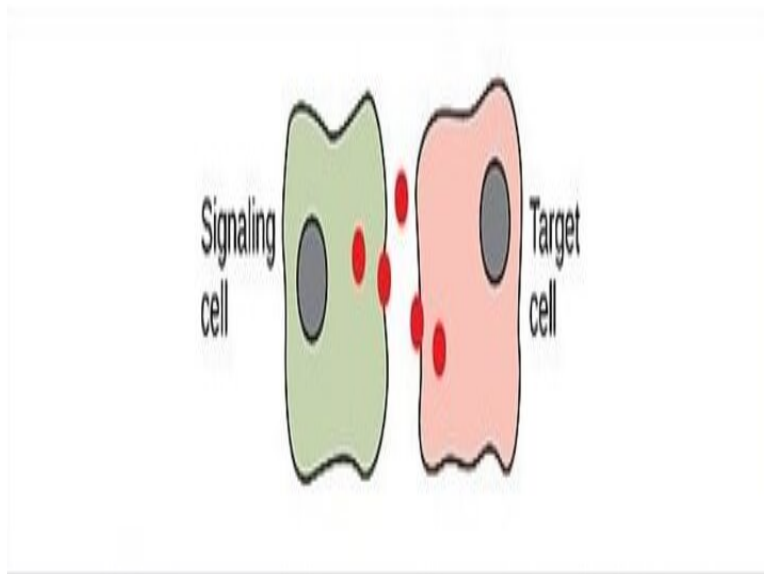
Question 2:

Which of the following mechanisms is seen in the baroreceptor reflex?

- a) Feedforward
- b) Positive feedback
- c) Negative feedback
- d) Adaptive control regulation

Question 3:

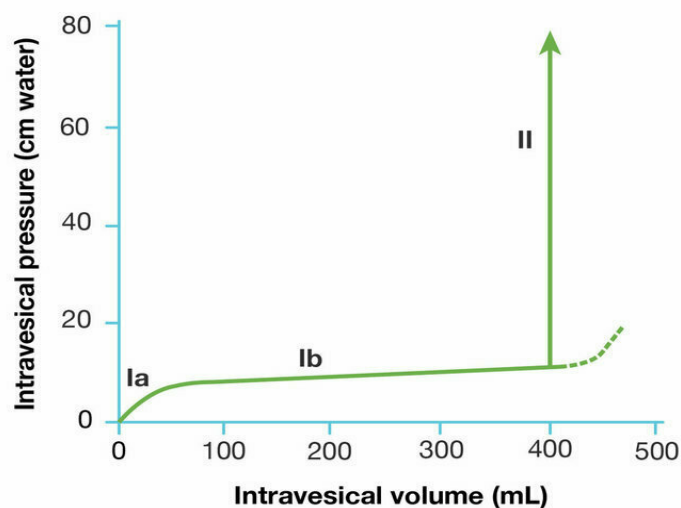
The image below depicts which of the following types of cell-to-cell signaling?



- a) Paracrine
- b) Autocrine
- c) Endocrine
- d) Merocrine

Question 4:

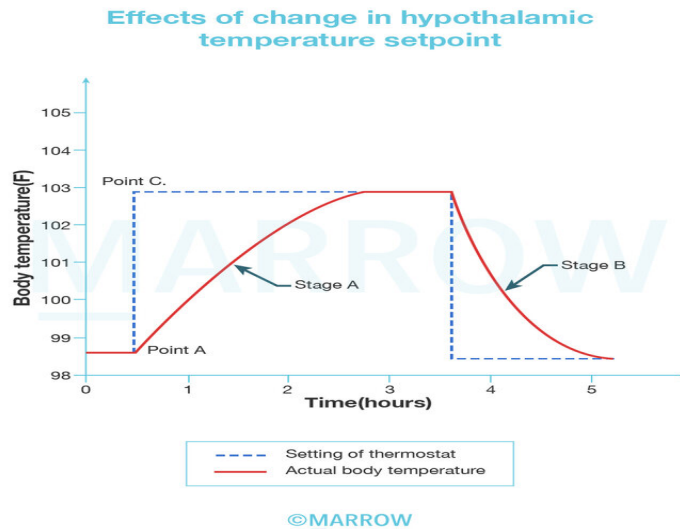
Which of the following statements is true regarding the given cystometrogram?



- a) Segment Ia is due to residual urine
- b) Segment Ib is due to Laplace law
- c) Micturition fails to happen in segment II
- d) The dotted line represents that micturition has occurred

Question 5:

In a patient, the hypothalamic thermostat was reset from point A to point C as shown below. Which of the following happens in stage A compared to stage B?



- a) Shivering
- b) Sweating
- c) Increased blood flow to skin
- d) Inhibition of chemical thermogenesis

Question 6:

A 65-year-old man suffered from a stroke 2 days ago. He now presents with involuntary, violent, and flinging movements of the limbs on one side. What is the likely site of lesion in this patient?

- a) Subthalamic nuclei
- b) Globus pallidus
- c) Putamen
- d) Caudate nucleus

Answer Key

Question No.	Correct Option
1	c
2	c
3	a

4	b
5	a
6	a

Detailed Explanations

Solution to Question 1:

In the above scenario, the patient has developed paresis (partial paralysis) but the pain sensation is intact (no numbness). This is because A fibers that carry motor, touch, and pressure sensations are more susceptible to pressure changes/compression than C fibers that carry pain sensations.

The compression of a large diameter nerve fiber leads to a reversible physiologic conduction block or neuropraxia.

The relative susceptibility of nerve fibers to conduction blocks produced by various agents:

Susceptibility	Most susceptible	Intermediate	Least susceptible
Local anesthetic	C	B	A
Hypoxia	B	A	C
Pressure	A	B	C

Solution to Question 2:

The negative feedback mechanism is seen in the baroreceptor reflex. If the activity of a particular system is increased, the controller system initiates the negative feedback mechanism, which prevents overactivity and ensures normal homeostasis.

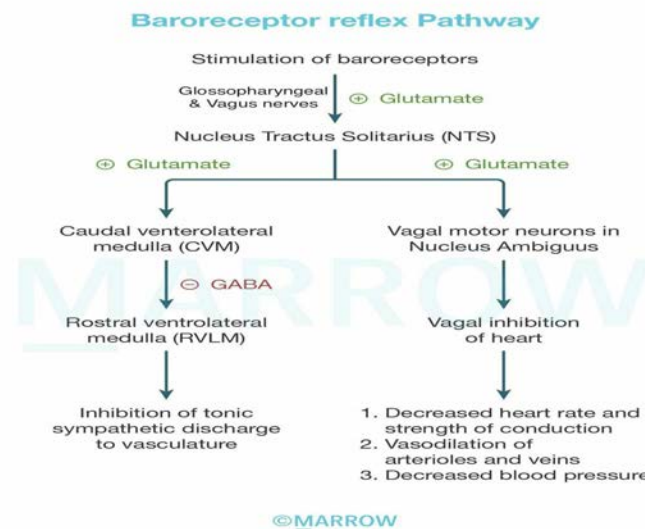
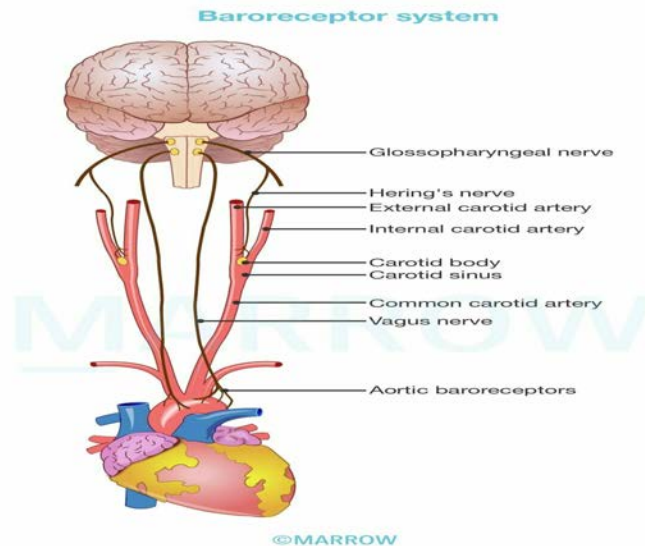
Baroreceptors are stretch receptors located in the heart and the adventitia of the blood vessels. They are stimulated by the stretching of the walls of the blood vessels when there is an increased volume or blood pressure in the vessel. They are important in the immediate management of blood pressure. They are of two types: carotid sinus and aortic arch receptors (high-pressure regions) and cardiopulmonary receptors (low-pressure regions).

An increase in blood pressure stimulates the baroreceptors. These signals are transmitted through small Hering's nerves to the glossopharyngeal nerves and then to the nucleus tractus solitarius (NTS) in the medulla. These signals release glutamate and activate the caudal ventrolateral medulla (CVLM). This in turn inhibits the rostral ventrolateral medulla (RVLM) considered the vasoconstrictor center for the brain and the heart.

Simultaneously, this pathway also excites the vagal parasympathetic center. The stimulation of the vagus results in a slowing of impulse generation in the SA node and

a slowing of conduction velocity in the AV node.

These neural changes produce vasodilation, venodilation, hypotension, bradycardia, and a decrease in cardiac output.



Other options:

Option A: The feedforward mechanism responds to the anticipation of a change in a regulated variable. For example, when a meal is in the digestive tract, a feedforward mechanism increases the secretion of insulin that promotes the cellular uptake and storage of ingested nutrients.

Option B: Positive feedback mechanism refers to the enhancement or amplification of the output so that it moves in the direction of the initial change. An example is seen in the luteinizing hormone (LH) surge where the estrogen positively regulates the further secretion of LH.

Option D: The adaptive control mechanism refers to the delayed type of negative feedback mechanism. When the feedforward mechanism has an error, the correction is made by a retrospective feedback mechanism called the adaptive control mechanism.

Solution to Question 3:

The given image shows the paracrine cell-to-cell signaling mechanism. In paracrine signaling, the signaling molecules released by a cell only affect target cells in close proximity to it.

An example of paracrine signaling is the release of endothelin-1, a potent vasoconstrictor released by the endothelial cells.

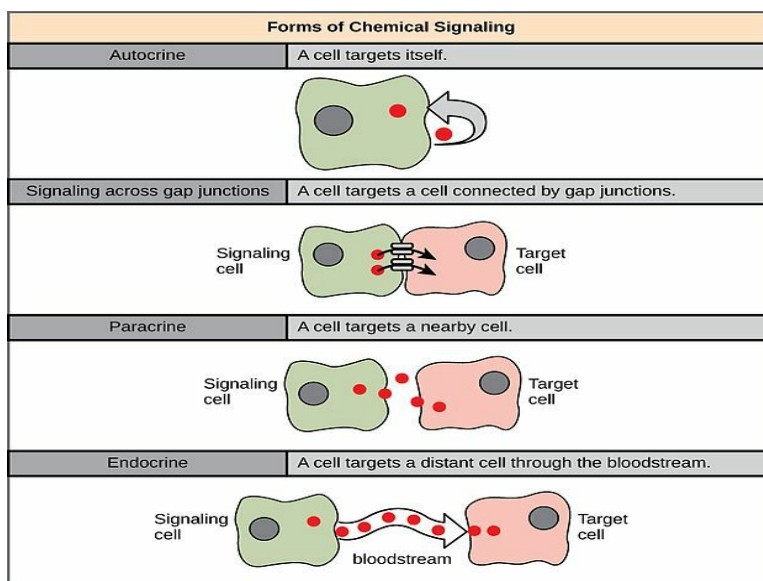
Synaptic signaling is a special case of paracrine signaling (for chemical synapses) between neurons and target cells.

Other options:

Option B: In the autocrine signaling mechanism, the cells respond to the signaling molecules that are released by themselves, which is the site of release and action within the same cell. Note that the signaling molecules are first secreted out of the cell and then act on the same cell through receptors. An example is the proliferation of activated T-cells by IL-2 released by the same cell after it interacts with antigen-presenting cells.

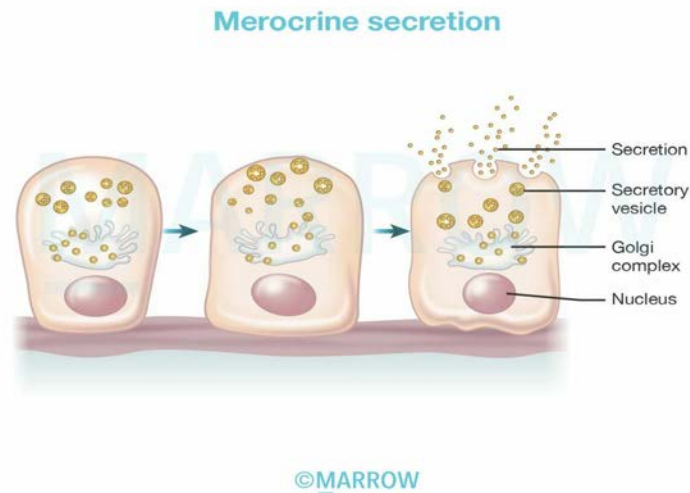
Option C: In endocrine signaling, a signaling molecule, a hormone (for example insulin, FSH, LH), is usually carried by the blood from its site of release to act on its target at a distance.

The below image shows the different types of extracellular cell-to-cell communication mechanisms.



Option D: Merocrine is a type of glandular secretion and not an extracellular signaling mechanism. In this, the secretions are poured out by exocytosis and the cell remains intact. An example of the merocrine gland is the goblet cell and eccrine sweat gland.

An illustration depicting this mechanism is given below.



Solution to Question 4:

The true statement regarding the given cystometrogram is that segment Ib is due to Laplace law.

Cystometrogram is a plot of intravesical pressure against the volume of fluid in the bladder. The curve shows three segments:

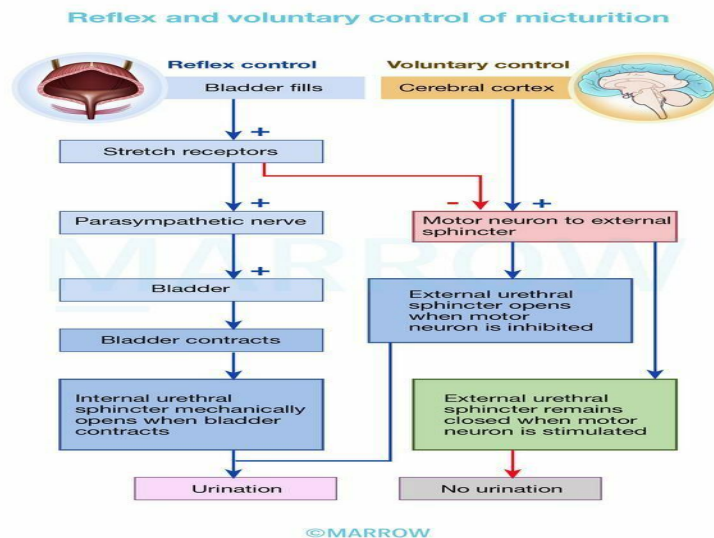
- Ia- initial slight rise in pressure when the first increments in the volume are produced
- Ib- long, nearly flat segment as further increments are produced
- II- sudden, sharp rise in pressure as the micturition reflex is triggered (not failure).
- The dotted line- pressure-volume relation if the micturition had not occurred.

The flatness of segment Ib is a demonstration of the law of Laplace. The law of Laplace states that the pressure (P) in a spherical viscus is equal to twice the wall tension (T) divided by the radius (R). This is expressed as $P = \frac{2T}{R}$. In the bladder, the tension increases as the bladder gets filled. Since the radius also increases there is only a slight increase in pressure until the bladder is relatively full.

The first urge to void is felt at a bladder volume of about 150 mL. A marked sense of fullness is felt at a bladder volume of 400mL.

Micturition reflex is the result of sensory signals from the stretch receptors in the bladder wall, which are conducted to the sacral segments of the cord through the pelvic nerves and then efferent signals conduct back again to the bladder through the parasympathetic nerve fibers.

The micturition reflex pathway is given below.



Solution to Question 5:

In the above scenario, the patient's hypothalamic thermostat has been reset from point A to a higher temperature (point C). The body responds to this change by using certain measures in stage A such as shivering.

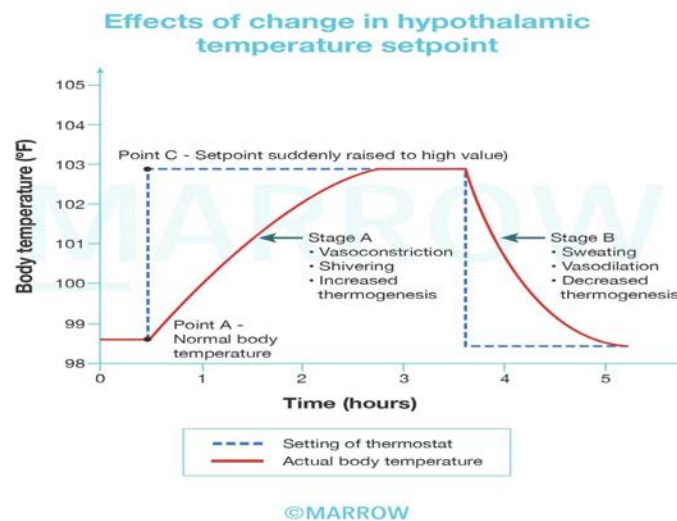
In stage A the body activates various heat gain mechanisms to match the hypothalamic temperature setpoint. These include:

- **Shivering**- the shivering center is located in the posterior hypothalamus. It is normally inhibited by signals from the heat center in the anterior hypothalamic-preoptic area but is excited by cold signals from the skin and spinal cord. ■A sudden increase in heat setpoint results in the activation of this center. It then transmits non-rhythmic signals through bilateral tracts through the lateral columns of the spinal cord, and finally to the anterior motor neurons. They increase the tone of the skeletal muscles throughout the body. As the tone rises above a certain critical level, shivering begins. During maximum shivering, body heat production can rise to four to five times normal.
- **Vasoconstriction** (decreased blood flow to skin) is mediated by increased sympathetic drive by the activation of the posterior hypothalamic sympathetic center.
- An increase in chemical thermogenesis in which heat production takes place in the brown fat by the mechanism of uncoupling of oxidative phosphorylation. This is the predominant mechanism seen in neonates.

In stage B, the hypothalamic temperature setpoint is lower than the normal body temperature. Hence body activates various heat loss mechanisms to match the hypothalamic temperature setpoint. It includes vasodilatation of the skin blood vessel, sweating, and decreased thermogenesis.

The temperature of the body is regulated majorly by a nervous feedback mechanism that operates through temperature-regulating centers located in the hypothalamus. Minor regulation is done by the thermal receptors in the skin and the deep thermal receptors present in the spinal

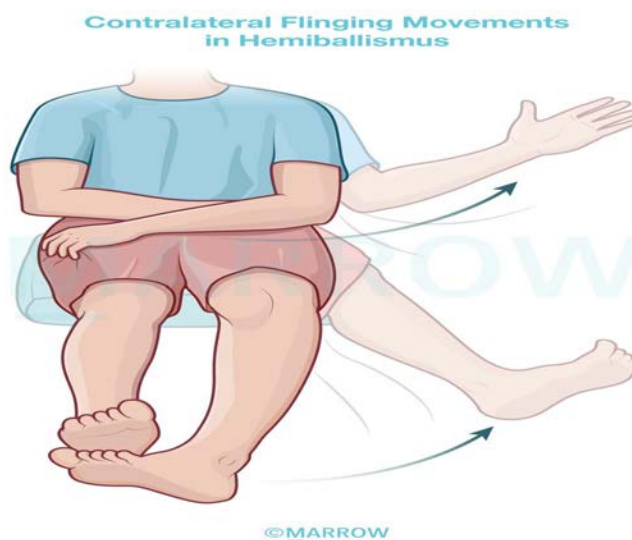
cord and abdominal viscera.



Solution to Question 6:

The above clinical scenario is suggestive of hemiballismus. A lesion in subthalamic nuclei is responsible for this condition.

A lesion in the subthalamic nucleus causes sudden, violent, involuntary, and flinging movements confined to one side of the body, commonly affecting the proximal muscles of the limbs, a condition called hemiballismus. The movements are seen contralateral to the side of the lesion, i.e., involuntary movements of the right arm and leg are seen in the lesion of the left subthalamic nucleus and vice versa.



Basal ganglia circuits have both direct and indirect pathways. The direct pathway stimulates the cortex to produce movement, whereas the indirect pathway inhibits it. The striatal activity in the direct pathway is mediated by D1 dopamine receptors and in the indirect pathway by D2 dopamine receptors.

The subthalamic nucleus is a part of the indirect pathway. Hence, a lesion at subthalamic nuclei leads to loss of inhibitory effect of the indirect pathway resulting in dopaminergic overactivity. This is responsible for features of hemiballismus.

This movement disorder is usually self-limiting and tends to resolve spontaneously after weeks or months. Dopamine antagonistic agents can be given to manage it as well.

Other options:

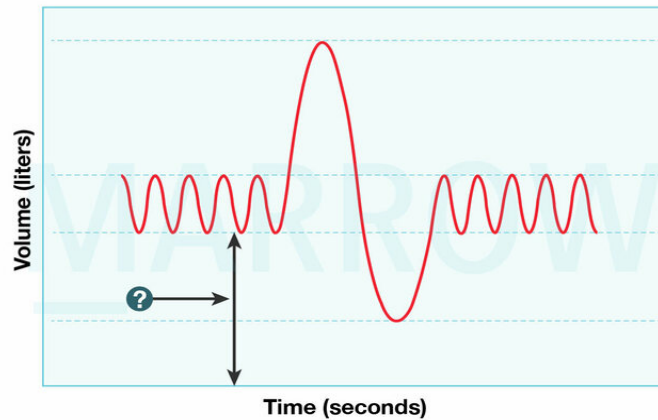
Option B: A lesion in the globus pallidus can give rise to athetosis. This involves spontaneous and continuous writhing movements of the hand, arm, neck, or face.

Options C and D: A lesion in the putamen and caudate nucleus can give rise to chorea. Chorea is defined as rapid, semi-purposeful, graceful, dance-like nonpatterned involuntary movements involving distal or proximal muscle groups.

Physiology NEET 2023

Question 1:

Identify the marked region in the given spirometry.



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- a) Functional residual capacity
- b) Tidal volume
- c) Expiratory reserve volume
- d) Expiratory capacity

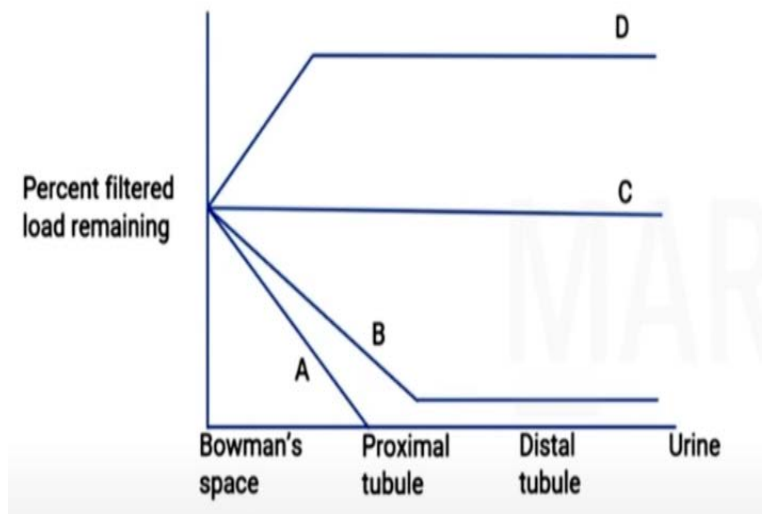
Question 2:

A man was brought to the emergency after he suddenly became unconscious while working in the field. On examination, his temperature was 105 degrees Fahrenheit and his skin turgor was decreased. Which of the following would not be seen in the patient?

- a) Tachypnea
- b) Hypotension
- c) Sweating
- d) Red hot skin

Question 3:

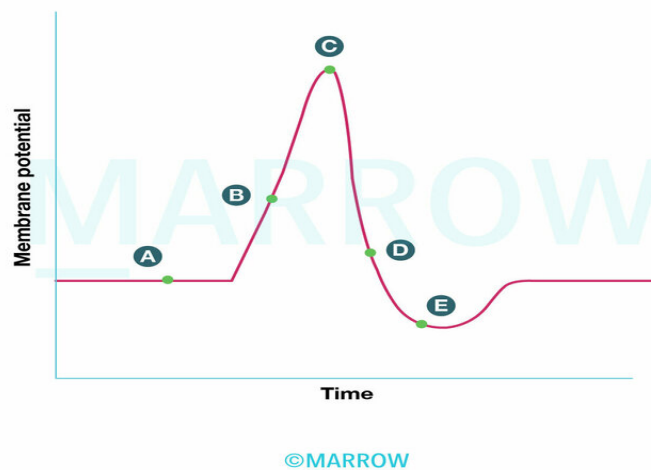
Identify the correctly matched pair of substances with their renal clearance from the graph given below.



- a) A - Glucose, B - PAH, C - Bicarbonate and D - Inulin
- b) A - Glucose, B - Bicarbonate, C - Inulin and D - PAH
- c) A - PAH, B - Inulin, C - Glucose and D - Bicarbonate
- d) A - Inulin, B - Glucose, C - Bicarbonate and D - PAH

Question 4:

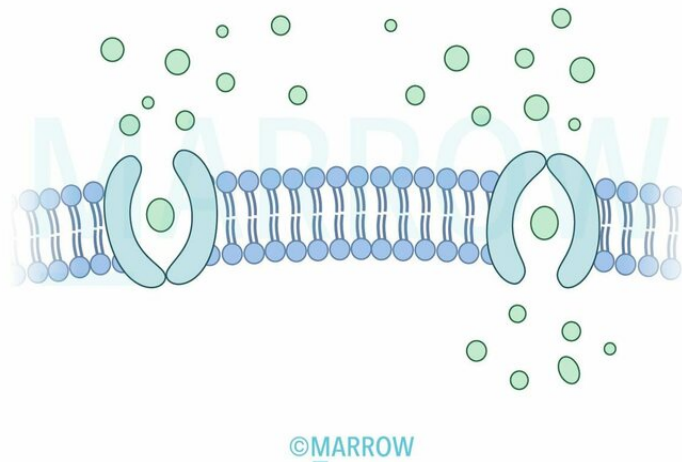
Identify the true statement regarding the given nerve action potential curve.



- a) Threshold point is at A
- b) At point E, the nerve is more excitable
- c) Point C to D is due to opening of Na^+ and closure of K^+ channels
- d) Point B to D is refractory period

Question 5:

Identify the type of transport across the cell membrane given below.



- a) Simple diffusion
- b) Facilitated diffusion
- c) Primary active transport
- d) Secondary active transport

Question 6:

A preterm baby who was delivered at 28 weeks developed respiratory distress syndrome. Which of the following is true about surface tension and compliance in this baby?

- a) Surface tension - decreased; Compliance - increased
- b) Surface tension - increased; Compliance - decreased
- c) Both surface tension and compliance decreased
- d) Both surface tension and compliance increased

Question 7:

A patient came to the hospital after a road traffic accident. He had severe muscle injury and his serum K^+ level was found to be 5.5 mEq/L. What will happen to the resting membrane potential in this patient?

- a) No change
- b) Becomes more negative
- c) Becomes more positive
- d) First becomes more positive then negative

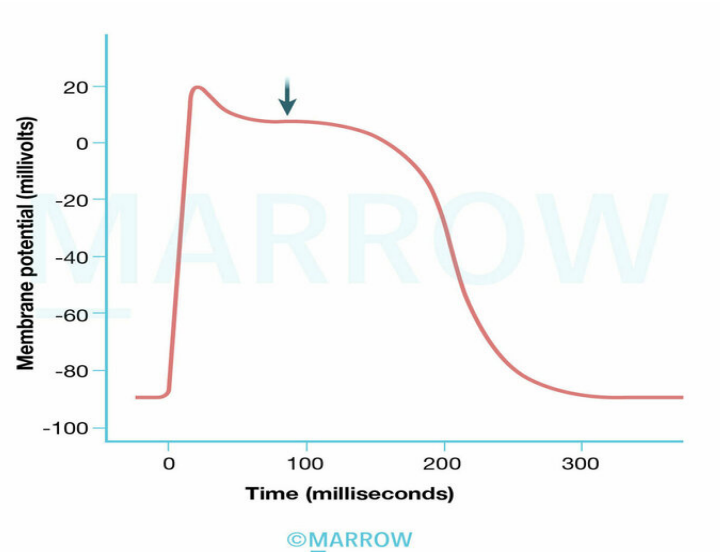
Question 8:

A patient met with a road traffic accident and developed a cervical spine injury. The fracture fragment had pierced the lateral aspect of the dorsal column tract. Which of the following findings is seen in this patient?

- a) Absence of ipsilateral lower limb proprioception
- b) Absence of fine motor movement of fingers
- c) Absence of ipsilateral arm proprioception
- d) Absence of contralateral lower limb proprioception

Question 9:

Identify the true statement regarding the point marked on the myocardial action potential curve.



- a) Due to opening of Na^+ and closure of fast K^+ channels
- b) Due to slow but prolonged opening of Ca^{2+} channels
- c) Due to closure of Na^+ and opening of slow K^+ channel
- d) Due to Na^+/K^+ ATPase

Answer Key

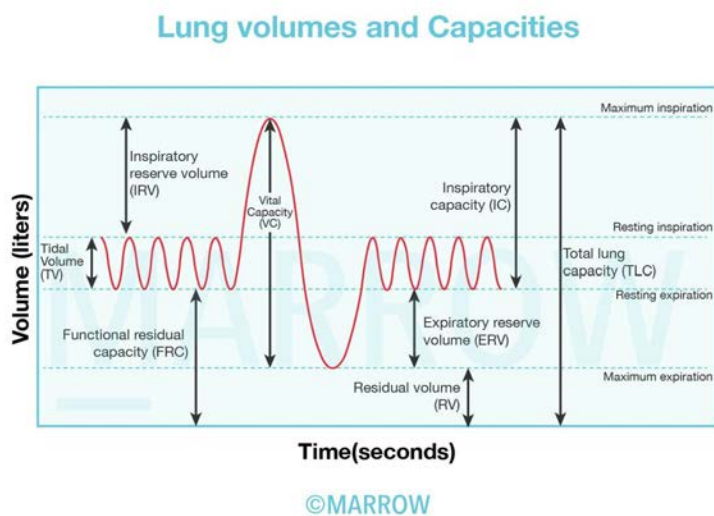
Question No.	Correct Option
1	a
2	c
3	b
4	d
5	b
6	b
7	c
8	c
9	b

Detailed Explanations

Solution to Question 1:

The marked area in the spirometry shows the total amount of air present in the lung after normal expiration. This is called functional residual capacity.

The various volumes and capacities in a spirometry are shown below.



Solution to Question 2:

The given clinical vignette in a patient with signs of dehydration and a body temperature of 105°F is highly suggestive of heatstroke. Sweating would not be seen in a patient with heatstroke.

The body has various mechanisms to maintain body temperature within a range of 36–37°C. When the body temperature is elevated by exposure to high environmental temperatures or physical exertion, the central control area in the medial preoptic and anterior hypothalamic nuclei is activated. This engages the autonomic pathways that promote sweating and cutaneous vasodilation, causing the heat to dissipate and bring the body to normal temperature.

Heatstroke occurs due to the failure of the heat regulation center of the hypothalamus. The person fails to lose heat by cutaneous circulation and sweating. It is likely to develop when the body temperature rises to a range of 105°F to 108°F. The predisposing factors include high temperature, increased humidity, minor infections, old age, and physical activity.

It is a medical emergency that leads to homeostatic imbalance. The symptoms include dizziness, abdominal distress, vomiting, delirium, and eventually leading to loss of consciousness. On examination, the skin is dry, hot, and flushed, with a complete absence of sweating (anhidrosis). Other signs seen in heat stroke are tachycardia, Kussmaul-type rapid deep breathing (tachypnea), hypotension, cardiac failure, dizziness, muscle cramping, confusion, and seizures.

Heatstroke can cause damage to multiple organ systems, including the brain, kidneys, and liver, which can lead to life-threatening complications. It is essential to initiate rapid cooling and supportive care measures to prevent permanent injury or death. The cooling methods used to treat heatstroke include:

- Cold water immersion: The patient is immersed in a tub of cold water, which is the most effective way to lower body temperature.
- Evaporative cooling: The patient's body is sprayed with water while fans blow air over the body to help facilitate evaporation and cooling.
- Applying cold packs: Applying ice packs to the neck, armpits, and groin can help cool the body by targeting areas where large blood vessels are located close to the skin surface.
- Misting and fanning: Spraying the patient with cool water and using a fan to circulate air can help promote evaporative cooling.

Preventive measures for heatstroke include staying hydrated, taking regular breaks in shaded areas, and avoiding strenuous activities during the hottest part of the day.

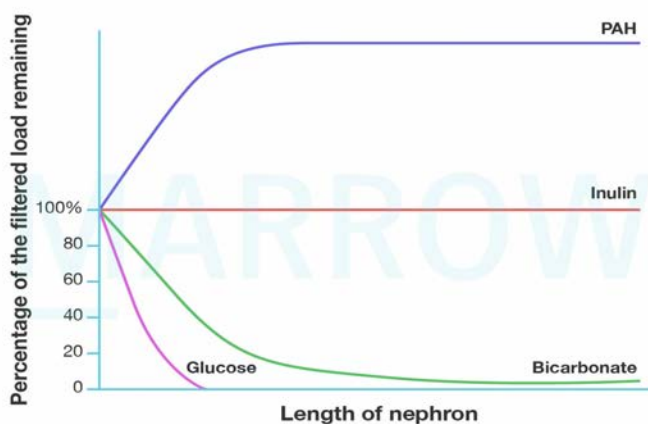
Other Options

- Option A: Tachypnea - Rapid, deep breathing (Kussmaul breathing) can be seen in heatstroke as the body tries to eliminate excess heat through increased respiratory rate.
- Option B: Hypotension - Dehydration and vasodilation in an attempt to dissipate heat can result in hypotension in heatstroke patients.
- Option D: Red hot skin - Heatstroke patients often present with hot, flushed skin as a result of increased blood flow to the skin surface and the body's inability to dissipate heat effectively.

Solution to Question 3:

The correctly matched pair of substances and their renal clearance is A - Glucose, B - Bicarbonate, C - Inulin and D - PAH.

The concentration of the substance changes as it passes through the tubules. The tubular cells may remove the substance from the filtrate (tubular reabsorption) or add more of the substance to the filtrate (tubular secretion) or both. The amount of the substance finally excreted equals the amount filtered plus the net amount transferred out by the tubules.



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PAH is filtered by the glomeruli and secreted by the tubular cells. Its concentration increases on moving through the length of the nephron due to secretion into tubules. It is represented by curve D.

Inulin is freely filtered by glomeruli and does not undergo any reabsorption or secretion. Hence, its concentration does not change throughout the length of the nephron and its excretion is constant and equal to the glomerular filtration rate. It is represented by curve C.

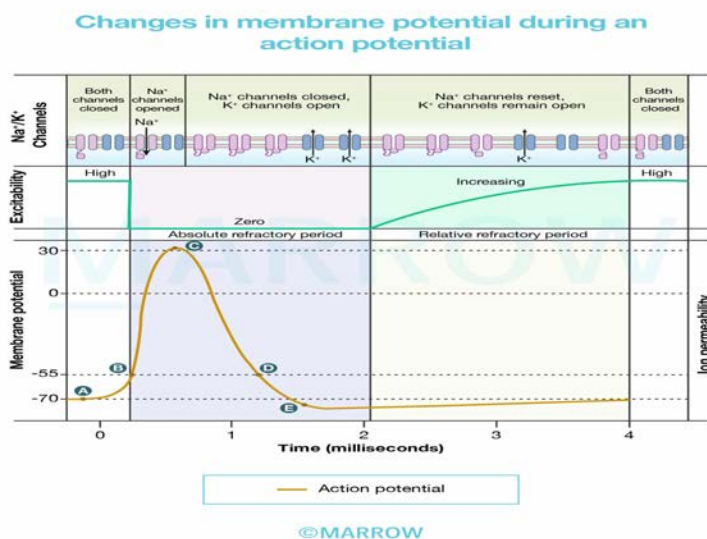
The majority of bicarbonate reabsorption occurs in the proximal tubule and a small amount of it flows into the distal tubules and collecting ducts. Its reabsorption occurs in all parts of the tubules, except the descending and ascending thin limbs of the loop of Henle. It is represented by curve B.

Substances like amino acids or glucose are reabsorbed along with sodium in the early portion of the proximal tubule. They have a graph that is sloping down as shown in curve A.

Solution to Question 4:

The correct statement regarding the given nerve action potential curve is point B to D is the refractory period.

The sequence of events occurring during an action potential is shown below



- Point A represents the resting state where both channels are closed.
- Na⁺ channels open in response to depolarizing stimulus and sodium enters the cell. The initiation of action potential occurs only after a certain potential is reached, which is called the threshold potential represented by point B. (Option A)
- After this, the sodium influx causes the opening of more Na⁺ channels through a positive feedback mechanism. Rapid depolarization occurs to reach point C.
- At point C, Na⁺ channels close and K⁺ channels open, causing repolarization due to K⁺ efflux. (Option C)

- Point D represents continued repolarization due to potassium efflux.
- At point E, Na^+ channels are reset and are ready to respond to another depolarizing stimulus. During this phase, the K^+ channels remain open and slowly return to the closed state, causing after-hyperpolarization. As a stronger stimulus is required to reach the threshold potential, the excitability of nerve fiber is reduced at point E. (Option B)
- As both channels close, the membrane returns to resting membrane potential back to point A.

Point B-D is a part of the refractory period. During this period, the application of even a stronger stimulus will not elicit an action potential.

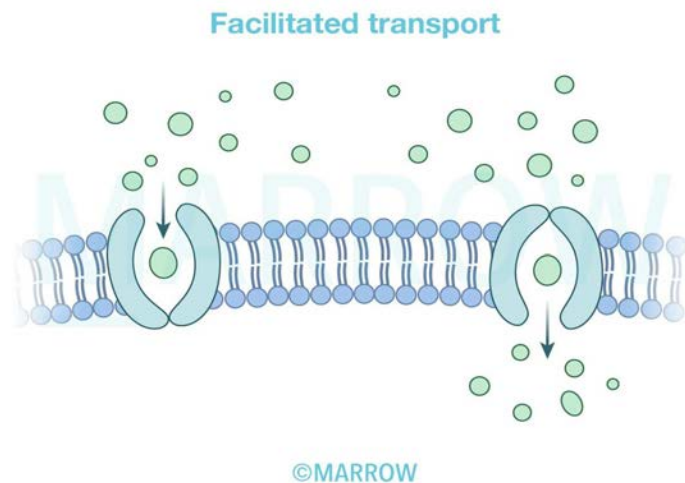
The refractory period is divided into an absolute refractory period and a relative refractory period. During the absolute refractory period, the nerve will not be excited no matter how strong a stimulus is given. However, during the relative refractory period, stronger than normal stimuli can cause excitation.

Solution to Question 5:

The given image shows the transport of a molecule across a membrane with the help of a carrier protein. This form of transport is called facilitated diffusion.

When carrier proteins help in the movement of substances in the direction of their chemical or electrical gradients (downhill movement), no energy input is required and the process is called facilitated diffusion.

Glucose transporters or GLUTs are examples of transport proteins in the cell membrane responsible for the transfer of glucose into the cells by facilitated diffusion. Facilitated diffusion involves the passage of molecules or ions inside the cell membrane by binding to a carrier/transport protein.



Other options -

Option A. Simple diffusion - It is along the concentration gradient but is not carrier-mediated. In the cell, examples of molecules that can use simple diffusion to travel in and out of the cell membrane are water, oxygen, carbon dioxide, and nitrogen. They pass directly through the cell membrane along the concentration gradient, without the use of energy.

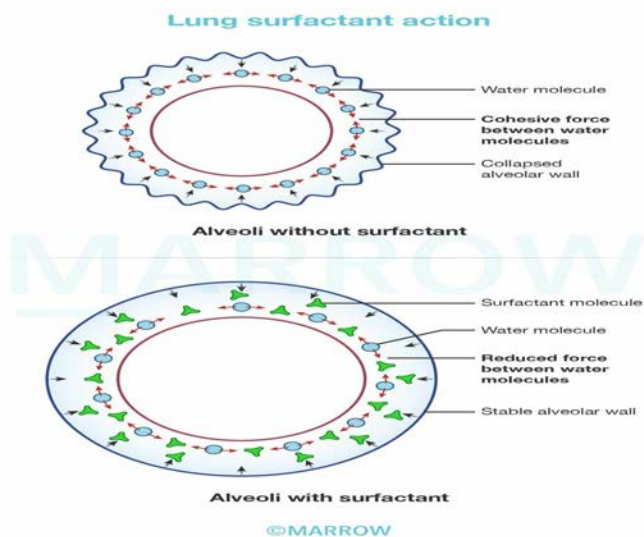
Option C. Primary active transport - It is a form of cellular transport requiring energy in the form of hydrolysis of ATP to ship and transfer molecules against gradient differences. The important examples of primary active transport include sodium-potassium ATPase, proton pump, and calcium ATPase.

Option D. Secondary active transport - In this form of transport, two or more substances are transported together across the membrane with help of a carrier molecule. As one of the substances (e.g., sodium) diffuses down its electrochemical gradient, the energy released is used to drive another substance (e.g., glucose) against the gradient. It does not require energy directly from ATP.

Solution to Question 6:

This preterm baby who developed respiratory distress syndrome has surfactant deficiency. In case of surfactant deficiency, the surface tension increases and compliance decreases.

Surfactant is secreted by the type 2 pneumocytes in the alveoli. Dipalmitoyl lecithin/dipalmitoylphosphatidylcholine (DPPC) is a major component of lung surfactant. It reduces surface tension inside the alveoli, as shown below.



Compliance is defined as the extent of lung expansion for each unit increase in pressure. An important factor that affects lung compliance is surface tension. As surface tension increases, the pressure required to keep the alveoli open also increases. As more pressure is required to expand the alveoli, lung compliance will be reduced.

Solution to Question 7:

In the given clinical scenario, the patient has hyperkalemia. It would result in the resting membrane potential becoming more positive.

The resting membrane potential represents an equilibrium situation where the concentration gradient of ions across the membrane is equal and opposite to the electrical gradients. At a normal potassium concentration of 3.5–5.0 mEq/L, the equilibrium potential for potassium is -90 mV. This is close to the resting membrane potential of neurons, which is about -70 mV.

A decrease in the extracellular sodium concentration reduces the size of the action potential but has little effect on the resting membrane potential. However, the intracellular and extracellular potassium (K^+) concentrations are the major determinants of the RMP.

This is because

- At rest, there are more open K^+ channels compared to Na^+ channels; therefore, the membrane permeability of K^+ is greater.
- The RMP is close to the equilibrium potential of K^+ .

In hyperkalemia, the extracellular level of K^+ is increased. Thus, the RMP moves closer to the threshold to elicit an action potential, leading to a more excitable nerve. This means the RMP has become more positive.

In hypokalemia, the extracellular level of K^+ is reduced. Thus, the RMP is reduced or becomes more negative and the nerve is hyperpolarized.

Hyperkalemia can lead to various clinical manifestations depending on the severity of the condition. Mild to moderate hyperkalemia may be asymptomatic or cause nonspecific symptoms such as fatigue, muscle weakness, and paresthesias. Severe hyperkalemia can lead to life-threatening complications such as cardiac arrhythmias and paralysis. The management of hyperkalemia includes treating the underlying cause, administering medications to stabilize the cardiac membrane (e.g., calcium gluconate), promoting the movement of potassium into cells (e.g., insulin and glucose, beta-agonists), and enhancing the removal of potassium from the body (e.g., diuretics, ion-exchange resins, or hemodialysis).

Solution to Question 8:

The above clinical scenario of transection of the spinal cord in the lateral aspect of the dorsal column would affect the fasciculus cuneatus. This lesion would result in absence of ipsilateral proprioception of the arm.

The fibers in the dorsal column are organized in such a way that the sacral segments are positioned most medially and those from the cervical segments are positioned most laterally.

These are called

- Fasciculus gracilis - carrying sensations from the ipsilateral lower limb
- Fasciculus cuneatus - carrying sensations from the ipsilateral upper limb.

This patient with an injury to the lateral part of the cervical cord's dorsal column tract will have damaged fasciculus cuneatus. This would result in loss of touch, vibration, and proprioception in

the arm of the same side of the lesion.

The extent of neurological deficits resulting from a spinal cord injury depends on the level of the injury and the severity of the damage. Injuries may be classified as complete or incomplete, depending on whether there is a total or partial loss of sensory and motor function below the level of injury. The key to solving this question lies in recognizing that the question mentions that the damage is to the lateral aspect of the dorsal column tract.

Other options:

Option A: Absence of ipsilateral lower limb proprioception - This would be seen if the fasciculus gracilis was affected, which is responsible for carrying sensations from the ipsilateral lower limb. However, in this case, the injury is to the lateral part of the dorsal column, affecting the fasciculus cuneatus.

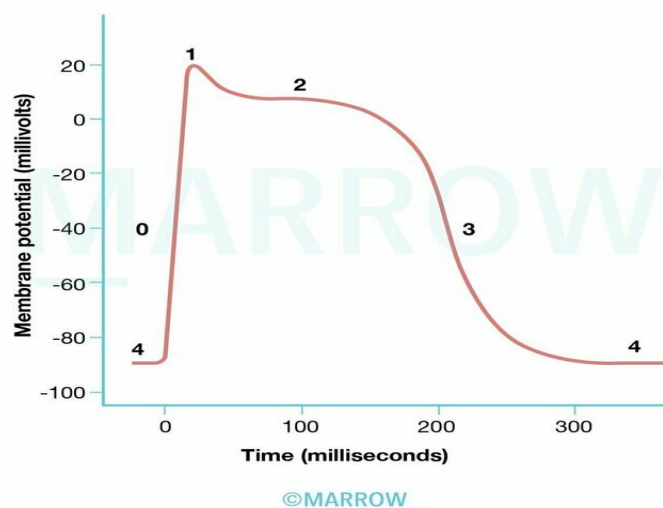
Option B: Absence of fine motor movement of fingers - This would be a result of damage to the corticospinal tract, which carries motor information. The question is specifically focused on the dorsal column tract, which is responsible for carrying sensory information.

Option D: Absence of contralateral lower limb proprioception - Proprioception is carried by the dorsal column pathway, and it is affected ipsilaterally. Damage to the lateral part of the dorsal column would not result in contralateral loss of proprioception.

Solution to Question 9:

The myocardial action potential curve marked above represents the plateau phase, which is due to the slower and prolonged opening of calcium channels.

Two types of voltage-gated Ca^{2+} channels (T- and L-types) are present in myocardial cells. The opening of the slower L-type Ca^{2+} channels is mostly responsible for changes in the action potential. The action potential curve of cardiac muscles consists of 5 phases, as shown below.



Phase 0: Depolarization - Opening of voltage-gated sodium channels (fast sodium channels) leads to sodium influx.

Phase 1: Initial repolarization - Closure of Na^+ channels and opening up of K^+ channels leads to potassium efflux.

Phase 2: Plateau - Slower but prolonged opening of voltage-gated Ca^{2+} channels, leading to calcium influx and reduced permeability of K^+ channels, causes the graph to plateau

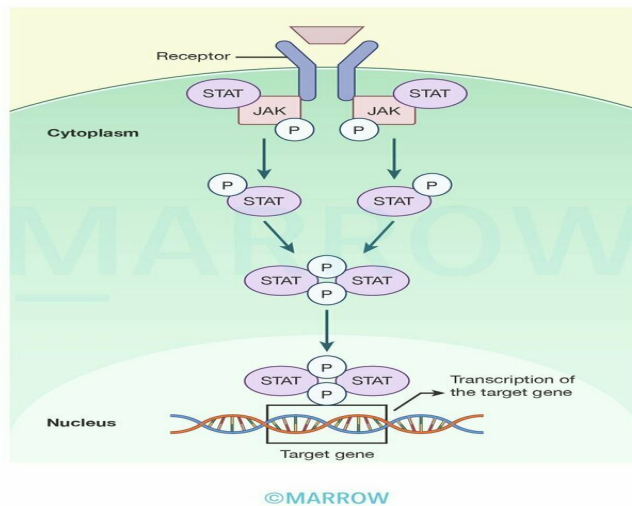
Phase 3: Rapid Repolarization - The closure of Ca^{2+} channels and opening of K^+ at the end of the plateau phase causes repolarization.

Phase 4: Resting membrane potential - Is reached at the end of phase 3

Physiology INI-CET 2023

Question 1:

Which of the following hormone acts via this pathway?



- a) Leptin
- b) Calcitonin
- c) Vasopressin
- d) Aldosterone

Question 2:

Which of the following options is false regarding the measurement of temperature in a febrile patient?

- a) Rectal temperature is ideal for measuring the core body temperature
- b) Rectal temperature is lower than axillary temperature
- c) Tympanic membrane temperature is lower than rectal temperature
- d) Rectal temperature is higher than oral temperature.

Question 3:

Which of the following statements is/are true about skeletal muscle?

- a) Only 2
- b) Only 3
- c) 3, 4
- d) 1, 4

Question 4:

Cells with prolific smooth endoplasmic reticulum secrete _____

- a) Eicosanoid hormones
- b) Steroid hormones
- c) Amine hormones
- d) Peptide hormones

Question 5:

At which volume or capacity, pressure measured at the upper airway becomes zero?

- a) Residual volume
- b) Functional residual capacity
- c) Total lung capacity
- d) Inspiratory capacity

Question 6:

Arrange the following bands in the sarcomere from the periphery to the center.

- a) 1,2,3,4
- b) 2,3,4,1
- c) 1,3,4,2
- d) 1,4,2,3

Question 7:

Which of the following are incorrectly paired?

- a) Gastrin - G cells, Antrum
- b) Secretin - S cells, Duodenum
- c) CCK - I cells, Duodenum and jejunum
- d) Motilin - M cells, Duodenum and Jejunum

Question 8:

Which of the following is responsible for fetal heart acceleration?

- a) Carotid body chemoreceptors
- b) Aortic baroreceptors
- c) Autonomic brain stem regulation
- d) ANP

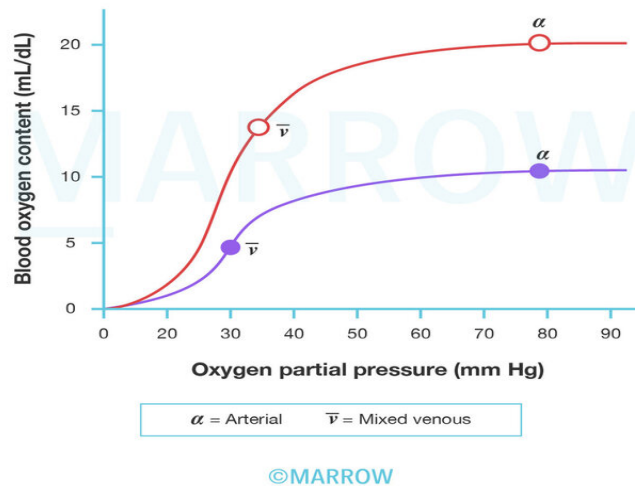
Question 9:

Our body was built to identify toxic substance by the recognition of bitter taste. Which of the following substances is responsible for the bitter taste?

- a) Amino acids
- b) Alkaloids
- c) Hydrogen ions
- d) Aldehydes

Question 10:

The type of hypoxia based on the graph is _____ (Consider the plot in red as normal)



- a) Stagnant hypoxia
- b) Histotoxic hypoxia
- c) Hypoxic hypoxia
- d) Anemic hypoxia

Question 11:

Assertion: RMP is dependent on proteins and phosphate ions. Reason: Diffusion potential can be calculated using Nernst Equation.

- a) The assertion is independently a true statement, but the reasons is independently a false statement.
- b) The assertion is independently a false statement, but the reasons is independently a true statement.
- c) Both assertion and reason are independently true and the reason is not the correct explanation for the assertion
- d) Both assertion and reason are independently true and the reason is the correct explanation for the assertion

Question 12:

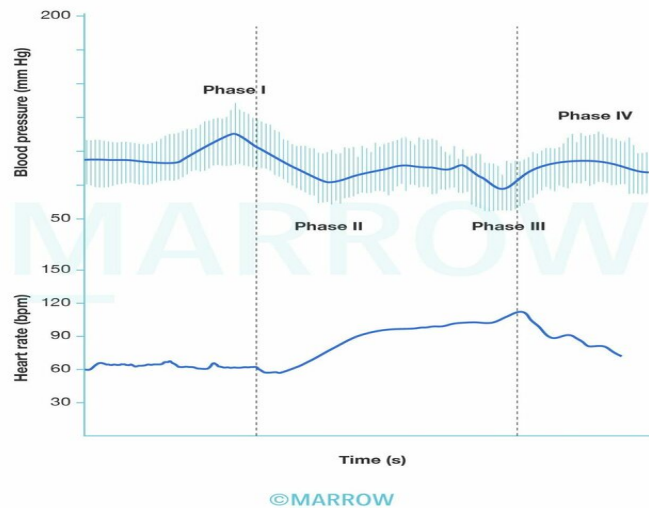
Windkessel effect is shown by all except _____

- a) Radial artery
- b) Renal Artery

- c) Aorta
- d) Carotid artery

Question 13:

Increase in blood pressure at the beginning of Valsalva maneuver is due to _____



- a) Increase in intrathoracic pressure adding to left ventricular pressure
- b) Increase in intrathoracic pressure adding to aortic pressure
- c) Increase in left atrial pressure
- d) Increase in left ventricular volume

Question 14:

Which of the following statements is false about male sexual development?

- a) At eight weeks the embryo is bipotent.
- b) Wolffian duct develops into epidymis and vas deferens on either side of the body.
- c) Mullerian inhibiting substance inhibits mullerian duct.
- d) Dihydrotestosterone leads to the development of secondary sexual characters and external genitalia.

Question 15:

The upstroke of action potential (phase 0) in the sinoatrial node is due to _____ ?

- a) Sodium influx via voltage gated sodium channels
- b) Calcium currents only
- c) Potassium efflux
- d) Influx of both sodium and calcium

Question 16:

Which of the following types of pain stimulus is associated with the Carrageenan-induced experimental pain model?

- a) Inflammatory stimulus
- b) Acute noxious stimulus
- c) Deep boring stimulus
- d) Neuropathic stimulus

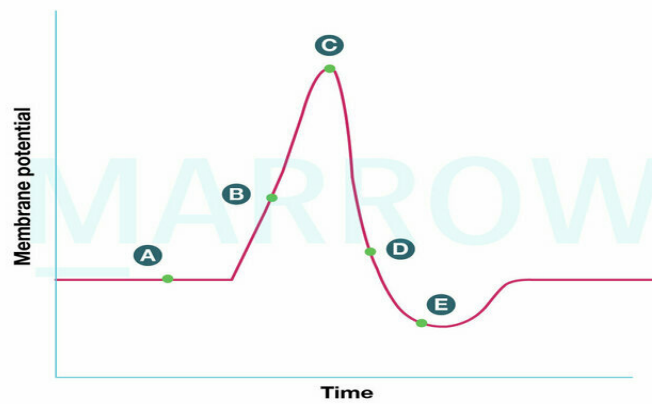
Question 17:

Insulin-independent glucose transporter is present in all of the following except?

- a) Brain
- b) Adipose tissue
- c) RBC
- d) Pancreas

Question 18:

Which of the following is responsible for the phase marked C to D in the given nerve action potential?

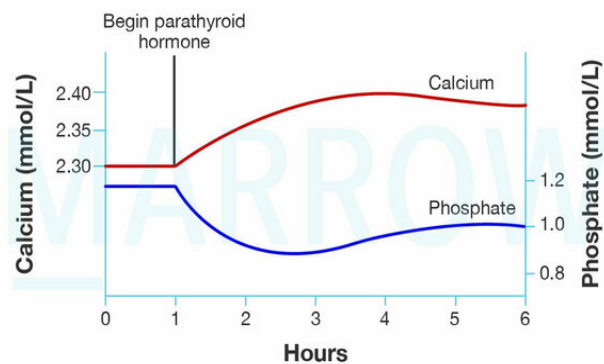


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- a) Na^+ influx
- b) K^+ influx
- c) Na^+ efflux
- d) K^+ efflux

Question 19:

The given graph shows the effect of PTH on blood calcium and phosphate levels in an experimental animal. How does PTH increase plasma calcium levels?



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- a) By increasing osteoclastic activity

- b) By decreasing renal excretion of calcium
- c) By increasing osteoblastic activity
- d) By increasing osteoclastic activity and decreasing renal excretion of calcium

Question 20:

In the given cross-section of a muscle spindle, the structure marked below is innervated by which of the following motor nerve fibres?



- a) Gamma
- b) Alpha
- c) Beta
- d) Delta

Question 21:

Which of the following is responsible for pneumoperitoneal bradycardia following CO₂ insufflation during laparoscopic surgery?

- a) Decreased afterload
- b) Vascular injury
- c) Air embolism
- d) Peritoneal stretching

Question 22:

Which of the following sodium transporters are present maximally in the early part of the PCT?

- a) 2,3,4
- b) 2,4
- c) 1, 2, 3
- d) 1,4

Question 23:

Which of the following statements is incorrect with respect to the baroreceptors present in the carotid sinus?

- a) They need the support of electromechanical transducers
- b) They are located in the deep smooth muscle layer
- c) Afferent fibres pass via cranial nerve IX
- d) They express several non-selective cation channels

Answer Key

Question No.	Correct Option
1	a
2	b
3	b
4	b
5	b
6	c
7	d
8	a
9	b
10	d
11	c
12	a
13	b

14	a
15	b
16	a
17	b
18	d
19	d
20	a
21	d
22	c
23	b

Detailed Explanations

Solution to Question 1:

The given image shows the JAK-STAT signaling pathway, which is associated with tyrosine kinase receptors. Among the given hormones leptin acts via this pathway.

Hormones that use receptor tyrosine kinase/ JAK-STAT signaling include the following:

- Growth hormone
- Fibroblast growth factor
- Hepatocyte growth factor
- Insulin
- Insulin-like growth factor-1
- Leptin
- Prolactin
- Vascular endothelial growth factor

Leptin is a hormone that is secreted by fat cells. The physiological effects of leptin are regulating appetite and energy balance. The leptin receptor acts via tyrosine kinase of the Janus kinase (JAK) family, JAK2. Leptin binds to the extracellular part of the leptin receptor, which causes phosphorylation (P) and activation of the intracellular JAK2. This causes phosphorylation of signal transducer and activation of transcription (STAT) proteins, which results in the transcription of target genes and the synthesis of proteins.

Phosphorylation of JAK2 also results in the activation of other intracellular enzyme pathways such as mitogen-activated protein kinases (MAPK) and phosphatidylinositol 3-kinase (PI3K).

Other options:

Hormones such as adrenal (Aldosterone) and gonadal steroid hormones, thyroid hormones, retinoid hormones, and vitamin D, bind with protein receptors inside the cell rather than on the

cell surface. These hormones are lipid soluble and they readily cross the cell membrane and interact with receptors in the cytoplasm or nucleus.

Vasopressin in the epithelial cells acts via V₂ receptor. They use adenylyl cyclase as their second messenger system. Vasopressin in the vascular smooth muscles acts via V₁ receptor that uses phospholipase C as the second messenger system.

Solution to Question 2:

With regards to the measurement of temperature in a febrile patient, rectal temperature is higher than axillary temperature.

For humans, the conventional normal value for oral temperature is 37°C (98.6°F).

Rectal temperature represents the core body temperature and is less affected by changes in environmental temperature. It remains relatively constant. In comparison, oral temperature is typically 0.5°C lower than rectal temperature. However, the oral temperature can be influenced by various factors such as consuming hot or cold fluids, chewing gum, smoking, and breathing through the mouth.

Tympanic membrane thermometers measure radiant heat emitted by the tympanic membrane and the nearby ear canal, providing an absolute value of temperature. Tympanic membrane readings are generally 0.8°C (1.6°F) lower than rectal temperatures. The axillary temperature measured is lower than the rectal temperature.

Solution to Question 3:

The true statement about skeletal muscle is that the calcium is released from sarcoplasmic reticulum.

Excitation-contraction coupling refers to the process by which the membrane depolarization of the muscle fiber (excitation) leads to the contraction of the muscle.

For the maximum contraction of the skeletal muscle, the current has to penetrate deeply into the muscle fiber to the vicinity of the separate myofibrils. This penetration of current is achieved by transmission of action potentials along transverse tubules (T tubules). The T tubules penetrate through the muscle fiber, from one side of the fiber to the other.

Sarcoplasmic reticulum contains calcium ions in high concentration within its vesicular tubules. These ions are released from each vesicle when an action potential occurs in the adjacent T tubule. (Statement 3)

This action potential when it reaches the T tubule, it is sensed by the dihydropyridine receptors linked to calcium release channels. Activation of this receptor leads to the opening of calcium release channels that releases the calcium ions into the sarcoplasm surrounding the myofibrils and causing contraction. (Statement 4)

In skeletal muscles, extracellular calcium supply partially contributes for the maintenance of skeletal muscle contraction. (Statement 1)

Solution to Question 4:

Cells with prolific smooth endoplasmic reticulum secrete steroid hormones.

The endoplasmic reticulum contains a complex series of tubules in the cytoplasm of the cell.

- Ribosomes are attached to the cytoplasmic side of the membrane in the rough(granular) endoplasmic reticulum. Protein synthesis and the initial folding of polypeptide chains with the formation of disulfide bonds are the functions of rough ER.
- Ribosomes are absent in the smooth (agranular) endoplasmic reticulum. Free ribosomes are found in the cytoplasm. It is the site of steroid synthesis in steroid-secreting cells and the site of detoxification processes in other cells.

The sarcoplasmic reticulum is a modified endoplasmic reticulum, that functions as an intracellular calcium store, which releases calcium into the cytoplasm stimulating cell contraction, proliferation, and migration.

Solution to Question 5:

Pressure measured at the upper airway becomes zero at functional residual capacity.

The curve of airway pressure obtained is plotted against the volume, which is the pressure-volume curve of the total respiratory system. This curve is used in understanding the fundamental relationship between the compliance of the lungs and the chest wall. It is measured by a spirometer.

The functional residual capacity is the volume of the air that remains in the lungs after normal expiration. The normal value is about 2.5 liters. At the lung volume of functional residual capacity (FRC), which is the relaxed state of the lungs, the pressure inside the lungs is effectively zero.

Other options:

Option A: Residual volume is the air left in the lungs after a maximal expiratory effort.

Option C: Total lung capacity is the sum of tidal volume, inspiratory reserve volume, expiratory reserve volume, and residual volume.

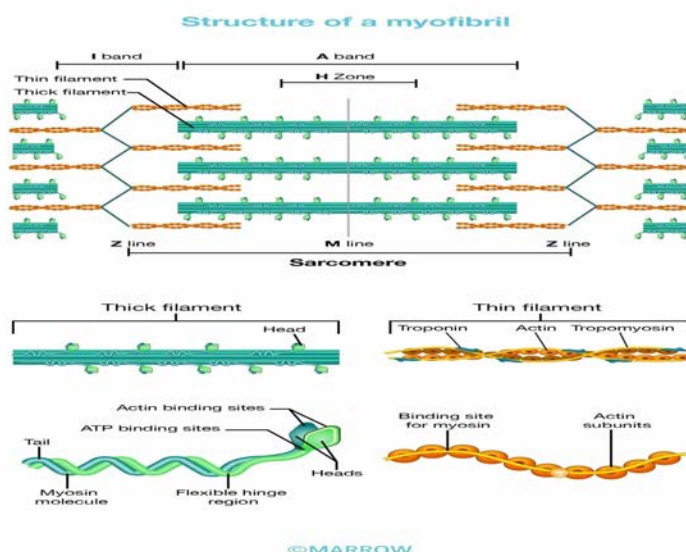
Option D: Inspiratory capacity is the maximum amount of air inspired from the end-expiratory level. It is the sum of tidal volume and inspiratory reserve volume.

Solution to Question 6:

The correct sequence of bands in the sarcomere from the periphery to the centre is: Z line→ A band→ H zone→ M line.

The sarcomere is the contractile unit of the skeletal muscle fibre.

- Z line provides attachment to actin filaments. The area between two adjacent Z lines is called the sarcomere. It is composed of thick and thin filaments:
- Thick filament: Myosin
- Thin filaments: Actin, tropomyosin and troponin
- I band is isotropic to light. It contains only actin filaments.
- A band is anisotropic to light. It contains myosin plus overlapped actin.
- H bands are the lighter areas in the centre of the A band. These are the areas where the thick filaments do not overlap the thin filaments, when the muscle is relaxed.
- M line is present in the centre of H band. There are slender cross-connections that hold the thick filaments in proper array.



Solution to Question 7:

Motilin is secreted by the Mo cells(not M cells) in the stomach, small intestine, and colon

Motilin is a gastrointestinal hormones that is secreted by enterochromaffin cells and Mo cells in the stomach, small intestine, and colon. It mediates its action via G-protein-coupled receptors on enteric neurons in the duodenum and colon. It produces contraction of smooth muscle in the stomach and intestines in the period between meals.

M cells are microfold cells are located in the epithelium covering mucosa-associated lymphoid tissues of the small intestine such as the Peyer's patches (PPs). Their function is to transport luminal antigens to the underlying lymphoid follicles to initiate an immune response.

Gastrin is secreted by G cells of the antrum, duodenum, and jejunum. It stimulates gastric acid secretion and mucosal growth.

Secretin is secreted by the S cells of the duodenum. Its primary function is to promote the secretion of bicarbonate-rich pancreatic juice by the ductal cells of the pancreas and duodenum. This secretion is poor in enzymes.

Cholecystokinin (CCK) is secreted by I cells of the duodenum, jejunum, and ileum. It stimulates pancreatic enzyme secretion, pancreatic bicarbonate secretion, gallbladder contraction, and growth of the exocrine pancreas. It also inhibits gastric emptying.

Solution to Question 8:

Amongst the given options, the best answer for the site responsible for Fetal heart acceleration is Carotid body chemoreceptors>Aortic baroreceptors>Autonomic brain stem regulation.

Though the brainstem is critical to mediating this reflex, the primary stimulus for initiating the changes in the heart rate is initiated by the carotid chemo and baro receptors, especially the chemoreceptor which is highly sensitive to the changes in O₂ levels in the blood. It plays a permissive role on fetal heart rate acceleration.

Arterial chemoreceptors in the carotid arteries>aortic arch, respond to low pH levels and decreased oxygen saturation. These chemoreceptors detect instances of significant fall in blood oxygen levels, setting off the peripheral chemoreflex and heightening both parasympathetic and sympathetic neural activities. The activation of the parasympathetic branch leads to a swift decline in Fetal Heart Rate (FHR), a clinical manifestation identified as intrapartum decelerations. Concurrently, the sympathetic branch prompts significant peripheral vasoconstriction across a range of vascular territories, which results in the elevation of arterial blood pressure, ensuring continuous perfusion to essential organs.

The permissive action on the FHR acceleration comes in instances of alleviated hypoxemia, detected by the chemoreceptors in the carotid body and aortic arch, suppressing the peripheral chemoreflex, leading to a reduction in parasympathetic and sympathetic neural activities. Deactivation of the parasympathetic branch results in a swift increase in Fetal Heart Rate (FHR), typically seen clinically as an increase in intrapartum accelerations.

Solution to Question 9:

Alkaloids cause an intensely bitter taste, that causes the rejection of the food.

Bitter taste sensations are caused by two particular classes of substances:

- Long-chain organic substances that contain nitrogen.
- Alkaloids

Alkaloids cause an intensely bitter taste, leading to the rejection of certain foods. They are found in many medicinal drugs such as quinine, caffeine, strychnine, and nicotine. The bitter taste sensation in many deadly toxins found in poisonous plants is often caused by alkaloids. This helps the person or animal to reject the food.

Sweet taste is caused by chemicals such as sugars, glycols, alcohols, aldehydes, ketones, amides, esters, some amino acids, some small proteins, sulfonic acids, halogenated acids, and inorganic salts of lead and beryllium.

Sour taste is caused by acids (Hydrogen ion concentration). The sour taste is directly proportional to the strength of the acid.

Solution to Question 10:

The graph represents the oxygen-hemoglobin dissociation curve that would be seen in anemic hypoxia and normal. The X-axis represents the partial pressure of oxygen (PO_2). The Y-axis represents the percentage saturation between the hemoglobin and the oxygen.

The graph indicates two lines: a red line representing normal hemoglobin and a purple line representing reduced hemoglobin levels. The red line corresponds to an arterial blood oxygen content of approximately 19 mL/dL, and venous blood oxygen content of 14 mL/dL. On the other hand, the purple line represents an arterial blood oxygen content of 9 mL/dL and venous blood oxygen content of 4 mL/dL.

This scenario demonstrates a 50% decrease in oxygen saturation of hemoglobin, while the partial pressure of oxygen (PO_2) remains within the normal range. This condition is characterized as anemic hypoxia.

In anemic hypoxia, the blood oxygen content is decreased. The patients might not experience any difficulty during rest because 2,3-DPG increases in the red blood cells. However, they may have difficulty during exercise because of a limited ability to increase O_2 delivery to the active tissues.

Conditions that cause a right shift in the oxygen hemoglobin curve are: (Conditions with the increased requirement of O_2 - metabolically active tissues)

- Hypoxia
- Hypercarbia
- Decreased pH
- Increased temperature
- Increased 2,3-biphosphoglycerate

Conditions that cause a left shift in the oxygen hemoglobin curve

- Increased oxygen
- Decreased carbon dioxide
- Increased pH
- Decreased temperature
- Decreased 2,3-biphosphoglycerate
- Fetal hemoglobin.

Solution to Question 11:

Both assertion and reason are independently true and the reason is not the correct explanation for the assertion.

Resting membrane potential (RMP) is the difference in the electrical potentials between the cytoplasm of the cell and the extracellular space. It represents an equilibrium situation where

the concentration gradient of ions across the membrane is equal and opposite to the electrical gradients.

Resting membrane potential is primarily influenced by sodium (Na⁺) and potassium (K⁺) ions, along with negatively charged intracellular proteins and organic phosphates. In neurons, the concentration of potassium is higher inside the cell, while sodium ions exhibit the opposite distribution. The establishment of this concentration difference is facilitated by the Na, K ATPase pump. Potassium ions passively move out of the cell, while sodium ions move into the cell.

The Goldman-Hodgkin-Katz equation is utilized to calculate the resting membrane potential since multiple ions are involved. Among these ions, potassium, which has the greatest conductance across the membrane at rest, becomes the primary contributor to the resting membrane potential.

Goldman-Hodgkin-Katz equation

$$EMF \text{ (millivolts)} = -61 \times \log \frac{C_{Na^+}^i P_{Na^+} + C_{K^+}^i P_{K^+} + C_{Cl^-}^i P_{Cl^-}}{C_{Na^+}^o P_{Na^+} + C_{K^+}^o P_{K^+} + C_{Cl^-}^o P_{Cl^-}}$$

The Nernst equation establishes a connection between the diffusion potential and the ion concentration difference across a membrane. The magnitude of the Nernst potential is determined by the ratio of the ion concentrations on the two sides of the membrane.

Nernst equation :

$$EMF(\text{millivolts}) = [61/z] \times \log [C_o / C_i]$$

- EMF = electromotive force (Nernst potential)
- z = charge of ion
- C_o = concentration outside
- C_i = concentration inside

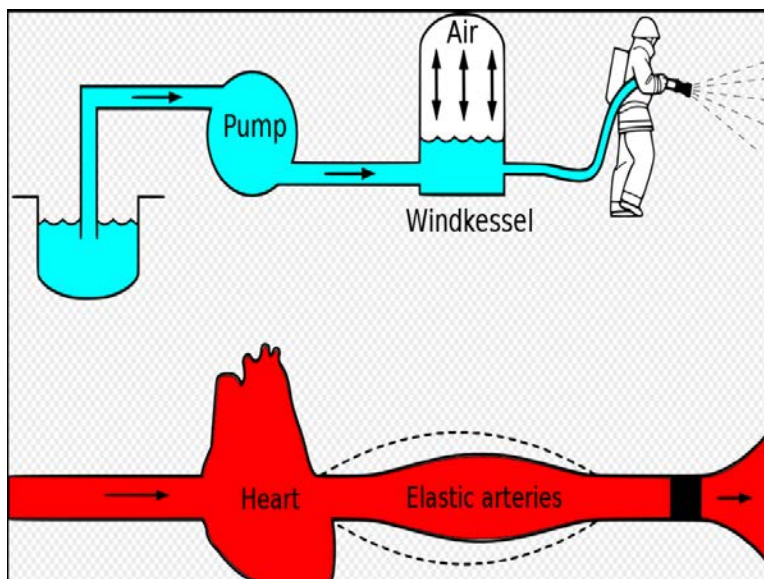
Solution to Question 12:

Windkessel effect is shown by all except the radial artery.

The windkessel effect is seen in the large elastic arteries like the aorta, carotid, and renal artery. The elasticity or windkessel effect is the exclusive property of large and elastic vessels. These vessels when stretched undergo elastic recoil.

The aorta is stretched during systole. This expansion causes temporary storage of blood, allowing more blood to enter the aorta. This buffer stock of blood is released during diastole, to maintain organ perfusion even during diastole.

The image given below shows the windkessel effect seen in the aorta.



Several methods can be used to evaluate the elastic properties of the aortic Windkessel in living humans. One simple approach is to measure the pulse wave velocity along the aorta, where a higher velocity indicates increased stiffness and elastic resistance. Alternative methods involve assessing the ratio between pulse pressure and changes in aortic volume ($\Delta P/\Delta V$). These methods can be noninvasively performed using ultrasound or tomographic techniques.

Solution to Question 13:

Increase in blood pressure at the beginning of the Valsalva maneuver is due to increase in intrathoracic pressure adding to aortic pressure.

Valsalva maneuver involves the performance of forced expiration against a closed glottis. It occurs during coughing, defecation, and heavy lifting. At the onset of straining, the blood pressure is raised because the increase in intrathoracic pressure is added to the pressure of the blood in the aorta. The pressure then falls as the high intrathoracic pressure compresses the veins and decreases the cardiac output.

When the glottis is opened and intrathoracic pressure returns to its normal state, cardiac output is regained, but there is constriction in the peripheral vessels. As a result, blood pressure elevates beyond the normal range, which triggers the baroreceptors, leading to bradycardia and a subsequent decrease in pressure back to normal levels.

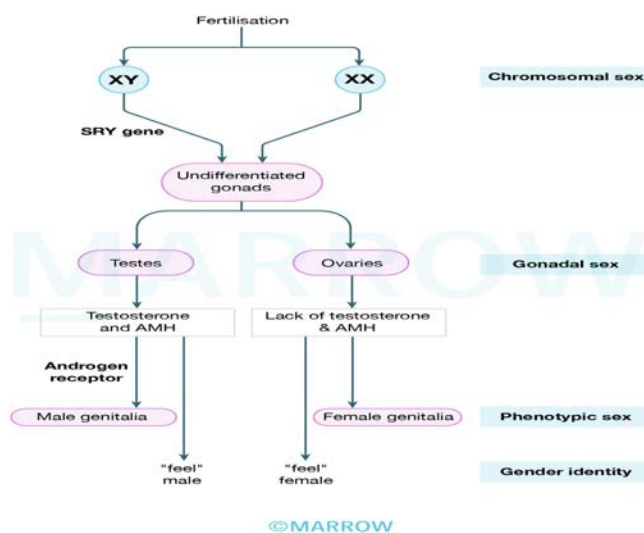
Solution to Question 14:

The embryo is bipotent till seven weeks of gestation, not eight.

At the seventh week, the embryo contains both male and female primordial genital ducts. In the case of a female fetus, uterine tubes and the uterus develops from the Mullerian duct. In case of a male fetus, the wolffian duct system on each side develops into the epididymis and vas deferens.

Testosterone is secreted by the Leydig cells of the fetal testis and MIS (Mullerian inhibiting substance) is secreted by the Sertoli cells. MIS is an amino acid that belongs to the family of transforming growth factor β (TGF- β). MIS causes apoptosis of the Mullerian ducts on the side on which it is secreted.

Dihydrotestosterone, a metabolite of testosterone is responsible for the formation of male external genitalia and male secondary sex characteristics. Development of the vas deferens, epididymis, and other structures from the wolffian ducts is due to the action of testosterone.



Solution to Question 15:

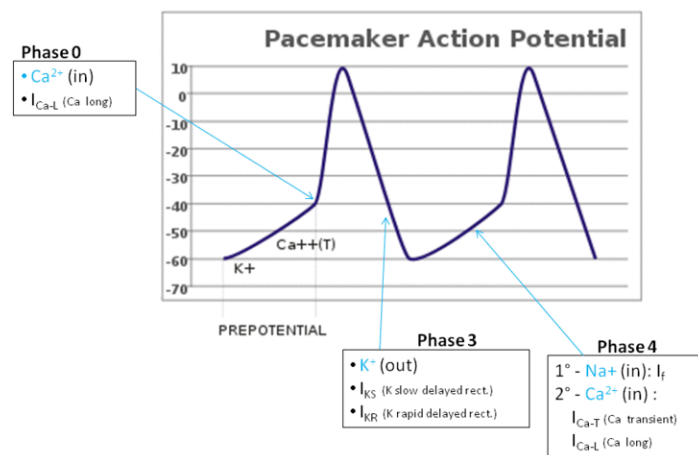
The upstroke of action potential (phase 0) in the sinoatrial node is due to calcium currents only. It occurs due to the influx of Ca^{2+} ions by the opening of voltage-gated L-type channels.

The action potential of the SA node differs from that of the atria, ventricles, and Purkinje fibres in the following ways: (1) In the absence of neural input, the SA node demonstrates automaticity and the ability to produce its own action potentials. (2) It has an unstable resting membrane potential. (3) It has no sustained plateau.

The resting membrane potential (RMP) of the SA node is -55 to -60 millivolts. Phases of this potential are:

- Phase 0 (Upstroke wave)- This is due to the influx of Ca^{2+} ions by the opening of voltage-gated L-type channels.
- Phase 3 (Downstroke wave)- This is due to the efflux of K^{+} leading to repolarization. (Option C)

- Phase 4 (Prepotential)- Because of ongoing K^+ efflux, hyperpolarization sets in. This makes the channel more permeable to Na^+ ions and since these Na^+ channels are activated during hyperpolarization they are referred to as funny currents (I_h / I_f). They form the first half of the pre-potential. The second half of the prepotential is completed by the opening of the voltage-gated T-(transient) type Ca^{2+} channel. The slope of Phase 4 in the SA node determines the heart rate.



Solution to Question 16:

The Carrageenan-induced pain model also called the Carrageenan-Induced Paw Edema model is a well-known model of acute inflammation.

It is employed to evaluate the anti-inflammatory properties of a variety of synthetic and natural substances. A non-antigenic phlogistic polysaccharide derived from red seaweed, carrageenan is frequently added to processed foods as a thickening, stabilising, gelling, and emulsifying agent. The activation of the complement system and inflammatory mediators is caused by the sulphated sugars found in carrageenan.

Carrageenan is injected subcutaneously into the plantar surface of the rat paw to cause inflammation. This process results in temporary inflammation and acute swelling, which peaks 3 to 5 hours after injection and subsides in 24 hours.

Postcapillary venules dilate due to the action of carrageenan, causing inflammatory cells and fluid to exude. During this process, a number of proinflammatory mediators are released. The cyclooxygenase pathway's activation is commonly associated with the carrageenan model. A biphasic curve represents the edema caused by carrageenan.

Part of the initial phase of inflammation caused by carrageenan is attributed to injection trauma and the release of acute phase mediators, particularly bradykinin, histamine, and serotonin. The second phase of carrageenan-induced inflammation, which happens about three hours after carrageenan is added, is mostly caused by prostaglandins.

Solution to Question 17:

Insulin-dependent (not independent) glucose transporter is found in the adipose tissue.

Glucose uptake by skeletal muscle, cardiac muscle and adipose tissues is via GLUT-4 which is insulin-dependent.

In the cytoplasm of insulin-sensitive cells, the vesicles contain a pool of GLUT-4 molecules. When the insulin receptors of these cells are activated, the vesicles move rapidly to the cell membrane and fuse with it, inserting the transporters into the cell membrane.

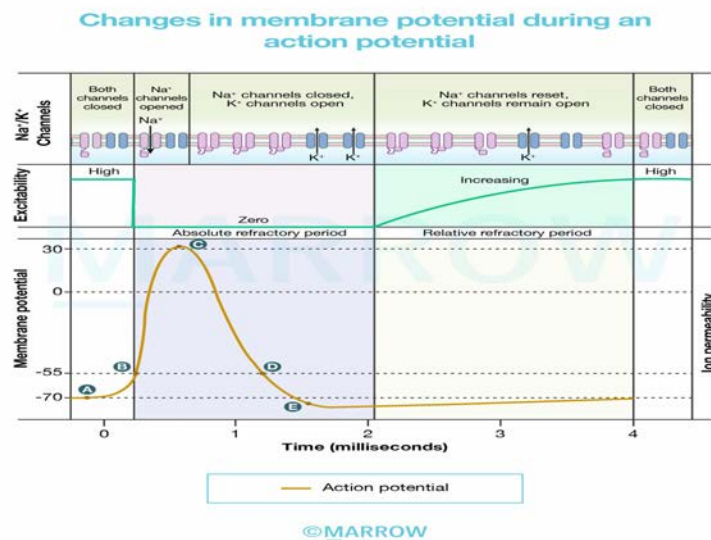
The membrane patches containing transporters are endocytosed when insulin action stops, preparing the vesicles for subsequent insulin exposure. Phosphoinositol 3-kinase is activated by the insulin receptor, which causes the vesicles to migrate to the cell membrane.

Most of the other GLUT transporters are not insulin-sensitive and are constitutively expressed in the cell membrane. Ex: GLUT-1 in RBC's and brain, GLUT-2 in the beta cells of the pancreas and GLUT-3 present in the brain and kidneys.

Solution to Question 18:

The phase C to D in the given nerve action potential is caused by K^+ efflux as a result of the opening of the K^+ channels, resulting in repolarization.

The sequence of events occurring during an action potential is shown below



- Point A represents the resting state where both channels are closed.
- Na^+ channels open in response to depolarizing stimulus resulting in Na^+ influx (option A). The initiation of action potential occurs only after a certain potential is reached, which is called the threshold potential represented by point B.
- After this, the sodium influx causes the opening of more Na^+ channels through a positive feedback mechanism. Rapid depolarization occurs to reach point C.

- At point C, Na⁺ channels close and K⁺ channels open, causing repolarization due to K⁺ efflux.
- Point D represents continued depolarization due to K⁺ efflux.
- At point E, Na⁺ channels are reset and are ready to respond to another depolarizing stimulus. During this phase, the K⁺ channels remain open and slowly return to the closed state, causing after-hyperpolarization. As a stronger stimulus is required to reach the threshold potential, the excitability of nerve fibre is reduced at point E.
- As both channels close, the membrane returns to resting membrane potential back to point A.

Point B-D is a part of the refractory period. During this period, the application of even a stronger stimulus will not elicit an action potential.

The refractory period is divided into an absolute refractory period and a relative refractory period. During the absolute refractory period, the nerve will not be excited no matter how strong a stimulus is given. However, during the relative refractory period, stronger than normal stimuli can cause excitation.

Solution to Question 19:

The parathyroid hormone increases the plasma calcium levels by increasing osteoclastic activity and decreasing renal excretion of calcium.

By regulating intestinal reabsorption, renal excretion, and the exchange of these ions between the extracellular fluid and bone, PTH offers a potent mechanism for controlling the concentrations of calcium and phosphate in extracellular fluid.

The parathormone (PTH) increases Ca⁺² concentration by:

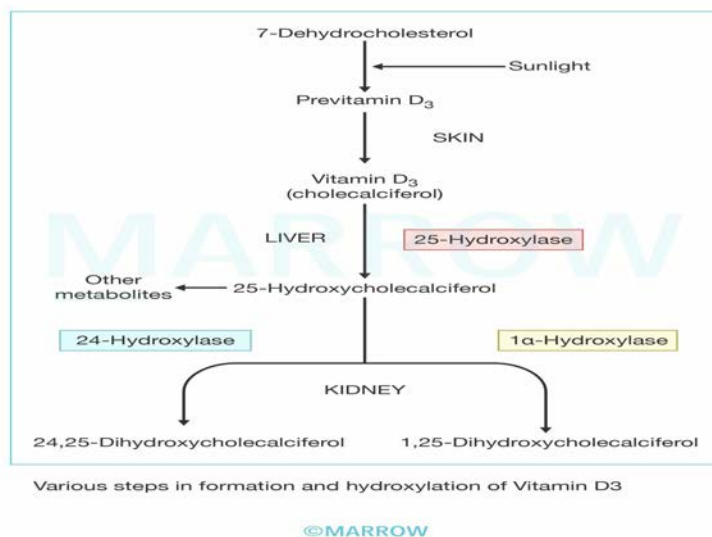
- Increased reabsorption of Ca⁺² by renal tubules, decreasing the urinary Ca⁺² excretion
- Increasing bone resorption, leading to Ca⁺² release into the ECF
- Decreased phosphate reabsorption by the renal tubules, increasing the urinary excretion of phosphate
- Conversion of 25-hydroxycholecalciferol to 1,25-dihydroxycholecalciferol which increases intestinal Ca²⁺ absorption

Activation of the osteoclasts occurs due to activated osteoblasts and osteocytes that send secondary signals to the osteoclasts. Preosteoclast cells become mature osteoclasts when their receptors are activated by RANKL, a significant secondary signal. Activation of the osteoclastic system occurs in two stages: (1) immediate activation of the osteoclasts that are already formed and (2) formation of new osteoclasts.

PTH administration decreases proximal tubular reabsorption of phosphate ions, which results in a rapid loss of phosphate in the urine. At the same time that PTH reduces phosphate reabsorption, it also increases calcium reabsorption through renal tubules. This ultimately leads to a decrease in the renal excretion of calcium.

Furthermore, it increases the reabsorption of hydrogen and magnesium ions while decreasing the reabsorption of ions from amino acids, potassium, and sodium in a manner similar to how it affects phosphate. Distal tubules and the thick ascending loop of the Henle are the primary sites

of the increased calcium reabsorption.



Solution to Question 20:

In the given cross-section of a muscle spindle, the structure marked shows intrafusal muscle fibres, which are innervated by gamma(γ) motor neurons.

Each muscle spindle is 3 to 10 mm long and consists of 3 to 12 tiny intrafusal muscle fibers that are pointed at their ends and attached to the glycocalyx of the surrounding large extrafusal muscle fibers. There are two kinds of sensory endings in each spindle, a single primary (group Ia) ending and up to eight secondary (group II) endings.

Intrafusal fibers are of 2 types: (1) nuclear bag fibers - subdivided as dynamic and static nuclear bag fibers and (2) nuclear chain fibers.

Each intrafusal muscle fiber is a tiny skeletal muscle fiber. Since there are either few or no actin and myosin filaments in the centre of each of these fibres, the centre does not contract when the ends do. It performs the role of a sensory receptor instead. The end portions that do contract are excited by small gamma(γ) motor nerve fibers which help control basic muscle tone.

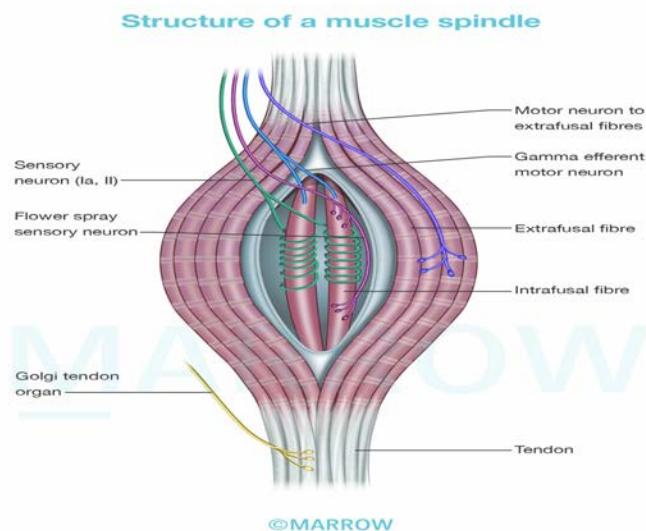
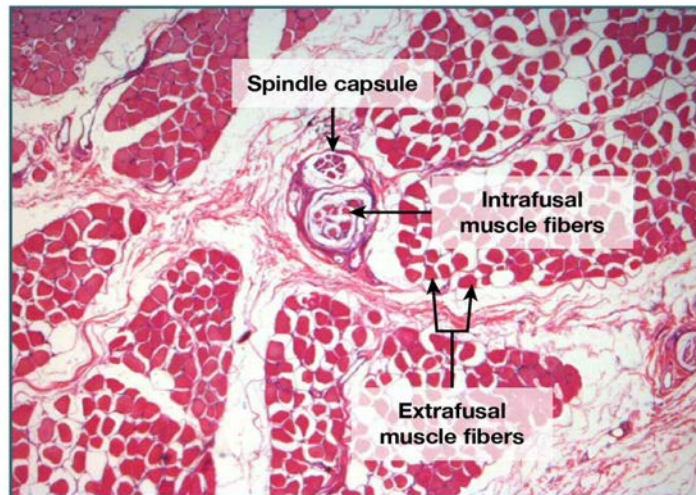
There are two types of γ motor neurons- (a) dynamic that supplies dynamic nuclear bag fibers and (b) static that supply static nuclear bag fibers and nuclear chain fibers.

Other options:

Option B: Alpha (α) motor fibers supply the extrafusal muscle fibers which are the regular contractile units of the muscle. They are on an average 14 micrometers in diameter and these fibers branch many times after they enter the muscle and innervate the large skeletal muscle fibers at the motor endplate. A single alpha nerve fibre can stimulate anywhere between three to several hundred skeletal muscle fibres, which are collectively called the motor unit.

Option C: Beta (β) motor fibers are for touch and pressure.

Option D: Delta (δ) motor fibers are for pain and temperature.



Solution to Question 21:

Vagal stimulation due to peritoneal stretching is responsible for bradycardia following CO₂ insufflation during laparoscopic surgery.

Carbon dioxide (CO₂) and nitrous oxide (N₂O) are used in pneumoperitoneum, a technique that enhances laparoscopic surgery by distending the abdominal cavity and splitting up its contents.

The physiologic effects of CO₂ pneumoperitoneum can be divided into local and systemic effects:

CO₂ pneumoperitoneum is associated with the risk of pathophysiological cardiovascular changes such as severe bradycardia, arrhythmia, and cardiac arrest requiring cardiopulmonary resuscitation. These changes are known to be associated with vagal-mediated cardiovascular reflexes triggered by rapid peritoneum distension due to insufflation.

This can be managed by desufflation of the abdomen, the use of vagolytic medications (such as atropine), and sufficient volume replacement.

Most people can tolerate the hemodynamic and metabolic effects of pneumoperitoneum for a brief duration, and most healthy people can tolerate them for a longer period of time. Long laparoscopic procedures can be challenging for patients whose cardiovascular systems are compromised.

In such cases, alternative gases like helium, neon and argon can be used. They do not cause any metabolic effects, but if they get direct access to the venous system, they can cause gas emboli because they are not as soluble in blood as CO₂ and N₂O.

Gas embolism is treated by positioning the patient in a left lateral decubitus position, head down, to trap the gas in the right ventricle's apex. The gas can then be aspirated out of the right ventricle using a rapidly placed central venous catheter.

Limiting intra-abdominal pressure (IAP) below 12-15 mmHg during insufflation is also effective in preventing pathophysiological changes of pneumoperitoneum.

Local Effects	Systemic effects
Peritoneal distention	Hypercarbia
Vagal reaction	Acidosis
Elevated diaphragm	Increased afterload
Altered venous return	Increased catecholamines

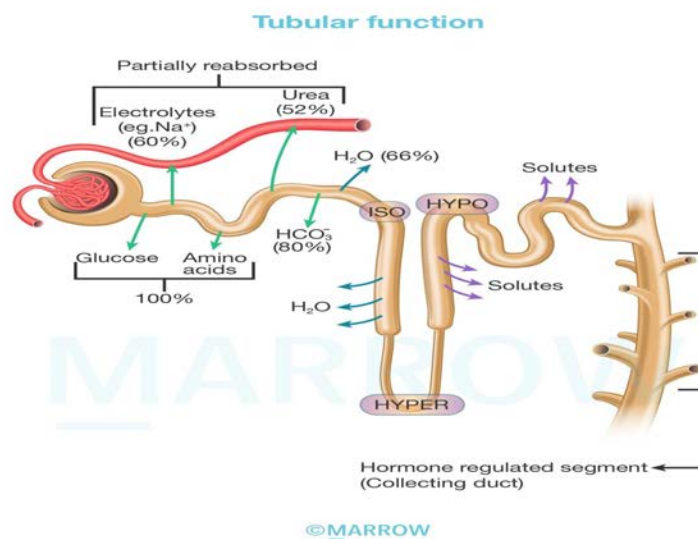
Solution to Question 22:

The sodium transporters present in the early part of PCT are Na-K-ATPase, Na⁺-glucose cotransporter (SGLT) and Na⁺-H⁺ antiporter.

Reabsorption of sodium in different parts of the nephron are:

- Proximal convoluted tubule (PCT):
- Nearly 60% of the filtered sodium is reabsorbed in the PCT. Solutes mainly absorbed in the proximal tubule are sodium, chloride, bicarbonate, phosphate, lactate, glucose, amino acids etc.
- Na⁺ is actively pumped from these cells into the interstitial space by Na-K ATPase in the basolateral membrane after travelling by cotransport or exchange via the tubular lumen into the tubular epithelial cells along its concentration and electrical gradients. It extrudes three Na⁺ in exchange for two K⁺ that are pumped into the cell.
- In early PCT, the main sodium channels used for reabsorption are Na⁺-glucose cotransporter (SGLT), Na⁺-H⁺ antiporter, Na⁺-amino acid cotransporter, Na⁺-phosphate cotransporter, etc.
- In distal PCT, due to high chloride concentration in the lumen, it favours NaCl reabsorption, driven by Cl⁻ gradient.

- The total solute and water reabsorption are proportional to each other. Thus, the entire proximal tubule also reabsorbs nearly 60% of the filtered water (isosmotic reabsorption).
- Loop of Henle:
 - The thin descending limb has no active transport of Na^+ outside the lumen but the loop is permeable to water, making the tubular fluid hypertonic.
 - A thick ascending limb reabsorbs nearly 30% of the Na^+ by $\text{Na}-2\text{Cl}-\text{K}$ cotransporter, an example of secondary active transport. This loop is impermeable to water. Since some of the solutes are reabsorbed and water is not reabsorbed, it dilutes the tubular fluid.
- Distal convoluted tubule: Approximately 7% of the filtered load of sodium chloride is reabsorbed in the early distal tubule. This occurs due to the presence of sodium chloride co-transporter. Even this segment dilutes the tubular fluid as it is impermeable to water.
- Cortical collecting tubule: The remaining 3% of the filtered Na^+ is absorbed via ENaCs (epithelial sodium channels) in this duct. This portion is regulated by aldosterone to maintain homeostasis in Na^+ balance.



Solution to Question 23:

Baroreceptors are not located in the deep smooth muscle layer. They are located in the adventitia layer.

Baroreceptors are mechanoreceptors which respond to stretch stimuli. They are located in the heart and adventitia of the blood vessels. There are two types of baroreceptors: high-pressure arterial baroreceptors and low-pressure volume receptors. Arterial baroreceptors are located within the carotid sinuses and the aortic arch.

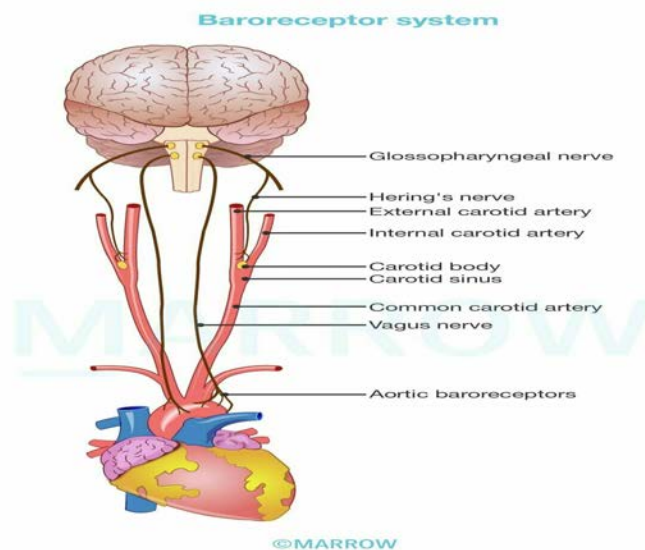
The carotid sinus is a small dilation of the internal carotid artery just above the bifurcation of the common carotid into external and internal carotid branches. The afferent nerve fibres from the carotid sinus form a distinct branch of the glossopharyngeal nerve (CN IX), the carotid sinus nerve.

Afferent signals from both the aortic and carotid sinus baroreceptors are sent to the nucleus tractus solitarius (NTS) and they create a negative feedback loop that keeps arterial blood pressure within normal limits.

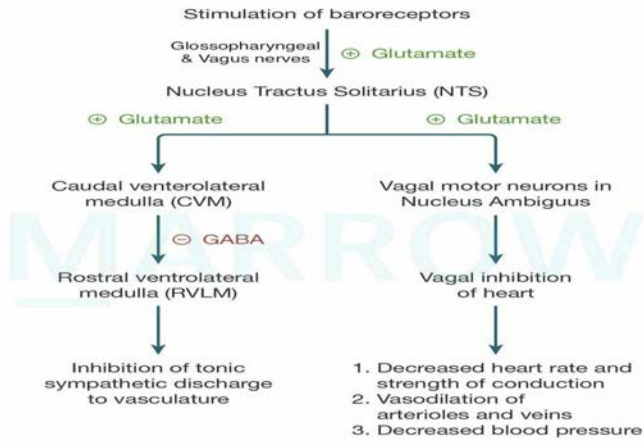
By stimulating these arterial baroreceptors in response to vascular distension, elevated arterial blood pressure can be reflexively brought back to normal levels by decreasing peripheral sympathetic outflow and activating peripheral parasympathetic outflow.

Three subsequent functional processes must take place in order to convert a mechanical change in the arterial vasculature into an excitatory signal:

- Vascular distension and the deformation of arterial baroreceptor terminals
- Mechanoelectrical transduction converts the mechanical deformation of arterial baroreceptor nerve terminals into baroreceptor membrane depolarization. Mechanosensitive ion channels such as the epithelial Na⁺ channels (ENaCs), acid-sensing ion channels (ASICs) and transient receptor potential (TRP) ion channels are responsible for mechanoelectrical transduction in arterial baroreceptor nerve terminals.
- Translation of membrane depolarization into action potentials via voltage-gated ion channels (Na⁺, Ca²⁺ and K⁺).



Baroreceptor reflex Pathway



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Physiology INI-CET 2024

Question 1:

Which of the following are properties of ventricular cardiac muscle?

- a) 1,2
- b) 1,2,3
- c) 1,2,4
- d) 1,2,3,4

Question 2:

Which of the following may lead to a decrease in GFR?

- a) Increase in renal blood flow
- b) Decrease in glomerular oncotic pressure
- c) Increase in glomerular hydrostatic pressure
- d) Decrease in bowman capsule oncotic pressure

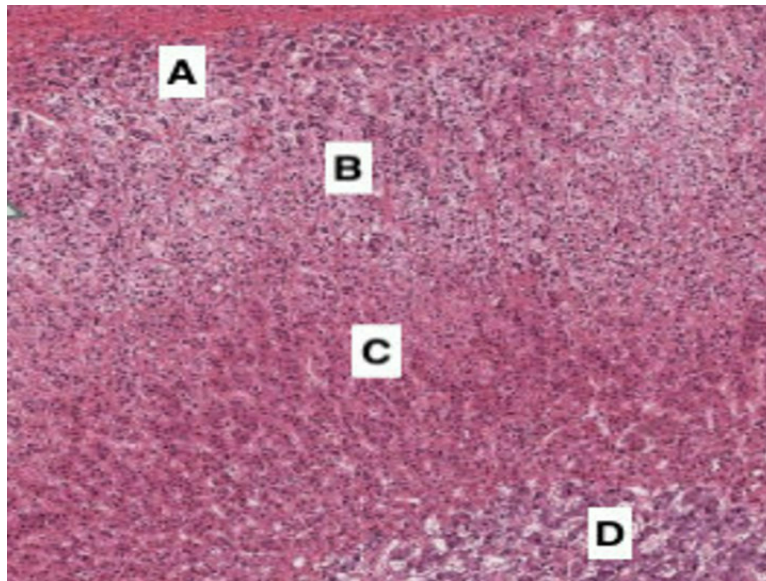
Question 3:

Which is the most potent naturally occurring estrogen?

- a) Estradiol
- b) Estriol
- c) Estrone
- d) Conjugated Estrogen

Question 4:

Which hormones are produced from the zones labelled in the image given?



- a) A-cortisol, B-aldosterone, C-androgens, D-epinephrine
- b) A-epinephrine, B-cortisol, C-aldosterone, D-androgens
- c) A-aldosterone, B-epinephrine, C-androgens, D-cortisol
- d) A-aldosterone, B-cortisol, C-androgens, D-epinephrine

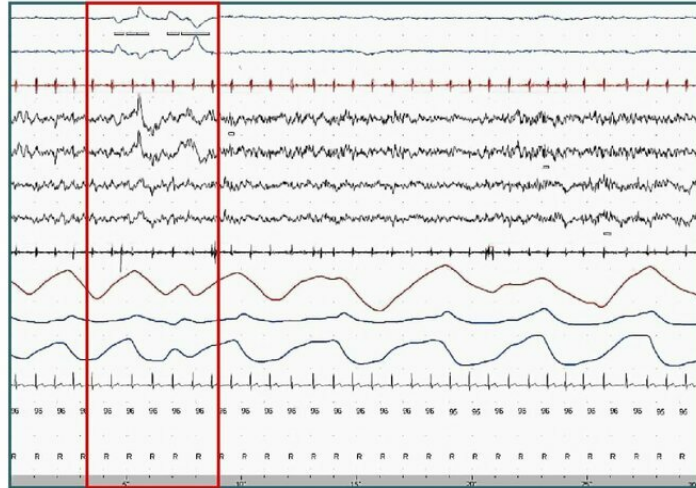
Question 5:

Erythropoietin is produced by:

- a) 1 only
- b) 2 only
- c) 1,2
- d) 1,2,3,4

Question 6:

EOG, EEG, and EMG of a patient are given respectively from above downwards. Which stage of sleep does the marked area represent?



- a) NREM I sleep
- b) NREM II sleep
- c) REM sleep
- d) NREM III sleep

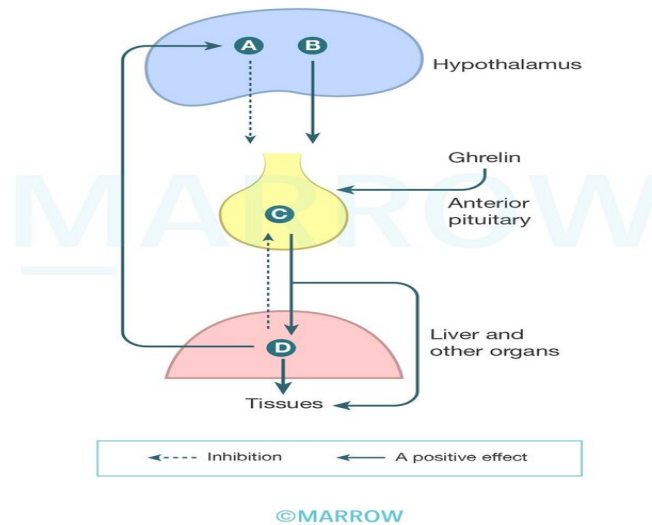
Question 7:

Which of the following statements is incorrect regarding ABO blood grouping?

- a) Antibodies are naturally present since birth
- b) The antigens are carbohydrate in nature
- c) The antibodies are predominantly IgM
- d) Antibodies are present only if antigens are absent

Question 8:

Identify the hormones marked in the image given



- a) A-Somatostatin, B-GHRH, C-GH, D-IGF-1
- b) A-GHRH, B-GH, C-IGF-1, D-Somatostatin
- c) A-GHRH, B-Somatostatin, C-GH, D-IGF-1
- d) A-GH, B-GHRH, C-IGF-1, D-Somatostatin

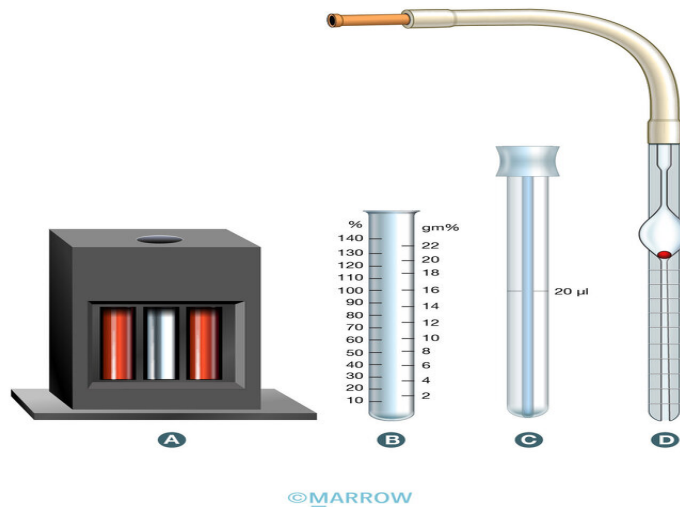
Question 9:

Identify the correct sequence.

- a) Protein kinase A » Adenylyl cyclase » cAMP
- b) cAMP » Adenylyl cyclase » Protein kinase A
- c) Adenylyl cyclase » cAMP » Protein kinase A
- d) Protein kinase A » cAMP » Adenylyl cyclase

Question 10:

Which among the following instruments is not used in the Sahli's method of hemoglobin estimation?



- a) A
- b) B
- c) C
- d) D

Question 11:

Which neurotransmitter is secreted by the SA node in response to increased blood pressure?

- a) Acetylcholine
- b) Epinephrine
- c) Norepinephrine
- d) Dopamine

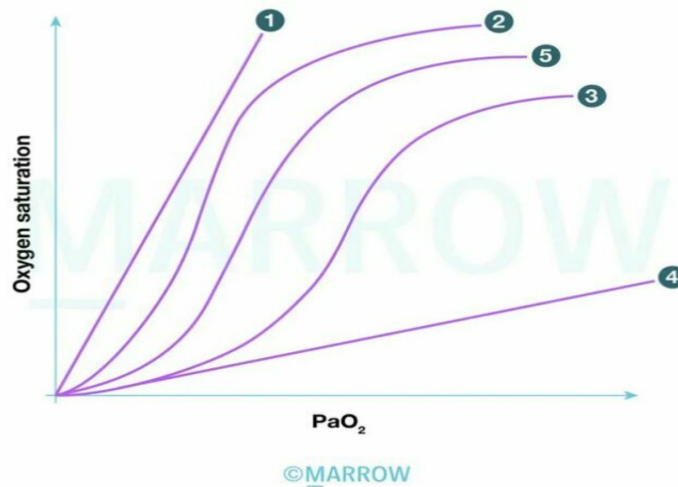
Question 12:

Match the following hormones with their receptors.

- a) A-1, B-2, C-3, D-4
- b) A-1, B-3, C-2, D-4
- c) A-3, B-4, C-1, D-2
- d) A-4, B-2, C-3, D-1

Question 13:

The O₂ saturation with respect to PO₂ is shown by the curve marked as 5. What happens to the oxygen Hb dissociation curve on transfusion of stored RBCs?



- a) 2
- b) 3
- c) 4
- d) 1

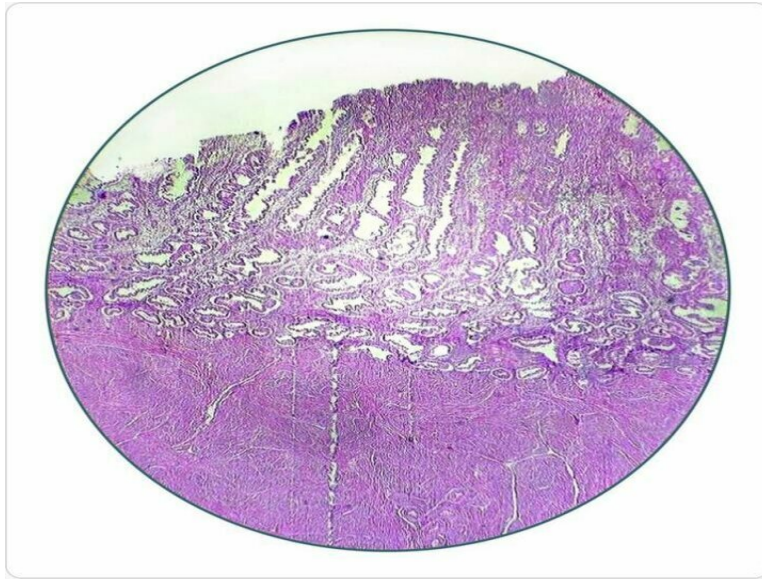
Question 14:

Not seen in decompression sickness

- a) Vision loss
- b) Hearing loss
- c) Hemodilution
- d) Coagulopathy

Question 15:

Which of the following ovarian structures is responsible for the following finding?



- a) Secondary follicle
- b) Theca interna
- c) Granulosa cell
- d) Corpus luteum

Question 16:

Which of the following does not form the inner component of photoreceptors?

- a) Mitochondria
- b) Membranous disc
- c) Golgi apparatus
- d) Endoplasmic reticulum

Question 17:

Which of the following does not stimulate catecholamine synthesis?

- a) Hyperglycemia
- b) Smoking
- c) Standing posture
- d) Exercise

Question 18:

While collecting a blood sample for serum calcium, which is easily alterable, which of the following should be prevented to measure it accurately?

- a) If a tourniquet is needed, tie and release as soon as the sample is collected.
- b) Collecting after exercise.
- c) Making a fist is not recommended.
- d) Not collecting samples from patients immobilized for long periods of time

Question 19:

Which receptor contributes to static 2 point discrimination?

- a) Merkel cell
- b) Meissner corpuscle
- c) Pacinian corpuscle
- d) All of them

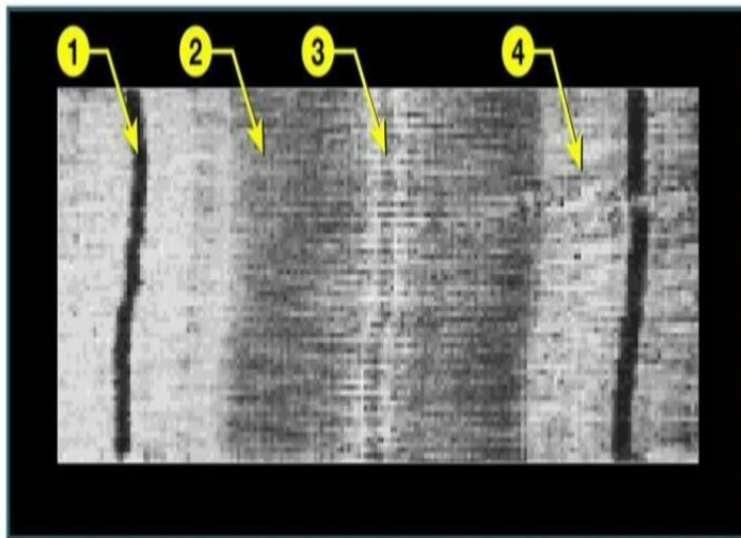
Question 20:

Mitochondrial sheaths are found in which part of the sperm?

- a) Tail
- b) Middle piece
- c) Principal piece
- d) Connecting piece

Question 21:

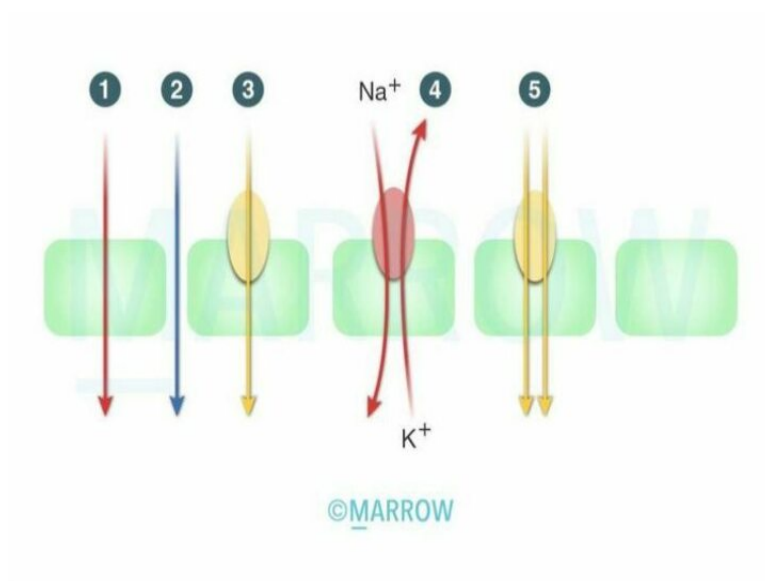
Identify the correct option



- a) 1-Z line, 2-A band, 3-H zone, 4-I band
- b) 1-A band, 2-Z line, 3-H zone, 4-I band
- c) 1-T tubule, 2-I band, 3-H zone, 4-A band
- d) 1-T tubule, 2- A band, 3-I band, 4-H zone

Question 22:

Which of these drug transport systems do not require ATP?



- a) 1,2,3
- b) 2,3,4

- c) 3,4,5
- d) 1,2,5

Question 23:

Choose the correct options from the given statements of the enteric nervous system-

- a) 1,2,3
- b) 1,4
- c) 1,3
- d) 2

Question 24:

What causes decrease in GFR?

- a) Decrease in glomerular coefficient
- b) Decrease in glomerular oncotic pressure
- c) Increase in glomerular hydrostatic pressure
- d) Decrease of Bowman's hydrostatic pressure

Question 25:

A 70-year-old man with a BP of 170/100mmHg. What is the reason for the increased pulse pressure in this patient?

- a) Decreased arterial compliance
- b) Increased cardiac output
- c) Increased sympathetic outflow
- d) Increased myocardial contractility

Question 26:

Physiological dead space :

- a) Can be measured by spirometry
- b) Is lesser than anatomical dead space

- c) Represents inadequately ventilated alveoli
- d) Represents inadequately perfused alveoli

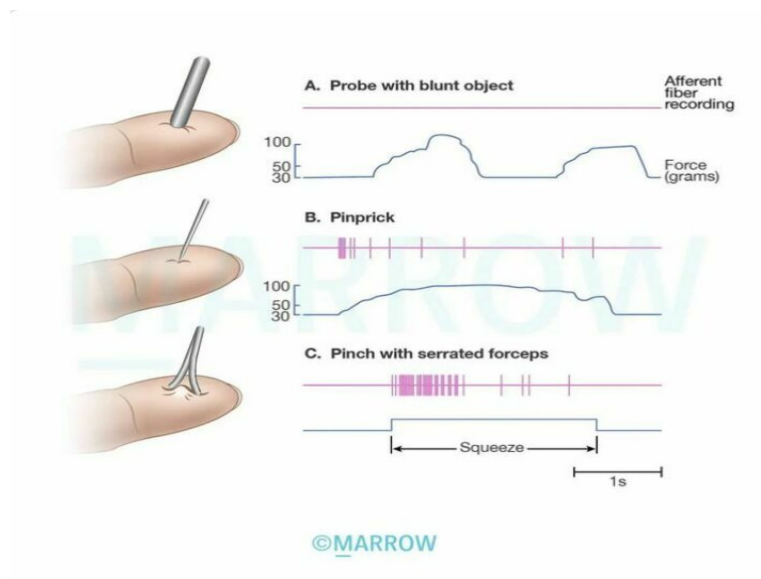
Question 27:

What contributes to the latent period of muscle twitch?

- a) 1,2,4
- b) 1,2,3
- c) 3,4
- d) 2,3

Question 28:

Mechanical nociceptors and their response to a probe with a blunt object, pinprick, and toothed forceps with serrations are recorded as an output of a force transducer coupled to the stimulator. What could be inferred from this?



- a) There are no responses from the nociceptors when stimulated by blunt forceps
- b) Pinching with toothed forceps causes stimulation of mechano-nociceptors, but no pain
- c) Pricking with a needle does not stimulate nociceptors
- d) Sharp pain sensation is carried by C fibres

Question 29:

Peripheral chemoreceptors are most sensitive to?

- a) Hypercapnia
- b) Hypoxia
- c) Acidosis
- d) Alkalosis

Question 30:

Identify the marked structure.



- a) Intercalated disc
- b) Connective tissue
- c) Cytoskeletal condensation
- d) Nerve fibre

Question 31:

Which blood component is not separated by centrifugation in the blood bank?

- a) Red blood cells
- b) Platelets
- c) Cryoprecipitate
- d) Albumin

Answer Key

Question No.	Correct Option
1	b
2	d
3	a
4	d
5	c
6	c
7	a
8	a
9	c
10	d
11	a
12	c
13	a
14	c
15	d
16	b
17	a
18	b
19	a
20	b
21	a
22	a
23	d
24	a
25	a
26	d
27	a
28	a
29	b
30	a
31	d

Detailed Explanations

Solution to Question 1:

All or none phenomenon, length tension relationship and long refractory period applies to cardiac muscles. Cardiac muscle cannot be tetanized due to long refractory period.

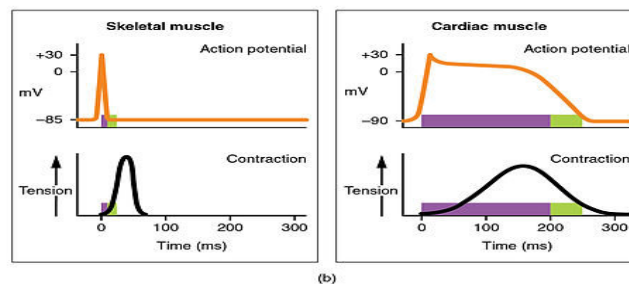
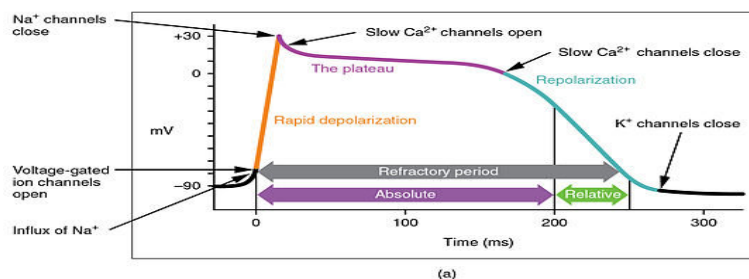
In cardiac myocytes, depolarization proceeds rapidly and an overshoot of the zero potential is present. This is followed by a plateau before the membrane potential reaches the RMP (-90mV). This plateau phase lasts for 200 msec or more. Repolarization is therefore not complete until contraction is 50% completed.

Phases of cardiac action potential

- 0 → Depolarisation and overshoot-Due to opening of voltage-gated Na^+ channels
- 1 → Initial rapid repolarization-Due to closure of Na^+ channels and opening up of K^+ channels
- 2 → Plateau phase - Due to slow but prolonged opening of Ca^{2+} channels
- 3 → Late rapid repolarization
- 4 → Baseline (RMP)

Phases 3 and 4 are due closure of Ca^{2+} channels and a slow, delayed increase of K^+ efflux through K^+ channels.

During phases 0 to 2 and about half of phase 3 of cardiac action potential, cardiac muscle cannot be excited again, that is, it is in its absolute refractory period. It remains relatively refractory until phase 4. Thus, cardiac muscle cannot be tetanized.



Solution to Question 2:

A decrease in bowman's capsule oncotic pressure would lead to a decrease in GFR.

Glomerular filtration rate (GFR) is the volume of fluid filtered from the renal glomerular capillaries into the Bowman's capsule per unit time.

The GFR is determined by sum of the hydrostatic and colloid osmotic forces across the glomerular membrane, which gives the net filtration pressure. It is dependant on Glomerular Kf (glomerular ultrafiltration coefficient) also.

Mathematically, it can be expressed as follows:

$$\text{GFR} = K_f [\text{PG} + \pi_B - \text{PB} - \pi_G]$$

Forces favoring Filtration

- PG: Glomerular hydrostatic pressure
- π_B : Bowman's capsule oncotic pressure

Forces opposing Filtration

- PB: Bowman's capsule hydrostatic pressure
- π_G : Glomerular capillary oncotic pressure

Factors that can increase the GFR are as follows:

- Increase in glomerular capillary filtration coefficient Kf: This coefficient depends on the hydraulic conductivity and surface area of the glomerular capillaries.
- Decrease in Bowman's capsule hydrostatic pressure: Conditions with urinary tract obstruction can lead to an increase in this pressure and a fall in GFR.
- Decrease in glomerular colloid osmotic pressure: Accumulation of plasma proteins that are not filtered from the blood contributes to this pressure and an increase reduces the further filtration and GFR. It is directly proportional to the following factors: Arterial plasma colloid osmotic pressure
- Filtration fraction

Solution to Question 3:

Estradiol is the most potent naturally occurring estrogen.

Classic estrogens are C18 steroid compounds containing a phenolic ring. Only three estrogens are present in significant quantities in the plasma of the human female— β -estradiol, estrone, and estriol.

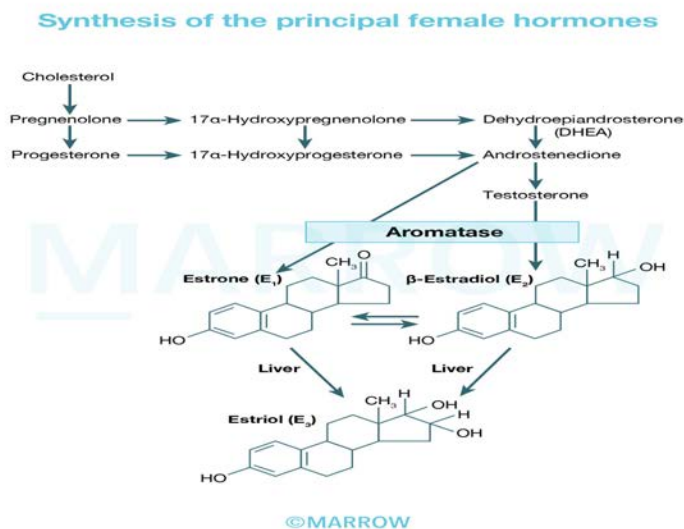
The principal estrogen secreted by the ovaries is β -estradiol.

Small amounts of estrone are also secreted, but most of this is formed in peripheral tissues from androgens secreted by the adrenal cortices and by ovarian thecal cells.

Estriol is a weak estrogen; it is an oxidative product derived from both estradiol and estrone, with the conversion occurring mainly in the liver.

The estrogenic potency of β -estradiol is 12 times that of estrone and 80 times that of estriol. Hence, estradiol is the most potent natural form of estrogen.

In the liver, estradiol, estrone, and estriol are converted to inactive glucuronide and sulfate conjugates. All these compounds, along with other metabolites, are excreted in the urine.



Solution to Question 4:

Option D is the correct option. This is a section through an adrenal gland showing both the medulla and the zones of the cortex. Following are the zones of the cortex and medulla, and the hormones they secrete.

- A – Zona Glomerulosa – secretes mineralocorticoids (aldosterone)
- B – Zona Fasciculata – secretes glucocorticoids (cortisol, corticosterone)
- C – Zona Reticularis – secretes androgens (dehydroepiandrosterone)
- D – Adrenal Medulla – Produces Catecholamines: Norepinephrine, Epinephrine, and Dopamine

On histological examination, the different layers are identified as:

- Outer zona glomerulosa - small, compact cells
- Central zona fasciculata - larger lipid-rich cells arranged in radial columns
- Inner zona reticularis - pigmented, compact cells
- Adrenal medulla - thin layer of large chromaffin cells

Solution to Question 5:

Erythropoietin is produced mainly by kidney and liver.

In adults, approximately 85% of erythropoietin is produced by the kidneys, while 15% is synthesized by the liver. Both of these organs possess the mRNA necessary for erythropoietin production. Although erythropoietin can be found in the spleen and salivary glands, these tissues

lack the mRNA required for its synthesis, indicating that they do not produce the hormone.

Erythropoietin is secreted by interstitial cells in the peri-tubular capillary bed of the kidneys and by perivenous hepatocytes in the liver. It is also generated in the brain, where it provides a safeguarding effect against excitotoxic harm caused by hypoxia; and in the uterus and oviducts, where it is stimulated by estrogen and seems to facilitate estrogen-driven angiogenesis.

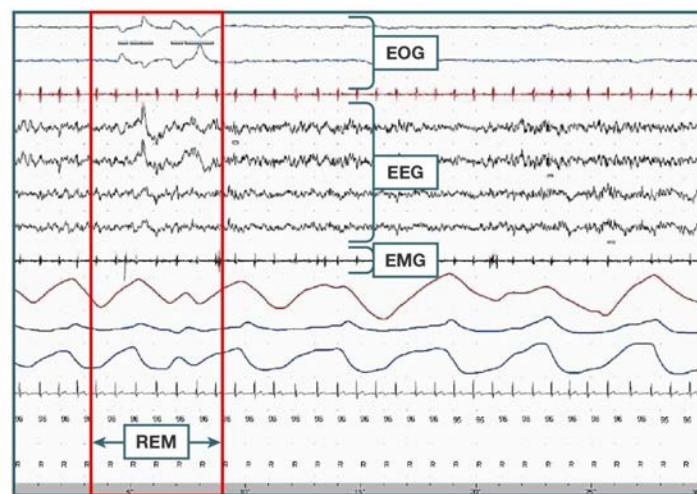
Solution to Question 6:

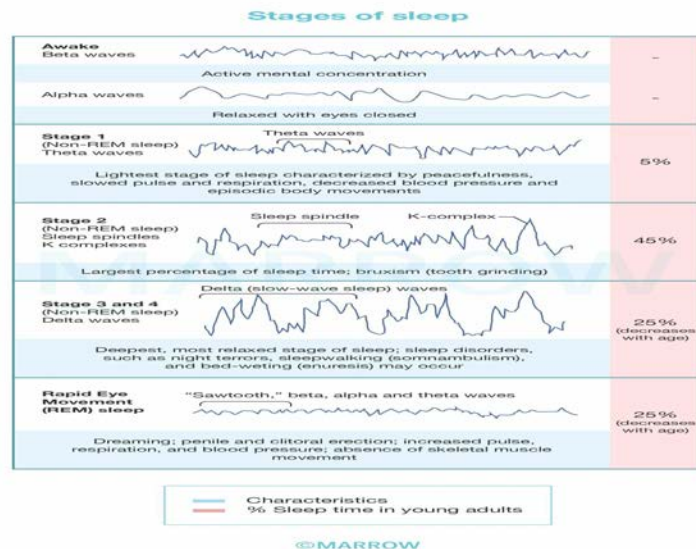
REM sleep is represented by the marked area in the given polysomnography.

The three zones in the given image represent EOG, EEG, and EMG, respectively. In the marked area,

- EOG shows Positive activity
- EEG shows Mixed waves
- EMG shows Minimal activity

This is characteristic of REM sleep.





Solution to Question 7:

Antibodies are not present since birth. At 2 to 8 months after birth, an infant begins to produce agglutinins (antibodies).

In the absence of type A agglutigen(antigen) in an individual's RBCs, antibodies called anti-A agglutinins form in the plasma. Similarly, when type B agglutigen is lacking in the RBCs, antibodies referred to as anti-B agglutinins develop in the plasma. Type A blood carries type A agglutinogens and anti-B agglutinins, while type B blood contains type B agglutinogens and anti-A agglutinins. Type AB blood possesses both A and B agglutinogens but no agglutinins. On the other hand, type O blood, which lacks agglutinogens, does have both anti-A and anti-B agglutinins.

The A and B antigens are complex oligosaccharides that differ in their terminal sugar. The agglutinins are gamma globulins produced by the bone marrow and lymph gland cells. Most of them are IgM and IgG immunoglobulin molecules.

Solution to Question 8:

In the given image, A: Somatostatin, B: GHRH, C: GH, D: IGF-1

- Growth hormone-releasing hormone (GHRH) secreted from the hypothalamus acts on the anterior pituitary to release growth hormone.
- Growth hormone (GH) increases the circulating levels of IGF -1 which exerts its effect on the tissues.
- Ghrelin promotes the secretion of growth hormone from the anterior pituitary.
- Somatostatin secreted from the hypothalamus is an inhibitory effect on the GH secretion from the anterior pituitary.

- Similarly, IGF -1 exerts negative feedback on GH secretion and positive feedback on somatostatin secretion, which in turn inhibits GH secretion.

Solution to Question 9:

The correct sequence in Gs is adenylyl cyclase » cAMP » Protein kinase A.

G protein-coupled receptors (GPCRs) are cell membrane receptors that span across the membrane 7 times. They also serve as receptors for various peptide hormones, neurotransmitters and amino acids. Subtypes of GPCRs are distinguished by their alpha subunits. The important subtypes of GPCRs are Gs, Gi, and Gq.

When hormones bind to the receptor, it enables the receptor to connect with a G protein. If the G protein activates the adenylyl cyclase-cAMP system, it is referred to as a Gs protein, indicating a stimulatory G protein. The Gs protein stimulates adenylyl cyclase, an enzyme located in the cell membrane, which then converts a small amount of cytoplasmic adenosine triphosphate into cAMP. This cAMP activates cAMP-dependent protein kinase, which phosphorylates specific proteins within the cell, initiating biochemical reactions that ultimately result in the cell's response to the hormone.

Solution to Question 10:

Option D is RBC pipette, it is not used in Sahli's method of hemoglobin estimation. The instruments used are comparator with brown glass standard (Option A), graduated hemoglobin tube (Option B), Sahli's pipette (Option C) and stirrer.

Sahli's method of estimation of hemoglobin is based on colorimetric measurement of the intensity of color developed on the addition of dilute hydrochloric acid (N/10 HCl) reagent to the blood.

In Sahli's method, hemoglobin is converted to acid hematin with the action of dilute hydrochloric acid (N/10 HCl). The brown color of the acid hematin is matched to its intensity with a standard brown glass comparator in a visual colorimeter called Sahli's colorimeter. It can be used to estimate oxyHb and reduced Hb.

Sahli's method is a simple bedside test and is cost-effective. However, it has some disadvantages, which include visual errors, the inability to measure other forms of hemoglobin like carboxyhemoglobin, methemoglobin, and sulfhemoglobin, fading of the comparator over time, and the color of acid hematin fading quickly.

The visual error can be reduced by taking the average of three readings; first, when the color is slightly darker than the standard; second, when the color exactly matches the standard; and third, when the color is slightly lighter than the standard.



Solution to Question 11:

Acetylcholine is released in response to increased blood pressure.

In response to raised BP, there is inhibition of sympathetic system and activation of parasympathetic system. Acetylcholine is the universal neurotransmitter of PNS.

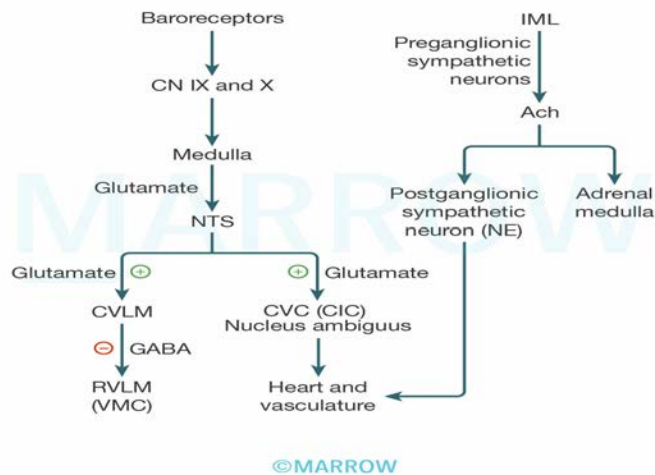
Baroreceptors are stretch receptors located in the heart and the adventitia of the blood vessels. They are stimulated by the stretching of the walls of the blood vessels when there is an increased volume or blood pressure in the vessel. They are important in the immediate management of blood pressure. They are of two types: carotid sinus and aortic arch receptors (high-pressure regions) and cardiopulmonary receptors (low-pressure regions).

An increase in blood pressure stimulates the baroreceptors. These signals are transmitted through small Hering's nerves to the glossopharyngeal nerves and then to the nucleus tractus solitarius (NTS) in the medulla. These signals release glutamate and activate the caudal ventrolateral medulla (CVLM). This in turn inhibits the rostral ventrolateral medulla (RVLM) considered the vasoconstrictor center for the brain and the heart.

Simultaneously, this pathway also excites the vagal parasympathetic center. The stimulation of the vagus results in a slowing of impulse generation in the SA node and a slowing of conduction velocity in the AV node.

These neural changes produce vasodilation, venodilation, hypotension, bradycardia, and a decrease in cardiac output.

Neural Pathway involved in regulation of BP



- NTS: Nucleus Tractus Solitarius
- CVLM: Caudal Ventrolateral Medulla
- RVLM : Rostral Ventrolateral Medulla (Vasomotor center)
- CVC : Cardiovagal center (Cardioinhibitory center)
- IML: intermediolateral nucleus

Solution to Question 12:

The correct answer is A-3, B-4, C-1, D-2.

Insulin binds to a receptor that has intrinsic tyrosine kinase activity. Upon activation, the receptor phosphorylates itself and downstream signaling proteins. (A-3)

Acetylcholine acts on nicotinic receptors, which are ligand-gated ion channels. Binding of acetylcholine opens the channel, allowing ion flow (e.g., Na⁺ or K⁺). (B-4)

Growth hormone binds to its receptor, which lacks intrinsic kinase activity but recruits Janus Kinase (JAK) to phosphorylate downstream proteins and activate the STAT (Signal Transducer and Activator of Transcription) pathway. (C-1)

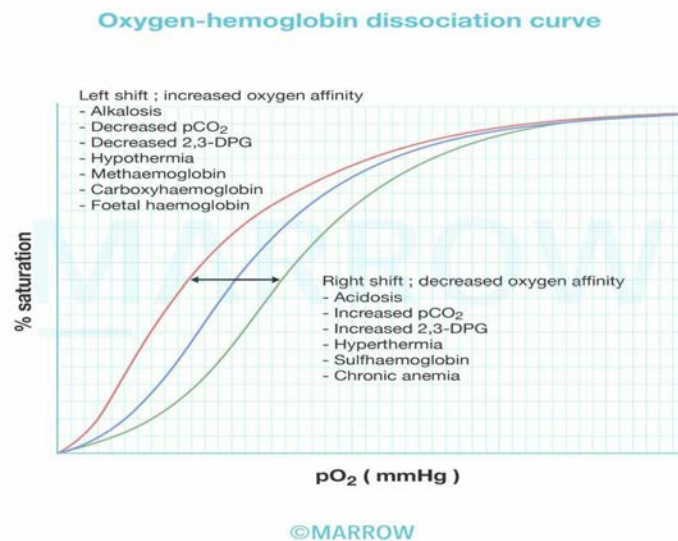
Thyroxine (and its active form, T₃) binds to intracellular nuclear receptors. These hormone-receptor complexes act as transcription factors, regulating gene expression directly in the nucleus. (D-2)

Hormone	Receptor
A. Insulin	3. Tyrosine Kinase
B. Acetylcholine	4. Ligand gated

Hormone	Receptor
C.Growth Hormone	1. JAK/STAT
D.Thyroxine	2. Nuclear Receptor

Solution to Question 13:

When stored RBCs are transfused, the oxygen-hemoglobin dissociation curve typically shifts to the left but still retains its sigmoid shape as represented by the curve marked as 2.



This is due to reduced levels of 2,3-bisphosphoglycerate (2,3-BPG) in stored red blood cells. 2,3-BPG is a metabolite in RBCs that binds to hemoglobin and reduces its affinity for oxygen.

During storage, metabolic processes in RBCs slow down, leading to a depletion of 2,3-BPG. Lower 2,3-BPG levels make hemoglobin hold onto oxygen more tightly, impairing oxygen delivery to tissues. Hence, this increases hemoglobin's affinity for oxygen, shifting the curve to the left.

Other Options:

Option B and C: show a right shift to the oxygen-hemoglobin dissociation curve. $\uparrow$ in 2,3-BPG seen in increased anaerobic glycolysis leads to the right shift of the curve. Storage of blood causes depletion of 2,3-BPG, hence leading to left shift instead of right.

Solution to Question 14:

Hemodilution is not seen in decompression sickness. Instead, hemoconcentration is seen in decompression sickness.

Hemodilution refers to the dilution of blood, usually due to excessive fluid intake or other medical conditions, but it is not a feature of decompression sickness.

Decompression sickness (DCS) is a condition that occurs when a person ascends too quickly after being exposed to high-pressure environments (such as deep-sea diving), causing nitrogen bubbles to form in the bloodstream and tissues.

Other options

Option A: Vision changes or loss can occur in DCS due to gas bubbles affecting the optic nerve or retinal circulation, leading to temporary or permanent vision impairment.

Option B: Gas bubbles in DCS can also affect the auditory system, leading to hearing loss or tinnitus, a common feature of the condition.

Option D: Decompression sickness can lead to coagulopathy, which refers to abnormal blood clotting due to the formation of nitrogen bubbles that disrupt the vascular system, potentially causing bleeding or clotting disorders.

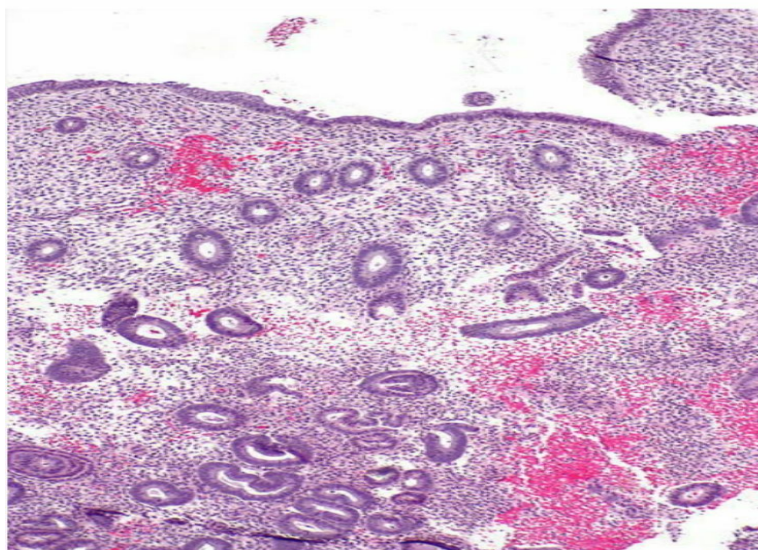
Solution to Question 15:

The ovarian structure responsible for the following finding is the corpus luteum.

The histological image shows a corkscrew appearance of the glands, and this is characteristic of the secretory phase of the endometrium in the menstrual cycle. This phase is stimulated by progesterone, which is secreted by the corpus luteum.

Endometrial Biopsy of the various stages of the ovarian cycle:

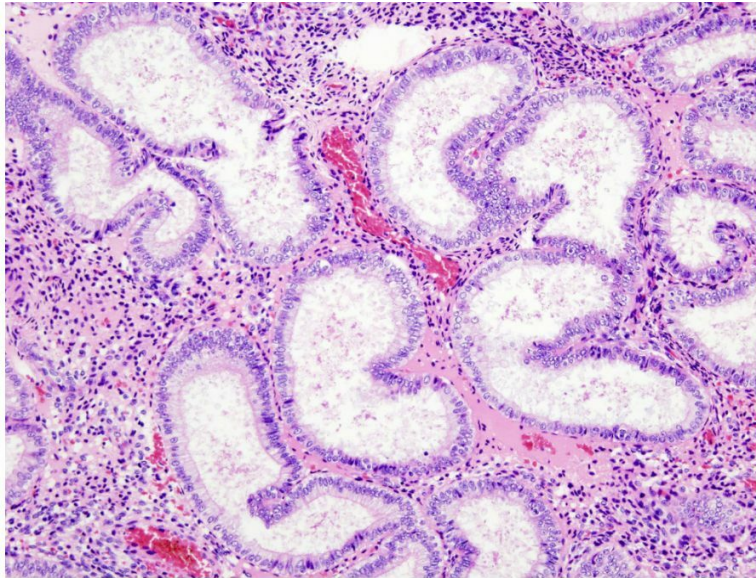
1. Proliferative phase: The glands are simple, straight, and tubular glands. The lining of the glands shows pseudostratified columnar epithelium with mitotic figures, indicating active cell proliferation. The stroma is dense and compact, with numerous cells undergoing mitosis.



2. Early secretory phase: The initially straight glands begin to coil as the endometrium transitions into the secretory phase. Subnuclear Vacuolation is seen in the epithelial cells of the endometrial

glands, caused by the accumulation of glycogen at the base of the cells.

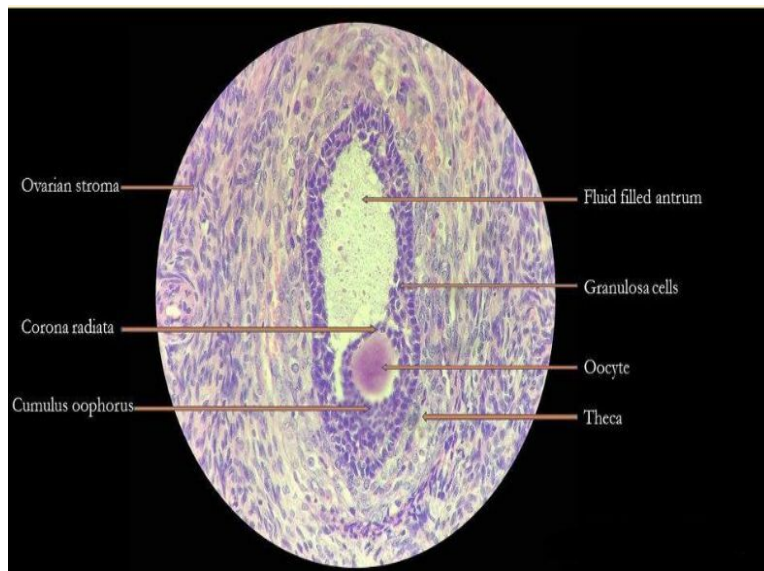
3. Late secretory phase: The glands become maximally tortuous and assume a distinctive "saw-tooth" appearance due to extreme coiling. They are filled with secretions rich in glycogen and glycoproteins, essential for supporting a potential embryo. The stroma is edematous.



Other Options:

Option A: Secondary follicle

The secondary follicle is a more advanced stage of follicular development following the primary follicle. It contains a fluid-filled cavity (antrum) surrounded by two layers: theca interna and theca externa.

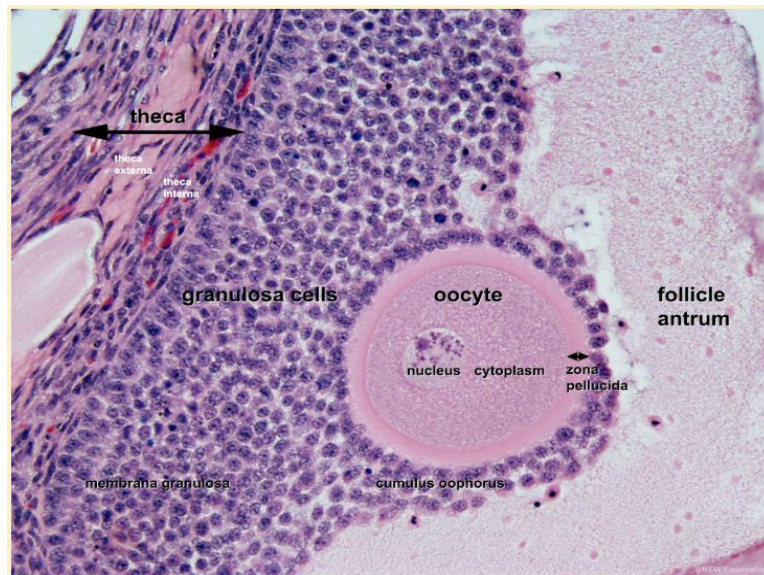


Option B and C: Theca interna and Granulosa cell

The theca interna is a layer of stromal cells just outside the basal lamina of the granulosa cells. They are composed of cuboidal endocrine cells with lipid droplets and abundant smooth

endoplasmic reticulum, reflecting its role in steroidogenesis.

Granulosa cells line the interior of the follicle. Cells are organized in layers, initially single-layered in primary follicles and multilayered in secondary follicles.



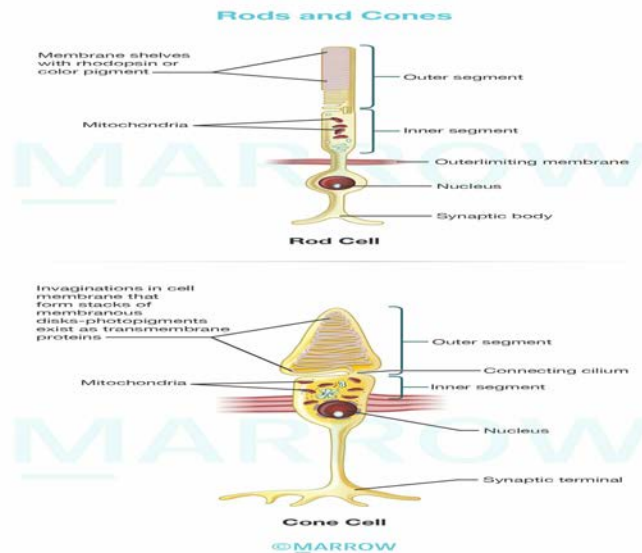
Solution to Question 16:

Among the given options, the membranous disc does not form the inner component of photoreceptors but instead forms the outer component.

The other three options i.e., mitochondria, Golgi apparatus, and endoplasmic reticulum form the inner component of photoreceptors.

Photoreceptors (rods and cones) are specialized neurons in the retina with distinct structural components:

- Outer segment: Contains membranous discs that have photopigments (like rhodopsin in rods and photopsins in cones) essential for light detection.
- Inner segment: Contains cellular organelles responsible for metabolism and protein synthesis, such as mitochondria, Golgi apparatus, and endoplasmic reticulum.



Solution to Question 17:

Hyperglycemia does not stimulate catecholamine synthesis. Instead, hypoglycemia stimulates catecholamine synthesis.

Catecholamine synthesis is primarily stimulated by various physiological stressors and factors that activate the sympathetic nervous system, such as physical exertion(exercise), changes in posture, or certain external stimuli like smoking.

Other options:

Option B: Smoking stimulates catecholamine release through the activation of the sympathetic nervous system, which increases catecholamines like adrenaline and norepinephrine.

Option C: When a person stands up, it can lead to orthostatic hypotension, which triggers the release of catecholamines to help increase blood pressure and maintain adequate perfusion.

Option D: Exercise stimulates the sympathetic nervous system and increases catecholamine production to support increased heart rate, blood pressure, and metabolic demands.

Solution to Question 18:

Blood collection after exercise should be prevented to measure serum calcium accurately.

Option A: A prolonged tourniquet is usually avoided while drawing blood for calcium measurements since hemoconcentration also leads to an increase in serum calcium levels. Hence, the tourniquet should be released as soon as the sample is collected.

Option C: Fist-clenching or forearm exercise can lead to falsely elevated ionized (free) calcium levels. Hence, making a fist while blood collection is not recommended.

Option D: Staying immobilised for long periods of time increases bone resorption and increases calcium levels. Hence, do not collect blood samples for serum calcium estimation in patients who are immobilised for long periods.

Solution to Question 19:

Merkel cell contributes to static 2 point discrimination.

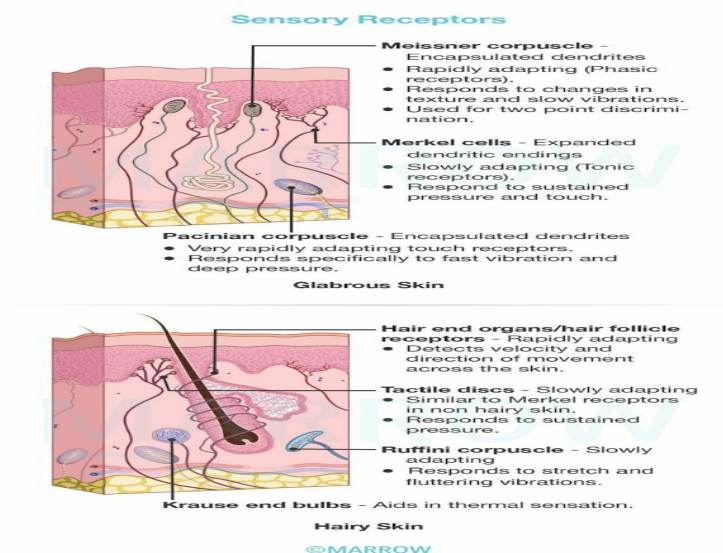
Merkel cells are responsible for static two-point discrimination. These cells are specialized for detecting pressure and texture and are involved in the detection of fine touch. They provide information about the spatial resolution of tactile stimuli, which is essential for static two-point discrimination, where the ability to differentiate between two distinct points of touch is tested.

Merkel cells are expanded dendritic endings in the epidermis of glabrous (non-hairy) skin. They respond to sustained pressure and touch. They are slowly adapting receptors.

Static vs. Dynamic Two-Point Discrimination

Static: Measures the ability to distinguish stationary points; mediated by Merkel cells (slow-adapting receptors).

Dynamic: Measures the ability to distinguish moving points; mediated by Meissner corpuscles (fast-adapting receptors).



Other options:

Option B: Meissner corpuscle: Meissner corpuscles are dendrites encapsulated in the connective tissue beneath the epidermis of glabrous (non-hairy) skin. They respond to tap and flutter (low-frequency vibration - 5-40Hz). They are rapidly adapting receptors.

Option C: Pacinian corpuscle: Pacinian corpuscles are the largest cutaneous mechanoreceptor. They are present in the dermis of glabrous (non-hairy) and hairy skin. A Pacinian corpuscle consists of a nerve ending encapsulated by concentric layers of connective tissue, giving it an onion-like appearance. They respond to fast vibration (60-500Hz) and deep pressure. They are

rapidly adapting receptors.

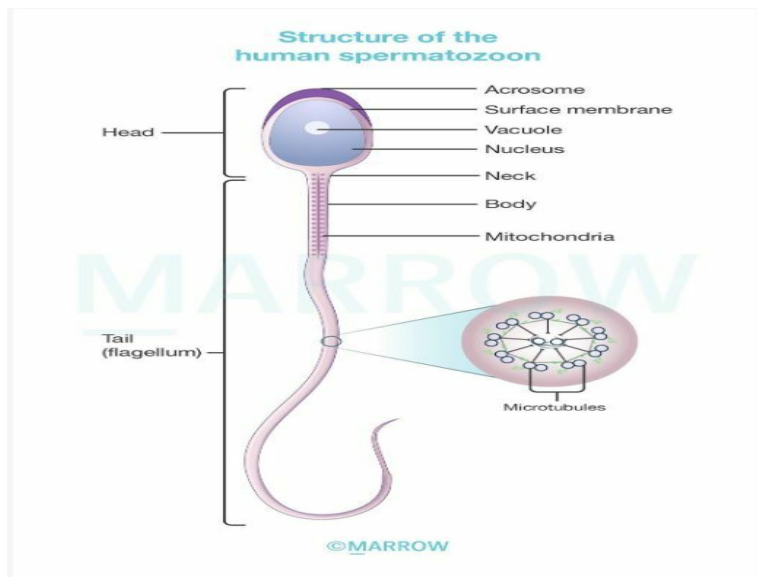
Solution to Question 20:

Mitochondrial sheaths are found in the middle piece of the sperm.

The middle piece is the segment of the sperm located just behind the head and neck.

It contains a spiral arrangement of mitochondria, forming the mitochondrial sheath around the central axoneme.

These mitochondria generate ATP, which provides the energy required for the flagellar movement of the sperm, enabling motility.



Other options:

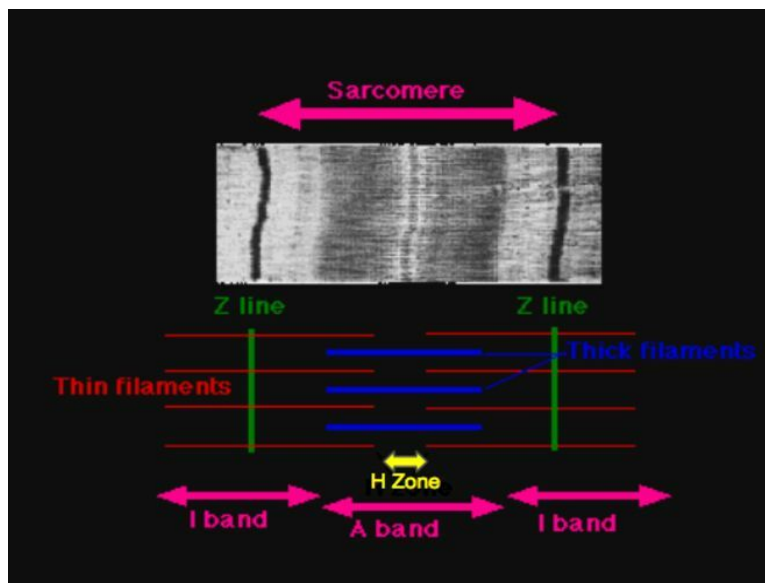
Option A: Tail: Comprises the middle piece, principal piece, and end piece and is primarily responsible for motility.

Option C: Principle piece: Longest part of the tail which contains fibrous sheaths around the axoneme but lacks mitochondria.

Option D: Connecting piece: Links the head of the sperm to the tail (neck region). Does not contain mitochondrial sheaths.

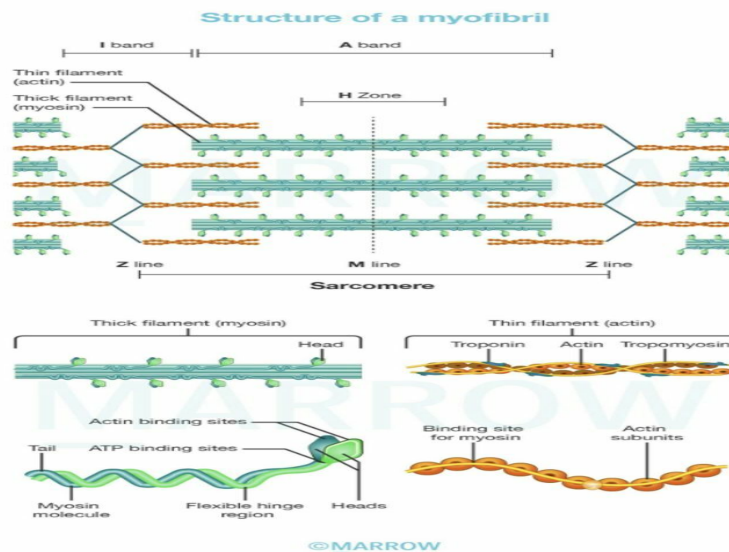
Solution to Question 21:

The correct marking is 1-Z line, 2-A band, 3-H zone, 4-I band.



H zone or H band denotes the less refractile region in the centre of A band.

- A band – Anisotropic / Dark band: Formed by thick filaments - Myosin
- I band – Isotropic / Light band: Formed by thin filaments – Actin, Troponin, Tropomyosin
- M line – Seen in the centre of H band: Due to central bulge of A band
- Z line – Seen in the centre of each I band: Narrow line with high refractile material appearing dark.



Solution to Question 22:

Paracellular transport, Simple Diffusion and Facilitated diffusion do not require ATP as they are all passive transport mechanisms.

- 1: Simple diffusion : Simple diffusion is a passive transport mechanism where molecules (water, carbon dioxide, nitrogen) move from an area of high concentration to an area of low concentration and is not carrier mediated.
- 2: Paracellular : Paracellular transport involves movement of molecules through tight junctions between cells without the need for energy (ATP).
- 3: Facilitated diffusion : This also occurs along a concentration gradient, but involves carrier proteins to help move the drug across the membrane. No ATP is used because the process is still passive (driven by the concentration gradient).
- 4: Antiport (Na/K ATPase): Antiport transport system actively moves ions (Na⁺, K⁺) against their concentration gradient using ATP in the opposite direction. It is a form of primary active transport.
- 5: Symport : Two substances are transported in the same direction across a membrane. ATP is required to move substances against their gradients, often via a secondary active transport mechanism.

Solution to Question 23:

The enteric nervous system functions independently of the brain and spinal Cord. (Statement 2)

Statement 2: Functions independent of brain and spinal cord: The enteric nervous system is often referred to as the "second brain" because it can function independently of the central nervous system (CNS). It has its own reflex circuits and can control gut motility, secretion, and blood flow without input from the brain or spinal cord, although it can be influenced by them.

Statement 1: The vagus is the only afferent from the gut: While the vagus nerve does provide significant afferent (sensory) input from the gut, other nerves, such as those from the pelvic splanchnic nerves, also contribute to afferent pathways. Therefore, the vagus is not the only afferent nerve.

Statement 3: ■Myenteric plexus in the lamina propria: The myenteric plexus is located between the circular and longitudinal muscle layers of the gut, not in the lamina propria (which is part of the mucosal layer of the gut).

Statement 4: Exclusively under the influence of nerves: The enteric nervous system is influenced by nerves, but it also responds to hormonal and immune system signals. Therefore, it is not exclusively under neural influence.

Statement 5: Only parasympathetic connects enteric to CNS: Both parasympathetic and sympathetic connect the enteric nervous system to the central nervous system. Not just the parasympathetic.

The enteric nervous system consists of submucosal and myenteric plexi. This Plexi can function independently and is often referred to as the 'little brain'. The enteric nervous system regulates gastrointestinal (GI) function and is connected to the central nervous system by parasympathetic and sympathetic fibers.

The myenteric (Auerbach's) plexus is present between the longitudinal and circular muscle layers. It innervates both these layers along the entire length of the GI tract. It primarily controls GI motility. On stimulation, it has the following effects:

- Increased tone of the GI wall
- Increased intensity and rate of rhythmic contractions
- Increased velocity of excitatory wave conduction

Some of the myenteric plexus neurons primarily release vasoactive inhibitory peptide (VIP), which causes the relaxation of the GI sphincters.

The submucosal (Meissner's) plexus is found between the mucosa and the circular muscle layers. It innervates the GI glands, endocrine cells, and submucosal blood vessels. It primarily controls GI secretions, absorption, and local blood flow.

The parasympathetic innervation of the plexi is from the vagus and pelvic nerves. They release either acetylcholine or substance P. It generally increases GI motility and secretion.

The sympathetic innervation is from the celiac, superior mesenteric, inferior mesenteric, and hypogastric sympathetic ganglia. It generally inhibits GI motility and stimulates sphincteric contraction.

Solution to Question 24:

Decrease in glomerular coefficient causes decrease in GFR.

The glomerular filtration coefficient (Kf) is a measure of how efficiently the glomerulus filters blood into the kidney tubules. It depends on two key factors:

- Surface Area for Filtration: Greater surface area means higher filtration.
- Permeability of the Glomerular Membrane: Higher permeability leads to greater filtration.

Glomerular filtration rate (GFR) is the volume of fluid filtered from the renal glomerular capillaries into the Bowman's capsule per unit time.

The GFR is determined by the sum of the hydrostatic and colloid osmotic forces across the glomerular membrane, which gives the net filtration pressure. It is dependent on Glomerular Kf (glomerular ultrafiltration coefficient) also.

Mathematically, it can be expressed as follows:

$$\text{GFR} = K_f [\text{PG} + \pi_B - \text{PB} - \pi_G]$$

Forces favoring Filtration

- PG: Glomerular hydrostatic pressure (Option C)
- π_B : Bowman's capsule oncotic pressure

Forces opposing Filtration

- PB: Bowman's capsule hydrostatic pressure (Option D)
- π_G : Glomerular capillary oncotic pressure (Option B)

The formula for glomerular filtration rate (GFR) incorporates Kf:

- A high Kf means efficient filtration (e.g., in healthy kidneys)-indicates high GFR

- A low K_f {occurs when the surface area is reduced (e.g., due to glomerular damage) or the membrane becomes less permeable (e.g., in diabetic nephropathy)}- indicates low GFR.

K_f is essential for maintaining proper kidney function, as it directly affects how well the kidneys filter blood to form urine.

Solution to Question 25:

Pulse pressure increases with age due to decreased arterial wall compliance.

Diastolic blood pressure falls after 60 years of age due to increased arterial stiffness and decreased compliance, often associated with arteriosclerosis. As the left ventricle contracts against stiffer, less compliant arteries, systolic blood pressure also increases.

As a result of these two changes, pulse pressure increases ($\uparrow PP = \uparrow SBP - \downarrow DBP$)

The factors that affect the pulse pressure:

- the stroke volume output of the heart
- the compliance or total distensibility of the arterial walls
- the character of ejection from the heart during systole (less important)

Pulse pressure = Stroke volume/ arterial compliance

Widened pulse pressure (> 40 mm Hg) occurs with several diseases such as:

- Aortic regurgitation
- Aortic sclerosis
- Severe iron deficiency anaemia
- Arteriosclerosis
- Hyperthyroidism

In the majority of these cases, systolic pressures increase while diastolic pressures remain near normal.

Solution to Question 26:

Physiological dead space represents inadequately perfused alveoli.

Physiological dead space is the part of the respiratory system where no gas exchange occurs. It includes:

- Anatomic dead space: Air in the conducting airways (nose, trachea, bronchi) that doesn't reach the alveoli.
- Alveolar dead space: Air in alveoli that are ventilated but not perfused by blood.

Together, it represents the total volume of air that doesn't participate in oxygen and carbon dioxide exchange.(Option B)

Anatomical dead space refers to the volume of all the space of the respiratory system other than the alveoli that do not participate in gas exchange. The anatomical dead space is not altered with a change in ventilation.

In healthy individuals:

- Anatomical dead space = 150 mL.
- Alveolar dead space is usually zero (increased in diseased conditions).

Alveolar Dead space represents inadequately perfused alveoli (wasted ventilation). (Option C)

In normal individuals, physiological dead space remains unaltered and is equal to the anatomical dead space (volume of all the space of the respiratory system other than the alveoli that do not participate in gas exchange).

Dead space cannot be measured using spirometry. (Option A)

Solution to Question 27:

Neuromuscular transmission, Excitation contraction coupling and transmission impulse from nerve coupling contribute to latent period of muscle twitch.

The latent period of a muscle twitch is the brief delay between the application of a stimulus (such as a nerve impulse) and the onset of muscle contraction. During this period, several physiological processes occur, but neuromuscular transmission and excitation-contraction coupling primarily contribute to this delay.

Neuromuscular transmission: The action potential travels down the motor neuron to the neuromuscular junction. Acetylcholine (ACh) is released into the synaptic cleft and binds to receptors on the muscle fiber's sarcolemma, triggering an action potential in the muscle fiber. (Statement 1)

Excitation-contraction coupling: The action potential propagates along the sarcolemma and into the transverse (T) tubules. This triggers the release of calcium ions from the sarcoplasmic reticulum into the cytosol. Calcium binds to troponin, initiating the interaction of actin and myosin filaments. (Statement 2)

Reuptake of calcium back into the sarcoplasmic reticulum: This occurs during the relaxation phase, not the latent period. (Statement 3)

Transmission of impulse from nerve: This is a part of neuromuscular transmission and is involved in the latent period. (Statement 4)

Solution to Question 28:

There are no responses from the nociceptors when stimulated by blunt forceps.

Mechanical nociceptors typically respond to noxious mechanical stimuli, such as sharp pressure or intense stretch. A blunt object, such as blunt forceps, may not provide a strong enough stimulus to activate these nociceptors, leading to no response.

The sharp painful sensation felt immediately upon injury is known as fast pain. The poorly localized dull aching pain is characteristic of slow pain. The nerve fibers carrying fast pain and slow pain are A δ and C respectively.

Nociceptors are innervated by both A δ and C fibers. A δ fibers are myelinated and larger in diameter with a faster conduction velocity. C fibers are unmyelinated and conduct impulses at a slower rate. When nociceptors are activated by a noxious stimulus, activated A δ fibers release glutamate and mediate fast pain. It is a rapid response that is responsible for localizing the noxious stimulus.

Other Options:

Option B: Pinching with toothed forceps causes stimulation of mechanonociceptors, but no pain: Pinching with toothed forceps typically applies more localized pressure, which can activate mechanical nociceptors. However, it may still cause a painful sensation, not just a mechanical response. It is more likely that pain is perceived, even if the response is primarily mechanical.

Option C: Pricking with a needle does not stimulate nociceptors: A needle prick is a sharp stimulus that activates nociceptors, particularly mechanical nociceptors in response to tissue damage.

Option D: Sharp pain sensation is carried by C fibers: Sharp pain (due to a needle prick) is typically carried by A-delta fibers, which transmit fast, localized pain sensations. C fibers are responsible for carrying dull, burning pain that occurs later after the initial sharp pain.

Solution to Question 29:

Peripheral chemoreceptors are most sensitive to hypoxia.

Peripheral Chemoreceptors are located in the carotid bodies (at the bifurcation of the common carotid arteries) and aortic bodies (near the aortic arch). They are sensitive to changes in pO₂, pCO₂, and pH of the blood. In response to changes in these parameters, they send impulses to the medullary respiratory centers and modify the duration and frequency of respiratory cycles.

Peripheral chemoreceptors are most sensitive to a drop in partial pressure of oxygen (PaO₂) in arterial blood.

Mechanism:

Type I glomus cells in the carotid and aortic bodies have O₂-sensitive K⁺ channels, which are inactivated in response to hypoxia. This reduces the K⁺ efflux, depolarizing the cell and causing Ca²⁺ influx, via L-type voltage-gated calcium channels. The Ca²⁺ influx triggers action potentials and consequent excitation of the afferent nerve endings in the peripheral chemoreceptors.

Hypoxia (PaO₂ < 60 mmHg) strongly stimulates these receptors, leading to an increase in respiratory rate and depth to improve oxygenation.

Other Options:

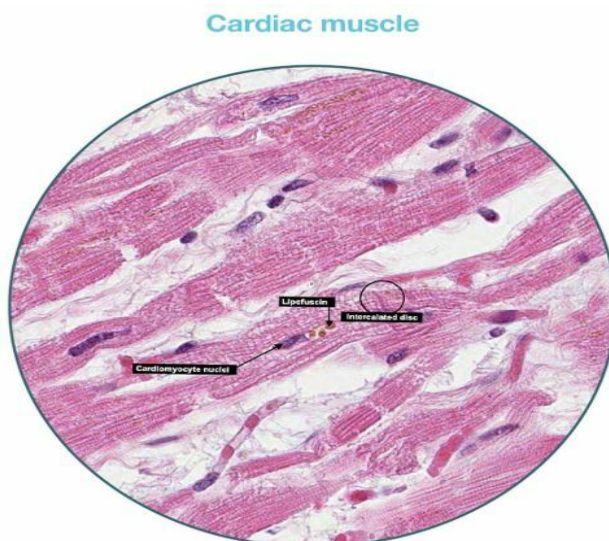
Peripheral chemoreceptors also respond to hypercapnia (Option A) and acidosis (Option C). However, these are primarily detected by central chemoreceptors, located in the medulla, which monitor CO₂ levels in cerebrospinal fluid.

Alkalosis (Option D) causes a weaker response from peripheral chemoreceptors compared to hypoxia.

Solution to Question 30:

The given section shows cardiac muscle and the marked structure is an intercalated disc.

Intercalated discs are specialized structures unique to cardiac muscle tissue. They are critical for the coordinated contraction of cardiac muscle cells. They are found at the junctions between adjacent cardiac muscle cells. They appear as dark, zigzagging lines under a microscope. They facilitate the rhythmic and coordinated contraction of the heart by ensuring efficient electrical and mechanical connections.



On microscopy, cardiac muscle can be identified by:

- Presence of striations similar to the skeletal muscle.
- Presence of single and centrally placed nucleus. A perinuclear halo may be seen surrounding the nucleus.
- The junctions between adjoining cardiac myocytes are seen as dark staining transverse lines running across the muscle fibre. These lines are called intercalated discs.
- Cells within one fibre branch and join the cells in adjacent fibres.

Other Options:

Option B: Connective Tissue: Found surrounding the cardiac muscle cells, providing structural support, but not at the cell junctions.

Option C: Cytoskeletal condensation: Refers to cytoskeletal organization within a cell and is not visible as distinct structures in cardiac muscle histology.

Option D: Nerve Fibre: Present in cardiac tissue to innervate the heart but appears as separate structures and not within cardiac muscle fibers.

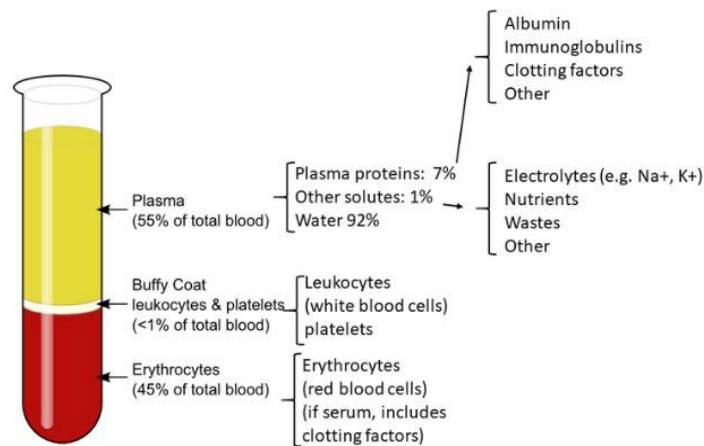
Solution to Question 31:

Albumin is not separated by centrifugation in the blood bank.

In blood banks, centrifugation is used to separate the main components of blood into distinct layers for storage and transfusion purposes. These components include:

- Red Blood Cells (RBCs): Separated by centrifugation and stored for patients with anemia or significant blood loss. (Option A)
- Platelets: Isolated through centrifugation of plasma-rich blood for use in patients with bleeding disorders. (Option B)
- Cryoprecipitate: Derived from plasma by controlled freezing and thawing, which allows the collection of clotting factors (e.g., Factor VIII, fibrinogen, von Willebrand factor). (Option C)

Albumin is a plasma protein that is extracted through a process called fractionation, not simple centrifugation. This involves the use of chemical, temperature, and pH adjustments to isolate albumin from plasma.



Physiology NEET 2024

Question 1:

What are the effects of posterior lesions in a Brodmann's area 22 in a stroke patient?

- a) Auditory function
- b) Receptive Speech Function
- c) Expressive Speech Function
- d) Visual functions

Question 2:

A 30-year-old female went on vacation and fell asleep while sitting on a beach. After some time she woke with sunburn. Later when she went back home, she experienced pain while taking a warm bath (40°C). Which of the following receptors is responsible for the pain?

- a) Innocuous thermal receptor - allodynia
- b) Innocuous thermal receptor - hyperalgesia
- c) Thermal receptor - allodynia
- d) Thermal receptor - hyperalgesia

Question 3:

To prevent polyspermy, which reaction occurs in the zona pellucida of the oocyte after fertilization?

- a) Acrosomal reaction
- b) Increased membrane potential of the oocyte
- c) Cortical reaction
- d) Decreased membrane potential of the oocyte

Question 4:

A 60-year-old patient came with a puncture wound on the left chest wall. Radiological investigations confirmed pneumothorax. Which of the following is correct regarding this

condition?

- a) Lung volume increases, thoracic volume increases
- b) Lung volume decreases, thoracic volume increases
- c) Lung volume increases, thoracic volume decreases
- d) lung volume decreases, thoracic volume decreases

Question 5:

Sour taste is mediated by which of the following receptors?

- a) T1R1 GPCR
- b) TRPV1 channel
- c) T1R3 GPCR
- d) Metabolic receptor

Question 6:

A patient who underwent right limb amputation 2 years back, now presents with severe pain in the limb (Phantom limb pain). On radiological investigations, What is the finding seen?

- a) Projection of adjacent fibres to overlap the right somatosensory cortex
- b) Expansion of left somatosensory cortex
- c) Projection of adjacent fibres to overlap the left somatosensory cortex
- d) Expansion of right somatosensory cortex

Question 7:

In a type 2 diabetes mellitus patient, serum insulin levels are found to be raised. He is advised to exercise to improve insulin resistance. Which of the following GLUT will help to improve insulin resistance after exercise ?

- a) GLUT 1
- b) GLUT 2
- c) GLUT 5
- d) GLUT 4

Question 8:

Which of the following is incorrect about iron absorption?

- a) Ferrireductase convert Fe^{2+} to Fe^{3+}
- b) RBC protophyrin is increased in deficiency
- c) Iron excretion is only through skin, gut and urothelial cells
- d) Body cannot compensate if more than 10 RBC/HPF per day is lost in urine

Question 9:

A 50-year-old patient presents with a chronic tingling sensation in the forearm. A neurological examination is done and appears to be normal. Which of the following receptors are correctly matched?

- a) Merkel cells-slow vibration
- b) Meissner's corpuscles-stretch
- c) Pacinian corpuscles-fast vibration
- d) Ruffini corpuscles-sustained pressure

Question 10:

The patient presented with alcoholic gait, and abnormal eye movements (nystagmus) after RTA. Which lobe of the cerebellum is likely affected?

- a) Flocculonodular Lobe
- b) Dentate nucleus
- c) Anterior lobe
- d) Interposed nucleus

Question 11:

Renal compensation in metabolic acidosis takes time because:

- a) Initially the pH of tubular fluid increases
- b) Low pH normally decreases bicarbonate excretion
- c) It takes hours to days for kidney to synthesize enzymes that produce NH_4^+
- d) Other ions in the tubule interfere with buffer action

Question 12:

The functioning of the thyroid involves the transport of iodine from the blood to the thyroid follicle. Iodine is transported from the follicle to the colloid. Which pump or channel is for transport of iodine from the follicle to the colloid?

- a) Sodium iodine symporter
- b) Calmodulin
- c) Claudin
- d) Pendrin

Question 13:

The resting membrane potential(RMP) of the nerve is -70mV. Increased permeability of which ion is responsible for the RMP?

- a) Calcium
- b) Sodium
- c) Potassium
- d) Chloride

Question 14:

Arrange the sequence of steps involved at neuromuscular junction endplate potential.

- a) Acetylcholine release - Calcium entry- Na⁺ entry
- b) Calcium entry- Acetylcholine release- Na⁺ entry
- c) Na⁺ entry - Calcium entry - Acetylcholine release
- d) Na⁺ entry - Acetylcholine release - Calcium entry

Answer Key

Question No.	Correct Option
1	b

2	a
3	c
4	b
5	b
6	c
7	d
8	a
9	c
10	a
11	c
12	d
13	c
14	b

Detailed Explanations

Solution to Question 1:

A posterior lesion in Brodmann's area 22, which is located in the superior temporal gyrus, typically affects receptive speech function. It is the sensory speech area.

The patient will most likely have Wernicke's aphasia due to the damage to Wernicke's area - area 22.

In Wernicke's aphasia, there is difficulty in finding words leading to poorly understood content for the listener. These patients also find it difficult to repeat sentences and name things properly. It is a type of fluent aphasia in which the speech is easy and fluent but there are difficulties related to the output of language like paraphasia (substitution of similar-sounding, non-words, or incorrect words). The speech is full of jargon and neologisms (non-existent words) that make little sense.

Other options:

Option A: Auditory Function is primarily mediated by Brodmann area 41, which is located in the primary auditory cortex within the superior temporal gyrus.

Option C: Expressive Speech Function is associated with Brodmann areas 44 and 45 (Broca's area), which are involved in language production and speech articulation. Lesions here can cause difficulty in producing coherent speech (Broca's aphasia).

Option D: Visual function primarily associated with Brodmann areas 17, 18, and 19 (the primary and secondary visual cortices).

Solution to Question 2:

The receptors responsible for this are innocuous thermal receptors which have perceived ordinarily non-noxious stimulus (i.e., warm bath) as severe pain also known as allodynia.

In this case, the warm bath at 40°C (typically not painful) causes pain due to sunburn, which sensitizes the skin. After sunburn, the skin becomes more sensitive, and thermal receptors that usually detect warmth now trigger pain signals in response to non-painful thermal stimuli, leading to allodynia.

Innocuous thermoreceptors, which detect non-harmful temperature changes (cold and warm), are specialized nerve endings found on the skin. Cold receptors are located on the dendritic endings of A δ and C fibers, while warm receptors are situated on C fibers. The specific thermosensitive molecular channels mediating innocuous heat stimuli is transient receptor potential (TRP) cation channel TRPV1 and TRPM8 cation channel for innocuous cold stimuli.

Thermal hyperalgesia (increased response to noxious stimulation) is mediated by activation of NMDA receptors. After tissue or nerve injury, C-fibers discharges release glutamate and neurokinins from primary afferent terminals, leading to the release of peptides (e.g., substance P; calcitonin gene-related peptide, CGRP). As a result, a voltage-dependent magnesium block on NMDA receptors is removed allowing for an influx of calcium that triggers a cascade of intracellular events leading to the development and maintenance of hyperalgesia. (Option D)

Solution to Question 3:

To prevent polyspermy, the reaction in the zona pellucida of the oocyte after fertilisation is the cortical reaction.

The chronological steps of fertilisation:

Acrosomal reaction: After a sperm reaches the oocyte and binds to the zona pellucida, the acrosome reaction occurs. The acrosome, an enzyme-rich organelle in the anterior of the sperm head, fuses at many points with the plasma membrane of the oocyte. Then those fused membranes are sloughed off from the head, exposing the acrosomal enzymes. The enzymes digest a path through the zona pellucida, enabling the sperm to advance.

Eventually, the sperm meets and fuses with the secondary oocyte plasma membrane.

This contact results in weak membrane depolarisation (decreased membrane potential) and a calcium wave.

Cortical reaction: The calcium wave triggers the release of lysosomal enzymes from the cortical granules lining the oolemma. These enzymes hydrolyse the ZP3 receptor on the zona pellucida. Digestion of ZP3 and alteration of zona pellucida (zona reaction) prevents other sperms from binding, thus creating a block to polyspermy.

Solution to Question 4:

In pneumothorax, lung volume is decreased and thoracic volume is increased.

Both lungs and the chest wall (thoracic cage) are elastic structures. Under normal conditions, the elastic tendency of the lungs to collapse is exactly balanced by the elastic tendency of the chest

wall to expand.

In pneumothorax, air is introduced into the pleural space, so the pleural pressure, which is normally subatmospheric, becomes equal to atmospheric pressure. The chest wall springs outward and the lungs collapse. This increases the thoracic volume and decreases the lung volume.

Solution to Question 5:

Sour taste is mediated by the TRPV1 channel. The TRPV1 (Transient Receptor Potential Vanilloid 1) channel is involved in detecting the acidity of foods, which contributes to the perception of sour taste.

It is a cation channel that allows the influx of various cations, including sodium (Na^+), calcium (Ca^{2+}), and magnesium (Mg^{2+}), in response to changes in temperature and the presence of acidic conditions or capsaicin

Taste Receptors

- Humans have five established basic tastes: sweet, salt, sour, bitter and umami.
- The taste receptors for these five basic tastes are of two major types:
- Ligand-gated channels (ionotropic receptors): for salt and sour taste
- G-Protein Coupled Receptors (metabotropic receptors): for sweet, bitter and umami
- Salt and sour taste are mediated through epithelial sodium channels (ENaC)
- Na^+ movement through ENaC $\rightarrow$ salt taste
- H^+ movement through ENaC $\rightarrow$ sour taste
- Sweet, bitter, and umami are sensed via GPCRs
- T1R2 and T1R3 families $\rightarrow$ sweet taste
- T2R family $\rightarrow$ bitter taste
- T1R1, T1R3, and Metabotropic glutamate receptor (mGluR4) $\rightarrow$ Umami (Option A and C)

Solution to Question 6:

In the given scenario of phantom limb pain, on the radiological investigation of the brain projection of adjacent fibres to overlap the left somatosensory cortex can be seen. This pain is based on cortical plasticity.

Cortical structures are capable of reorganization after the loss of a body part. The cortical representation of the neighbouring parts spreads into the cortical area that was formerly occupied by the representation of the amputated part. This is called cortical plasticity leads to the phenomenon of phantom limb pain. The sensory cortical area occupied by the right limb is the left somatosensory cortex.

Cortical plasticity / Phantom limb: It is the sensation that an amputated or missing limb is still attached. Plasticity is the ability of the brain to adapt to such a condition by reorganization of thalamic and cortical structures.

The converse also holds true: if the cortical area representing a limb is removed, the somatosensory map of the limb moves to the surrounding cortex.

Diagnosis and Management :

- PET scan is used to document these plastic changes in cortex.
- Use of epidural analgesia during amputation surgery decreases incidence of phantom pain.
- Spinal cord stimulation also helps in reducing phantom pain. (interferes with ascending impulses to brain)

Solution to Question 7:

There is an increase in the number of GLUT 4 receptors in response to exercise. This increases the insulin dependent glucose uptake into the cells, contributing to a decrease in insulin resistance.

GLUT-4 is an insulin-dependent GLUT. Other GLUTs are not insulin-sensitive and are constitutively expressed in the cell membrane.

Insulin resistance is a condition in which the cells in muscles, fat, and liver are less responsive to insulin and have difficulty absorbing glucose from the blood. The pancreas then produces more insulin to facilitate the entry of glucose into the cells.

An arbitrary but clinically useful benchmark considers patients insulin resistant if they require greater than 1 unit/kilogram/day of exogenous insulin to maintain glycemic control. Patients requiring greater than 200 units of exogenous insulin per day are considered severely insulin resistant.

Other options

Option A: GLUT 1 receptors are present in the red blood cell, brain, kidney, colon, retina, and placenta. It is responsible for glucose uptake.

Option B: GLUT-2 is found in the liver, pancreatic β cell, small intestine, and kidney. Its function is the rapid uptake or release of glucose.

Option C: GLUT-5 is present in the small intestine and is involved in the absorption of fructose.

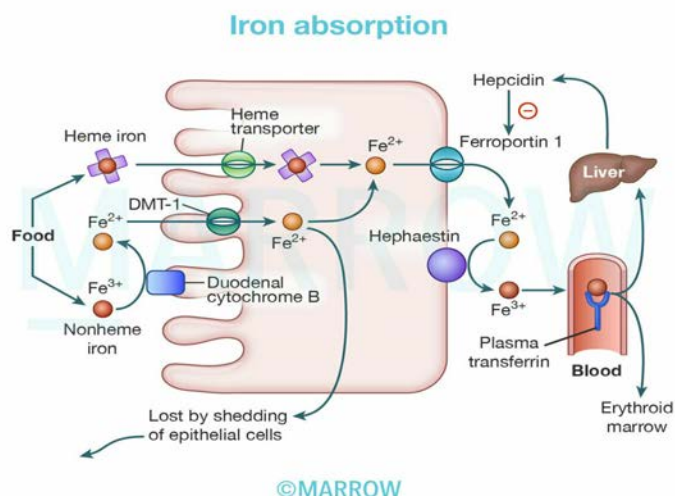
Solution to Question 8:

Ferrireductase is an enzyme that converts Fe^{3+} (ferric iron) to Fe^{2+} (ferrous iron), not the other way around. Ferrous is the form of iron that is more readily absorbed in the gut.

Almost all the iron absorption occurs in the duodenum.

Most of the iron in the diet is in the ferric (Fe^{3+}) form, whereas it is the ferrous (Fe^{2+}) form that is absorbed.

Body iron stores are controlled by intestinal absorption as there is no exit to iron from the body. Divalent metal transporter 1 (DMT1) transports non-heme iron into the enterocyte. Iron absorption in the duodenum is regulated by hepcidin release from the liver in response to increases in intrahepatic iron levels. It binds to ferroportin leading to endocytosis and degradation within the enterocyte. This prevents the transfer of iron from enterocyte to plasma resulting in iron trapping within duodenal cells in the form of mucosal ferritin. The trapped iron is lost as these cells are sloughed. The iron transported by ferroportin into the plasma is transferred to other parts with the help of transferrin and stored as serum ferritin and hemosiderin. The image depicting iron absorption and storage is given below:



Other options:

Option B: RBC protoporphyrin is increased in deficiency, increased protoporphyrin levels in red blood cells can indicate iron deficiency or disorders of heme synthesis.

Option C: Iron is primarily excreted through the gut, skin, and urinary tract, but the body generally has very limited mechanisms for iron excretion compared to iron absorption and recycling.

Option D: The body's iron homeostasis is finely tuned, and a significant loss of red blood cells, such as more than 10 ml per day, can overwhelm the body's ability to compensate and maintain iron balance.

Solution to Question 9:

Pacian corpuscles are responsible for sensing fast vibration. The chronic tingling sensation is mediated by pressure-sensitive A β fibers and low-conduction velocity C fibers.

Pacian corpuscles are the most sensitive mechanoreceptors in the somatosensory system. They are responsive to high-frequency (30–500 Hz) vibratory stimuli and deep pressure. They are also examples of rapidly adapting receptors.

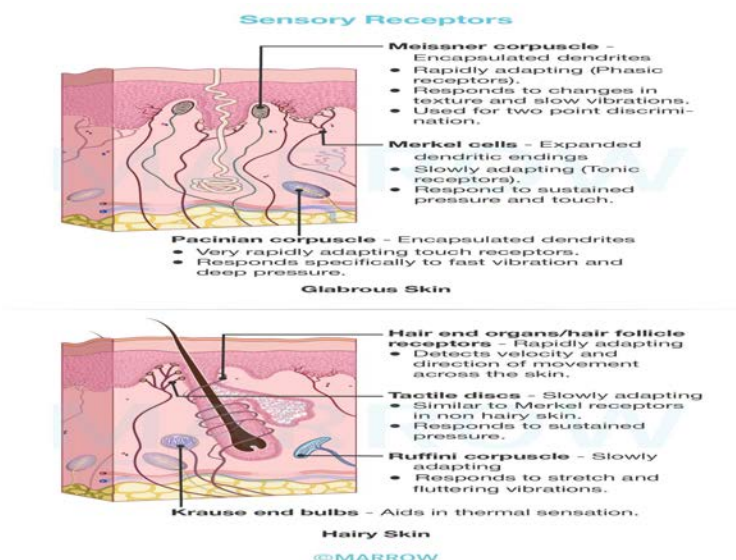
Chronic tingling causing conditions include Diabetes, carpal tunnel syndrome, ulnar nerve palsy, radial nerve palsy, tarsal tunnel syndrome, sciatica, vitamin B1,B6,B12 deficiencies, and alcoholism.

Other options

Option A: The Merkel cells are expanded dendritic endings in the epidermis of glabrous skin and respond to sustained pressure and touch.

Option B: Meissner's corpuscle senses tap and flutter. They are present in dermal papillae. They are rapidly adapting receptors.

Option D: Ruffini's corpuscles are enlarged dendritic endings with elongated capsules in the dermis of glabrous and hairy skin. They respond to stretch and fluttering vibrations. They are slowly adapting receptors.



Solution to Question 10:

The flocculonodular lobe is involved in this patient with an alcoholic gait and abnormal eye movements (nystagmus) following an RTA.

The cerebellum has three parts based on function.

- **Vestibulocerebellum** (or flocculonodular lobe): The oldest part of the cerebellum which contains nodulus in the vermis and the flanking flocculus. It has vestibular connections and is concerned with equilibrium and gait. Loss of it can cause disturbance of equilibrium and postural movements, especially during rapid movements.
- **Spinocerebellum**: The region that receives proprioceptive input from the body as well as a copy of the “motor plan” from the motor cortex. By comparing the plan with performance, it smooths and coordinates movements that are ongoing. The superior vermis is a part of the spinocerebellum. Loss of it can cause problems with coordination.
- **Cerebrocerebellum**: They are the newest from a phylogenetic point of view, reaching their greatest development in humans. They interact with the motor cortex in planning and

programming movements.

Other options:

Option B: The Dentate nucleus (largest) is one of the 4 deep cerebellar nuclei others being, Emboliform, Globose, and Fastigial nuclei. These receive input from climbing fibers and are the only motor output from the cerebellar cortex.

Option C: Anterior lobe is a part of the Paleo/Spinocerebellum responsible for the coordination of movement and regulation of muscle tone whose lesion results in hypotonia.

Option D: Interposed nuclei / Nucleus interpositus is the term used to describe the cerebellar nuclei, Emboliform, and Globose.

Solution to Question 11:

Compensation in metabolic acidosis happens late in the kidney because it takes hours to days for it to synthesize enzymes that produce NH_4^+ (ammonium ions).

Acid-base balance equation: $\text{CO}_2 + \text{H}_2\text{O} \rightleftharpoons \text{H}_2\text{CO}_3 \rightleftharpoons \text{H}^+$ (acid) and HCO_3^- (base)

- First line of defence (in seconds): Buffers (phosphate and ammonium)
- Second line of defence (in Minutes): Lungs
- Third line of defence (in hours - days): Kidney

The kidneys regulate the acid-base balance by two major mechanisms: the reabsorption of bicarbonate and the generation of new bicarbonate via ammonia production and distal nephron acid secretion. Urinary NH_4^+ is produced primarily in the proximal tubule cells by converting glutamine to glutamate.

Renal failure leads to a buildup of acids due to loss of excretory function of the kidney. The decreased glomerular filtration rate also leads to diminished excretion of the renal buffers phosphate and ammonium. Thus, bicarbonate reabsorption decreases, resulting in metabolic acidosis.

Solution to Question 12:

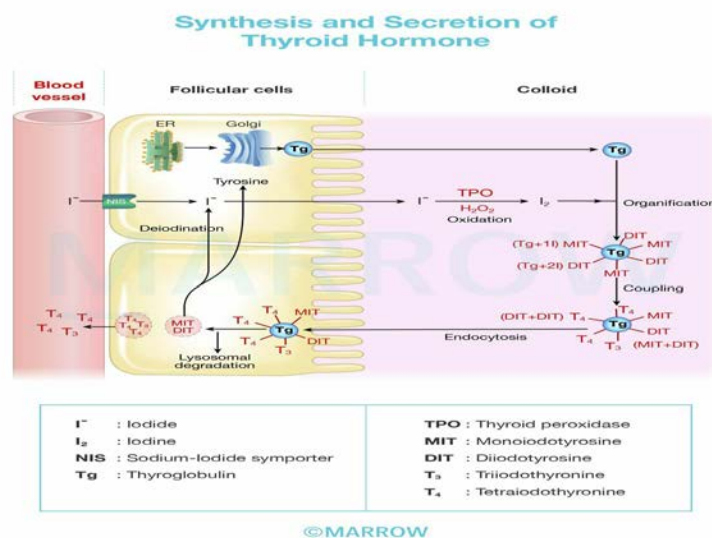
The channel needed for the transport of iodine from the follicle to the colloid is Pendrin.

Steps in synthesis and secretion of thyroid hormone:

- Iodide uptake: The basolateral sodium-iodide symporter (NIS) actively takes up I^- (iodide) from the circulation into the follicular cells. It is then transported into the colloid through a pendrin (chloride-iodide) transporter
- Thyroglobulin synthesis: Thyroglobulin (Tg), is a large glycoprotein rich in tyrosine residues. It is synthesized by follicular cells.
- Iodination of Tg: This is regulated by the thyroid peroxidase (TPO) enzyme. TPO has three functions:

- Oxidation: TPO oxidizes I⁻ to I₂ (iodine) in the presence of H₂O₂
- Organification: TPO links tyrosine residues of thyroglobulin with iodine (I₂) to form monoiodotyrosine (MIT), diiodotyrosine (DIT)
- Coupling reactions: TPO combines iodinated tyrosine residues to make MIT + DIT = T₃ (triiodothyronine), DIT + DIT = T₄ (tetraiodothyronine)
- Storage: Thyroid hormones are bound to thyroglobulin and stored in the follicular lumen as colloids. The colloid has hormone reserves for 2-3 months
- Release: Iodinated thyroglobulin is taken up by the thyroid follicles via endocytosis and fuses with lysosomes. It is cleaved by lysosomal proteolytic enzymes into MIT, DIT, T₃, and T₄. T₄ (80%) and T₃ (20%) are released into the blood
- Peripheral T₃ conversion: T₄ undergoes deiodination in extrathyroidal tissue to form T₃, which is the active form of the hormone.

Pendred syndrome: Pendrin is present in the thyroid gland and inner ear. The mutation causes goiter and sensorineural hearing loss.



Other options

Option A: Sodium-iodide symporter (NIS) is responsible for transporting iodine from the bloodstream into the thyroid follicular cell, not into the colloid. It works by bringing in iodine along with sodium from the bloodstream into the follicle.

Option B: Calmodulin is structurally and functionally related to troponin. In place of troponin, smooth muscle cells contain a large amount of the regulatory protein calmodulin; it is different in the manner in which it initiates contraction. Calmodulin initiates contraction by activating the myosin light chain kinase.

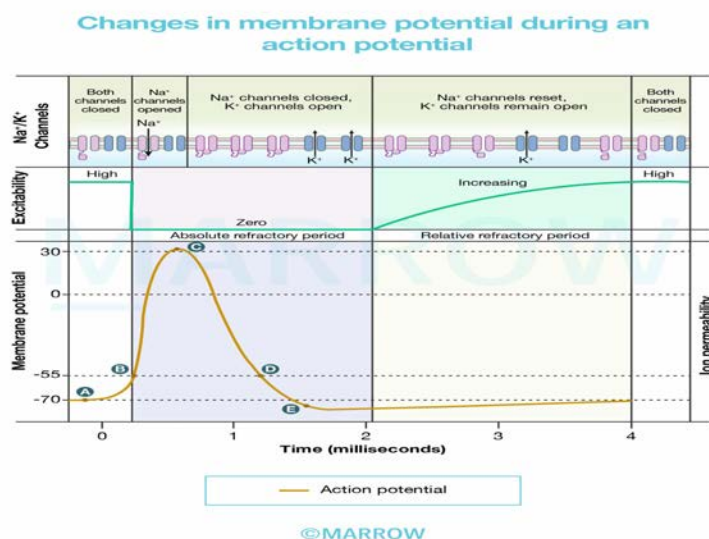
Option C: Claudin is a transmembrane protein involved in the formation of tight junctions.

Solution to Question 13:

Increased permeability of potassium ions (K^+) is responsible for the RMP of a nerve as there are more K^+ channels open at rest, facilitating the flow of potassium ions across the membrane.

Resting membrane potential (RMP) is the difference in the electrical potentials between the cytoplasm of the cell and the extracellular space.

At rest, there are more open K^+ channels compared to Na^+ channels, thus the membrane permeability of K^+ is greater. As a result of this, the intracellular and extracellular K^+ concentrations are the major determinants of the RMP. Furthermore, the RMP is close to the equilibrium potential of K^+ in general.



Clinical Significance: In hyperkalemia, the extracellular level of K^+ is increased. Thus the RMP moves closer to the threshold to elicit an action potential leading to a more excitable nerve. The excitable nerve, stimulates the muscle repeatedly causing fasciculations, exhaustion of muscle, and weakness. In hypokalemia, the extracellular level of K^+ is reduced. Thus the RMP is reduced and the neuron is hyperpolarized.

Solution to Question 14:

The sequence of steps involved at neuromuscular junction endplate potential is Calcium entry- Acetylcholine release- Na^+ entry.

Events occurring at the Chemical Synapse:

The action potential arriving at the end of the motor neuron increases the permeability of its endings to Ca^{2+}

↓

Ca^{2+} enters the endings and triggers a marked increase in exocytosis of the acetylcholine-containing synaptic vesicles

↓

The binding of acetylcholine to nicotinic receptors increases the Na^+ and K^+ conductance

↓

The resultant influx of Na^+ produces a local depolarizing potential, the endplate potential

↓

The current created by this local potential depolarizes the adjacent muscle membrane to its firing level

↓

Action potentials are generated on either side of the endplate and are conducted away from the endplate in both directions along the muscle fiber. This muscle action potential, in turn, initiates muscle contraction.

Acetylcholine is then removed from the synaptic cleft by acetylcholinesterase, which is present in high concentration at the neuromuscular junction.

